Recapture Analsyis: Final

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Contents

Recapture Analysis: Combined

In this markdown, we'll combine the results from `COLONY` and `PLINK2` to identify recaptures across the samples processed by RAPiD Genomics / LGC and GTSeek. We'll also estimate naive genotyping error rates, assess kinship coefficients between technical replicates, evaluate concordance in sex inferences from genotypic data, assign clone IDs for all samples, and assign final sexes for all unique genotypes. We also incorporate one additional filter in this markdown, removing individuals with excessively high heterozygosity (*i.e.*, $F_{IS} < -0.5$), which may be indicative of sample contamination.

The outputs from this step will be used subsequently to create a capture history object for SCR modeling and to subsample to one sample per unique genotype for population genetic analyses.

The strategy used to combine inferences from `COLONY` and `PLINK2` is as follows:

1. Start with `COLONY` clones as recaptures - most (if not all) of these are also recovered by `PLINK`
2. For each unique `COLONY` clone, ask if `PLINK` provides compelling evidence to move the clone to a larger group
    - Mean kinship across pairwise comparisons of clone A and clone B needs to be > 0.45 (based on the range of kinship coefficients seen in technical replicates)
    - At least one pairwise comparison between the clones needs to be based on a substantial number of shared markers (> 50)

Load COLONY and PLINK2 Results

First, load the files with results from COLONY and PLINK2.

```
#Read the metadata
smd <- read.csv("../sampleinfo/sampleMetadata_sexGT.csv",
                            header = TRUE)

#Read the COLONY output
col.out <- read.table("CGA.BestClone", header = TRUE)

#Read the PLINK output
kingrel <- read.table("recapPLINK.kin0", header = FALSE)
colnames(kingrel) <- c("id1","id2","nsnp","hehet","ibs0","kinship")

nrow(kingrel) #Number of rows (pairwise comparisons) in PLINK output...
```

```
## [1] 1365378
```

```
choose(sum(!smd$FILT),2) # should match choose(# retained samples, 2)
```

```
## [1] 1365378
```

Evaluate Technical Replicates - Error Rate, Kinship

First pull the technical replicate samples out of the larger data file. Start by extracting genotypes (in 012 format from the combined file that we used for the recapture analysis.

```
ulimit -n 3000
vcftools --vcf ../snpfiltering/finalSCR.recode.vcf --012 --out finalSCR
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf ../snpfiltering/finalSCR.recode.vcf
    --012
    --out finalSCR

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
```

```
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
After filtering, kept 1653 out of 1653 Individuals
Writing 012 matrix files ... Done.
After filtering, kept 925 out of a possible 925 Sites
Run Time = 1.00 seconds
```

Now read this into R and subset to the technical replicate samples

```r
reps <- smd[which(smd$is.rep == TRUE),]
length(unique(sub("R$","",reps$ugaID)))
```

```
## [1] 60
```

```r
nrow(reps) #60 replicated samples = 120 rows
```

```
## [1] 120
```

```r
genos <- read.table("finalSCR.012", header = FALSE)
genos.ind <- read.table("finalSCR.012.indv", header = FALSE)
genos[,1] <- genos.ind[,1]

repl.out <- data.frame(ugaID = unique(sub("R$","",reps$ugaID)),
                       vcfID1=NA, vcfID2=NA,
                       nSNPgt1=NA, nSNPgt2=NA,
                       filt1=NA, filt2=NA,
                       colclone1=NA, colclone2=NA,
                       is.clonePLINK=NA, is.cloneCOLONY=NA,
                       nSNPshared=NA, nSNPmatch=NA,
                       kinship=NA, GTservice=NA)
for(i in 1:nrow(repl.out)) {
  repl.out$GTservice[i] <- reps$GTservice[which(reps$ugaID == repl.out$ugaID[i])][1]

  #Replicates use the same UGA ID in LGC data...
  if(repl.out$GTservice[i] == "LGC") {
    #Record vcfIDs
    repl.out$vcfID1[i] <- reps$vcfID[which(reps$ugaID==repl.out$ugaID[i])][1]
    repl.out$vcfID2[i] <- reps$vcfID[which(reps$ugaID==repl.out$ugaID[i])][2]
  #But append an "R" in GTSeek data
  } else if(repl.out$GTservice[i] == "GTSeek") {
    repl.out$vcfID1[i] <- reps$vcfID[which(reps$ugaID==repl.out$ugaID[i])]
    repl.out$vcfID2[i] <- reps$vcfID[which(reps$ugaID==paste0(repl.out$ugaID[i],"R"))]
  }


  #Record number of SNPs genotyped
  repl.out$nSNPgt1[i] <- smd$nSNPgt[which(smd$vcfID == repl.out$vcfID1[i])]
  repl.out$nSNPgt2[i] <- smd$nSNPgt[which(smd$vcfID == repl.out$vcfID2[i])]
```

```r
#Record filtered status - if TRUE, no need to look for matches
repl.out$filt1[i] <- smd$FILT[which(smd$vcfID == repl.out$vcfID1[i])]
repl.out$filt2[i] <- smd$FILT[which(smd$vcfID == repl.out$vcfID2[i])]

if(repl.out$GTservice[i] == "LGC") {
  #Record clone IDs
  if(repl.out$filt1[i] == FALSE) {
    repl.out$colclone1[i] <- col.out$CloneIndex[grep(repl.out$vcfID1[i],
                                                     col.out[,3], fixed=TRUE)]
  }
  if(repl.out$filt2[i] == FALSE) {
    repl.out$colclone2[i] <- col.out$CloneIndex[grep(repl.out$vcfID2[i],
                                                     col.out[,3], fixed = TRUE)]
  }
} else if(repl.out$GTservice[i] == "GTSeek") { #Need to account for the overlap in vcfIDs for reps
  if(repl.out$filt2[i] == FALSE) {
    repl.out$colclone2[i] <- col.out$CloneIndex[grep(repl.out$vcfID2[i],
                                                     col.out[,3], fixed = TRUE)]
  }
  if(repl.out$filt1[i] == FALSE) {
    tmp <- col.out$CloneIndex[grep(repl.out$vcfID1[i],
                                   col.out[,3], fixed = TRUE)]
    if(length(tmp) > 1) {
      repl.out$colclone1[i] <- tmp[which(!tmp %in% repl.out$colclone2[i])]
      rm(tmp)
    } else {
      repl.out$colclone1[i] <- tmp
      rm(tmp)
    }
  }
}

if(repl.out$filt1[i] == repl.out$filt2[i] & repl.out$filt1[i] == FALSE) {
  #Record kinship estimate
  repl.out$kinship[i] <- kingrel$kinship[kingrel$id1 %in% c(repl.out$vcfID1[i],
                                                            repl.out$vcfID2[i]) &
                                         kingrel$id2 %in% c(repl.out$vcfID1[i],
                                                            repl.out$vcfID2[i])]

  repl.out$nSNPshared[i] <- kingrel$nsnp[kingrel$id1 %in% c(repl.out$vcfID1[i],
                                                          repl.out$vcfID2[i]) &
                                       kingrel$id2 %in% c(repl.out$vcfID1[i],
                                                          repl.out$vcfID2[i])]

  #Check if identified as recapture via PLINK
  if(!is.na(repl.out$kinship[i])) {
    if(repl.out$kinship[i] > 0.354) {
      repl.out$is.clonePLINK[i] <- TRUE
    } else {
      repl.out$is.clonePLINK[i] <- FALSE
    }
  } else {
    repl.out$is.clonePLINK[i] <- NA
```

```r
    }

    #Check if identified as recapture via COLONY
    if(repl.out$colclone1[i] == repl.out$colclone2[i]) {
      repl.out$is.cloneCOLONY[i] <- TRUE
    } else {
      repl.out$is.cloneCOLONY[i] <- FALSE
    }

    tmp.geno <- genos[genos$V1 %in% c(repl.out$vcfID1[i], repl.out$vcfID2[i]),-1]
    ovrlp <- which(colSums(tmp.geno == -1) == 0)
    if(length(ovrlp) > 0) {
      #If as.matrix(), even single SNP overlaps are OK!
      tmp.geno <- as.matrix(tmp.geno[,ovrlp])
      repl.out$nSNPmatch[i] <- sum(tmp.geno[1,] == tmp.geno[2,])
    } else {
      repl.out$nSNPmatch[i] <- NA
    }

    rm(tmp.geno, ovrlp)
  }

}

#Some high-level summaries for the pairs of replicated samples
sum(is.na(repl.out$is.clonePLINK))
```

```
## [1] 17
```

```r
sum(is.na(repl.out$is.cloneCOLONY))
```

```
## [1] 17
```

```r
sum(is.na(repl.out$is.clonePLINK) & is.na(repl.out$is.cloneCOLONY))
```

```
## [1] 17
```

```r
repl.out[is.na(repl.out$is.clonePLINK),]
```

```
##      ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1
## 2   121-23 UGA_173101_P015_WB11 UGA_173101_P016_WA12       1      72  TRUE
## 7   306-23 UGA_173101_P015_WH09 UGA_173101_P015_WH12       0       0  TRUE
## 14    14-C UGA_173101_P018_WA02 UGA_173101_P018_WG06     904       1 FALSE
## 17  280-23 UGA_173101_P018_WD05 UGA_173101_P019_WC10      22      19  TRUE
## 21  134-24 UGA_173101_P019_WE07 UGA_173101_P019_WE09       0     849  TRUE
## 26  434-24 UGA_173101_P019_WG10 UGA_173101_P019_WH01       1      46  TRUE
## 37  517-24           GTS_517-24          GTS_517-24R      NA      NA  TRUE
## 38  716-23           GTS_716-23          GTS_716-23R      NA      NA  TRUE
## 39  726-24           GTS_726-24          GTS_726-24R      NA      NA  TRUE
## 46 1384-24          GTS_1384-24         GTS_1384-24R      NA      NA  TRUE
## 52 2926-24          GTS_2926-24         GTS_2926-24R      NA      NA  TRUE
```

```
## 54  396-24           GTS_396-24          GTS_396-24R     113       NA FALSE
## 55  424-23           GTS_424-23          GTS_424-23R      NA       NA  TRUE
## 56  191-24           GTS_191-24          GTS_191-24R      NA       NA  TRUE
## 57  466-23           GTS_466-23          GTS_466-23R      NA       NA  TRUE
## 58  286-23           GTS_286-23          GTS_286-23R       3       NA  TRUE
## 60  311-23           GTS_311-23          GTS_311-23R      69       NA  TRUE
##    filt2 colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 2   TRUE        NA        NA            NA             NA         NA        NA
## 7   TRUE        NA        NA            NA             NA         NA        NA
## 14  TRUE        16        NA            NA             NA         NA        NA
## 17  TRUE        NA        NA            NA             NA         NA        NA
## 21 FALSE        NA        32            NA             NA         NA        NA
## 26  TRUE        NA        NA            NA             NA         NA        NA
## 37  TRUE        NA        NA            NA             NA         NA        NA
## 38  TRUE        NA        NA            NA             NA         NA        NA
## 39  TRUE        NA        NA            NA             NA         NA        NA
## 46  TRUE        NA        NA            NA             NA         NA        NA
## 52  TRUE        NA        NA            NA             NA         NA        NA
## 54  TRUE       257        NA            NA             NA         NA        NA
## 55  TRUE        NA        NA            NA             NA         NA        NA
## 56  TRUE        NA        NA            NA             NA         NA        NA
## 57  TRUE        NA        NA            NA             NA         NA        NA
## 58  TRUE        NA        NA            NA             NA         NA        NA
## 60  TRUE        NA        NA            NA             NA         NA        NA
##    kinship GTservice
## 2       NA       LGC
## 7       NA       LGC
## 14      NA       LGC
## 17      NA       LGC
## 21      NA       LGC
## 26      NA       LGC
## 37      NA    GTSeek
## 38      NA    GTSeek
## 39      NA    GTSeek
## 46      NA    GTSeek
## 52      NA    GTSeek
## 54      NA    GTSeek
## 55      NA    GTSeek
## 56      NA    GTSeek
## 57      NA    GTSeek
## 58      NA    GTSeek
## 60      NA    GTSeek
```

```r
#NAs come from cases where one or both members of a pair is filtered from the dataset

#PLINK recognizes samples as clones?
table(repl.out$is.clonePLINK)
```

```
##
## FALSE  TRUE
##     1    42
```

```r
repl.out[which(repl.out$is.clonePLINK == FALSE),]
```

```
##    ugaID                vcfID1                vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1 35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     557     754 FALSE FALSE
##   colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 1        72       114         FALSE          FALSE        489       371
##    kinship GTservice
## 1 0.307325       LGC
```

```r
#All but 35-SGA

#COLONY recognizes samples as clones?
table(repl.out$is.cloneCOLONY)
```

```
##
## FALSE  TRUE
##     4    39
```

```r
repl.out[which(repl.out$is.cloneCOLONY == FALSE),]
```

```
##     ugaID                vcfID1                vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1  35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     557     754 FALSE FALSE
## 29 227-23 UGA_173101_P021_WH06 UGA_173101_P023_WC12     896     894 FALSE FALSE
## 41 935-23           GTS_935-23           GTS_935-23R     123     123 FALSE FALSE
## 53 151-24           GTS_151-24           GTS_151-24R     122     118 FALSE FALSE
##    colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 1         72       114         FALSE          FALSE        489       371
## 29        14       185          TRUE          FALSE        880       763
## 41       255       289          TRUE          FALSE        122       122
## 53       255        33          TRUE          FALSE        117       117
##     kinship GTservice
## 1  0.307325       LGC
## 29 0.425949       LGC
## 41 0.500000    GTSeek
## 53 0.500000    GTSeek
```

```r
#35-SGA, 227-23, 935-23, and 151-24

#How often do the programs agree??
table(repl.out$is.clonePLINK[which(repl.out$is.cloneCOLONY == TRUE)])
```

```
##
## TRUE
##   39
```

```r
table(repl.out$is.cloneCOLONY[which(repl.out$is.clonePLINK == TRUE)])
```

```
##
## FALSE  TRUE
##     3    39
```

```
repl.out[which(repl.out$is.cloneCOLONY == FALSE &
                 repl.out$is.clonePLINK == TRUE),]
```

```
##     ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 29 227-23 UGA_173101_P021_WH06 UGA_173101_P023_WC12     896     894 FALSE FALSE
## 41 935-23           GTS_935-23          GTS_935-23R     123     123 FALSE FALSE
## 53 151-24           GTS_151-24          GTS_151-24R     122     118 FALSE FALSE
##    colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 29        14       185          TRUE          FALSE        880       763
## 41       255       289          TRUE          FALSE        122       122
## 53       255        33          TRUE          FALSE        117       117
##     kinship GTservice
## 29 0.425949       LGC
## 41 0.500000    GTSeek
## 53 0.500000    GTSeek
```

So all `COLONY` matches in this set are confirmed by `PLINK`, but `PLINK` finds a handful of matches that COLONY does not. Obviously, all (or almost all) of the matches in this set are real. These come from duplicated samples. It's somewhat worrisome that `COLONY` does not match pairs of samples with matching genotypes at 117 of 117 and 122 of 122 mutually genotyped loci (see the comparisons above). It's less surprising that COLONY doesn't match the `227-23` sample, given the number of genotype mismatches there. Note that `PLINK` *does* recover the former two sample pairs (`151-24` and `935-23`) as matches, so the combined approach we're using would classify them as members of the same clone!

With the mismatches we see for `227-23` and `35-SGA`, it seems best to exclude those samples from the dataset.

Mismatching Genotypes?

Now let's look at the number of mismatches between replicated genotype calls. This will give us a crude estimate of the genotyping error rate.

```
repl.out$mismatch <- repl.out$nSNPshared-repl.out$nSNPmatch
repl.out$errorRate <- repl.out$mismatch/repl.out$nSNPshared

range(repl.out$mismatch, na.rm = TRUE)
```

```
## [1]   0 118
```

```
range(repl.out$errorRate, na.rm = TRUE)
```

```
## [1] 0.0000000 0.2413088
```

```
#Naive estimate of genotyping error rate is the average proportion of mismatches?
mean(repl.out$errorRate, na.rm = TRUE)
```

```
## [1] 0.01729965
```

```
# 1.73% including all samples
mean(repl.out$errorRate[which(repl.out$GTservice == "LGC")], na.rm=TRUE)
```

```
## [1] 0.0294302
```

```r
# 2.94% for LGC samples
mean(repl.out$errorRate[which(repl.out$GTservice == "GTSeek")], na.rm = TRUE)
```

```
## [1] 0.0004516712
```

```r
# < 0.1% for GTSeek

#Excluding samples that are suspicious (where COLONY doesn't find a match)
mean(repl.out$errorRate[which(repl.out$is.cloneCOLONY==TRUE)])
```

```
## [1] 0.009477482
```

```r
# 0.94% with only "good" pairs
mean(repl.out$errorRate[which(repl.out$is.cloneCOLONY==TRUE &
                                repl.out$GTservice == "LGC")])
```

```
## [1] 0.01571703
```

```r
# 1.6% for LGC matching samples
mean(repl.out$errorRate[which(repl.out$is.cloneCOLONY==TRUE &
                                repl.out$GTservice == "GTSeek")])
```

```
## [1] 0.0005081301
```

```r
# again <0.1% for GTSeek
```

Error rates are 2-3% for the capture-seq (LGC) dataset, but substantially lower for GT-seq (GTSeek) data. The interesting portions of the table comparing technical replicates…

```r
repl.out[,c("ugaID","nSNPgt1","nSNPgt2","filt1","filt2","nSNPshared",
            "nSNPmatch","kinship","mismatch","errorRate")]
```

```
##       ugaID nSNPgt1 nSNPgt2 filt1 filt2 nSNPshared nSNPmatch  kinship mismatch
## 1    35-SGA     557     754 FALSE FALSE        489       371 0.307325      118
## 2    121-23       1      72  TRUE  TRUE         NA        NA       NA       NA
## 3      8-23     918     911 FALSE FALSE        910       908 0.498737        2
## 4    47-SGA     875     912 FALSE FALSE        873       865 0.493355        8
## 5      23-C     904     917 FALSE FALSE        903       898 0.496753        5
## 6    58-SGA     886     888 FALSE FALSE        864       848 0.485866       16
## 7    306-23       0       0  TRUE  TRUE         NA        NA       NA       NA
## 8    166-24     917     917 FALSE FALSE        915       913 0.498810        2
## 9    68-NGA     918     915 FALSE FALSE        914       912 0.498392        2
## 10    21-23     747     776 FALSE FALSE        660       627 0.465909       33
## 11   251-23     920     921 FALSE FALSE        920       919 0.499359        1
## 12   331-23     566     914 FALSE FALSE        566       544 0.476496       22
## 13   409-23     745     774 FALSE FALSE        656       624 0.473333       32
## 14     14-C     904       1 FALSE  TRUE         NA        NA       NA       NA
## 15   241-23     919     920 FALSE FALSE        918       915 0.498037        3
## 16   358-24     922     922 FALSE FALSE        922       922 0.500000        0
## 17   280-23      22      19  TRUE  TRUE         NA        NA       NA       NA
## 18    50-23     911     921 FALSE FALSE        911       908 0.498125        3
## 19   358-23     921     921 FALSE FALSE        919       916 0.497232        3
## 20     5-24     838     754 FALSE FALSE        705       673 0.473770       32
## 21   134-24       0     849  TRUE FALSE         NA        NA       NA       NA
## 22   219-24     921     902 FALSE FALSE        902       894 0.495717        8
## 23   277-24     920     915 FALSE FALSE        914       913 0.499258        1
## 24   397-24     913     914 FALSE FALSE        911       902 0.492902        9
## 25   423-24     885     901 FALSE FALSE        871       858 0.491667       13
## 26   434-24       1      46  TRUE  TRUE         NA        NA       NA       NA
## 27      1-C     902     915 FALSE FALSE        898       894 0.497579        4
## 28     17-C     877     900 FALSE FALSE        865       844 0.483696       21
## 29   227-23     896     894 FALSE FALSE        880       763 0.425949      117
## 30   389-24     911     586 FALSE FALSE        586       549 0.454878       37
## 31   306-24     902     916 FALSE FALSE        902       901 0.499403        1
## 32   930-23     122     123 FALSE FALSE        122       122 0.500000        0
## 33   564-24     122     122 FALSE FALSE        122       122 0.500000        0
## 34   781-23     123     123 FALSE FALSE        122       122 0.500000        0
## 35   798-23     122     123 FALSE FALSE        122       122 0.500000        0
```

```
## 36  651-23     122      123 FALSE FALSE        122       122 0.500000        0
## 37  517-24      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 38  716-23      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 39  726-24      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 40  980-24     125      124 FALSE FALSE        124       124 0.500000        0
## 41  935-23     123      123 FALSE FALSE        122       122 0.500000        0
## 42 1306-24     123      123 FALSE FALSE        122       122 0.500000        0
## 43  567-24     122      123 FALSE FALSE        122       122 0.500000        0
## 44  881-24     123      118 FALSE FALSE        118       118 0.500000        0
## 45 2972-24     122      123 FALSE FALSE        120       120 0.500000        0
## 46 1384-24      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 47  933-24     120      121 FALSE FALSE        118       118 0.500000        0
## 48 1522-24     124      123 FALSE FALSE        123       122 0.492188        1
## 49 3058-24     123      123 FALSE FALSE        122       122 0.500000        0
## 50 2665-24     122      123 FALSE FALSE        121       121 0.500000        0
## 51 2869-24     125      124 FALSE FALSE        124       124 0.500000        0
## 52 2926-24      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 53  151-24     122      118 FALSE FALSE        117       117 0.500000        0
## 54  396-24     113       NA FALSE  TRUE         NA        NA       NA       NA
## 55  424-23      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 56  191-24      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 57  466-23      NA       NA  TRUE  TRUE         NA        NA       NA       NA
## 58  286-23       3       NA  TRUE  TRUE         NA        NA       NA       NA
## 59  500-24     122      123 FALSE FALSE        122       122 0.500000        0
## 60  311-23      69       NA  TRUE  TRUE         NA        NA       NA       NA
##      errorRate
## 1  0.241308793
## 2           NA
## 3  0.002197802
## 4  0.009163803
## 5  0.005537099
## 6  0.018518519
## 7           NA
## 8  0.002185792
## 9  0.002188184
## 10 0.050000000
## 11 0.001086957
## 12 0.038869258
## 13 0.048780488
## 14          NA
## 15 0.003267974
## 16 0.000000000
## 17          NA
## 18 0.003293085
## 19 0.003264418
## 20 0.045390071
## 21          NA
## 22 0.008869180
## 23 0.001094092
## 24 0.009879254
## 25 0.014925373
## 26          NA
## 27 0.004454343
## 28 0.024277457
## 29 0.132954545
## 30 0.063139932
## 31 0.001108647
## 32 0.000000000
## 33 0.000000000
## 34 0.000000000
## 35 0.000000000
## 36 0.000000000
## 37          NA
## 38          NA
## 39          NA
## 40 0.000000000
## 41 0.000000000
## 42 0.000000000
## 43 0.000000000
## 44 0.000000000
## 45 0.000000000
## 46          NA
## 47 0.000000000
## 48 0.008130081
## 49 0.000000000
## 50 0.000000000
## 51 0.000000000
## 52          NA
## 53 0.000000000
## 54          NA
## 55          NA
## 56          NA
## 57          NA
## 58          NA
## 59 0.000000000
## 60          NA
```

Kinship?

The range of kinship coefficients for (good) technical replicates will be used below as a threshold for combining `COLONY` and `PLINK` inferences. Let's calculate that now.

```r
#Range of kinship values between replicates
range(repl.out$kinship, na.rm = TRUE)
```

```
## [1] 0.307325 0.500000
```

```r
#Average kinship of "good" pairs
mean(repl.out$kinship[which(repl.out$is.cloneCOLONY==TRUE)])
```

```
## [1] 0.4938836
```

```r
range(repl.out$kinship[which(repl.out$is.cloneCOLONY==TRUE)])
```

```
## [1] 0.454878 0.500000
```

```r
#So we can set the filter pretty high... >0.45 for recaps
```

Inferred as Recaptures?

Ideally, all technical replicates would be identified as such and called recaptures in our analysis. This isn't the case, a couple of the samples don't seem all that similar. Perhaps some contamination?

Suggest removing these samples from the dataset:

i. `35-SGA` - high error rate in replicated sample (24.6%), not identified as a match by `COLONY` *or* `PLINK2`. Unsure which genotype is accurate so best to drop them both.

ii. `227-23` - error rate his high again (13.6%), sample not identified as a match by `COLONY`. Kinship estimates are high, but the lowest of all pairs of replicated samples identified by `PLINK` as recaptures. Note that this pair of samples showed up in our **`sexSamples`** markdown too. One replicate is inferred to be male, the other inferred to be female. Best to exclude both samples.

Meanwhile, we should keep the match from `PLINK` for `935-23` and `151-24` and their replicates. They match at all 122 and 117 shared SNP loci, surprising that `COLONY` would not identify this pair as a match!

```r
table(repl.out$is.cloneCOLONY)
```

```
##
## FALSE  TRUE
##     4    39
```

```r
table(repl.out$is.clonePLINK)
```

```
##
## FALSE  TRUE
##     1    42
```

```r
repl.out[which(repl.out$is.cloneCOLONY == FALSE),]
```

```
##     ugaID                vcfID1                vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1  35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     557     754 FALSE FALSE
## 29 227-23 UGA_173101_P021_WH06 UGA_173101_P023_WC12     896     894 FALSE FALSE
## 41 935-23           GTS_935-23           GTS_935-23R     123     123 FALSE FALSE
## 53 151-24           GTS_151-24           GTS_151-24R     122     118 FALSE FALSE
##    colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
```

```
## 1          72       114         FALSE          FALSE        489       371
## 29         14       185          TRUE          FALSE        880       763
## 41        255       289          TRUE          FALSE        122       122
## 53        255        33          TRUE          FALSE        117       117
##     kinship GTservice mismatch errorRate
## 1  0.307325       LGC      118 0.2413088
## 29 0.425949       LGC      117 0.1329545
## 41 0.500000    GTSeek        0 0.0000000
## 53 0.500000    GTSeek        0 0.0000000
```

```
repl.out[which(repl.out$is.cloneCOLONY == FALSE & repl.out$is.clonePLINK == FALSE),]
```

```
##    ugaID               vcfID1               vcfID2 nSNPgt1 nSNPgt2 filt1 filt2
## 1 35-SGA UGA_173101_P015_WB01 UGA_173101_P015_WB03     557     754 FALSE FALSE
##   colclone1 colclone2 is.clonePLINK is.cloneCOLONY nSNPshared nSNPmatch
## 1        72       114         FALSE          FALSE        489       371
##    kinship GTservice mismatch errorRate
## 1 0.307325       LGC      118 0.2413088
```

```
#Drop samples by setting FILT to TRUE
which(smd$ugaID == "35-SGA")
```

```
## [1] 13 15
```

```
smd$FILT[which(smd$ugaID == "35-SGA")] <- TRUE
which(smd$ugaID == "227-23")
```

```
## [1] 648 786
```

```
smd$FILT[which(smd$ugaID == "227-23")] <- TRUE
```

Check for Odd Patterns

Beyond the technical replicates, we have some other information that will be useful for evaluating the quality of the inferred recaptures. For instance, some bears were directly captured and have known sex information. We wouldn't want to mistake males for females in these samples. Space is also useful, a CGA bear is unlikely to be sampled in NGA and vice versa.

```
nSex <- nSexGT <- nRegion <- c()
for(i in 1:nrow(col.out)) {
  tmpclone <- strsplit(col.out[i,3], split = ",")[[1]]
  tmp <- smd[which(smd$vcfID %in% tmpclone),]
  nSex[i] <- length(unique(tmp$knownSEX[tmp$knownSEX != ""]))
  nSexGT[i] <- length(unique(tmp$sexGT[tmp$sexGT != "undet"]))
  nRegion[i] <- length(unique(tmp$region))
  rm(tmp, tmpclone)
}
```

Known Sex

We should not have multiple known sexes per clone

```r
nSex
```

```
##   [1] 0 0 1 2 1 0 0 0 0 1 1 1 1 0 2 1 1 1 1 2 1 0 1 1 1 0 1 0 0 1 0 1 1 0 1 0 1
##  [38] 1 1 1 0 0 0 2 2 1 1 1 0 1 1 1 1 1 0 1 0 1 1 0 1 0 0 0 1 1 0 2 2 1 2 1 1 1
##  [75] 0 1 1 1 2 1 0 1 1 1 1 1 1 1 1 1 2 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1
## [112] 1 2 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 2 0 0 1 1 1 1 1 1 1 1 1 0 1 0 1 1 1
## [149] 0 1 2 1 1 1 1 0 1 1 0 1 1 1 1 0 1 1 0 1 1 0 0 1 0 1 2 0 1 0 1 0 1 1 2 1 0
## [186] 0 1 1 0 1 2 0 0 0 0 0 0 1 0 0 1 0 1 0 0 0 0 1 0 1 0 0 1 0 0 0 0 0 2 1 0 1
## [223] 1 1 0 1 0 1 1 1 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [260] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [297] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [334] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```r
max(nSex)
```

```
## [1] 2
```

```r
#Does genotype-inferred sex match known sex of samples?
table(smd$sexGT[smd$knownSEX == "F"])
```

```
## 
##     F undet 
##    64     3
```

```r
table(smd$sexGT[smd$knownSEX == "M"])
```

```
## 
##     F     M undet 
##     1    82     4
```

```r
#The single case of a mismatch (within a sample) between a known sex
# and genotype-inferred sex
smd[which(smd$knownSEX=="M" & smd$sexGT == "F"),]
```

```
##                    extID ugaID is.SCR is.rep week year snare utmSnare
## 71 UGA_173101_P015_WF11 63-DC  FALSE  FALSE <NA> 2024  <NA>     <NA>
##                                                                      fqfile
## 71 RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
##    knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice sexGT
## 71        M  Den Check    CGA    913 UGA_173101_P015_WF11 FALSE       LGC     F
```

```r
bigMD <- read.csv("../input/Finalized Hair Samples w Metadata_Carr.csv", header = TRUE)
bigMD[bigMD$Hair.ID == smd$ugaID[which(smd$knownSEX=="M" & smd$sexGT == "F")],]
```

```
##      Sample.. RandomID Hair.ID    Rapid.Genomics.ID Site.ID Week Check.Date
## 1112     1063     1063   63-DC UGA_173101_P012_WC09
```

```
##      Strand Sample.Quality DNR.ID          Database.ID     PIT.Tag     Bear Sex
## 1112                              CGBB_M_2024_C_87815B 606-306-842 87815 C2   M
##      Year   Date  County Region      What Plate Follicles Abnormalitiy.
## 1112 2024 3/2/24 Houston    CGA Den Check    12         14
##                       Comments        Address Latitude Longitude X X.1
## 1112 87815 Biological Cub 2 87815 Den 2024 32.52144 -83.44958
```

```r
#This was a captured cub, identified as male in the field
#  we saw this individual before - see the `sexSamples` markdown for more detail
```

Genotype-inferred Sex

Similarly, our genotype-inferred sexes should not vary within clones.

```r
nSexGT
```

```
##   [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [112] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [149] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [186] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1
## [223] 1 0 1 1 1 1 1 1 1 0 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1
## [260] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [297] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [334] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```r
max(nSexGT)
```

```
## [1] 2
```

```r
which(nSexGT > 1)
```

```
## [1]  14  30  45 238 255
```

```r
#They do in a few cases...
# 14 30 45 238 255

#Clone 14 - 30 samples, 26 female, 2 male, 2 undet - no direct capture to confirm majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 14,3], ",")[[1]]])
```

```
##
##     F     M undet
##    26     2     2
```

```r
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 14,3], ",")[[1]],]
```

```
##                      extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 91  UGA_173101_P015_WH07 432-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 138 UGA_173101_P016_WD06 179-23   TRUE  FALSE    7 2023 HS 027 264464x3592495
```

```
## 186 UGA_173101_P016_WH06 312-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 188 UGA_173101_P016_WH08 330-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 194 UGA_173101_P017_WA02 393-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 256 UGA_173101_P017_WF04 468-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 260 UGA_173101_P017_WF08 409-23   TRUE   TRUE    5 2023 HS 027 264464x3592495
## 285 UGA_173101_P017_WH09 409-23   TRUE   TRUE    5 2023 HS 027 264464x3592495
## 338 UGA_173101_P018_WE02  52-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 345 UGA_173101_P018_WE09 206-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 351 UGA_173101_P018_WF03  35-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 392 UGA_173101_P019_WC02 188-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 396 UGA_173101_P019_WC06 200-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 406 UGA_173101_P019_WD04 362-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 467 UGA_173101_P020_WA05 209-23   TRUE  FALSE    7 2023 HS 025 265276x3591787
## 469 UGA_173101_P020_WA07  38-23   TRUE  FALSE    3 2023 HS 027 264464x3592495
## 471 UGA_173101_P020_WA09 319-23   TRUE  FALSE    6 2023 HS 032 263672x3593292
## 540 UGA_173101_P020_WG06 277-23   TRUE  FALSE    5 2023 HS 025 265276x3591787
## 541 UGA_173101_P020_WG07 443-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 549 UGA_173101_P020_WH03 212-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 550 UGA_173101_P020_WH04 271-23   TRUE  FALSE    6 2023 HS 025 265276x3591787
## 557 UGA_173101_P020_WH11 447-23   TRUE  FALSE    7 2023 HS 027 264464x3592495
## 583 UGA_173101_P021_WC01  91-23   TRUE  FALSE    5 2023 HS 027 264464x3592495
## 648 UGA_173101_P021_WH06 227-23   TRUE   TRUE    5 2023 HS 027 264464x3592495
## 666 UGA_173101_P022_WA12 160-23   TRUE  FALSE    6 2023 HS 027 264464x3592495
## 735 UGA_173101_P022_WG09 471-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 766 UGA_173101_P023_WB04  37-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 768 UGA_173101_P023_WB06 190-23   TRUE  FALSE    5 2023 HS 025 265276x3591787
## 773 UGA_173101_P023_WB11   7-23   TRUE  FALSE    4 2023 HS 027 264464x3592495
## 826 UGA_173101_P023_WG04 136-23   TRUE  FALSE    6 2023 HS 025 265276x3591787
##                                                                             fqfile
## 91    RAPiD-Genomics_F534_UGA_173101_P015_WH07_i5-502_i7-1_S14829_L003_R1_001.fastq
## 138 RAPiD-Genomics_F534_UGA_173101_P016_WD06_i5-504_i7-138_S14876_L003_R1_001.fastq
## 186 RAPiD-Genomics_F534_UGA_173101_P016_WH06_i5-504_i7-186_S14923_L003_R1_001.fastq
## 188 RAPiD-Genomics_F534_UGA_173101_P016_WH08_i5-504_i7-188_S14925_L003_R1_001.fastq
## 194  RAPiD-Genomics_F534_UGA_173101_P017_WA02_i5-512_i7-98_S14931_L003_R1_001.fastq
## 256 RAPiD-Genomics_F534_UGA_173101_P017_WF04_i5-512_i7-160_S14993_L003_R1_001.fastq
## 260 RAPiD-Genomics_F534_UGA_173101_P017_WF08_i5-512_i7-164_S14997_L003_R1_001.fastq
## 285 RAPiD-Genomics_F534_UGA_173101_P017_WH09_i5-512_i7-189_S15022_L003_R1_001.fastq
## 338 RAPiD-Genomics_F534_UGA_173101_P018_WE02_i5-537_i7-146_S15075_L003_R1_001.fastq
## 345 RAPiD-Genomics_F534_UGA_173101_P018_WE09_i5-537_i7-153_S15084_L003_R1_001.fastq
## 351 RAPiD-Genomics_F534_UGA_173101_P018_WF03_i5-537_i7-159_S15090_L003_R1_001.fastq
## 392 RAPiD-Genomics_F534_UGA_173101_P019_WC02_i5-1102_i7-67_S15131_L003_R1_001.fastq
## 396  RAPiD-Genomics_F534_UGA_173101_P019_WC06_i5-1102_i7-3_S15135_L003_R1_001.fastq
## 406 RAPiD-Genomics_F534_UGA_173101_P019_WD04_i5-1102_i7-91_S15145_L003_R1_001.fastq
## 467   RAPiD-Genomics_F566_UGA_173101_P020_WA05_i5-505_i7-38_S3895_L001_R1_001.fastq
## 469   RAPiD-Genomics_F566_UGA_173101_P020_WA07_i5-505_i7-77_S3897_L001_R1_001.fastq
## 471   RAPiD-Genomics_F566_UGA_173101_P020_WA09_i5-505_i7-36_S3899_L001_R1_001.fastq
## 540   RAPiD-Genomics_F566_UGA_173101_P020_WG06_i5-505_i7-14_S3968_L001_R1_001.fastq
## 541   RAPiD-Genomics_F566_UGA_173101_P020_WG07_i5-505_i7-18_S3969_L001_R1_001.fastq
## 549   RAPiD-Genomics_F566_UGA_173101_P020_WH03_i5-505_i7-88_S3977_L001_R1_001.fastq
## 550   RAPiD-Genomics_F566_UGA_173101_P020_WH04_i5-505_i7-73_S3978_L001_R1_001.fastq
## 557   RAPiD-Genomics_F566_UGA_173101_P020_WH11_i5-505_i7-95_S3985_L001_R1_001.fastq
## 583     RAPiD-Genomics_F566_UGA_173101_P021_WC01_i5-501_i7-65_S25_L003_R1_001.fastq
## 648     RAPiD-Genomics_F566_UGA_173101_P021_WH06_i5-501_i7-49_S90_L003_R1_001.fastq
## 666  RAPiD-Genomics_F566_UGA_173101_P022_WA12_i5-1098_i7-23_S7594_L003_R1_001.fastq
```

```
## 735  RAPiD-Genomics_F566_UGA_173101_P022_WG09_i5-1098_i7-41_S7663_L003_R1_001.fastq
## 766  RAPiD-Genomics_F566_UGA_173101_P023_WB04_i5-1099_i7-20_S7694_L003_R1_001.fastq
## 768   RAPiD-Genomics_F566_UGA_173101_P023_WB06_i5-1099_i7-6_S7696_L003_R1_001.fastq
## 773   RAPiD-Genomics_F566_UGA_173101_P023_WB11_i5-1099_i7-5_S7701_L003_R1_001.fastq
## 826  RAPiD-Genomics_F566_UGA_173101_P023_WG04_i5-1099_i7-10_S7754_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 91           Hair Snare    CGA    911 UGA_173101_P015_WH07 FALSE       LGC
## 138          Hair Snare    CGA    853 UGA_173101_P016_WD06 FALSE       LGC
## 186          Hair Snare    CGA    782 UGA_173101_P016_WH06 FALSE       LGC
## 188          Hair Snare    CGA    870 UGA_173101_P016_WH08 FALSE       LGC
## 194          Hair Snare    CGA    916 UGA_173101_P017_WA02 FALSE       LGC
## 256          Hair Snare    CGA    917 UGA_173101_P017_WF04 FALSE       LGC
## 260          Hair Snare    CGA    745 UGA_173101_P017_WF08 FALSE       LGC
## 285          Hair Snare    CGA    774 UGA_173101_P017_WH09 FALSE       LGC
## 338          Hair Snare    CGA    915 UGA_173101_P018_WE02 FALSE       LGC
## 345          Hair Snare    CGA    884 UGA_173101_P018_WE09 FALSE       LGC
## 351          Hair Snare    CGA    899 UGA_173101_P018_WF03 FALSE       LGC
## 392          Hair Snare    CGA    919 UGA_173101_P019_WC02 FALSE       LGC
## 396          Hair Snare    CGA    919 UGA_173101_P019_WC06 FALSE       LGC
## 406          Hair Snare    CGA    922 UGA_173101_P019_WD04 FALSE       LGC
## 467          Hair Snare    CGA    913 UGA_173101_P020_WA05 FALSE       LGC
## 469          Hair Snare    CGA    892 UGA_173101_P020_WA07 FALSE       LGC
## 471          Hair Snare    CGA    852 UGA_173101_P020_WA09 FALSE       LGC
## 540          Hair Snare    CGA    910 UGA_173101_P020_WG06 FALSE       LGC
## 541          Hair Snare    CGA    608 UGA_173101_P020_WG07 FALSE       LGC
## 549          Hair Snare    CGA    918 UGA_173101_P020_WH03 FALSE       LGC
## 550          Hair Snare    CGA    916 UGA_173101_P020_WH04 FALSE       LGC
## 557          Hair Snare    CGA    918 UGA_173101_P020_WH11 FALSE       LGC
## 583          Hair Snare    CGA    895 UGA_173101_P021_WC01 FALSE       LGC
## 648          Hair Snare    CGA    896 UGA_173101_P021_WH06  TRUE       LGC
## 666          Hair Snare    CGA    865 UGA_173101_P022_WA12 FALSE       LGC
## 735          Hair Snare    CGA    915 UGA_173101_P022_WG09 FALSE       LGC
## 766          Hair Snare    CGA    908 UGA_173101_P023_WB04 FALSE       LGC
## 768          Hair Snare    CGA    897 UGA_173101_P023_WB06 FALSE       LGC
## 773          Hair Snare    CGA    905 UGA_173101_P023_WB11 FALSE       LGC
## 826          Hair Snare    CGA    914 UGA_173101_P023_WG04 FALSE       LGC
##     sexGT
## 91      F
## 138     F
## 186     F
## 188     F
## 194     F
## 256 undet
## 260     F
## 285     F
## 338     F
## 345     F
## 351     F
## 392     F
## 396     F
## 406     F
## 467     F
## 469     F
## 471     F
```

```
## 540     F
## 541     F
## 549     M
## 550     M
## 557     F
## 583     F
## 648     F
## 666 undet
## 735     F
## 766     F
## 768     F
## 773     F
## 826     F
```

```r
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 14,3], ",")[[1]]]
```

```
##  [1] "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" ""
## [26] "" "" "" "" ""
```

```r
#Clone 30 - 6 samples, 5 female, 1 male - direct capture confirms majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 30,3], ",")[[1]]])
```

```
##
## F M
## 5 1
```

```r
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 30,3], ",")[[1]],]
```

```
##                    extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 86   UGA_173101_P015_WH02   45-C  FALSE  FALSE <NA> 2024   <NA>           <NA>
## 169 UGA_173101_P016_WG01 448-24   TRUE  FALSE    5 2024 HS 051 260705x3599596
## 209 UGA_173101_P017_WB05 497-24   TRUE  FALSE    2 2024 HS 048 261267x3598930
## 426 UGA_173101_P019_WE12 154-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 698 UGA_173101_P022_WD08 341-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 784 UGA_173101_P023_WC10 205-23   TRUE  FALSE    7 2023 HS 051 260705x3599596
##                                                                           fqfile
## 86   RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169 RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209 RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426 RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
## 698  RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
## 784  RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 86         F    Capture    CGA    882 UGA_173101_P015_WH02 FALSE       LGC
## 169           Hair Snare    CGA    621 UGA_173101_P016_WG01 FALSE       LGC
## 209           Hair Snare    CGA    923 UGA_173101_P017_WB05 FALSE       LGC
## 426           Hair Snare    CGA    920 UGA_173101_P019_WE12 FALSE       LGC
## 698           Hair Snare    CGA    862 UGA_173101_P022_WD08 FALSE       LGC
## 784           Hair Snare    CGA    908 UGA_173101_P023_WC10 FALSE       LGC
##     sexGT
## 86      F
## 169     F
```

```
## 209      M
## 426      F
## 698      F
## 784      F
```

```r
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 30,3], ",")[[1]]]
```

```
## [1] "F" ""  ""  ""  ""  ""
```

```r
#Clone 45 - 6 samples, 5 female, 1 male - direct capture confirms majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 45,3], ",")[[1]]])
```

```
##
## F M
## 5 1
```

```r
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 45,3], ",")[[1]],]
```

```
##                                     extID   ugaID is.SCR is.rep week year
## 49                   UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                  UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                  UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>           <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                   fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                        <NA>
## 1598                                                                        <NA>
## 1819                                                                        <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F     Capture    CGA    892 UGA_173101_P015_WE01 FALSE       LGC
## 492               Hair Snare    CGA    915 UGA_173101_P020_WC06 FALSE       LGC
## 629               Hair Snare    CGA    913 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120          GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123         GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124         GTS_2920-24 FALSE    GTSeek
##      sexGT
## 49       F
## 492      F
## 629      M
## 1048     F
## 1598     F
## 1819     F
```

```r
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 45,3], ",")[[1]]]
```

```
## [1] "F" ""  ""  NA  NA  NA
```

```r
#Clone 238 - 9 samples, 8 male, 1 female - no direct capture to confirm majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 238,3], ",")[[1]]])
```

```
##
## F M
## 1 8
```

```r
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 238,3], ",")[[1]],]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 828     GTseek_024_384i5_001_P098.001_5-23    5-23   TRUE  FALSE    6 2023
## 1213  GTseek_025_384i5_003_P098.005_738-24  738-24   TRUE  FALSE    7 2024
## 1451 GTseek_025_384i5_241_P098.006_2962-24 2962-24   TRUE  FALSE    4 2024
## 1615 GTseek_026_384i5_025_P098.010_2748-24 2748-24   TRUE  FALSE    7 2024
## 1715 GTseek_026_384i5_163_P098.009_2320-24 2320-24   TRUE  FALSE    7 2024
## 1760 GTseek_026_384i5_253_P098.010_2864-24 2864-24   TRUE  FALSE    7 2024
## 1792 GTseek_026_384i5_317_P098.010_2894-24 2894-24   TRUE  FALSE    4 2024
## 1816 GTseek_026_384i5_365_P098.009_2729-24 2729-24   TRUE  FALSE    7 2024
## 2095 GTseek_028_384i5_270_P098.UGA3_260-24  260-24   TRUE  FALSE    4 2024
##       snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt
## 828  HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    104
## 1213 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    119
## 1451 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    119
## 1615 HS 008 256165x3589588   <NA>     <NA> Hair Sample    CGA    122
## 1715 HS 008 256165x3589588   <NA>     <NA> Hair Sample    CGA    124
## 1760 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    124
## 1792 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    124
## 1816 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    121
## 2095 HS 001 248187x3586614   <NA>     <NA> Hair Sample    CGA    112
##            vcfID  FILT GTservice sexGT
## 828     GTS_5-23 FALSE    GTSeek     F
## 1213  GTS_738-24 FALSE    GTSeek     M
## 1451 GTS_2962-24 FALSE    GTSeek     M
## 1615 GTS_2748-24 FALSE    GTSeek     M
## 1715 GTS_2320-24 FALSE    GTSeek     M
## 1760 GTS_2864-24 FALSE    GTSeek     M
## 1792 GTS_2894-24 FALSE    GTSeek     M
## 1816 GTS_2729-24 FALSE    GTSeek     M
## 2095  GTS_260-24 FALSE    GTSeek     M
```

```r
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 238,3], ",")[[1]]]
```

```
## [1] NA NA NA NA NA NA NA NA NA
```

```r
#Clone 255 - 28 samples, 25 male, 1 female, 2 undet - no direct capture to confirm majority
table(smd$sexGT[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 255,3], ",")[[1]]])
```

```
##
##     F     M undet
##     1    25     2
```

```r
smd[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 255,3], ",")[[1]],]
```

```
##                                       extID   ugaID is.SCR is.rep week year
## 873   GTseek_024_384i5_046_P098.003_949-23  949-23   TRUE  FALSE    6 2023
## 936   GTseek_024_384i5_109_P098.001_646-23  646-23   TRUE  FALSE    7 2023
## 962   GTseek_024_384i5_135_P098.001_655-23  655-23   TRUE  FALSE    6 2023
## 963   GTseek_024_384i5_136_P098.003_983-23  983-23   TRUE  FALSE    3 2023
## 974   GTseek_024_384i5_148_P098.004_630-24  630-24   TRUE  FALSE    3 2024
## 993   GTseek_024_384i5_167_P098.001_665-23  665-23   TRUE  FALSE    7 2023
## 998   GTseek_024_384i5_172_P098.003_997-23  997-23   TRUE  FALSE    7 2023
## 1013  GTseek_024_384i5_187_P098.002_818-23  818-23   TRUE  FALSE    6 2023
## 1029  GTseek_024_384i5_203_P098.001_677-23  677-23   TRUE  FALSE    3 2023
## 1374  GTseek_025_384i5_164_P098.007_935-23  935-23   TRUE   TRUE    7 2023
## 1411  GTseek_025_384i5_201_P098.005_855-24  855-24   TRUE  FALSE    3 2024
## 1426 GTseek_025_384i5_216_P098.008_1344-24 1344-24   TRUE  FALSE    3 2024
## 1473  GTseek_025_384i5_263_P098.005_886-24  886-24   TRUE  FALSE    3 2024
## 1490 GTseek_025_384i5_280_P098.008_1403-24 1403-24   TRUE  FALSE    3 2024
## 1513  GTseek_025_384i5_303_P098.005_931-24  931-24   TRUE  FALSE    3 2024
## 1518 GTseek_025_384i5_308_P098.008_1428-24 1428-24   TRUE  FALSE    2 2024
## 1527 GTseek_025_384i5_317_P098.006_3012-24 3012-24   TRUE  FALSE    6 2024
## 1548 GTseek_025_384i5_338_P098.008_1451-24 1451-24   TRUE  FALSE    3 2024
## 1563  GTseek_025_384i5_353_P098.005_955-24  955-24   TRUE  FALSE    2 2024
## 1641 GTseek_026_384i5_061_P098.010_2777-24 2777-24   TRUE  FALSE    3 2024
## 1691 GTseek_026_384i5_129_P098.009_1672-24 1672-24   TRUE  FALSE    3 2024
## 1717 GTseek_026_384i5_167_P098.009_2376-24 2376-24   TRUE  FALSE    6 2024
## 1747 GTseek_026_384i5_227_P098.009_2669-24 2669-24   TRUE  FALSE    6 2024
## 1756 GTseek_026_384i5_245_P098.010_2857-24 2857-24   TRUE  FALSE    3 2024
## 1782 GTseek_026_384i5_297_P098.009_2697-24 2697-24   TRUE  FALSE    2 2024
## 1805 GTseek_026_384i5_343_P098.010_2906-24 2906-24   TRUE  FALSE    6 2024
## 1903 GTseek_028_384i5_078_P098.UGA3_151-24  151-24   TRUE   TRUE    3 2024
## 1920 GTseek_028_384i5_095_P098.UGA2_426-23  426-23   TRUE  FALSE    6 2023
##        snare       utmSnare fqfile knownSEX  sampSOURCE region nSNPgt
## 873  HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    124
## 936  HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    120
## 962  HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    116
## 963  HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    121
## 974  HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    119
## 993  HS 002 255814x3587061   <NA>     <NA> Hair Sample    CGA    122
## 998  HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    123
## 1013 HS 009 259209x3585507   <NA>     <NA> Hair Sample    CGA    123
## 1029 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    119
## 1374 HS 002 255814x3587061   <NA>     <NA> Hair Sample    CGA    123
## 1411 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    123
## 1426 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    124
## 1473 HS 017 260417x3588089   <NA>     <NA> Hair Sample    CGA    124
## 1490 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    122
## 1513 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    119
## 1518 HS 014 261409x3586557   <NA>     <NA> Hair Sample    CGA    123
## 1527 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    122
## 1548 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    123
```

```
## 1563 HS 014 261409x3586557   <NA>     <NA> Hair Sample    CGA    122
## 1641 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    121
## 1691 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    121
## 1717 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    123
## 1747 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    123
## 1756 HS 011 257416x3586308   <NA>     <NA> Hair Sample    CGA    124
## 1782 HS 014 261409x3586557   <NA>     <NA> Hair Sample    CGA    122
## 1805 HS 009 259092x3585434   <NA>     <NA> Hair Sample    CGA    124
## 1903 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    122
## 1920 HS 012 258412x3586648   <NA>     <NA> Hair Sample    CGA    119
##            vcfID  FILT GTservice sexGT
## 873   GTS_949-23 FALSE    GTSeek     M
## 936   GTS_646-23 FALSE    GTSeek undet
## 962   GTS_655-23 FALSE    GTSeek     F
## 963   GTS_983-23 FALSE    GTSeek     M
## 974   GTS_630-24 FALSE    GTSeek     M
## 993   GTS_665-23 FALSE    GTSeek undet
## 998   GTS_997-23 FALSE    GTSeek     M
## 1013  GTS_818-23 FALSE    GTSeek     M
## 1029  GTS_677-23 FALSE    GTSeek     M
## 1374  GTS_935-23 FALSE    GTSeek     M
## 1411  GTS_855-24 FALSE    GTSeek     M
## 1426 GTS_1344-24 FALSE    GTSeek     M
## 1473  GTS_886-24 FALSE    GTSeek     M
## 1490 GTS_1403-24 FALSE    GTSeek     M
## 1513  GTS_931-24 FALSE    GTSeek     M
## 1518 GTS_1428-24 FALSE    GTSeek     M
## 1527 GTS_3012-24 FALSE    GTSeek     M
## 1548 GTS_1451-24 FALSE    GTSeek     M
## 1563  GTS_955-24 FALSE    GTSeek     M
## 1641 GTS_2777-24 FALSE    GTSeek     M
## 1691 GTS_1672-24 FALSE    GTSeek     M
## 1717 GTS_2376-24 FALSE    GTSeek     M
## 1747 GTS_2669-24 FALSE    GTSeek     M
## 1756 GTS_2857-24 FALSE    GTSeek     M
## 1782 GTS_2697-24 FALSE    GTSeek     M
## 1805 GTS_2906-24 FALSE    GTSeek     M
## 1903  GTS_151-24 FALSE    GTSeek     M
## 1920  GTS_426-23 FALSE    GTSeek     M
```

```
smd$knownSEX[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex == 255,3], ",")[[1]]]
```

```
##  [1] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## [26] NA NA NA
```

So sex is not inferred perfectly. That's not surprising, given the pretty lenient threshold we're using to identify males (just 5 reads required at 2 SNPs within 3 Y-specific scaffolds). Contamination could easily lead to us calling a sample a male erroneously. Nonetheless, instances with multiple sexes are relatively rare, and in all cases there is either a clear majority from the set of samples or a direct capture that confirms the consensus from the set.

Region

We shouldn't have multiple regions for any clones…

```
nRegion
```

```
##   [1] 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [112] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [149] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [186] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [223] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [260] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [297] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [334] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```
which(nRegion > 1)
```

```
## [1]   3 282
```

```
smd$region[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex==3,3],",")[[1]]]
```

```
##  [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA-H" "CGA"   "CGA"   "CGA"
```

```
#Not really a separate region, just a historic hair sample (from 2012)

smd$region[smd$vcfID %in% strsplit(col.out[col.out$CloneIndex==282,3],",")[[1]]]
```

```
## [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA-H"
```

```
#Not really a separate region, just a historic hair sample (from 2012)
```

So no instances of recaptures identified across regions. That's good!

Known vs. Genotype-inferred Sex

Finally, comparing the known sex of unique genotypes (from direct captures) with that inferred for all samples within a clone. . .

```
#check for discrepancies between known and genotyped sex for all clones
discrepancy <- multknown <- c()
#for(i in 1:max(sampleMetadata$cloneID, na.rm = TRUE)) {
for(i in 1:nrow(col.out)) {
  tmpclone <- strsplit(col.out[i,3], split = ",")[[1]]
  tmp <- smd[which(smd$vcfID %in% tmpclone),]
  tmp.knownSEX <- unique(tmp$knownSEX[tmp$knownSEX != "" & !is.na(tmp$knownSEX)])
  if(length(tmp.knownSEX) > 0) {
    if(length(tmp.knownSEX) > 1) {
      multknown[i] <- TRUE
      discrepancy[i] <- NA
    } else {
```

```r
      multknown[i] <- FALSE
      if(sum(tmp$sexGT[tmp$sexGT != "undet"] != tmp.knownSEX) == 0) {
        discrepancy[i] <- FALSE
      } else {
        discrepancy[i] <- TRUE
      }
    }
  }
  rm(tmpclone, tmp, tmp.knownSEX)
}

sum(discrepancy, na.rm = TRUE) #3 cases with discrepancies...
```

```
## [1] 3
```

```r
sum(multknown, na.rm = TRUE) #0 cases with multiple "known" sexes
```

```
## [1] 0
```

```r
#Check the discrepancies...
which(discrepancy == TRUE)
```

```
## [1]  30  45 140
```

```r
#Clone 30 - seen above, capture agrees with majority of samples
smd[smd$vcfID %in% strsplit(col.out[which(col.out$CloneIndex == 30),3],",")[[1]],]
```

```
##                     extID  ugaID is.SCR is.rep week year  snare       utmSnare
## 86  UGA_173101_P015_WH02   45-C  FALSE  FALSE <NA> 2024   <NA>           <NA>
## 169 UGA_173101_P016_WG01 448-24   TRUE  FALSE    5 2024 HS 051 260705x3599596
## 209 UGA_173101_P017_WB05 497-24   TRUE  FALSE    2 2024 HS 048 261267x3598930
## 426 UGA_173101_P019_WE12 154-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 698 UGA_173101_P022_WD08 341-24   TRUE  FALSE    3 2024 HS 048 261267x3598930
## 784 UGA_173101_P023_WC10 205-23   TRUE  FALSE    7 2023 HS 051 260705x3599596
##                                                                                  fqfile
## 86   RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169 RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209 RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426 RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
## 698  RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
## 784  RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
##     knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 86         F    Capture    CGA    882 UGA_173101_P015_WH02 FALSE       LGC
## 169           Hair Snare    CGA    621 UGA_173101_P016_WG01 FALSE       LGC
## 209           Hair Snare    CGA    923 UGA_173101_P017_WB05 FALSE       LGC
## 426           Hair Snare    CGA    920 UGA_173101_P019_WE12 FALSE       LGC
## 698           Hair Snare    CGA    862 UGA_173101_P022_WD08 FALSE       LGC
## 784           Hair Snare    CGA    908 UGA_173101_P023_WC10 FALSE       LGC
##     sexGT
## 86      F
## 169     F
```

```
## 209     M
## 426     F
## 698     F
## 784     F
```

```
#Clone 45 - seen above, capture agrees with majority of samples
smd[smd$vcfID %in% strsplit(col.out[which(col.out$CloneIndex == 45),3],",")[[1]],]
```

```
##                                      extID   ugaID is.SCR is.rep week year
## 49                     UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                    UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                    UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>           <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                         fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                      <NA>
## 1598                                                                      <NA>
## 1819                                                                      <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F     Capture    CGA    892 UGA_173101_P015_WE01 FALSE       LGC
## 492             Hair Snare    CGA    915 UGA_173101_P020_WC06 FALSE       LGC
## 629             Hair Snare    CGA    913 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120          GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123         GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124         GTS_2920-24 FALSE    GTSeek
##      sexGT
## 49       F
## 492      F
## 629      M
## 1048     F
## 1598     F
## 1819     F
```

```
#Clone 140 - noted previously, cub recorded as M in the field, genotype looks F
smd[smd$vcfID %in% strsplit(col.out[which(col.out$CloneIndex == 140),3],",")[[1]],]
```

```
##                   extID ugaID is.SCR is.rep week year snare utmSnare
## 71 UGA_173101_P015_WF11 63-DC  FALSE  FALSE <NA> 2024  <NA>     <NA>
##                                                                       fqfile
## 71 RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
##    knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice sexGT
## 71        M  Den Check    CGA    913 UGA_173101_P015_WF11 FALSE       LGC     F
```

Compare Inferred Recaptures

Now we create a new data frame to compare inferences (same individual / not) for each pair of samples. Note that, for PLINK, we're using 0.354 as a threshold for KING relatedness coefficients of duplicated samples. This is based off of information provided in the PLINK2 manual. Quoting directly (emphasis added)...

Note that KING kinship coefficients are scaled such that duplicate samples have kinship 0.5, not 1. First-degree relations (parent-child, full siblings) correspond to ~0.25, second-degree relations correspond to ~0.125, etc. ***It is conventional to use a cutoff of ~0.354 (the geometric mean of 0.5 and 0.25) to screen for monozygotic twins and duplicate samples****, ~0.177 to add first-degree relations, etc.*

However, based on our replicated genotypes, we will increase this threshold substantially when combining inferences from PLINK and COLONY (to a mean kinship - across samples within clones - of at least 0.45).

Overall summaries of the matches

First let's see how many pairs of samples are identified as the same individual in our anlayses...

```
#Create a new data frame with info for each pair...
compare.out <- data.frame(id1 = kingrel$id1,
                          id2 = kingrel$id2,
                          kinship = kingrel$kinship,
                          nsnp = kingrel$nsnp,
                          colclone1 = NA,
                          colclone2 = NA,
                          isclonePLINK = NA,
                          iscloneCOL = NA)

#This loop takes a little bit of time (~50 minutes)
T1 <- Sys.time()
for(r in 1:nrow(kingrel)) {
  tmp.grep1 <- grep(compare.out$id1[r], col.out[,3])
  if(length(tmp.grep1) > 1) {
    for(i in 1:length(tmp.grep1)) {
      tmp.split <- strsplit(col.out[tmp.grep1[i],3], ",")[[1]]
      if(compare.out$id1[r] %in% tmp.split) {
        compare.out$colclone1[r] <- col.out$CloneIndex[tmp.grep1[i]]
      }
      rm(tmp.split)
    }
  } else {
    compare.out$colclone1[r] <- col.out$CloneIndex[tmp.grep1]
  }

  tmp.grep2 <- grep(compare.out$id2[r], col.out[,3])
  if(length(tmp.grep2) > 1) {
    for(i in 1:length(tmp.grep2)) {
      tmp.split <- strsplit(col.out[tmp.grep2[i],3], ",")[[1]]
      if(compare.out$id2[r] %in% tmp.split) {
        compare.out$colclone2[r] <- col.out$CloneIndex[tmp.grep2[i]]
      }
      rm(tmp.split)
    }
  } else {
```

```
    compare.out$colclone2[r] <- col.out$CloneIndex[tmp.grep2]
  }

  rm(tmp.grep1, tmp.grep2)

  if(compare.out$colclone1[r] == compare.out$colclone2[r]) {
    compare.out$iscloneCOL[r] <- TRUE
  } else {
    compare.out$iscloneCOL[r] <- FALSE
  }

  if(!is.na(compare.out$kinship[r])) {
    if(compare.out$kinship[r] >= 0.354) {
      compare.out$isclonePLINK[r] <- TRUE
    } else {
      compare.out$isclonePLINK[r] <- FALSE
    }
  } else {
    compare.out$isclonePLINK[r] <- FALSE
  }

}
Sys.time()-T1
```

```
## Time difference of 3.406621 hours
```

```
#How many pairs identified by COLONY are also identified by PLINK?
table(compare.out$isclonePLINK[compare.out$iscloneCOL == TRUE])
```

```
##
##  TRUE
## 15876
```

```
#How many pairs identified by PLINK are also identified by COLONY?
table(compare.out$iscloneCOL[compare.out$isclonePLINK == TRUE])
```

```
##
## FALSE  TRUE
## 19367 15876
```

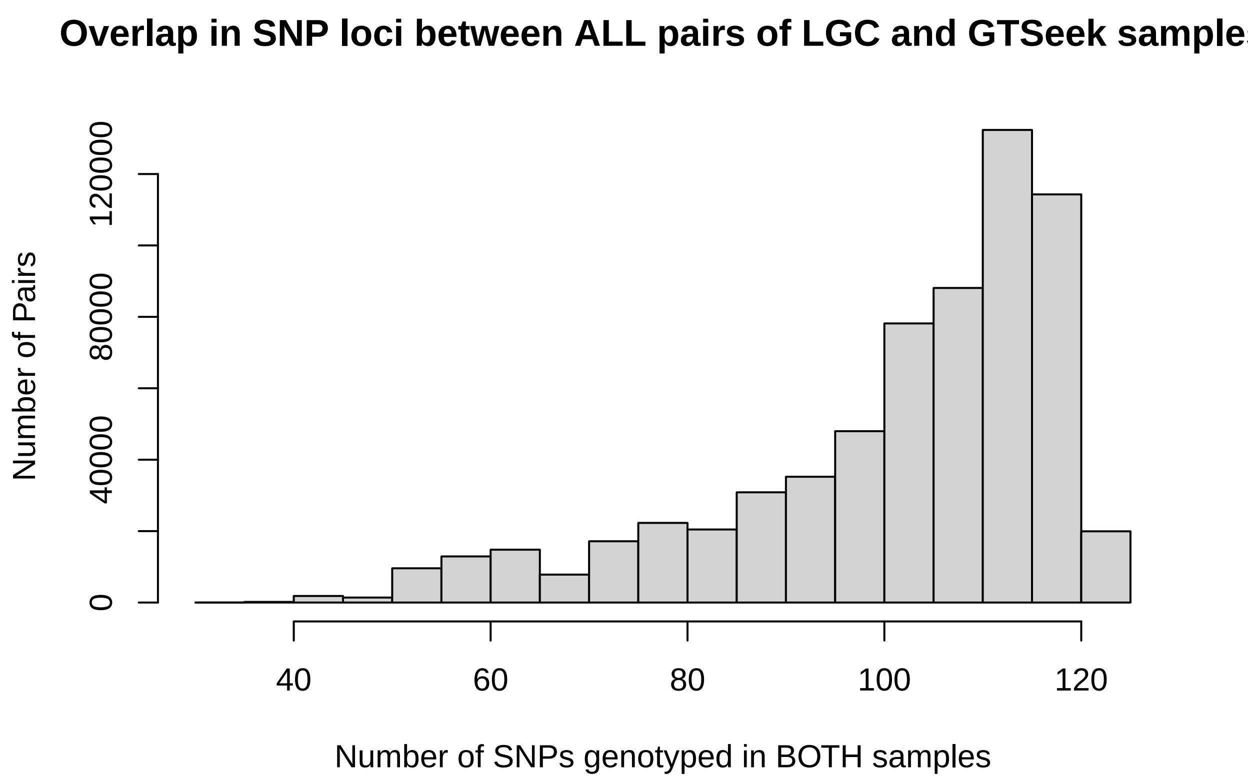
From a total of 1365378 pairwise comparisons, `COLONY` identifies 15876 matches, while `PLINK` identifies 35243 matches. There are 19367 comparisons where `PLINK` finds a match but `COLONY` does not and 0 comparisons where `COLONY` finds a match but `PLINK` does not. In other words 100% of `COLONY` matches are confirmed by `PLINK`, while 45% of `PLINK` matches are confirmed by `COLONY`.

Let's look more closely at some of the discrepancies between the two analyses. First those from the `COLONY` analysis *only*..

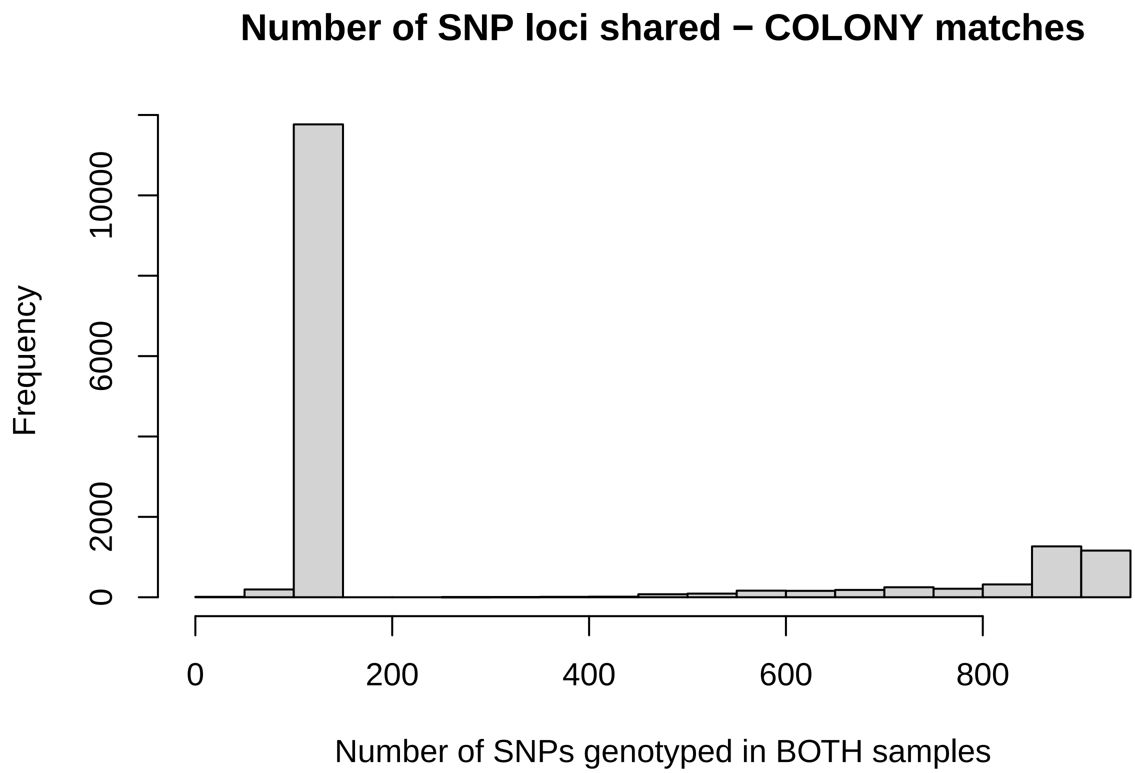
```
compare.out[which(compare.out$iscloneCOL == TRUE & compare.out$isclonePLINK == FALSE),]
```

```
## [1] id1          id2          kinship      nsnp         colclone1
## [6] colclone2    isclonePLINK iscloneCOL
## <0 rows> (or 0-length row.names)
```

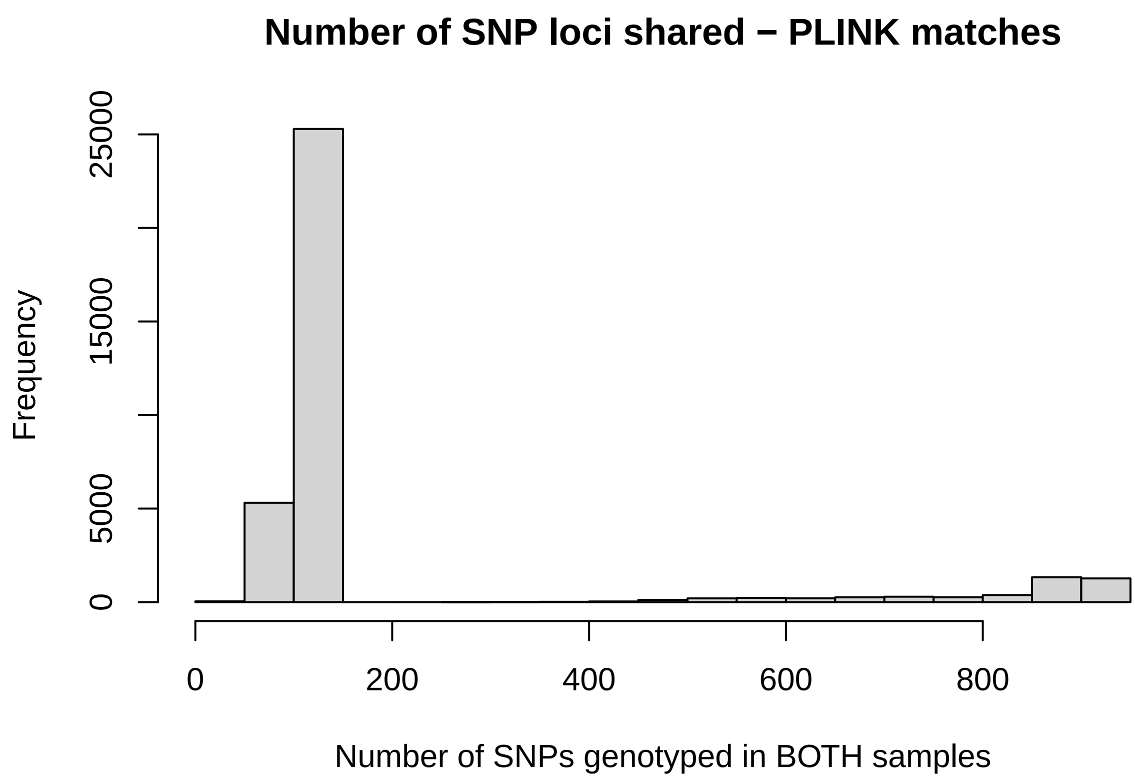
```r
hist(compare.out$nsnp[which(compare.out$id1 %in% smd$vcfID[which(smd$GTservice == "GTSeek")] &
                            compare.out$id2 %in% smd$vcfID[which(smd$GTservice == "LGC")] |
                            compare.out$id1 %in% smd$vcfID[which(smd$GTservice == "LGC")] &
                            compare.out$id2 %in% smd$vcfID[which(smd$GTservice == "GTSeek")])],
     main = "Overlap in SNP loci between ALL pairs of LGC and GTSeek samples",
     xlab = "Number of SNPs genotyped in BOTH samples", ylab = "Number of Pairs")
```




```r
hist(compare.out$nsnp[which(compare.out$iscloneCOL == TRUE)],
     main = "Number of SNP loci shared - COLONY matches",
     xlab = "Number of SNPs genotyped in BOTH samples")
```




```r
hist(compare.out$nsnp[which(compare.out$iscl onePLINK == TRUE)],
     main = "Number of SNP loci shared - PLINK matches",
     xlab = "Number of SNPs genotyped in BOTH samples")
```




```r
#Range of shared SNPs for COLONY matches b/w GTSeek and LGC
colMatches <- compare.out[which(compare.out$iscloneCOL==TRUE),]
range(colMatches$nsnp[which(colMatches$id1 %in% smd$vcfID[which(smd$GTservice == "GTSeek")] &
                            colMatches$id2 %in% smd$vcfID[which(smd$GTservice == "LGC")] |
                            colMatches$id1 %in% smd$vcfID[which(smd$GTservice == "LGC")] &
                            colMatches$id2 %in% smd$vcfID[which(smd$GTservice == "GTSeek")])])
```

```
## [1]  41 122
```

```r
#Same for PLINK
plinkMatches <- compare.out[which(compare.out$isclonePLINK == TRUE),]
range(plinkMatches$nsnp[which(plinkMatches$id1 %in% smd$vcfID[which(smd$GTservice == "GTSeek")] &
                            plinkMatches$id2 %in% smd$vcfID[which(smd$GTservice == "LGC")] |
                            plinkMatches$id1 %in% smd$vcfID[which(smd$GTservice == "LGC")] &
                            plinkMatches$id2 %in% smd$vcfID[which(smd$GTservice == "GTSeek")])])
```

```
## [1]  40 125
```

The filters for LGC samples (at least 50% of SNPs genotyped) keep us from matching samples based on *very* few loci.

Final Recapture Calls

Now to combine the output from the two programs using the scheme described above. `COLONY` must be missing a few genotype matches. For each `COLONY` clone, are there samples that would have been combined by PLINK? If so, are *all* members of those clones also identified by PLINK as recaptures of the focal group? What is the mean kinship coefficient estimated between the samples associated with the two COLONY clones?

Below, we set two thresholds for re-assigning a sample to a different COLONY clone. These include:

i. Mean kinship to samples in the recipient clone ≥ 0.45 (based on range of kinship estimates for technical replicates)
ii. At least one of the comparisons to samples in the other clone needs to be based on a minimum of 50 SNP loci (this number will show up again below)

Now to reassign samples to clones...

```
## Set up the object
totClones <- max(c(compare.out$colclone1, compare.out$colclone2))
finalCloneDF <- data.frame(initial = c(1:totClones),
                           init.size = NA,
                           prob = NA,
                           final = c(1:totClones),
                           possibles = NA,
                           is.moved = FALSE,
                           is.recip = FALSE)

## Fill in init.size & prob
for(i in 1:totClones) {
  tmpclone <- strsplit(col.out[which(col.out$CloneIndex == i),3], ",")[[1]]
  finalCloneDF$init.size[i] <- length(tmpclone)
  finalCloneDF$prob[i] <- col.out$Prob[which(col.out$CloneIndex == i)]
  rm(tmpclone)
}

## Loop over the clones and identify all possible matches
for(i in 1:totClones) {
  tmp <- compare.out[which(compare.out$colclone1 == i & compare.out$colclone2 != i |
                             compare.out$colclone2 == i & compare.out$colclone1 != i),]
  ## If no other clones are a possible match for the focal clone
  ##  set the final clone ID, it doesn't move, and it's not a recipient
  if(sum(tmp$isclonePLINK) == 0) {
    finalCloneDF$final[i] <- i
  } else {
    ## If some clones do match...
    ##  identify the set of clones that match the focal clone
    tmp.match <- unique(c(tmp$colclone1[tmp$isclonePLINK==TRUE],
                          tmp$colclone2[tmp$isclonePLINK==TRUE]))
    tmp.match <- tmp.match[-which(tmp.match == i)]
    ##  then continue adding possible matches to all of these other clones
    ##  do this until the list of possibilities stops growing!!
    growing <- TRUE
    while(growing) {
      tmp.tmp.match <- unique(
        c(compare.out$colclone1[which(compare.out$colclone1 %in% tmp.match &
                                        compare.out$colclone2 %in% tmp.match &
                                        compare.out$iscloneCOL == FALSE &
                                        compare.out$isclonePLINK == TRUE)],
          compare.out$colclone2[which(compare.out$colclone1 %in% tmp.match &
```

```
                                        compare.out$colclone2 %in% tmp.match &
                                        compare.out$iscloneCOL == FALSE &
                                        compare.out$isclonePLINK == TRUE)]))
    tmp.tmp.match <- tmp.tmp.match[-which(tmp.tmp.match == i)]
    if(length(tmp.tmp.match) > length(tmp.match)) {
      tmp.match <- tmp.tmp.match
    } else {
      growing <- FALSE
    }
    rm(tmp.tmp.match)
  }

  #Record the possible matches...
  finalCloneDF$possibles[i] <- paste0(tmp.match, collapse="_")

  ## If any of these clones have been moved to a new clone...
  ##  replace them in the tmp.match list with their new clone
  for(j in tmp.match) {
    if(finalCloneDF$final[which(finalCloneDF$initial == j)] != j) {
      tmp.match[which(tmp.match == j)] <- finalCloneDF$final[which(finalCloneDF$initial == j)]
    }
  }
  tmp.match <- unique(tmp.match[which(tmp.match != i)])
  ## This means we'll need to refer to the final column when calculating
  ##  mean kinship values for the pairs below...

  ## Now check each of these to ask if it should be moved...
  real.matches <- i #The focal clone, which can not have moved yet...

  for(j in tmp.match) {
    i.set <- unique(c(
      real.matches, #The focal clone plus any earlier in the list that matched
      #Other clones previously moved into the focal clone
      finalCloneDF$initial[which(finalCloneDF$final == i)]
    ))
    j.set <- unique(c(
      j, #The possible match
      #If j has moved, it's final clone
      finalCloneDF$final[which(finalCloneDF$initial == j)],
      #Other clones that previously moved into j's final clone
      finalCloneDF$initial[which(finalCloneDF$final ==
                                  finalCloneDF$final[which(finalCloneDF$initial ==
                                                            j)])]
    ))

    subtmp <- compare.out[which(compare.out$colclone1 %in% i.set &
                                  compare.out$colclone2 %in% j.set |
                                compare.out$colclone1 %in% j.set &
                                  compare.out$colclone2 %in% i.set),]

    ## Focus on comparisons involving the possible mover - j
    #subtmp <- subtmp[which(subtmp$colclone1 == j | subtmp$colclone2 == j),]
```

```r
      ## Check mean kinship
      if(mean(subtmp$kinship) >= 0.45 & max(subtmp$nsnp) >= 50) {
        real.matches <- c(real.matches, j)
      }
      rm(subtmp, i.set, j.set)
    }

    ## Now if there are real matches, figure out where to put the set of samples
    if(length(real.matches) > 1) {
      tmp.sizes <- data.frame(finalClone=unique(finalCloneDF$final[real.matches]),
                              finalSize = NA)
      for(r in 1:nrow(tmp.sizes)) {
        tmp.sizes$finalSize[r] <- sum(
          finalCloneDF$init.size[which(finalCloneDF$final ==
                                         tmp.sizes$finalClone[r])]
          )
      }
      bigclone <- tmp.sizes$finalClone[which(tmp.sizes$finalSize ==
                                                max(tmp.sizes$finalSize))]
      if(length(bigclone) > 1) {
        bigclone <- min(bigclone)
      }

      finalCloneDF$final[real.matches] <- bigclone

      rm(tmp.sizes, bigclone)
    }
    rm(tmp.match)
  }
}

#Did clone move? Was it a recipient of movers?
for(i in 1:totClones) {
  if(finalCloneDF$initial[i] != finalCloneDF$final[i]) {
    finalCloneDF$is.moved[i] <- TRUE
  }
  if(length(which(finalCloneDF$final==i)) > 1) {
    finalCloneDF$is.recip[i] <- TRUE
  }
}

#Structure of the object
head(finalCloneDF)
```

```
##   initial init.size  prob final                  possibles is.moved is.recip
## 1       1         6 1.000     1                        245    FALSE     TRUE
## 2       2         1 0.221   248  32_199_214_225_244_248_259     TRUE    FALSE
## 3       3        13 0.887     3                    235_240    FALSE     TRUE
## 4       4        43 0.947     4 6_64_67_205_217_221_242_263    FALSE     TRUE
## 5       5         3 1.000     5                        307    FALSE    FALSE
## 6       6         1 1.000     6 4_64_67_159_195_209_242_263    FALSE    FALSE
```

```r
#How many clones total from COLONY
nrow(finalCloneDF)
```

```
## [1] 356
```

```r
#How many survive?
length(unique(finalCloneDF$final))
```

```
## [1] 298
```

```r
#Difference
length(unique(finalCloneDF$final)) - nrow(finalCloneDF)
```

```
## [1] -58
```

```r
#How many moved?
sum(finalCloneDF$initial != finalCloneDF$final)
```

```
## [1] 58
```

```r
sum(finalCloneDF$is.moved)
```

```
## [1] 58
```

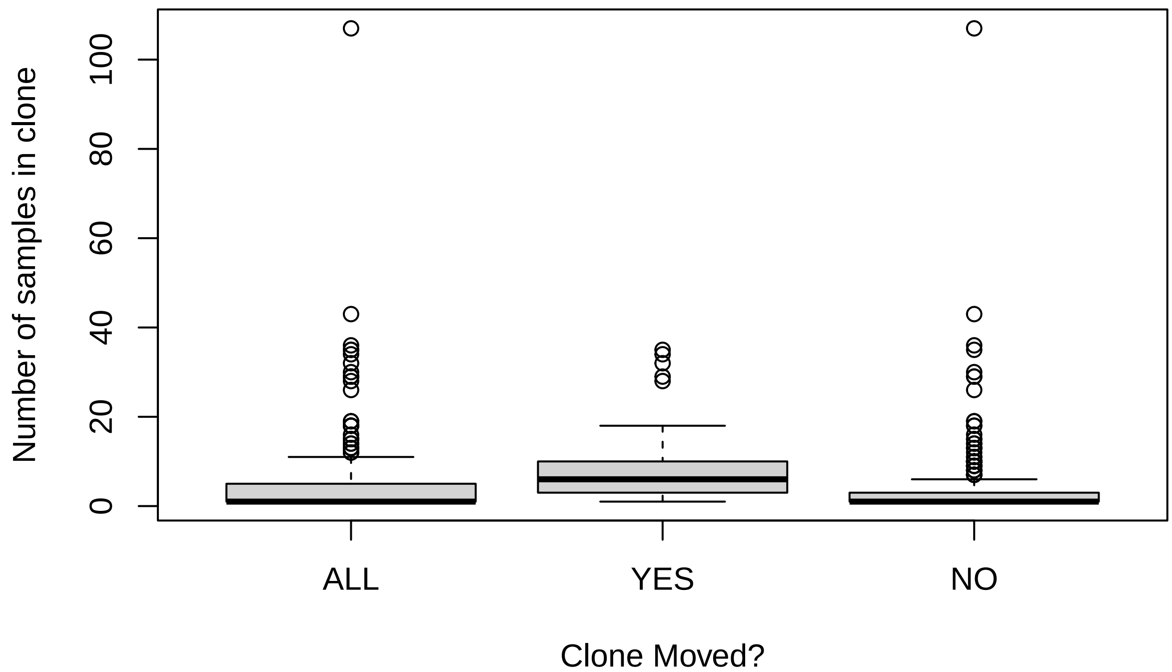
```r
#Into how many recipients?
sum(finalCloneDF$is.recip)
```

```
## [1] 37
```

```r
#Shouldn't be any recipients that were moved
table(finalCloneDF$is.moved[which(finalCloneDF$is.recip == TRUE)])
```

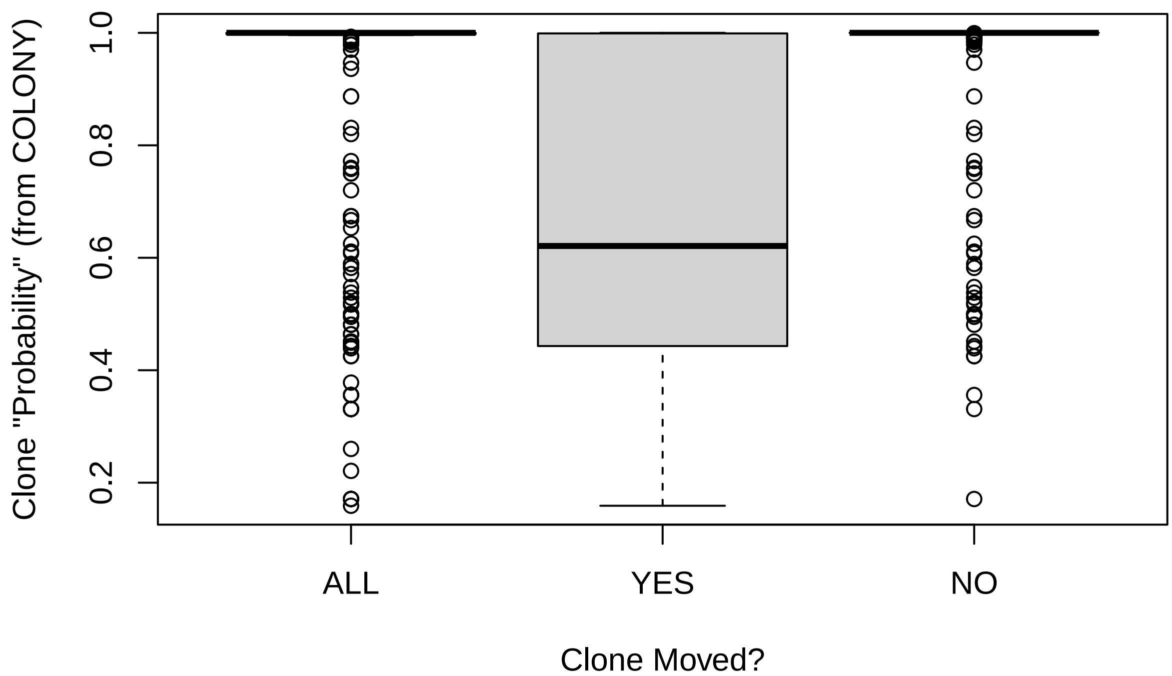
```
##
## FALSE
##    37
```

```r
#Did clones reassigned to other clones tend to be sparse?
boxplot(finalCloneDF$init.size,
        finalCloneDF$init.size[finalCloneDF$is.moved],
        finalCloneDF$init.size[!finalCloneDF$is.moved],
        names = c("ALL","YES","NO"),
        xlab = "Clone Moved?",
        ylab = "Number of samples in clone")
```




```
#It doesn't appear that they were

#Did clones that were moved tend to have lower probabilities from COLONY?
boxplot(finalCloneDF$prob,
        finalCloneDF$prob[finalCloneDF$is.moved],
        finalCloneDF$prob[!finalCloneDF$is.moved],
        names = c("ALL","YES","NO"),
        xlab = "Clone Moved?",
        ylab = 'Clone "Probability" (from COLONY)')
```




```
#Yes, COLONY tended to be less confident about clones that were reassigned...
```

Interested to look more closely at clones that were moved by this combination. In those cases, there should be more than one row of the dataset (initial `COLONY` clone) that gets assigned to a particular final clone.

```
#37 clones were the recipients of other clones
table(finalCloneDF$final)
```

```
##
##   1   3   4   5   6   7   8  11  14  15  17  18  19  20  21  22  24  29  33  34
##   2   2   5   1   1   3   2   1   3   1   1   1   1   1   1   1   1   1   5   1
##  35  37  38  40  41  42  43  44  45  47  48  49  50  51  52  53  54  56  57  58
##   1   1   1   1   1   1   2   1   1   1   1   1   1   1   1   1   1   1   1   1
##  59  61  63  65  66  67  68  69  70  71  72  73  74  76  77  78  79  80  81  82
##   1   1   1   1   1   1   2   1   1   1   1   1   1   1   1   1   1   1   1   1
##  83  84  85  86  87  88  90  91  92  94  95  97  98  99 101 102 103 104 105 106
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 146 147
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 168 169 170 172 173 175 176 177 179 180 181 182 183 184 185 186 187 188 189 190
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
```

```
## 191 192 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 212 213 214 215 216 218 219 220 221 222 223 224 226 227 229 231 232 233 234 235
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 236 237 239 241 243 246 247 248 249 251 252 254 256 257 260 262 265 266 267 268
##   1   1   2   2   3   1   2   6   1   2   1   2   2   4   2   2   2   2   2   2
## 270 272 273 274 277 278 279 280 283 284 285 286 287 288 290 292 293 294 295 296
##   2   2   1   4   3   3   1   1   2   3   2   1   1   1   3   1   2   1   2   1
## 297 298 299 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318
##   1   1   1   2   1   1   1   1   1   2   1   1   1   1   1   1   1   1   1   1
## 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
## 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1
```

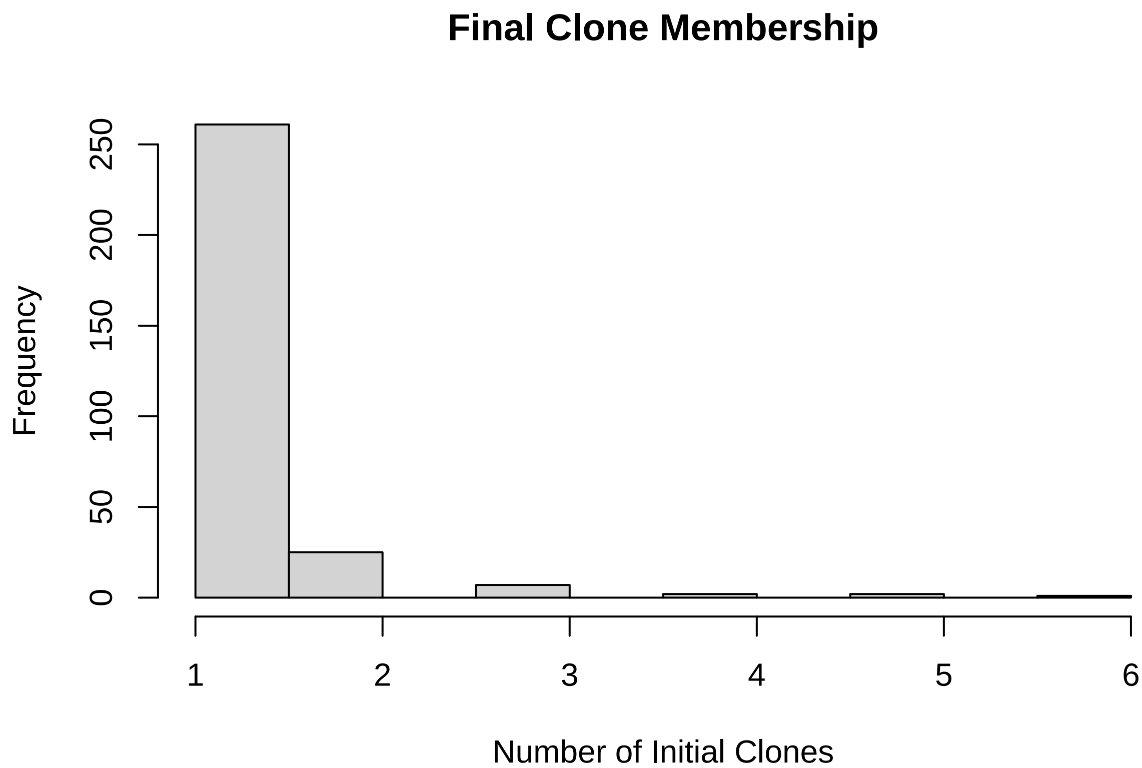
```r
sum(table(finalCloneDF$final) > 1)
```

```
## [1] 37
```

```r
names(which(table(finalCloneDF$final) > 1))
```

```
##  [1] "1"   "3"   "4"   "7"   "8"   "14"  "33"  "43"  "68"  "239" "241" "243"
## [13] "247" "248" "251" "254" "256" "257" "260" "262" "265" "266" "267" "268"
## [25] "270" "272" "274" "277" "278" "283" "284" "285" "290" "293" "295" "302"
## [37] "308"
```

```r
hist(table(finalCloneDF$final), main = "Final Clone Membership",
     xlab = "Number of Initial Clones")
```




```r
#Spot checking moved clones...
#Final Clone 4 - 4 other clones moved into clone 4
cloneInQ <- 4
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

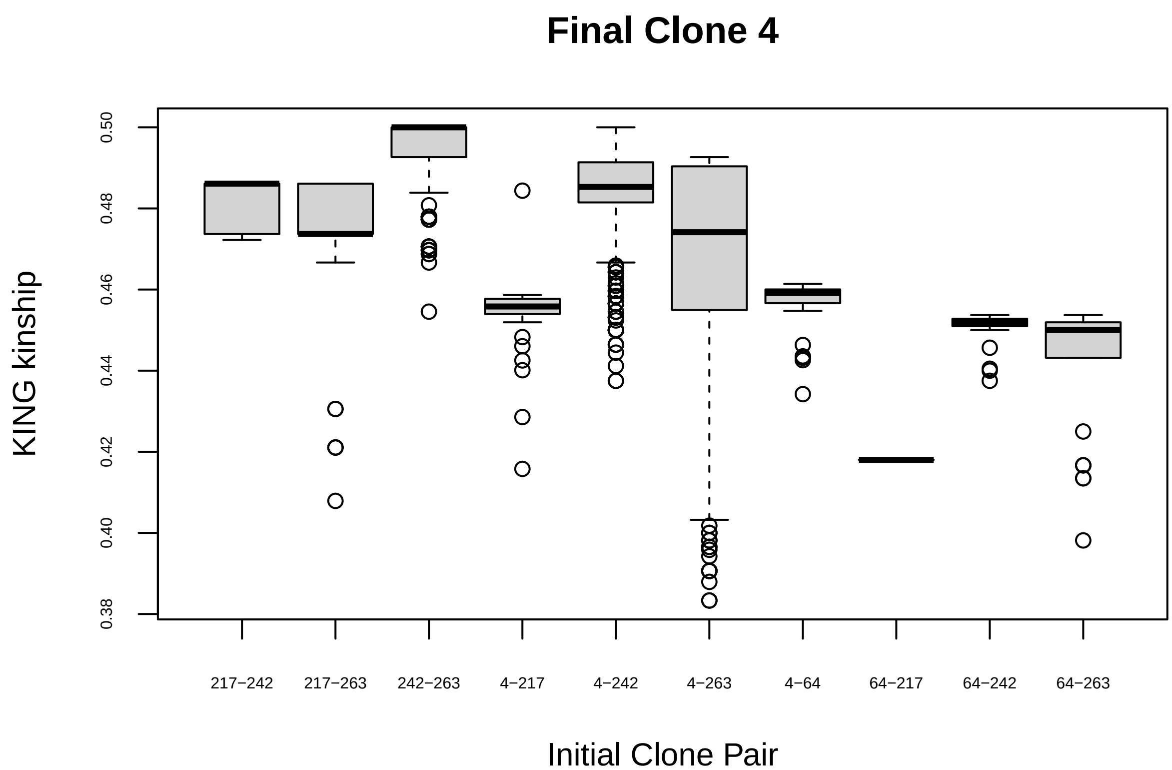
```
##     initial init.size  prob final                      possibles is.moved is.recip
## 4         4        43 0.947     4 6_64_67_205_217_221_242_263    FALSE     TRUE
## 64       64         1 0.571     4  4_67_205_217_221_242_263_6     TRUE    FALSE
## 217     217         1 0.260     4        221_4_242_263_64_205     TRUE    FALSE
## 242     242        35 0.464     4   263_4_6_64_67_205_217_221     TRUE    FALSE
## 263     263        29 0.464     4   242_4_64_67_205_217_221_6     TRUE    FALSE
```

```r
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final                      possibles is.moved is.recip
## 4         4        43 0.947     4 6_64_67_205_217_221_242_263    FALSE     TRUE
## 6         6         1 1.000     6 4_64_67_159_195_209_242_263    FALSE    FALSE
## 64       64         1 0.571     4  4_67_205_217_221_242_263_6     TRUE    FALSE
## 67       67         1 1.000    67  4_159_195_209_242_263_6_64    FALSE    FALSE
## 205     205         1 0.611   205        4_217_221_242_263_64    FALSE    FALSE
## 217     217         1 0.260     4        221_4_242_263_64_205     TRUE    FALSE
## 221     221         1 0.451   221        4_242_263_64_205_217    FALSE    FALSE
## 242     242        35 0.464     4   263_4_6_64_67_205_217_221     TRUE    FALSE
```

```
## 263      263         29 0.464      4    242_4_64_67_205_217_221_6      TRUE     FALSE
```

```
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```



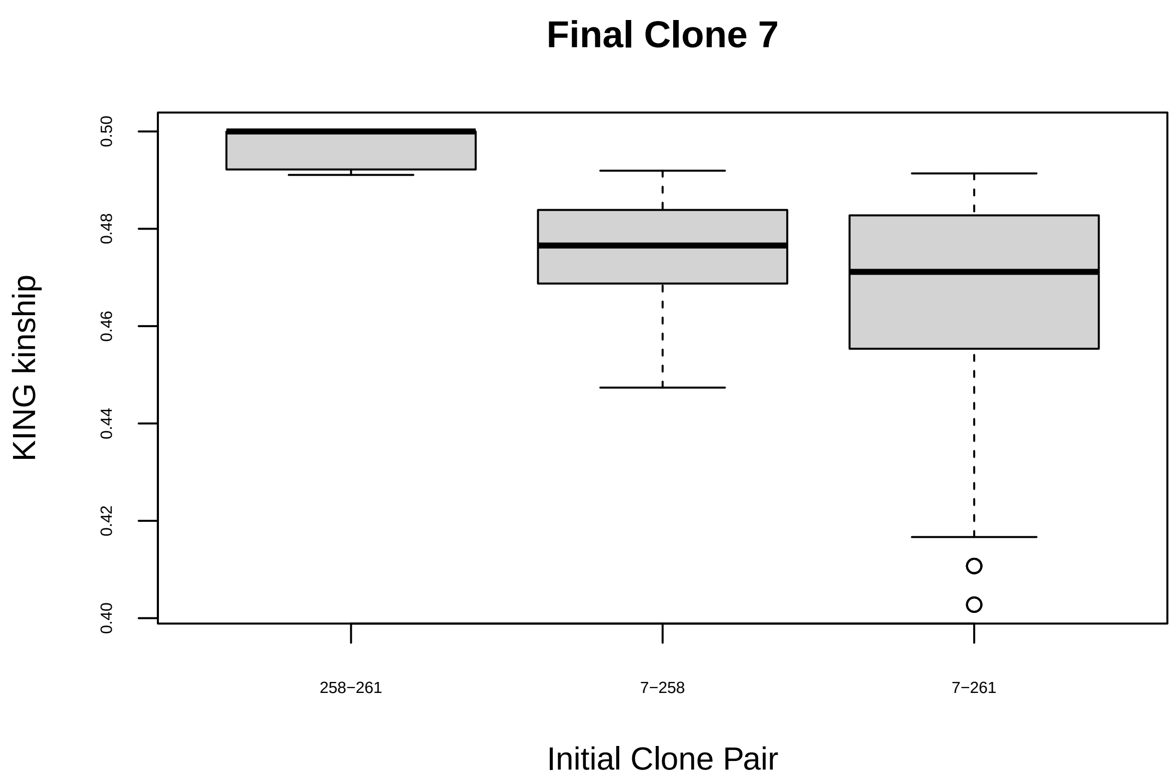

```
#Final Clone 7 - 2 other clones moved into clone 7
cloneInQ <- 7
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

```
##     initial init.size  prob final possibles is.moved is.recip
## 7         7        29 0.991     7   258_261    FALSE     TRUE
## 258     258         9 0.991     7     261_7     TRUE    FALSE
## 261     261        15 0.991     7     258_7     TRUE    FALSE
```

```r
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final possibles is.moved is.recip
## 7         7        29 0.991     7   258_261    FALSE     TRUE
## 258     258         9 0.991     7     261_7     TRUE    FALSE
## 261     261        15 0.991     7     258_7     TRUE    FALSE
```

```r
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```



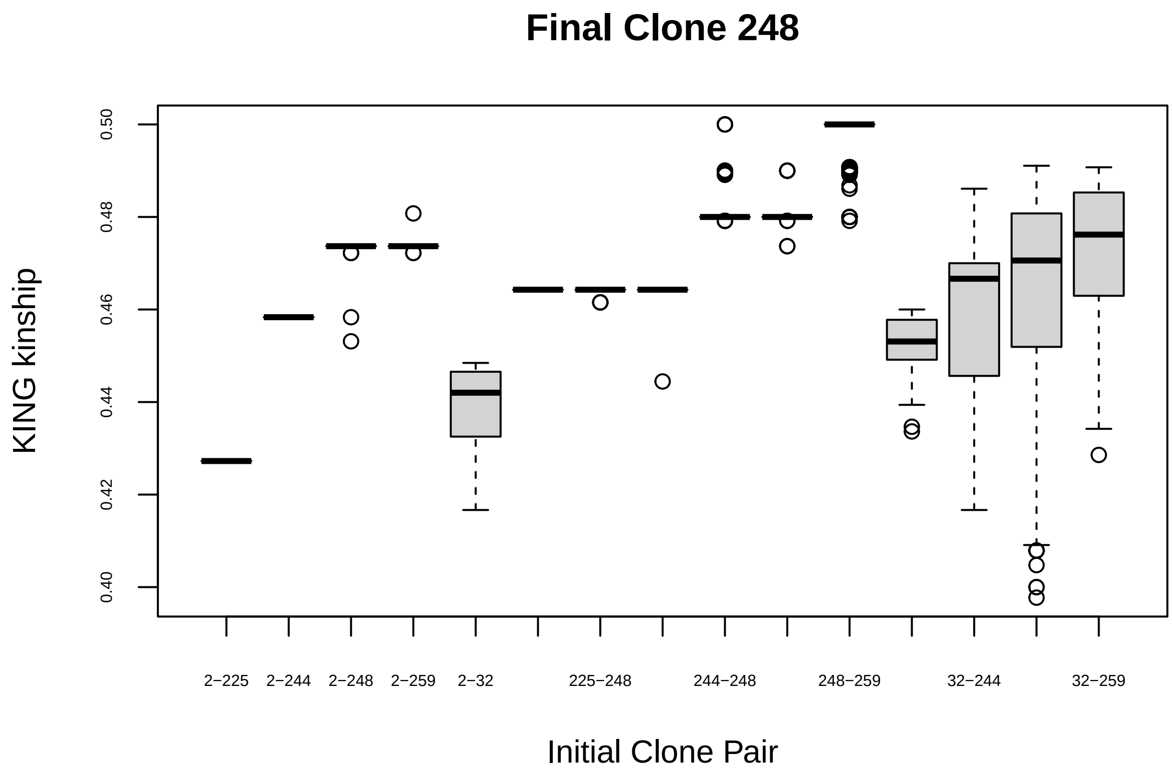

```r
#Final Clone 248 - 5 other clones moved into clone 248
cloneInQ <- 248
finalCloneDF[which(finalCloneDF$final == cloneInQ),]
```

```
##     initial init.size  prob final                  possibles is.moved is.recip
## 2         2         1 0.221   248 32_199_214_225_244_248_259     TRUE    FALSE
## 32       32        34 0.356   248  199_214_225_244_248_259_2     TRUE    FALSE
## 225     225         1 0.451   248   32_244_248_259_2_199_214     TRUE    FALSE
## 244     244         2 0.356   248   248_259_2_32_199_214_225     TRUE    FALSE
## 248     248        36 0.356   248   259_244_2_32_199_214_225    FALSE     TRUE
## 259     259        18 0.356   248   248_244_2_32_199_214_225     TRUE    FALSE
```

```
finalCloneDF[finalCloneDF$initial %in%
               unique(unlist(strsplit(
                 finalCloneDF$possibles[finalCloneDF$final == cloneInQ],"_"))),]
```

```
##     initial init.size  prob final                  possibles is.moved is.recip
## 2         2         1 0.221   248 32_199_214_225_244_248_259     TRUE    FALSE
## 32       32        34 0.356   248  199_214_225_244_248_259_2     TRUE    FALSE
## 199     199         1 0.987   199      32_225_244_248_259_2    FALSE    FALSE
## 214     214         1 0.495   214      225_32_244_248_259_2    FALSE    FALSE
## 225     225         1 0.451   248   32_244_248_259_2_199_214     TRUE    FALSE
## 244     244         2 0.356   248   248_259_2_32_199_214_225     TRUE    FALSE
## 248     248        36 0.356   248   259_244_2_32_199_214_225    FALSE     TRUE
## 259     259        18 0.356   248   248_244_2_32_199_214_225     TRUE    FALSE
```

```
tmp.comp <- compare.out[which(
  compare.out$colclone1 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone2 %in%
    finalCloneDF$initial[which(finalCloneDF$final == cloneInQ)] &
    compare.out$colclone1 != compare.out$colclone2),]
boxlabs <- apply(tmp.comp[,c("colclone1","colclone2")], 1,
                 FUN=function(x) paste(sort(x), collapse = "-"))
boxplot(tmp.comp$kinship ~ boxlabs,
        ylab = "KING kinship", xlab = "Initial Clone Pair",
        main = paste("Final Clone",cloneInQ), cex.axis = 0.5)
```



Overall this looks pretty reasonble. There were some low-kinship outliers within some of these clones, though.

Another check, do we ever move samples into a clone that has already been removed from the population? In other words, clone B is identified as a recapture of clone A and moved accordingly. Subsequently, clone C is identified as a recapture of clone B (but not clone A) and moved into an "empty" clone. Does this happen?

```
#Any cases where a clone gets emptied out (moved to another, larger clone)
# but then another clone is assigned to the emptied clone?

#This should show up as rows where initial != final, but is.recip = TRUE
which(finalCloneDF$is.recip == TRUE &
        finalCloneDF$initial != finalCloneDF$final)
```

```
## integer(0)
```

```
# None... good!
```

Check for Odd Patterns in the FINAL clone assignments

Want to revisit the checks above, given our new clone assignments.

```
#Add COLclone to smd
smd$COLclone <- NA
smd$finalClone <- NA
```

```r
for(i in 1:nrow(smd)){
  thisclone <- col.out$CloneIndex[grep(smd$vcfID[i], col.out[,3])]

  if(length(thisclone) > 1) {
    this.this <- which(sapply(strsplit(col.out[thisclone,3],","), FUN=function(x) sum(x==smd$vcfID[i]))
    thisclone <- thisclone[this.this]
    rm(this.this)
  }

  if(length(thisclone) > 0) {
    smd$COLclone[i] <- thisclone
    smd$finalClone[i] <- finalCloneDF$final[finalCloneDF$initial == smd$COLclone[i]]
  }
  rm(thisclone)

  if(smd$FILT[i] == TRUE) {
    smd$finalClone[i] <- NA
  }
}
#Samples without a clone ID were those that were filtered from the dataset
sum(is.na(smd$finalClone))
```

```
## [1] 556
```

```r
sum(smd$FILT)
```

```
## [1] 556
```

```r
length(which(is.na(smd$finalClone) & smd$FILT))
```

```
## [1] 556
```

```r
nSex2 <- nSexGT2 <- nRegion2 <- c()
for(i in 1:max(smd$finalClone, na.rm = TRUE)) {
  thisclone <- which(smd$finalClone == i)
  if(length(thisclone) == 0) {
    nSex2[i] <- NA
    nSexGT2[i] <- NA
    nRegion2[i] <- NA
  } else {
    tmp <- smd[thisclone,]
    nSex2[i] <- length(unique(tmp$knownSEX[tmp$knownSEX != "" & !is.na(tmp$knownSEX)]))
    nSexGT2[i] <- length(unique(tmp$sexGT[tmp$sexGT != "undet" & !is.na(tmp$sexGT)]))
    nRegion2[i] <- length(unique(tmp$region))
    rm(tmp)
  }
}
```

Replicated samples

We don't want to have any replicates that are assigned to different final clones!

```r
repIDs <- unique(sub("R","",smd$ugaID[which(smd$is.rep == TRUE)]))
check.rep.clone <- data.frame(ugaID = repIDs,
                              clone1 = NA,
                              clone2 = NA)
for(i in 1:length(repIDs)) {
  tmp.match <- c(repIDs[i], paste0(repIDs[i],"R"))
  check.rep.clone[i,-1] <- smd$finalClone[which(smd$ugaID %in% tmp.match)]
  rm(tmp.match)
}
check.rep.clone
```

```
##       ugaID clone1 clone2
## 1    35-SGA     NA     NA
## 2    121-23     NA     NA
## 3      8-23     68     68
## 4    47-SGA     77     77
## 5      23-C      3      3
## 6    58-SGA     92     92
## 7    306-23     NA     NA
## 8    166-24    257    257
## 9    68-NGA     17     17
## 10    21-23     43     43
## 11   251-23     38     38
## 12   331-23    251    251
## 13   409-23     14     14
## 14     14-C    278     NA
## 15   241-23      1      1
## 16   358-24     33     33
## 17   280-23     NA     NA
## 18    50-23      4      4
## 19   358-23     37     37
## 20     5-24    248    248
## 21   134-24     NA    248
## 22   219-24     29     29
## 23   277-24    295    295
## 24   397-24     24     24
## 25   423-24    267    267
## 26   434-24     NA     NA
## 27      1-C     20     20
## 28     17-C    128    128
## 29   227-23     NA     NA
## 30   389-24    278    278
## 31   306-24    257    257
## 32   930-23    239    239
## 33   564-24     24     24
## 34   781-23     14     14
## 35   798-23    278    278
## 36   651-23     14     14
## 37   517-24     NA     NA
## 38   716-23     NA     NA
## 39   726-24     NA     NA
## 40   980-24    290    290
## 41   935-23     33     33
```

```
## 42 1306-24    302    302
## 43  567-24     24     24
## 44  881-24    274    274
## 45 2972-24    278    278
## 46 1384-24     NA     NA
## 47  933-24    131    131
## 48 1522-24    262    262
## 49 3058-24    302    302
## 50 2665-24    344    344
## 51 2869-24    257    257
## 52 2926-24     NA     NA
## 53  151-24     33     33
## 54  396-24    257     NA
## 55  424-23     NA     NA
## 56  191-24     NA     NA
## 57  466-23     NA     NA
## 58  286-23     NA     NA
## 59  500-24    302    302
## 60  311-23     NA     NA
```

```r
noNAs <- check.rep.clone[which(rowSums(is.na(check.rep.clone))==0),]
nrow(noNAs)
```

```
## [1] 41
```

```r
sum(noNAs$clone1 == noNAs$clone2)
```

```
## [1] 41
```

Good! We don't assign any replicated samples to separate clones!

Known Sex

As above, we should not have multiple known sexes per clone.

```r
nSex2
```

```
##   [1]  0 NA  1  1  1  0  0  0 NA NA  1 NA NA  0  1 NA  1  1  1  1  1  0 NA  0 NA
##  [26] NA NA NA  0 NA NA NA  0  0  0 NA  0  0 NA  0  0  0  0  1  1 NA  1  1  0  1
##  [51]  0  1  1  1 NA  1  0  1  1 NA  1 NA  0 NA  1  1  0  1  1  1  1 NA  1  1 NA
##  [76]  1  1  1  1  1  0  1  1  1  1  1  1  1 NA  1  1  1 NA  1  1 NA  1  1  0 NA
## [101]  0  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1
## [126]  0  0  1  1  1  1  0  0  1  1  1  1  1  1  1  1  1  0  0 NA  1  1  1  0  0
## [151]  1  1  1  0  1  0  1  1  0  0  1  1  1  0  0  1  0  1  1  0 NA  0  0 NA  1
## [176]  0  1 NA  1  0  0  1  1  1 NA  0  1  0  0  0  1  0 NA  0  0  0  0  0  0  0
## [201]  0  0  1  0  0  0  0  0  0  1  0  0  0  0  0  0 NA  0  1  0  0  0  1  1 NA
## [226]  0  0 NA  0 NA  0  0  0  0  0  0  0 NA  0 NA  0 NA  0 NA NA  0  1  1  0 NA
## [251]  0  0 NA  1 NA  0  1 NA NA  0 NA  0 NA NA  0  0  1  1 NA  1 NA  0  0  0 NA
## [276] NA  0  1  0  0 NA NA  1  1  1  0  0  0 NA  0 NA  0  0  0  0  0  0  0  0 NA
## [301] NA  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [326]  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## [351]  0  0  0  0  0  0
```

```r
sum(!is.na(nSex2))
```

```
## [1] 295
```

```r
max(nSex2, na.rm = TRUE)
```

```
## [1] 1
```

Genotype-inferred Sex

And again, we would hope that there were relatively few cases of multiple genotype-inferred sexes within clones.

```r
nSexGT2
```

```
##   [1]  1 NA  1  1  1  1  1  2 NA NA  1 NA NA  2  1 NA  1  1  1  1  1  1 NA  1 NA
##  [26] NA NA NA  1 NA NA NA  2  1  1 NA  1  1 NA  1  1  1  1  1  2 NA  1  1  1  1
##  [51]  1  1  1  1 NA  1  0  1  1 NA  1 NA  1 NA  1  1  1  1  1  1  1 NA  1  1 NA
##  [76]  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1 NA  1  1 NA  1  1  1 NA
## [101]  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1
## [126]  1  1  1  1  1  1  0  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1
## [151]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1 NA  1
## [176]  1  1 NA  1  1  1  1  1  1 NA  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1
## [201]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  0  1  1  0 NA
## [226]  1  1 NA  1 NA  1  0  1  1  1  1  1 NA  1 NA  1 NA  1 NA NA  1  1  1  1 NA
## [251]  1  1 NA  1 NA  1  1 NA NA  1 NA  1 NA NA  1  1  1  1 NA  1 NA  1  1  1 NA
## [276] NA  1  1  1  1 NA NA  1  2  1  1  1  1 NA  1 NA  1  1  1  1  1  1  1  1 NA
## [301] NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [326]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [351]  1  1  1  1  1  1
```

```r
sum(!is.na(nSexGT2))
```

```
## [1] 295
```

```r
max(nSexGT2, na.rm = TRUE)
```

```
## [1] 2
```

```r
which(nSexGT2 > 1)
```

```
## [1]   8  14  33  45 284
```

```r
#They do in a few cases...
#8, 14, 33, 45, 284

#All of these were noted previously...
#Clone 8 - 17 M 1 F, no direct captures
table(smd$sexGT[smd$finalClone == 8])
```

```
##
##  F  M
##  1 17
```

```r
smd[which(smd$finalClone == 8),]
```

```
##                                     extID   ugaID is.SCR is.rep week year
## 258                  UGA_173101_P017_WF06   30-23   TRUE  FALSE    6 2023
## 356                  UGA_173101_P018_WF08   61-24   TRUE  FALSE    7 2024
## 365                  UGA_173101_P018_WG05  139-24   TRUE  FALSE    7 2024
## 464                  UGA_173101_P020_WA02  135-24   TRUE  FALSE    7 2024
## 480                  UGA_173101_P020_WB06  481-24   TRUE  FALSE    7 2024
## 526                  UGA_173101_P020_WF04  482-24   TRUE  FALSE    7 2024
## 725                  UGA_173101_P022_WF11  257-24   TRUE  FALSE    7 2024
## 751                  UGA_173101_P023_WA01  285-24   TRUE  FALSE    4 2024
## 825                  UGA_173101_P023_WG03  326-24   TRUE  FALSE    4 2024
## 828     GTseek_024_384i5_001_P098.001_5-23    5-23   TRUE  FALSE    6 2023
## 1213  GTseek_025_384i5_003_P098.005_738-24  738-24   TRUE  FALSE    7 2024
## 1451 GTseek_025_384i5_241_P098.006_2962-24 2962-24   TRUE  FALSE    4 2024
## 1615 GTseek_026_384i5_025_P098.010_2748-24 2748-24   TRUE  FALSE    7 2024
## 1715 GTseek_026_384i5_163_P098.009_2320-24 2320-24   TRUE  FALSE    7 2024
## 1760 GTseek_026_384i5_253_P098.010_2864-24 2864-24   TRUE  FALSE    7 2024
## 1792 GTseek_026_384i5_317_P098.010_2894-24 2894-24   TRUE  FALSE    4 2024
## 1816 GTseek_026_384i5_365_P098.009_2729-24 2729-24   TRUE  FALSE    7 2024
## 2095 GTseek_028_384i5_270_P098.UGA3_260-24  260-24   TRUE  FALSE    4 2024
##       snare       utmSnare
## 258  HS 012 258412x3586648
## 356  HS 012 258412x3586648
## 365  HS 012 258412x3586648
## 464  HS 008 256165x3589588
## 480  HS 012 258412x3586648
## 526  HS 008 256165x3589588
## 725  HS 012 258412x3586648
## 751  HS 001 248187x3586614
## 825  HS 001 248187x3586614
## 828  HS 012 258412x3586648
## 1213 HS 001 248187x3586614
## 1451 HS 001 248187x3586614
## 1615 HS 008 256165x3589588
## 1715 HS 008 256165x3589588
## 1760 HS 012 258412x3586648
## 1792 HS 001 248187x3586614
## 1816 HS 012 258412x3586648
## 2095 HS 001 248187x3586614
##                                                                        fqfile
## 258  RAPiD-Genomics_F534_UGA_173101_P017_WF06_i5-512_i7-162_S14995_L003_R1_001.fastq
## 356  RAPiD-Genomics_F534_UGA_173101_P018_WF08_i5-537_i7-164_S15095_L003_R1_001.fastq
## 365  RAPiD-Genomics_F534_UGA_173101_P018_WG05_i5-537_i7-173_S15104_L003_R1_001.fastq
## 464    RAPiD-Genomics_F566_UGA_173101_P020_WA02_i5-505_i7-27_S3892_L001_R1_001.fastq
## 480     RAPiD-Genomics_F566_UGA_173101_P020_WB06_i5-505_i7-6_S3908_L001_R1_001.fastq
## 526    RAPiD-Genomics_F566_UGA_173101_P020_WF04_i5-505_i7-45_S3954_L001_R1_001.fastq
## 725   RAPiD-Genomics_F566_UGA_173101_P022_WF11_i5-1098_i7-51_S7653_L003_R1_001.fastq
## 751   RAPiD-Genomics_F566_UGA_173101_P023_WA01_i5-1099_i7-59_S7679_L003_R1_001.fastq
## 825   RAPiD-Genomics_F566_UGA_173101_P023_WG03_i5-1099_i7-22_S7753_L003_R1_001.fastq
```

```
## 828                                                                                      <NA>
## 1213                                                                                     <NA>
## 1451                                                                                     <NA>
## 1615                                                                                     <NA>
## 1715                                                                                     <NA>
## 1760                                                                                     <NA>
## 1792                                                                                     <NA>
## 1816                                                                                     <NA>
## 2095                                                                                     <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 258            Hair Snare    CGA    913 UGA_173101_P017_WF06 FALSE       LGC
## 356            Hair Snare    CGA    913 UGA_173101_P018_WF08 FALSE       LGC
## 365            Hair Snare    CGA    919 UGA_173101_P018_WG05 FALSE       LGC
## 464            Hair Snare    CGA    902 UGA_173101_P020_WA02 FALSE       LGC
## 480            Hair Snare    CGA    916 UGA_173101_P020_WB06 FALSE       LGC
## 526            Hair Snare    CGA    885 UGA_173101_P020_WF04 FALSE       LGC
## 725            Hair Snare    CGA    909 UGA_173101_P022_WF11 FALSE       LGC
## 751            Hair Snare    CGA    822 UGA_173101_P023_WA01 FALSE       LGC
## 825            Hair Snare    CGA    518 UGA_173101_P023_WG03 FALSE       LGC
## 828      <NA> Hair Sample    CGA    104             GTS_5-23 FALSE    GTSeek
## 1213     <NA> Hair Sample    CGA    119           GTS_738-24 FALSE    GTSeek
## 1451     <NA> Hair Sample    CGA    119          GTS_2962-24 FALSE    GTSeek
## 1615     <NA> Hair Sample    CGA    122          GTS_2748-24 FALSE    GTSeek
## 1715     <NA> Hair Sample    CGA    124          GTS_2320-24 FALSE    GTSeek
## 1760     <NA> Hair Sample    CGA    124          GTS_2864-24 FALSE    GTSeek
## 1792     <NA> Hair Sample    CGA    124          GTS_2894-24 FALSE    GTSeek
## 1816     <NA> Hair Sample    CGA    121          GTS_2729-24 FALSE    GTSeek
## 2095     <NA> Hair Sample    CGA    112           GTS_260-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 258      M        8          8
## 356      M        8          8
## 365      M        8          8
## 464      M        8          8
## 480      M        8          8
## 526      M        8          8
## 725      M        8          8
## 751      M        8          8
## 825      M        8          8
## 828      F      238          8
## 1213     M      238          8
## 1451     M      238          8
## 1615     M      238          8
## 1715     M      238          8
## 1760     M      238          8
## 1792     M      238          8
## 1816     M      238          8
## 2095     M      238          8
```

```r
unique(smd$COLclone[which(smd$finalClone == 8)])
```

```
## [1]   8 238
```

```r
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 8)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 8)] &
              compare.out$iscloneCOL==FALSE,]
```

```
##                 id1                 id2  kinship nsnp colclone1 colclone2
## 217478     GTS_5-23 UGA_173101_P020_WB06 0.434783   97       238         8
## 217537     GTS_5-23 UGA_173101_P018_WG05 0.437500  100       238         8
## 217541     GTS_5-23 UGA_173101_P018_WF08 0.427083   97       238         8
## 217716     GTS_5-23 UGA_173101_P017_WF06 0.423913   96       238         8
## 217817     GTS_5-23 UGA_173101_P020_WA02 0.443182   87       238         8
## 217881     GTS_5-23 UGA_173101_P023_WG03 0.473684   50       238         8
## 217899     GTS_5-23 UGA_173101_P023_WA01 0.416667   68       238         8
## 217912     GTS_5-23 UGA_173101_P022_WF11 0.420455   91       238         8
## 218041     GTS_5-23 UGA_173101_P020_WF04 0.431818   91       238         8
## 455543   GTS_738-24 UGA_173101_P020_WB06 0.491667  112       238         8
## 455602   GTS_738-24 UGA_173101_P018_WG05 0.491935  115       238         8
## 455606   GTS_738-24 UGA_173101_P018_WF08 0.483333  110       238         8
## 455781   GTS_738-24 UGA_173101_P017_WF06 0.483333  110       238         8
## 455882   GTS_738-24 UGA_173101_P020_WA02 0.491379  102       238         8
## 455946   GTS_738-24 UGA_173101_P023_WG03 0.467391   60       238         8
## 455964   GTS_738-24 UGA_173101_P023_WA01 0.477273   78       238         8
## 455977   GTS_738-24 UGA_173101_P022_WF11 0.482759  106       238         8
## 456106   GTS_738-24 UGA_173101_P020_WF04 0.482143  105       238         8
## 651519  GTS_2962-24 UGA_173101_P020_WB06 0.500000  112       238         8
## 651578  GTS_2962-24 UGA_173101_P018_WG05 0.500000  114       238         8
## 651582  GTS_2962-24 UGA_173101_P018_WF08 0.491071  110       238         8
## 651757  GTS_2962-24 UGA_173101_P017_WF06 0.491071  109       238         8
## 651858  GTS_2962-24 UGA_173101_P020_WA02 0.490385  101       238         8
## 651922  GTS_2962-24 UGA_173101_P023_WG03 0.465909   60       238         8
## 651940  GTS_2962-24 UGA_173101_P023_WA01 0.488095   78       238         8
## 651953  GTS_2962-24 UGA_173101_P022_WF11 0.490385  106       238         8
## 652082  GTS_2962-24 UGA_173101_P020_WF04 0.480769  105       238         8
## 810909  GTS_2748-24 UGA_173101_P020_WB06 0.467742  115       238         8
## 810968  GTS_2748-24 UGA_173101_P018_WG05 0.468750  117       238         8
## 810972  GTS_2748-24 UGA_173101_P018_WF08 0.458333  112       238         8
## 811147  GTS_2748-24 UGA_173101_P017_WF06 0.459677  112       238         8
## 811248  GTS_2748-24 UGA_173101_P020_WA02 0.458333  105       238         8
## 811312  GTS_2748-24 UGA_173101_P023_WG03 0.468750   61       238         8
## 811330  GTS_2748-24 UGA_173101_P023_WA01 0.443182   79       238         8
## 811343  GTS_2748-24 UGA_173101_P022_WF11 0.456897  109       238         8
## 811472  GTS_2748-24 UGA_173101_P020_WF04 0.448276  108       238         8
## 914636  GTS_2320-24 UGA_173101_P020_WB06 0.468750  117       238         8
## 914695  GTS_2320-24 UGA_173101_P018_WG05 0.469697  119       238         8
## 914699  GTS_2320-24 UGA_173101_P018_WF08 0.459677  114       238         8
## 914874  GTS_2320-24 UGA_173101_P017_WF06 0.460938  114       238         8
## 914975  GTS_2320-24 UGA_173101_P020_WA02 0.458333  106       238         8
## 915039  GTS_2320-24 UGA_173101_P023_WG03 0.468750   62       238         8
## 915057  GTS_2320-24 UGA_173101_P023_WA01 0.445652   81       238         8
## 915070  GTS_2320-24 UGA_173101_P022_WF11 0.458333  111       238         8
## 915199  GTS_2320-24 UGA_173101_P020_WF04 0.448276  109       238         8
## 962586  GTS_2864-24 UGA_173101_P020_WB06 0.492188  117       238         8
## 962645  GTS_2864-24 UGA_173101_P018_WG05 0.492424  119       238         8
## 962649  GTS_2864-24 UGA_173101_P018_WF08 0.484375  114       238         8
```

```
## 962824  GTS_2864-24 UGA_173101_P017_WF06 0.484375  114       238         8
## 962925  GTS_2864-24 UGA_173101_P020_WA02 0.483333  106       238         8
## 962989  GTS_2864-24 UGA_173101_P023_WG03 0.460000   63       238         8
## 963007  GTS_2864-24 UGA_173101_P023_WA01 0.479167   81       238         8
## 963020  GTS_2864-24 UGA_173101_P022_WF11 0.483871  111       238         8
## 963149  GTS_2864-24 UGA_173101_P020_WF04 0.474138  109       238         8
## 998999  GTS_2894-24 UGA_173101_P020_WB06 0.492188  117       238         8
## 999058  GTS_2894-24 UGA_173101_P018_WG05 0.492424  119       238         8
## 999062  GTS_2894-24 UGA_173101_P018_WF08 0.483871  114       238         8
## 999237  GTS_2894-24 UGA_173101_P017_WF06 0.484848  114       238         8
## 999338  GTS_2894-24 UGA_173101_P020_WA02 0.483871  106       238         8
## 999402  GTS_2894-24 UGA_173101_P023_WG03 0.468750   62       238         8
## 999420  GTS_2894-24 UGA_173101_P023_WA01 0.478261   81       238         8
## 999433  GTS_2894-24 UGA_173101_P022_WF11 0.483333  111       238         8
## 999562  GTS_2894-24 UGA_173101_P020_WF04 0.474138  109       238         8
## 1024604 GTS_2729-24 UGA_173101_P020_WB06 0.483871  114       238         8
## 1024663 GTS_2729-24 UGA_173101_P018_WG05 0.484375  117       238         8
## 1024667 GTS_2729-24 UGA_173101_P018_WF08 0.475000  112       238         8
## 1024842 GTS_2729-24 UGA_173101_P017_WF06 0.475806  112       238         8
## 1024943 GTS_2729-24 UGA_173101_P020_WA02 0.475000  105       238         8
## 1025007 GTS_2729-24 UGA_173101_P023_WG03 0.458333   62       238         8
## 1025025 GTS_2729-24 UGA_173101_P023_WA01 0.465909   80       238         8
## 1025038 GTS_2729-24 UGA_173101_P022_WF11 0.474138  108       238         8
## 1025167 GTS_2729-24 UGA_173101_P020_WF04 0.465517  108       238         8
## 1266444  GTS_260-24 UGA_173101_P020_WB06 0.461538  106       238         8
## 1266503  GTS_260-24 UGA_173101_P018_WG05 0.462963  108       238         8
## 1266507  GTS_260-24 UGA_173101_P018_WF08 0.453704  104       238         8
## 1266682  GTS_260-24 UGA_173101_P017_WF06 0.451923  102       238         8
## 1266783  GTS_260-24 UGA_173101_P020_WA02 0.447917   96       238         8
## 1266847  GTS_260-24 UGA_173101_P023_WG03 0.462500   57       238         8
## 1266865  GTS_260-24 UGA_173101_P023_WA01 0.452381   73       238         8
## 1266878  GTS_260-24 UGA_173101_P022_WF11 0.450000   99       238         8
## 1267007  GTS_260-24 UGA_173101_P020_WF04 0.458333   99       238         8
##         isclonePLINK iscloneCOL
## 217478          TRUE      FALSE
## 217537          TRUE      FALSE
## 217541          TRUE      FALSE
## 217716          TRUE      FALSE
## 217817          TRUE      FALSE
## 217881          TRUE      FALSE
## 217899          TRUE      FALSE
## 217912          TRUE      FALSE
## 218041          TRUE      FALSE
## 455543          TRUE      FALSE
## 455602          TRUE      FALSE
## 455606          TRUE      FALSE
## 455781          TRUE      FALSE
## 455882          TRUE      FALSE
## 455946          TRUE      FALSE
## 455964          TRUE      FALSE
## 455977          TRUE      FALSE
## 456106          TRUE      FALSE
## 651519          TRUE      FALSE
## 651578          TRUE      FALSE
```

```
## 651582            TRUE      FALSE
## 651757            TRUE      FALSE
## 651858            TRUE      FALSE
## 651922            TRUE      FALSE
## 651940            TRUE      FALSE
## 651953            TRUE      FALSE
## 652082            TRUE      FALSE
## 810909            TRUE      FALSE
## 810968            TRUE      FALSE
## 810972            TRUE      FALSE
## 811147            TRUE      FALSE
## 811248            TRUE      FALSE
## 811312            TRUE      FALSE
## 811330            TRUE      FALSE
## 811343            TRUE      FALSE
## 811472            TRUE      FALSE
## 914636            TRUE      FALSE
## 914695            TRUE      FALSE
## 914699            TRUE      FALSE
## 914874            TRUE      FALSE
## 914975            TRUE      FALSE
## 915039            TRUE      FALSE
## 915057            TRUE      FALSE
## 915070            TRUE      FALSE
## 915199            TRUE      FALSE
## 962586            TRUE      FALSE
## 962645            TRUE      FALSE
## 962649            TRUE      FALSE
## 962824            TRUE      FALSE
## 962925            TRUE      FALSE
## 962989            TRUE      FALSE
## 963007            TRUE      FALSE
## 963020            TRUE      FALSE
## 963149            TRUE      FALSE
## 998999            TRUE      FALSE
## 999058            TRUE      FALSE
## 999062            TRUE      FALSE
## 999237            TRUE      FALSE
## 999338            TRUE      FALSE
## 999402            TRUE      FALSE
## 999420            TRUE      FALSE
## 999433            TRUE      FALSE
## 999562            TRUE      FALSE
## 1024604           TRUE      FALSE
## 1024663           TRUE      FALSE
## 1024667           TRUE      FALSE
## 1024842           TRUE      FALSE
## 1024943           TRUE      FALSE
## 1025007           TRUE      FALSE
## 1025025           TRUE      FALSE
## 1025038           TRUE      FALSE
## 1025167           TRUE      FALSE
## 1266444           TRUE      FALSE
## 1266503           TRUE      FALSE
```

```
## 1266507          TRUE      FALSE
## 1266682          TRUE      FALSE
## 1266783          TRUE      FALSE
## 1266847          TRUE      FALSE
## 1266865          TRUE      FALSE
## 1266878          TRUE      FALSE
## 1267007          TRUE      FALSE
```

```
#Clone 14 - 56 F, 2 M, 2 undet, no direct captures
table(smd$sexGT[smd$finalClone == 14])
```

```
##
##     F     M undet
##    56     2     2
```

```
smd[which(smd$finalClone == 14),]
```

```
##                                         extID   ugaID is.SCR is.rep week year
## 91                      UGA_173101_P015_WH07  432-23   TRUE  FALSE    4 2023
## 138                     UGA_173101_P016_WD06  179-23   TRUE  FALSE    7 2023
## 186                     UGA_173101_P016_WH06  312-23   TRUE  FALSE    6 2023
## 188                     UGA_173101_P016_WH08  330-23   TRUE  FALSE    6 2023
## 194                     UGA_173101_P017_WA02  393-23   TRUE  FALSE    5 2023
## 256                     UGA_173101_P017_WF04  468-23   TRUE  FALSE    5 2023
## 260                     UGA_173101_P017_WF08  409-23   TRUE   TRUE    5 2023
## 285                     UGA_173101_P017_WH09  409-23   TRUE   TRUE    5 2023
## 338                     UGA_173101_P018_WE02   52-23   TRUE  FALSE    4 2023
## 345                     UGA_173101_P018_WE09  206-23   TRUE  FALSE    5 2023
## 351                     UGA_173101_P018_WF03   35-23   TRUE  FALSE    6 2023
## 392                     UGA_173101_P019_WC02  188-23   TRUE  FALSE    4 2023
## 396                     UGA_173101_P019_WC06  200-23   TRUE  FALSE    5 2023
## 406                     UGA_173101_P019_WD04  362-23   TRUE  FALSE    5 2023
## 467                     UGA_173101_P020_WA05  209-23   TRUE  FALSE    7 2023
## 469                     UGA_173101_P020_WA07   38-23   TRUE  FALSE    3 2023
## 471                     UGA_173101_P020_WA09  319-23   TRUE  FALSE    6 2023
## 540                     UGA_173101_P020_WG06  277-23   TRUE  FALSE    5 2023
## 541                     UGA_173101_P020_WG07  443-23   TRUE  FALSE    5 2023
## 549                     UGA_173101_P020_WH03  212-23   TRUE  FALSE    6 2023
## 550                     UGA_173101_P020_WH04  271-23   TRUE  FALSE    6 2023
## 557                     UGA_173101_P020_WH11  447-23   TRUE  FALSE    7 2023
## 583                     UGA_173101_P021_WC01   91-23   TRUE  FALSE    5 2023
## 666                     UGA_173101_P022_WA12  160-23   TRUE  FALSE    6 2023
## 735                     UGA_173101_P022_WG09  471-23   TRUE  FALSE    4 2023
## 766                     UGA_173101_P023_WB04   37-23   TRUE  FALSE    4 2023
## 768                     UGA_173101_P023_WB06  190-23   TRUE  FALSE    5 2023
## 773                     UGA_173101_P023_WB11    7-23   TRUE  FALSE    4 2023
## 826                     UGA_173101_P023_WG04  136-23   TRUE  FALSE    6 2023
## 861   GTseek_024_384i5_034_P098.003_938-23  938-23   TRUE  FALSE    5 2023
## 867   GTseek_024_384i5_040_P098.003_944-23  944-23   TRUE  FALSE    5 2023
## 902   GTseek_024_384i5_075_P098.001_631-23  631-23   TRUE  FALSE    5 2023
## 910   GTseek_024_384i5_083_P098.002_781-23  781-23   TRUE   TRUE    4 2023
## 922  GTseek_024_384i5_095_P098.002_781-23R 781-23R   TRUE   TRUE    4 2023
## 926   GTseek_024_384i5_099_P098.001_638-23  638-23   TRUE  FALSE    3 2023
```

```
## 935   GTseek_024_384i5_108_P098.003_972-23  972-23   TRUE  FALSE    6 2023
## 960   GTseek_024_384i5_133_P098.001_651-23  651-23   TRUE   TRUE    4 2023
## 961   GTseek_024_384i5_134_P098.003_982-23  982-23   TRUE  FALSE    4 2023
## 970  GTseek_024_384i5_143_P098.001_651-23R 651-23R   TRUE   TRUE    4 2023
## 977   GTseek_024_384i5_151_P098.002_803-23  803-23   TRUE  FALSE    6 2023
## 1005  GTseek_024_384i5_179_P098.002_809-23  809-23   TRUE  FALSE    5 2023
## 1026 GTseek_024_384i5_200_P098.003_1009-23 1009-23   TRUE  FALSE    6 2023
## 1032 GTseek_024_384i5_206_P098.003_1013-23 1013-23   TRUE  FALSE    5 2023
## 1113  GTseek_024_384i5_287_P098.002_871-23  871-23   TRUE  FALSE    3 2023
## 1119  GTseek_024_384i5_293_P098.001_713-23  713-23   TRUE  FALSE    6 2023
## 1155  GTseek_024_384i5_329_P098.001_732-23  732-23   TRUE  FALSE    7 2023
## 1201  GTseek_024_384i5_375_P098.002_919-23  919-23   TRUE  FALSE    4 2023
## 1224  GTseek_025_384i5_014_P098.007_671-23  671-23   TRUE  FALSE    4 2023
## 1256  GTseek_025_384i5_046_P098.007_706-23  706-23   TRUE  FALSE    5 2023
## 1286  GTseek_025_384i5_076_P098.007_776-23  776-23   TRUE  FALSE    6 2023
## 1312  GTseek_025_384i5_102_P098.007_847-23  847-23   TRUE  FALSE    5 2023
## 1340  GTseek_025_384i5_130_P098.007_881-23  881-23   TRUE  FALSE    5 2023
## 1344  GTseek_025_384i5_134_P098.007_890-23  890-23   TRUE  FALSE    5 2023
## 1406 GTseek_025_384i5_196_P098.007_1002-23 1002-23   TRUE  FALSE    5 2023
## 1914 GTseek_028_384i5_089_P098.UGA2_399-23  399-23   TRUE  FALSE    3 2023
## 1922 GTseek_028_384i5_097_P098.UGA1_104-23  104-23   TRUE  FALSE    5 2023
## 1956 GTseek_028_384i5_131_P098.UGA1_139-23  139-23   TRUE  FALSE    4 2023
## 1968 GTseek_028_384i5_143_P098.UGA1_150-23  150-23   TRUE  FALSE    5 2023
## 1998 GTseek_028_384i5_173_P098.UGA1_181-23  181-23   TRUE  FALSE    6 2023
## 2018 GTseek_028_384i5_193_P098.UGA1_187-23  187-23   TRUE  FALSE    5 2023
##       snare       utmSnare
## 91   HS 027 264464x3592495
## 138  HS 027 264464x3592495
## 186  HS 027 264464x3592495
## 188  HS 027 264464x3592495
## 194  HS 027 264464x3592495
## 256  HS 027 264464x3592495
## 260  HS 027 264464x3592495
## 285  HS 027 264464x3592495
## 338  HS 027 264464x3592495
## 345  HS 027 264464x3592495
## 351  HS 027 264464x3592495
## 392  HS 027 264464x3592495
## 396  HS 027 264464x3592495
## 406  HS 027 264464x3592495
## 467  HS 025 265276x3591787
## 469  HS 027 264464x3592495
## 471  HS 032 263672x3593292
## 540  HS 025 265276x3591787
## 541  HS 027 264464x3592495
## 549  HS 027 264464x3592495
## 550  HS 025 265276x3591787
## 557  HS 027 264464x3592495
## 583  HS 027 264464x3592495
## 666  HS 027 264464x3592495
## 735  HS 027 264464x3592495
## 766  HS 027 264464x3592495
## 768  HS 025 265276x3591787
## 773  HS 027 264464x3592495
```

```
## 826  HS 025 265276x3591787
## 861  HS 027 264464x3592495
## 867  HS 027 264464x3592495
## 902  HS 027 264464x3592495
## 910  HS 027 264464x3592495
## 922  HS 027 264464x3592495
## 926  HS 027 264464x3592495
## 935  HS 027 264464x3592495
## 960  HS 027 264464x3592495
## 961  HS 027 264464x3592495
## 970  HS 027 264464x3592495
## 977  HS 027 264464x3592495
## 1005 HS 027 264464x3592495
## 1026 HS 032 263672x3593292
## 1032 HS 027 264464x3592495
## 1113 HS 027 264464x3592495
## 1119 HS 025 265276x3591787
## 1155 HS 025 265276x3591787
## 1201 HS 027 264464x3592495
## 1224 HS 027 264464x3592495
## 1256 HS 027 264464x3592495
## 1286 HS 027 264464x3592495
## 1312 HS 027 264464x3592495
## 1340 HS 027 264464x3592495
## 1344 HS 027 264464x3592495
## 1406 HS 027 264464x3592495
## 1914 HS 027 264464x3592495
## 1922 HS 025 265276x3591787
## 1956 HS 027 264464x3592495
## 1968 HS 027 264464x3592495
## 1998 HS 032 263672x3593292
## 2018 HS 027 264464x3592495
##                                                                                   fqfile
## 91      RAPiD-Genomics_F534_UGA_173101_P015_WH07_i5-502_i7-1_S14829_L003_R1_001.fastq
## 138  RAPiD-Genomics_F534_UGA_173101_P016_WD06_i5-504_i7-138_S14876_L003_R1_001.fastq
## 186  RAPiD-Genomics_F534_UGA_173101_P016_WH06_i5-504_i7-186_S14923_L003_R1_001.fastq
## 188  RAPiD-Genomics_F534_UGA_173101_P016_WH08_i5-504_i7-188_S14925_L003_R1_001.fastq
## 194   RAPiD-Genomics_F534_UGA_173101_P017_WA02_i5-512_i7-98_S14931_L003_R1_001.fastq
## 256  RAPiD-Genomics_F534_UGA_173101_P017_WF04_i5-512_i7-160_S14993_L003_R1_001.fastq
## 260  RAPiD-Genomics_F534_UGA_173101_P017_WF08_i5-512_i7-164_S14997_L003_R1_001.fastq
## 285  RAPiD-Genomics_F534_UGA_173101_P017_WH09_i5-512_i7-189_S15022_L003_R1_001.fastq
## 338  RAPiD-Genomics_F534_UGA_173101_P018_WE02_i5-537_i7-146_S15075_L003_R1_001.fastq
## 345  RAPiD-Genomics_F534_UGA_173101_P018_WE09_i5-537_i7-153_S15084_L003_R1_001.fastq
## 351  RAPiD-Genomics_F534_UGA_173101_P018_WF03_i5-537_i7-159_S15090_L003_R1_001.fastq
## 392  RAPiD-Genomics_F534_UGA_173101_P019_WC02_i5-1102_i7-67_S15131_L003_R1_001.fastq
## 396   RAPiD-Genomics_F534_UGA_173101_P019_WC06_i5-1102_i7-3_S15135_L003_R1_001.fastq
## 406  RAPiD-Genomics_F534_UGA_173101_P019_WD04_i5-1102_i7-91_S15145_L003_R1_001.fastq
## 467     RAPiD-Genomics_F566_UGA_173101_P020_WA05_i5-505_i7-38_S3895_L001_R1_001.fastq
## 469     RAPiD-Genomics_F566_UGA_173101_P020_WA07_i5-505_i7-77_S3897_L001_R1_001.fastq
## 471     RAPiD-Genomics_F566_UGA_173101_P020_WA09_i5-505_i7-36_S3899_L001_R1_001.fastq
## 540     RAPiD-Genomics_F566_UGA_173101_P020_WG06_i5-505_i7-14_S3968_L001_R1_001.fastq
## 541     RAPiD-Genomics_F566_UGA_173101_P020_WG07_i5-505_i7-18_S3969_L001_R1_001.fastq
## 549     RAPiD-Genomics_F566_UGA_173101_P020_WH03_i5-505_i7-88_S3977_L001_R1_001.fastq
## 550     RAPiD-Genomics_F566_UGA_173101_P020_WH04_i5-505_i7-73_S3978_L001_R1_001.fastq
```

```
## 557    RAPiD-Genomics_F566_UGA_173101_P020_WH11_i5-505_i7-95_S3985_L001_R1_001.fastq
## 583      RAPiD-Genomics_F566_UGA_173101_P021_WC01_i5-501_i7-65_S25_L003_R1_001.fastq
## 666   RAPiD-Genomics_F566_UGA_173101_P022_WA12_i5-1098_i7-23_S7594_L003_R1_001.fastq
## 735   RAPiD-Genomics_F566_UGA_173101_P022_WG09_i5-1098_i7-41_S7663_L003_R1_001.fastq
## 766   RAPiD-Genomics_F566_UGA_173101_P023_WB04_i5-1099_i7-20_S7694_L003_R1_001.fastq
## 768    RAPiD-Genomics_F566_UGA_173101_P023_WB06_i5-1099_i7-6_S7696_L003_R1_001.fastq
## 773    RAPiD-Genomics_F566_UGA_173101_P023_WB11_i5-1099_i7-5_S7701_L003_R1_001.fastq
## 826   RAPiD-Genomics_F566_UGA_173101_P023_WG04_i5-1099_i7-10_S7754_L003_R1_001.fastq
## 861                                                                             <NA>
## 867                                                                             <NA>
## 902                                                                             <NA>
## 910                                                                             <NA>
## 922                                                                             <NA>
## 926                                                                             <NA>
## 935                                                                             <NA>
## 960                                                                             <NA>
## 961                                                                             <NA>
## 970                                                                             <NA>
## 977                                                                             <NA>
## 1005                                                                            <NA>
## 1026                                                                            <NA>
## 1032                                                                            <NA>
## 1113                                                                            <NA>
## 1119                                                                            <NA>
## 1155                                                                            <NA>
## 1201                                                                            <NA>
## 1224                                                                            <NA>
## 1256                                                                            <NA>
## 1286                                                                            <NA>
## 1312                                                                            <NA>
## 1340                                                                            <NA>
## 1344                                                                            <NA>
## 1406                                                                            <NA>
## 1914                                                                            <NA>
## 1922                                                                            <NA>
## 1956                                                                            <NA>
## 1968                                                                            <NA>
## 1998                                                                            <NA>
## 2018                                                                            <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 91             Hair Snare    CGA    911 UGA_173101_P015_WH07 FALSE       LGC
## 138            Hair Snare    CGA    853 UGA_173101_P016_WD06 FALSE       LGC
## 186            Hair Snare    CGA    782 UGA_173101_P016_WH06 FALSE       LGC
## 188            Hair Snare    CGA    870 UGA_173101_P016_WH08 FALSE       LGC
## 194            Hair Snare    CGA    916 UGA_173101_P017_WA02 FALSE       LGC
## 256            Hair Snare    CGA    917 UGA_173101_P017_WF04 FALSE       LGC
## 260            Hair Snare    CGA    745 UGA_173101_P017_WF08 FALSE       LGC
## 285            Hair Snare    CGA    774 UGA_173101_P017_WH09 FALSE       LGC
## 338            Hair Snare    CGA    915 UGA_173101_P018_WE02 FALSE       LGC
## 345            Hair Snare    CGA    884 UGA_173101_P018_WE09 FALSE       LGC
## 351            Hair Snare    CGA    899 UGA_173101_P018_WF03 FALSE       LGC
## 392            Hair Snare    CGA    919 UGA_173101_P019_WC02 FALSE       LGC
## 396            Hair Snare    CGA    919 UGA_173101_P019_WC06 FALSE       LGC
## 406            Hair Snare    CGA    922 UGA_173101_P019_WD04 FALSE       LGC
```

```
## 467           Hair Snare    CGA    913 UGA_173101_P020_WA05 FALSE       LGC
## 469           Hair Snare    CGA    892 UGA_173101_P020_WA07 FALSE       LGC
## 471           Hair Snare    CGA    852 UGA_173101_P020_WA09 FALSE       LGC
## 540           Hair Snare    CGA    910 UGA_173101_P020_WG06 FALSE       LGC
## 541           Hair Snare    CGA    608 UGA_173101_P020_WG07 FALSE       LGC
## 549           Hair Snare    CGA    918 UGA_173101_P020_WH03 FALSE       LGC
## 550           Hair Snare    CGA    916 UGA_173101_P020_WH04 FALSE       LGC
## 557           Hair Snare    CGA    918 UGA_173101_P020_WH11 FALSE       LGC
## 583           Hair Snare    CGA    895 UGA_173101_P021_WC01 FALSE       LGC
## 666           Hair Snare    CGA    865 UGA_173101_P022_WA12 FALSE       LGC
## 735           Hair Snare    CGA    915 UGA_173101_P022_WG09 FALSE       LGC
## 766           Hair Snare    CGA    908 UGA_173101_P023_WB04 FALSE       LGC
## 768           Hair Snare    CGA    897 UGA_173101_P023_WB06 FALSE       LGC
## 773           Hair Snare    CGA    905 UGA_173101_P023_WB11 FALSE       LGC
## 826           Hair Snare    CGA    914 UGA_173101_P023_WG04 FALSE       LGC
## 861      <NA> Hair Sample   CGA    123           GTS_938-23 FALSE    GTSeek
## 867      <NA> Hair Sample   CGA    121           GTS_944-23 FALSE    GTSeek
## 902      <NA> Hair Sample   CGA    121           GTS_631-23 FALSE    GTSeek
## 910      <NA> Hair Sample   CGA    123           GTS_781-23 FALSE    GTSeek
## 922      <NA> Hair Sample   CGA    123          GTS_781-23R FALSE    GTSeek
## 926      <NA> Hair Sample   CGA    120           GTS_638-23 FALSE    GTSeek
## 935      <NA> Hair Sample   CGA    121           GTS_972-23 FALSE    GTSeek
## 960      <NA> Hair Sample   CGA    122           GTS_651-23 FALSE    GTSeek
## 961      <NA> Hair Sample   CGA    122           GTS_982-23 FALSE    GTSeek
## 970      <NA> Hair Sample   CGA    123          GTS_651-23R FALSE    GTSeek
## 977      <NA> Hair Sample   CGA    123           GTS_803-23 FALSE    GTSeek
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## 1026     <NA> Hair Sample   CGA    122          GTS_1009-23 FALSE    GTSeek
## 1032     <NA> Hair Sample   CGA    123          GTS_1013-23 FALSE    GTSeek
## 1113     <NA> Hair Sample   CGA    122           GTS_871-23 FALSE    GTSeek
## 1119     <NA> Hair Sample   CGA    121           GTS_713-23 FALSE    GTSeek
## 1155     <NA> Hair Sample   CGA    121           GTS_732-23 FALSE    GTSeek
## 1201     <NA> Hair Sample   CGA    123           GTS_919-23 FALSE    GTSeek
## 1224     <NA> Hair Sample   CGA    121           GTS_671-23 FALSE    GTSeek
## 1256     <NA> Hair Sample   CGA    122           GTS_706-23 FALSE    GTSeek
## 1286     <NA> Hair Sample   CGA    124           GTS_776-23 FALSE    GTSeek
## 1312     <NA> Hair Sample   CGA    124           GTS_847-23 FALSE    GTSeek
## 1340     <NA> Hair Sample   CGA    122           GTS_881-23 FALSE    GTSeek
## 1344     <NA> Hair Sample   CGA    124           GTS_890-23 FALSE    GTSeek
## 1406     <NA> Hair Sample   CGA    123          GTS_1002-23 FALSE    GTSeek
## 1914     <NA> Hair Sample   CGA    124           GTS_399-23 FALSE    GTSeek
## 1922     <NA> Hair Sample   CGA    109           GTS_104-23 FALSE    GTSeek
## 1956     <NA> Hair Sample   CGA    124           GTS_139-23 FALSE    GTSeek
## 1968     <NA> Hair Sample   CGA    114           GTS_150-23 FALSE    GTSeek
## 1998     <NA> Hair Sample   CGA    124           GTS_181-23 FALSE    GTSeek
## 2018     <NA> Hair Sample   CGA    120           GTS_187-23 FALSE    GTSeek
##      sexGT COLclone finalClone
## 91       F       14         14
## 138      F       14         14
## 186      F       14         14
## 188      F       14         14
## 194      F       14         14
## 256  undet       14         14
## 260      F       14         14
```

```
## 285      F      14          14
## 338      F      14          14
## 345      F      14          14
## 351      F      14          14
## 392      F      14          14
## 396      F      14          14
## 406      F      14          14
## 467      F      14          14
## 469      F      14          14
## 471      F      14          14
## 540      F      14          14
## 541      F      14          14
## 549      M      14          14
## 550      M      14          14
## 557      F      14          14
## 583      F      14          14
## 666  undet      14          14
## 735      F      14          14
## 766      F      14          14
## 768      F      14          14
## 773      F      14          14
## 826      F      14          14
## 861      F     250          14
## 867      F     250          14
## 902      F     250          14
## 910      F     269          14
## 922      F     269          14
## 926      F     250          14
## 935      F     269          14
## 960      F     250          14
## 961      F     250          14
## 970      F     250          14
## 977      F     269          14
## 1005     F     269          14
## 1026     F     250          14
## 1032     F     250          14
## 1113     F     269          14
## 1119     F     250          14
## 1155     F     250          14
## 1201     F     269          14
## 1224     F     250          14
## 1256     F     250          14
## 1286     F     250          14
## 1312     F     250          14
## 1340     F     250          14
## 1344     F     250          14
## 1406     F     250          14
## 1914     F     269          14
## 1922     F     269          14
## 1956     F     269          14
## 1968     F     269          14
## 1998     F     269          14
## 2018     F     269          14
```

```r
unique(smd$COLclone[which(smd$finalClone == 14)])
```

```
## [1]  14 250 269
```

```r
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 14)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 14)],]
```

```
##                          id1                  id2  kinship nsnp colclone1
## 1240    UGA_173101_P019_WC06 UGA_173101_P020_WA09 0.497449  852        14
## 1393    UGA_173101_P019_WC02 UGA_173101_P020_WA09 0.496803  852        14
## 1429    UGA_173101_P019_WC02 UGA_173101_P019_WC06 0.498201  918        14
## 1726    UGA_173101_P019_WD04 UGA_173101_P020_WA09 0.496811  852        14
## 1762    UGA_173101_P019_WD04 UGA_173101_P019_WC06 0.497625  919        14
## 1765    UGA_173101_P019_WD04 UGA_173101_P019_WC02 0.498206  919        14
## 2643    UGA_173101_P018_WF03 UGA_173101_P020_WA09 0.491425  837        14
## 2679    UGA_173101_P018_WF03 UGA_173101_P019_WC06 0.493210  899        14
## 2682    UGA_173101_P018_WF03 UGA_173101_P019_WC02 0.493827  899        14
## 2688    UGA_173101_P018_WF03 UGA_173101_P019_WD04 0.493210  899        14
## 2941    UGA_173101_P018_WE09 UGA_173101_P020_WA09 0.492167  828        14
## 2977    UGA_173101_P018_WE09 UGA_173101_P019_WC06 0.493842  884        14
## 2980    UGA_173101_P018_WE09 UGA_173101_P019_WC02 0.495690  884        14
## 2986    UGA_173101_P018_WE09 UGA_173101_P019_WD04 0.495074  884        14
## 3000    UGA_173101_P018_WE09 UGA_173101_P018_WF03 0.489241  869        14
## 4965    UGA_173101_P020_WA07 UGA_173101_P020_WA09 0.494751  837        14
## 5001    UGA_173101_P020_WA07 UGA_173101_P019_WC06 0.494431  892        14
## 5004    UGA_173101_P020_WA07 UGA_173101_P019_WC02 0.496287  892        14
## 5010    UGA_173101_P020_WA07 UGA_173101_P019_WD04 0.494431  892        14
## 5024    UGA_173101_P020_WA07 UGA_173101_P018_WF03 0.489924  878        14
## 5028    UGA_173101_P020_WA07 UGA_173101_P018_WE09 0.492424  863        14
## 5166    UGA_173101_P016_WD06 UGA_173101_P020_WA09 0.486450  802        14
## 5202    UGA_173101_P016_WD06 UGA_173101_P019_WC06 0.488520  853        14
## 5205    UGA_173101_P016_WD06 UGA_173101_P019_WC02 0.489130  853        14
## 5211    UGA_173101_P016_WD06 UGA_173101_P019_WD04 0.489158  853        14
## 5225    UGA_173101_P016_WD06 UGA_173101_P018_WF03 0.483553  838        14
## 5229    UGA_173101_P016_WD06 UGA_173101_P018_WE09 0.485640  831        14
## 5252    UGA_173101_P016_WD06 UGA_173101_P020_WA07 0.486220  836        14
## 8793    UGA_173101_P017_WA02 UGA_173101_P020_WA09 0.496173  852        14
## 8829    UGA_173101_P017_WA02 UGA_173101_P019_WC06 0.497608  915        14
## 8832    UGA_173101_P017_WA02 UGA_173101_P019_WC02 0.499400  915        14
## 8838    UGA_173101_P017_WA02 UGA_173101_P019_WD04 0.498804  916        14
## 8852    UGA_173101_P017_WA02 UGA_173101_P018_WF03 0.493210  897        14
## 8856    UGA_173101_P017_WA02 UGA_173101_P018_WE09 0.496305  883        14
## 8879    UGA_173101_P017_WA02 UGA_173101_P020_WA07 0.495668  892        14
## 8881    UGA_173101_P017_WA02 UGA_173101_P016_WD06 0.489796  852        14
## 10600   UGA_173101_P015_WH07 UGA_173101_P020_WA09 0.493573  848        14
## 10636   UGA_173101_P015_WH07 UGA_173101_P019_WC06 0.495192  910        14
## 10639   UGA_173101_P015_WH07 UGA_173101_P019_WC02 0.494591  910        14
## 10645   UGA_173101_P015_WH07 UGA_173101_P019_WD04 0.495204  911        14
## 10659   UGA_173101_P015_WH07 UGA_173101_P018_WF03 0.489506  895        14
## 10663   UGA_173101_P015_WH07 UGA_173101_P018_WE09 0.491358  878        14
## 10686   UGA_173101_P015_WH07 UGA_173101_P020_WA07 0.490695  887        14
## 10688   UGA_173101_P015_WH07 UGA_173101_P016_WD06 0.487179  849        14
```

```
## 10719   UGA_173101_P015_WH07 UGA_173101_P017_WA02 0.495181  907         14
## 14721   UGA_173101_P020_WA05 UGA_173101_P020_WA09 0.498087  852         14
## 14757   UGA_173101_P020_WA05 UGA_173101_P019_WC06 0.497625  912         14
## 14760   UGA_173101_P020_WA05 UGA_173101_P019_WC02 0.497010  912         14
## 14766   UGA_173101_P020_WA05 UGA_173101_P019_WD04 0.497630  913         14
## 14780   UGA_173101_P020_WA05 UGA_173101_P018_WF03 0.491975  895         14
## 14784   UGA_173101_P020_WA05 UGA_173101_P018_WE09 0.494458  881         14
## 14807   UGA_173101_P020_WA05 UGA_173101_P020_WA07 0.495668  890         14
## 14809   UGA_173101_P020_WA05 UGA_173101_P016_WD06 0.488520  853         14
## 14840   UGA_173101_P020_WA05 UGA_173101_P017_WA02 0.497608  910         14
## 14853   UGA_173101_P020_WA05 UGA_173101_P015_WH07 0.495204  908         14
## 27276   UGA_173101_P018_WE02 UGA_173101_P020_WA09 0.495536  852         14
## 27312   UGA_173101_P018_WE02 UGA_173101_P019_WC06 0.497017  915         14
## 27315   UGA_173101_P018_WE02 UGA_173101_P019_WC02 0.498801  915         14
## 27321   UGA_173101_P018_WE02 UGA_173101_P019_WD04 0.498210  915         14
## 27335   UGA_173101_P018_WE02 UGA_173101_P018_WF03 0.492593  898         14
## 27339   UGA_173101_P018_WE02 UGA_173101_P018_WE09 0.495690  881         14
## 27362   UGA_173101_P018_WE02 UGA_173101_P020_WA07 0.495050  892         14
## 27364   UGA_173101_P018_WE02 UGA_173101_P016_WD06 0.489158  853         14
## 27395   UGA_173101_P018_WE02 UGA_173101_P017_WA02 0.499402  912         14
## 27408   UGA_173101_P018_WE02 UGA_173101_P015_WH07 0.494578  909         14
## 27434   UGA_173101_P018_WE02 UGA_173101_P020_WA05 0.497017  910         14
## 43086   UGA_173101_P016_WH06 UGA_173101_P020_WA09 0.487500  737         14
## 43122   UGA_173101_P016_WH06 UGA_173101_P019_WC06 0.489583  782         14
## 43125   UGA_173101_P016_WH06 UGA_173101_P019_WC02 0.489583  782         14
## 43131   UGA_173101_P016_WH06 UGA_173101_P019_WD04 0.490278  782         14
## 43145   UGA_173101_P016_WH06 UGA_173101_P018_WF03 0.483848  773         14
## 43149   UGA_173101_P016_WH06 UGA_173101_P018_WE09 0.486506  763         14
## 43172   UGA_173101_P016_WH06 UGA_173101_P020_WA07 0.485915  770         14
## 43174   UGA_173101_P016_WH06 UGA_173101_P016_WD06 0.482353  742         14
## 43205   UGA_173101_P016_WH06 UGA_173101_P017_WA02 0.490278  782         14
## 43218   UGA_173101_P016_WH06 UGA_173101_P015_WH07 0.488162  780         14
## 43244   UGA_173101_P016_WH06 UGA_173101_P020_WA05 0.490278  781         14
## 43306   UGA_173101_P016_WH06 UGA_173101_P018_WE02 0.489583  782         14
## 43675   UGA_173101_P016_WH08 UGA_173101_P020_WA09 0.487299  818         14
## 43711   UGA_173101_P016_WH08 UGA_173101_P019_WC06 0.488665  870         14
## 43714   UGA_173101_P016_WH08 UGA_173101_P019_WC02 0.490554  870         14
## 43720   UGA_173101_P016_WH08 UGA_173101_P019_WD04 0.488665  870         14
## 43734   UGA_173101_P016_WH08 UGA_173101_P018_WF03 0.484576  855         14
## 43738   UGA_173101_P016_WH08 UGA_173101_P018_WE09 0.487821  843         14
## 43761   UGA_173101_P016_WH08 UGA_173101_P020_WA07 0.489046  850         14
## 43763   UGA_173101_P016_WH08 UGA_173101_P016_WD06 0.480458  815         14
## 43794   UGA_173101_P016_WH08 UGA_173101_P017_WA02 0.489924  870         14
## 43807   UGA_173101_P016_WH08 UGA_173101_P015_WH07 0.486111  867         14
## 43833   UGA_173101_P016_WH08 UGA_173101_P020_WA05 0.488035  868         14
## 43895   UGA_173101_P016_WH08 UGA_173101_P018_WE02 0.489295  870         14
## 43955   UGA_173101_P016_WH08 UGA_173101_P016_WH06 0.481322  754         14
## 50418   UGA_173101_P017_WF04 UGA_173101_P020_WA09 0.494260  851         14
## 50454   UGA_173101_P017_WF04 UGA_173101_P019_WC06 0.495238  916         14
## 50457   UGA_173101_P017_WF04 UGA_173101_P019_WC02 0.495813  917         14
## 50463   UGA_173101_P017_WF04 UGA_173101_P019_WD04 0.496437  917         14
## 50477   UGA_173101_P017_WF04 UGA_173101_P018_WF03 0.490741  898         14
## 50481   UGA_173101_P017_WF04 UGA_173101_P018_WE09 0.493227  882         14
## 50504   UGA_173101_P017_WF04 UGA_173101_P020_WA07 0.490718  891         14
```

```
## 50506   UGA_173101_P017_WF04 UGA_173101_P016_WD06 0.485969  852         14
## 50537   UGA_173101_P017_WF04 UGA_173101_P017_WA02 0.496411  913         14
## 50550   UGA_173101_P017_WF04 UGA_173101_P015_WH07 0.492788  910         14
## 50576   UGA_173101_P017_WF04 UGA_173101_P020_WA05 0.494062  911         14
## 50638   UGA_173101_P017_WF04 UGA_173101_P018_WE02 0.495823  914         14
## 50698   UGA_173101_P017_WF04 UGA_173101_P016_WH06 0.488889  782         14
## 50700   UGA_173101_P017_WF04 UGA_173101_P016_WH08 0.487406  870         14
## 59011   UGA_173101_P017_WF08 UGA_173101_P020_WA09 0.482031  704         14
## 59047   UGA_173101_P017_WF08 UGA_173101_P019_WC06 0.481287  745         14
## 59050   UGA_173101_P017_WF08 UGA_173101_P019_WC02 0.483533  745         14
## 59056   UGA_173101_P017_WF08 UGA_173101_P019_WD04 0.482784  745         14
## 59070   UGA_173101_P017_WF08 UGA_173101_P018_WF03 0.478030  735         14
## 59074   UGA_173101_P017_WF08 UGA_173101_P018_WE09 0.478788  729         14
## 59097   UGA_173101_P017_WF08 UGA_173101_P020_WA07 0.482576  730         14
## 59099   UGA_173101_P017_WF08 UGA_173101_P016_WD06 0.474219  712         14
## 59130   UGA_173101_P017_WF08 UGA_173101_P017_WA02 0.484281  744         14
## 59143   UGA_173101_P017_WF08 UGA_173101_P015_WH07 0.479790  742         14
## 59169   UGA_173101_P017_WF08 UGA_173101_P020_WA05 0.482036  744         14
## 59231   UGA_173101_P017_WF08 UGA_173101_P018_WE02 0.483533  745         14
## 59291   UGA_173101_P017_WF08 UGA_173101_P016_WH06 0.477349  649         14
## 59293   UGA_173101_P017_WF08 UGA_173101_P016_WH08 0.474006  721         14
## 59315   UGA_173101_P017_WF08 UGA_173101_P017_WF04 0.481287  744         14
## 66081   UGA_173101_P017_WH09 UGA_173101_P020_WA09 0.482887  732         14
## 66117   UGA_173101_P017_WH09 UGA_173101_P019_WC06 0.484419  774         14
## 66120   UGA_173101_P017_WH09 UGA_173101_P019_WC02 0.483003  774         14
## 66126   UGA_173101_P017_WH09 UGA_173101_P019_WD04 0.484419  774         14
## 66140   UGA_173101_P017_WH09 UGA_173101_P018_WF03 0.476945  763         14
## 66144   UGA_173101_P017_WH09 UGA_173101_P018_WE09 0.479286  761         14
## 66167   UGA_173101_P017_WH09 UGA_173101_P020_WA07 0.479046  758         14
## 66169   UGA_173101_P017_WH09 UGA_173101_P016_WD06 0.475904  737         14
## 66200   UGA_173101_P017_WH09 UGA_173101_P017_WA02 0.483711  774         14
## 66213   UGA_173101_P017_WH09 UGA_173101_P015_WH07 0.484419  772         14
## 66239   UGA_173101_P017_WH09 UGA_173101_P020_WA05 0.484419  772         14
## 66301   UGA_173101_P017_WH09 UGA_173101_P018_WE02 0.483003  774         14
## 66361   UGA_173101_P017_WH09 UGA_173101_P016_WH06 0.475962  678         14
## 66363   UGA_173101_P017_WH09 UGA_173101_P016_WH08 0.474415  747         14
## 66385   UGA_173101_P017_WH09 UGA_173101_P017_WF04 0.482295  774         14
## 66411   UGA_173101_P017_WH09 UGA_173101_P017_WF08 0.473333  656         14
## 83451   UGA_173101_P023_WG04 UGA_173101_P020_WA09 0.496811  850         14
## 83487   UGA_173101_P023_WG04 UGA_173101_P019_WC06 0.498219  913         14
## 83490   UGA_173101_P023_WG04 UGA_173101_P019_WC02 0.497608  913         14
## 83496   UGA_173101_P023_WG04 UGA_173101_P019_WD04 0.498223  914         14
## 83510   UGA_173101_P023_WG04 UGA_173101_P018_WF03 0.493827  893         14
## 83514   UGA_173101_P023_WG04 UGA_173101_P018_WE09 0.494458  880         14
## 83537   UGA_173101_P023_WG04 UGA_173101_P020_WA07 0.493812  889         14
## 83539   UGA_173101_P023_WG04 UGA_173101_P016_WD06 0.489158  851         14
## 83570   UGA_173101_P023_WG04 UGA_173101_P017_WA02 0.498206  911         14
## 83583   UGA_173101_P023_WG04 UGA_173101_P015_WH07 0.495803  907         14
## 83609   UGA_173101_P023_WG04 UGA_173101_P020_WA05 0.498223  909         14
## 83671   UGA_173101_P023_WG04 UGA_173101_P018_WE02 0.497613  910         14
## 83731   UGA_173101_P023_WG04 UGA_173101_P016_WH06 0.490278  779         14
## 83733   UGA_173101_P023_WG04 UGA_173101_P016_WH08 0.488035  867         14
## 83755   UGA_173101_P023_WG04 UGA_173101_P017_WF04 0.495853  911         14
## 83781   UGA_173101_P023_WG04 UGA_173101_P017_WF08 0.482036  743         14
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## 83801   UGA_173101_P023_WG04 UGA_173101_P017_WH09 0.484419  772        14
## 88005   UGA_173101_P023_WB11 UGA_173101_P020_WA09 0.495490  848        14
## 88041   UGA_173101_P023_WB11 UGA_173101_P019_WC06 0.495763  904        14
## 88044   UGA_173101_P023_WB11 UGA_173101_P019_WC02 0.496377  905        14
## 88050   UGA_173101_P023_WB11 UGA_173101_P019_WD04 0.496377  905        14
## 88064   UGA_173101_P023_WB11 UGA_173101_P018_WF03 0.491315  889        14
## 88068   UGA_173101_P023_WB11 UGA_173101_P018_WE09 0.492593  875        14
## 88091   UGA_173101_P023_WB11 UGA_173101_P020_WA07 0.493797  884        14
## 88093   UGA_173101_P023_WB11 UGA_173101_P016_WD06 0.485180  847        14
## 88124   UGA_173101_P023_WB11 UGA_173101_P017_WA02 0.495763  903        14
## 88137   UGA_173101_P023_WB11 UGA_173101_P015_WH07 0.492131  899        14
## 88163   UGA_173101_P023_WB11 UGA_173101_P020_WA05 0.495773  903        14
## 88225   UGA_173101_P023_WB11 UGA_173101_P018_WE02 0.495157  903        14
## 88285   UGA_173101_P023_WB11 UGA_173101_P016_WH06 0.486111  777        14
## 88287   UGA_173101_P023_WB11 UGA_173101_P016_WH08 0.486709  860        14
## 88309   UGA_173101_P023_WB11 UGA_173101_P017_WF04 0.493357  904        14
## 88335   UGA_173101_P023_WB11 UGA_173101_P017_WF08 0.478979  740        14
## 88355   UGA_173101_P023_WB11 UGA_173101_P017_WH09 0.480769  769        14
## 88400   UGA_173101_P023_WB11 UGA_173101_P023_WG04 0.495169  903        14
## 89268   UGA_173101_P023_WB06 UGA_173101_P020_WA09 0.485849  839        14
## 89304   UGA_173101_P023_WB06 UGA_173101_P019_WC06 0.487342  895        14
## 89307   UGA_173101_P023_WB06 UGA_173101_P019_WC02 0.486742  896        14
## 89313   UGA_173101_P023_WB06 UGA_173101_P019_WD04 0.487374  897        14
## 89327   UGA_173101_P023_WB06 UGA_173101_P018_WF03 0.480179  881        14
## 89331   UGA_173101_P023_WB06 UGA_173101_P018_WE09 0.483161  864        14
## 89354   UGA_173101_P023_WB06 UGA_173101_P020_WA07 0.485256  876        14
## 89356   UGA_173101_P023_WB06 UGA_173101_P016_WD06 0.477027  836        14
## 89387   UGA_173101_P023_WB06 UGA_173101_P017_WA02 0.487342  894        14
## 89400   UGA_173101_P023_WB06 UGA_173101_P015_WH07 0.484810  891        14
## 89426   UGA_173101_P023_WB06 UGA_173101_P020_WA05 0.487374  892        14
## 89488   UGA_173101_P023_WB06 UGA_173101_P018_WE02 0.486709  895        14
## 89548   UGA_173101_P023_WB06 UGA_173101_P016_WH06 0.480714  767        14
## 89550   UGA_173101_P023_WB06 UGA_173101_P016_WH08 0.477513  855        14
## 89572   UGA_173101_P023_WB06 UGA_173101_P017_WF04 0.484848  895        14
## 89598   UGA_173101_P023_WB06 UGA_173101_P017_WF08 0.472644  730        14
## 89618   UGA_173101_P023_WB06 UGA_173101_P017_WH09 0.470326  759        14
## 89663   UGA_173101_P023_WB06 UGA_173101_P023_WG04 0.488005  895        14
## 89674   UGA_173101_P023_WB06 UGA_173101_P023_WB11 0.484137  888        14
## 91821   UGA_173101_P023_WB04 UGA_173101_P020_WA09 0.496164  849        14
## 91857   UGA_173101_P023_WB04 UGA_173101_P019_WC06 0.495803  907        14
## 91860   UGA_173101_P023_WB04 UGA_173101_P019_WC02 0.495181  908        14
## 91866   UGA_173101_P023_WB04 UGA_173101_P019_WD04 0.495813  908        14
## 91880   UGA_173101_P023_WB04 UGA_173101_P018_WF03 0.490050  888        14
## 91884   UGA_173101_P023_WB04 UGA_173101_P018_WE09 0.491975  878        14
## 91907   UGA_173101_P023_WB04 UGA_173101_P020_WA07 0.492556  885        14
## 91909   UGA_173101_P023_WB04 UGA_173101_P016_WD06 0.487852  849        14
## 91940   UGA_173101_P023_WB04 UGA_173101_P017_WA02 0.495783  905        14
## 91953   UGA_173101_P023_WB04 UGA_173101_P015_WH07 0.493341  899        14
## 91979   UGA_173101_P023_WB04 UGA_173101_P020_WA05 0.497010  904        14
## 92041   UGA_173101_P023_WB04 UGA_173101_P018_WE02 0.495192  904        14
## 92101   UGA_173101_P023_WB04 UGA_173101_P016_WH06 0.487465  776        14
## 92103   UGA_173101_P023_WB04 UGA_173101_P016_WH08 0.485406  861        14
## 92125   UGA_173101_P023_WB04 UGA_173101_P017_WF04 0.493437  906        14
## 92151   UGA_173101_P023_WB04 UGA_173101_P017_WF08 0.480539  741        14
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## 92171   UGA_173101_P023_WB04 UGA_173101_P017_WH09 0.482906  767         14
## 92216   UGA_173101_P023_WB04 UGA_173101_P023_WG04 0.496420  906         14
## 92227   UGA_173101_P023_WB04 UGA_173101_P023_WB11 0.493932  901         14
## 92230   UGA_173101_P023_WB04 UGA_173101_P023_WB06 0.485406  890         14
## 113065  UGA_173101_P020_WG06 UGA_173101_P020_WA09 0.497449  849         14
## 113101  UGA_173101_P020_WG06 UGA_173101_P019_WC06 0.497619  909         14
## 113104  UGA_173101_P020_WG06 UGA_173101_P019_WC02 0.497002  909         14
## 113110  UGA_173101_P020_WG06 UGA_173101_P019_WD04 0.497625  910         14
## 113124  UGA_173101_P020_WG06 UGA_173101_P018_WF03 0.491955  890         14
## 113128  UGA_173101_P020_WG06 UGA_173101_P018_WE09 0.493827  877         14
## 113151  UGA_173101_P020_WG06 UGA_173101_P020_WA07 0.494431  887         14
## 113153  UGA_173101_P020_WG06 UGA_173101_P016_WD06 0.489770  849         14
## 113184  UGA_173101_P020_WG06 UGA_173101_P017_WA02 0.497602  907         14
## 113197  UGA_173101_P020_WG06 UGA_173101_P015_WH07 0.495192  903         14
## 113223  UGA_173101_P020_WG06 UGA_173101_P020_WA05 0.498812  907         14
## 113285  UGA_173101_P020_WG06 UGA_173101_P018_WE02 0.497010  906         14
## 113345  UGA_173101_P020_WG06 UGA_173101_P016_WH06 0.489583  777         14
## 113347  UGA_173101_P020_WG06 UGA_173101_P016_WH08 0.487374  863         14
## 113369  UGA_173101_P020_WG06 UGA_173101_P017_WF04 0.495261  908         14
## 113395  UGA_173101_P020_WG06 UGA_173101_P017_WF08 0.482036  740         14
## 113415  UGA_173101_P020_WG06 UGA_173101_P017_WH09 0.483711  771         14
## 113460  UGA_173101_P020_WG06 UGA_173101_P023_WG04 0.498223  909         14
## 113471  UGA_173101_P020_WG06 UGA_173101_P023_WB11 0.495763  901         14
## 113474  UGA_173101_P020_WG06 UGA_173101_P023_WB06 0.487374  892         14
## 113480  UGA_173101_P020_WG06 UGA_173101_P023_WB04 0.497017  904         14
## 127780  UGA_173101_P020_WH04 UGA_173101_P020_WA09 0.491071  850         14
## 127816  UGA_173101_P020_WH04 UGA_173101_P019_WC06 0.491686  914         14
## 127819  UGA_173101_P020_WH04 UGA_173101_P019_WC02 0.491029  914         14
## 127825  UGA_173101_P020_WH04 UGA_173101_P019_WD04 0.491706  916         14
## 127839  UGA_173101_P020_WH04 UGA_173101_P018_WF03 0.485802  894         14
## 127843  UGA_173101_P020_WH04 UGA_173101_P018_WE09 0.487685  880         14
## 127866  UGA_173101_P020_WH04 UGA_173101_P020_WA07 0.488243  888         14
## 127868  UGA_173101_P020_WH04 UGA_173101_P016_WD06 0.482143  851         14
## 127899  UGA_173101_P020_WH04 UGA_173101_P017_WA02 0.491627  912         14
## 127912  UGA_173101_P020_WH04 UGA_173101_P015_WH07 0.489209  906         14
## 127938  UGA_173101_P020_WH04 UGA_173101_P020_WA05 0.492891  909         14
## 128000  UGA_173101_P020_WH04 UGA_173101_P018_WE02 0.491050  910         14
## 128060  UGA_173101_P020_WH04 UGA_173101_P016_WH06 0.482639  778         14
## 128062  UGA_173101_P020_WH04 UGA_173101_P016_WH08 0.481108  866         14
## 128084  UGA_173101_P020_WH04 UGA_173101_P017_WF04 0.490544  912         14
## 128110  UGA_173101_P020_WH04 UGA_173101_P017_WF08 0.475299  742         14
## 128130  UGA_173101_P020_WH04 UGA_173101_P017_WH09 0.479462  772         14
## 128175  UGA_173101_P020_WH04 UGA_173101_P023_WG04 0.492317  913         14
## 128186  UGA_173101_P020_WH04 UGA_173101_P023_WB11 0.489734  904         14
## 128189  UGA_173101_P020_WH04 UGA_173101_P023_WB06 0.481061  896         14
## 128195  UGA_173101_P020_WH04 UGA_173101_P023_WB04 0.491050  907         14
## 128242  UGA_173101_P020_WH04 UGA_173101_P020_WG06 0.492908  910         14
## 133918  UGA_173101_P020_WG07 UGA_173101_P020_WA09 0.474453  595         14
## 133954  UGA_173101_P020_WG07 UGA_173101_P019_WC06 0.475719  608         14
## 133957  UGA_173101_P020_WG07 UGA_173101_P019_WC02 0.476619  608         14
## 133963  UGA_173101_P020_WG07 UGA_173101_P019_WD04 0.478417  608         14
## 133977  UGA_173101_P020_WG07 UGA_173101_P018_WF03 0.474265  601         14
## 133981  UGA_173101_P020_WG07 UGA_173101_P018_WE09 0.472727  595         14
## 134004  UGA_173101_P020_WG07 UGA_173101_P020_WA07 0.474170  601         14
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## 134006  UGA_173101_P020_WG07 UGA_173101_P016_WD06 0.470755  584         14
## 134037  UGA_173101_P020_WG07 UGA_173101_P017_WA02 0.477518  608         14
## 134050  UGA_173101_P020_WG07 UGA_173101_P015_WH07 0.476619  607         14
## 134076  UGA_173101_P020_WG07 UGA_173101_P020_WA05 0.477518  608         14
## 134138  UGA_173101_P020_WG07 UGA_173101_P018_WE02 0.476619  608         14
## 134198  UGA_173101_P020_WG07 UGA_173101_P016_WH06 0.465625  537         14
## 134200  UGA_173101_P020_WG07 UGA_173101_P016_WH08 0.463619  589         14
## 134222  UGA_173101_P020_WG07 UGA_173101_P017_WF04 0.476619  608         14
## 134248  UGA_173101_P020_WG07 UGA_173101_P017_WF08 0.464592  513         14
## 134268  UGA_173101_P020_WG07 UGA_173101_P017_WH09 0.463265  540         14
## 134313  UGA_173101_P020_WG07 UGA_173101_P023_WG04 0.477518  608         14
## 134324  UGA_173101_P020_WG07 UGA_173101_P023_WB11 0.471818  608         14
## 134327  UGA_173101_P020_WG07 UGA_173101_P023_WB06 0.468401  603         14
## 134333  UGA_173101_P020_WG07 UGA_173101_P023_WB04 0.476619  608         14
## 134380  UGA_173101_P020_WG07 UGA_173101_P020_WG06 0.476619  608         14
## 134410  UGA_173101_P020_WG07 UGA_173101_P020_WH04 0.469424  608         14
## 134955  UGA_173101_P020_WH11 UGA_173101_P020_WA09 0.498087  850         14
## 134991  UGA_173101_P020_WH11 UGA_173101_P019_WC06 0.498219  915         14
## 134994  UGA_173101_P020_WH11 UGA_173101_P019_WC02 0.497608  915         14
## 135000  UGA_173101_P020_WH11 UGA_173101_P019_WD04 0.498223  918         14
## 135014  UGA_173101_P020_WH11 UGA_173101_P018_WF03 0.492593  895         14
## 135018  UGA_173101_P020_WH11 UGA_173101_P018_WE09 0.494458  881         14
## 135041  UGA_173101_P020_WH11 UGA_173101_P020_WA07 0.495050  889         14
## 135043  UGA_173101_P020_WH11 UGA_173101_P016_WD06 0.489158  851         14
## 135074  UGA_173101_P020_WH11 UGA_173101_P017_WA02 0.498206  913         14
## 135087  UGA_173101_P020_WH11 UGA_173101_P015_WH07 0.495803  907         14
## 135113  UGA_173101_P020_WH11 UGA_173101_P020_WA05 0.499408  910         14
## 135175  UGA_173101_P020_WH11 UGA_173101_P018_WE02 0.497613  911         14
## 135235  UGA_173101_P020_WH11 UGA_173101_P016_WH06 0.490278  779         14
## 135237  UGA_173101_P020_WH11 UGA_173101_P016_WH08 0.488035  867         14
## 135259  UGA_173101_P020_WH11 UGA_173101_P017_WF04 0.495853  913         14
## 135285  UGA_173101_P020_WH11 UGA_173101_P017_WF08 0.482036  743         14
## 135305  UGA_173101_P020_WH11 UGA_173101_P017_WH09 0.484419  772         14
## 135350  UGA_173101_P020_WH11 UGA_173101_P023_WG04 0.498818  914         14
## 135361  UGA_173101_P020_WH11 UGA_173101_P023_WB11 0.496377  905         14
## 135364  UGA_173101_P020_WH11 UGA_173101_P023_WB06 0.488005  897         14
## 135370  UGA_173101_P020_WH11 UGA_173101_P023_WB04 0.497613  908         14
## 135417  UGA_173101_P020_WH11 UGA_173101_P020_WG06 0.499408  910         14
## 135447  UGA_173101_P020_WH11 UGA_173101_P020_WH04 0.493499  916         14
## 135459  UGA_173101_P020_WH11 UGA_173101_P020_WG07 0.477518  608         14
## 147711  UGA_173101_P022_WG09 UGA_173101_P020_WA09 0.498087  850         14
## 147747  UGA_173101_P022_WG09 UGA_173101_P019_WC06 0.498219  913         14
## 147750  UGA_173101_P022_WG09 UGA_173101_P019_WC02 0.497608  913         14
## 147756  UGA_173101_P022_WG09 UGA_173101_P019_WD04 0.498223  914         14
## 147770  UGA_173101_P022_WG09 UGA_173101_P018_WF03 0.492593  893         14
## 147774  UGA_173101_P022_WG09 UGA_173101_P018_WE09 0.494458  880         14
## 147797  UGA_173101_P022_WG09 UGA_173101_P020_WA07 0.495050  888         14
## 147799  UGA_173101_P022_WG09 UGA_173101_P016_WD06 0.489158  851         14
## 147830  UGA_173101_P022_WG09 UGA_173101_P017_WA02 0.498206  910         14
## 147843  UGA_173101_P022_WG09 UGA_173101_P015_WH07 0.495803  905         14
## 147869  UGA_173101_P022_WG09 UGA_173101_P020_WA05 0.499408  910         14
## 147931  UGA_173101_P022_WG09 UGA_173101_P018_WE02 0.497613  909         14
## 147991  UGA_173101_P022_WG09 UGA_173101_P016_WH06 0.490278  778         14
## 147993  UGA_173101_P022_WG09 UGA_173101_P016_WH08 0.488035  866         14
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## 148015  UGA_173101_P022_WG09 UGA_173101_P017_WF04 0.495853  911         14
## 148041  UGA_173101_P022_WG09 UGA_173101_P017_WF08 0.482036  742         14
## 148061  UGA_173101_P022_WG09 UGA_173101_P017_WH09 0.484419  770         14
## 148106  UGA_173101_P022_WG09 UGA_173101_P023_WG04 0.498818  912         14
## 148117  UGA_173101_P022_WG09 UGA_173101_P023_WB11 0.496377  904         14
## 148120  UGA_173101_P022_WG09 UGA_173101_P023_WB06 0.488005  894         14
## 148126  UGA_173101_P022_WG09 UGA_173101_P023_WB04 0.497613  908         14
## 148173  UGA_173101_P022_WG09 UGA_173101_P020_WG06 0.499408  909         14
## 148203  UGA_173101_P022_WG09 UGA_173101_P020_WH04 0.493499  913         14
## 148215  UGA_173101_P022_WG09 UGA_173101_P020_WG07 0.477518  608         14
## 148217  UGA_173101_P022_WG09 UGA_173101_P020_WH11 0.500000  914         14
## 153750  UGA_173101_P022_WA12 UGA_173101_P020_WA09 0.481233  821         14
## 153786  UGA_173101_P022_WA12 UGA_173101_P019_WC06 0.482234  865         14
## 153789  UGA_173101_P022_WA12 UGA_173101_P019_WC02 0.481552  864         14
## 153795  UGA_173101_P022_WA12 UGA_173101_P019_WD04 0.482234  865         14
## 153809  UGA_173101_P022_WA12 UGA_173101_P018_WF03 0.473168  849         14
## 153813  UGA_173101_P022_WA12 UGA_173101_P018_WE09 0.479275  841         14
## 153836  UGA_173101_P022_WA12 UGA_173101_P020_WA07 0.480469  848         14
## 153838  UGA_173101_P022_WA12 UGA_173101_P016_WD06 0.473649  813         14
## 153869  UGA_173101_P022_WA12 UGA_173101_P017_WA02 0.482188  864         14
## 153882  UGA_173101_P022_WA12 UGA_173101_P015_WH07 0.481552  861         14
## 153908  UGA_173101_P022_WA12 UGA_173101_P020_WA05 0.484137  865         14
## 153970  UGA_173101_P022_WA12 UGA_173101_P018_WE02 0.481552  864         14
## 154030  UGA_173101_P022_WA12 UGA_173101_P016_WH06 0.475877  746         14
## 154032  UGA_173101_P022_WA12 UGA_173101_P016_WH08 0.474206  828         14
## 154054  UGA_173101_P022_WA12 UGA_173101_P017_WF04 0.480280  863         14
## 154080  UGA_173101_P022_WA12 UGA_173101_P017_WF08 0.468750  711         14
## 154100  UGA_173101_P022_WA12 UGA_173101_P017_WH09 0.468750  738         14
## 154145  UGA_173101_P022_WA12 UGA_173101_P023_WG04 0.482868  865         14
## 154156  UGA_173101_P022_WA12 UGA_173101_P023_WB11 0.480179  861         14
## 154159  UGA_173101_P022_WA12 UGA_173101_P023_WB06 0.473333  851         14
## 154165  UGA_173101_P022_WA12 UGA_173101_P023_WB04 0.482824  864         14
## 154212  UGA_173101_P022_WA12 UGA_173101_P020_WG06 0.483503  863         14
## 154242  UGA_173101_P022_WA12 UGA_173101_P020_WH04 0.480330  864         14
## 154254  UGA_173101_P022_WA12 UGA_173101_P020_WG07 0.462454  595         14
## 154256  UGA_173101_P022_WA12 UGA_173101_P020_WH11 0.484137  865         14
## 154280  UGA_173101_P022_WA12 UGA_173101_P022_WG09 0.484137  865         14
## 166768  UGA_173101_P021_WC01 UGA_173101_P020_WA09 0.494859  842         14
## 166804  UGA_173101_P021_WC01 UGA_173101_P019_WC06 0.494578  895         14
## 166807  UGA_173101_P021_WC01 UGA_173101_P019_WC02 0.493932  894         14
## 166813  UGA_173101_P021_WC01 UGA_173101_P019_WD04 0.494578  895         14
## 166827  UGA_173101_P021_WC01 UGA_173101_P018_WF03 0.488750  878         14
## 166831  UGA_173101_P021_WC01 UGA_173101_P018_WE09 0.491337  869         14
## 166854  UGA_173101_P021_WC01 UGA_173101_P020_WA07 0.491272  877         14
## 166856  UGA_173101_P021_WC01 UGA_173101_P016_WD06 0.485897  841         14
## 166887  UGA_173101_P021_WC01 UGA_173101_P017_WA02 0.494552  893         14
## 166900  UGA_173101_P021_WC01 UGA_173101_P015_WH07 0.492092  890         14
## 166926  UGA_173101_P021_WC01 UGA_173101_P020_WA05 0.495783  894         14
## 166988  UGA_173101_P021_WC01 UGA_173101_P018_WE02 0.493961  894         14
## 167048  UGA_173101_P021_WC01 UGA_173101_P016_WH06 0.485994  769         14
## 167050  UGA_173101_P021_WC01 UGA_173101_P016_WH08 0.484097  854         14
## 167072  UGA_173101_P021_WC01 UGA_173101_P017_WF04 0.492150  893         14
## 167098  UGA_173101_P021_WC01 UGA_173101_P017_WF08 0.478979  733         14
## 167118  UGA_173101_P021_WC01 UGA_173101_P017_WH09 0.480714  764         14
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## 167163  UGA_173101_P021_WC01 UGA_173101_P023_WG04 0.495181  895         14
## 167174  UGA_173101_P021_WC01 UGA_173101_P023_WB11 0.492665  889         14
## 167177  UGA_173101_P021_WC01 UGA_173101_P023_WB06 0.483974  879         14
## 167183  UGA_173101_P021_WC01 UGA_173101_P023_WB04 0.495157  892         14
## 167230  UGA_173101_P021_WC01 UGA_173101_P020_WG06 0.495773  892         14
## 167260  UGA_173101_P021_WC01 UGA_173101_P020_WH04 0.489759  894         14
## 167272  UGA_173101_P021_WC01 UGA_173101_P020_WG07 0.474729  606         14
## 167274  UGA_173101_P021_WC01 UGA_173101_P020_WH11 0.496386  895         14
## 167298  UGA_173101_P021_WC01 UGA_173101_P022_WG09 0.496386  895         14
## 167309  UGA_173101_P021_WC01 UGA_173101_P022_WA12 0.481458  856         14
## 182725  UGA_173101_P021_WH06 UGA_173101_P020_WA09 0.488064  841         14
## 182761  UGA_173101_P021_WH06 UGA_173101_P019_WC06 0.488125  895         14
## 182764  UGA_173101_P021_WH06 UGA_173101_P019_WC02 0.488750  895         14
## 182770  UGA_173101_P021_WH06 UGA_173101_P019_WD04 0.488750  895         14
## 182784  UGA_173101_P021_WH06 UGA_173101_P018_WF03 0.483503  877         14
## 182788  UGA_173101_P021_WH06 UGA_173101_P018_WE09 0.484056  866         14
## 182811  UGA_173101_P021_WH06 UGA_173101_P020_WA07 0.486709  877         14
## 182813  UGA_173101_P021_WH06 UGA_173101_P016_WD06 0.476729  839         14
## 182844  UGA_173101_P021_WH06 UGA_173101_P017_WA02 0.488125  895         14
## 182857  UGA_173101_P021_WH06 UGA_173101_P015_WH07 0.484336  889         14
## 182883  UGA_173101_P021_WH06 UGA_173101_P020_WA05 0.488125  890         14
## 182945  UGA_173101_P021_WH06 UGA_173101_P018_WE02 0.487500  893         14
## 183005  UGA_173101_P021_WH06 UGA_173101_P016_WH06 0.480337  769         14
## 183007  UGA_173101_P021_WH06 UGA_173101_P016_WH08 0.477214  855         14
## 183029  UGA_173101_P021_WH06 UGA_173101_P017_WF04 0.485625  894         14
## 183055  UGA_173101_P021_WH06 UGA_173101_P017_WF08 0.472973  730         14
## 183075  UGA_173101_P021_WH06 UGA_173101_P017_WH09 0.476812  762         14
## 183120  UGA_173101_P021_WH06 UGA_173101_P023_WG04 0.487500  895         14
## 183131  UGA_173101_P021_WH06 UGA_173101_P023_WB11 0.487469  888         14
## 183134  UGA_173101_P021_WH06 UGA_173101_P023_WB06 0.475064  879         14
## 183140  UGA_173101_P021_WH06 UGA_173101_P023_WB04 0.486250  891         14
## 183187  UGA_173101_P021_WH06 UGA_173101_P020_WG06 0.488125  891         14
## 183217  UGA_173101_P021_WH06 UGA_173101_P020_WH04 0.481875  894         14
## 183229  UGA_173101_P021_WH06 UGA_173101_P020_WG07 0.472527  605         14
## 183231  UGA_173101_P021_WH06 UGA_173101_P020_WH11 0.488750  895         14
## 183255  UGA_173101_P021_WH06 UGA_173101_P022_WG09 0.488750  894         14
## 183266  UGA_173101_P021_WH06 UGA_173101_P022_WA12 0.473822  854         14
## 183289  UGA_173101_P021_WH06 UGA_173101_P021_WC01 0.484925  882         14
## 203856  UGA_173101_P020_WH03 UGA_173101_P020_WA09 0.492985  850         14
## 203892  UGA_173101_P020_WH03 UGA_173101_P019_WC06 0.493468  915         14
## 203895  UGA_173101_P020_WH03 UGA_173101_P019_WC02 0.492823  915         14
## 203901  UGA_173101_P020_WH03 UGA_173101_P019_WD04 0.492299  918         14
## 203915  UGA_173101_P020_WH03 UGA_173101_P018_WF03 0.487654  895         14
## 203919  UGA_173101_P020_WH03 UGA_173101_P018_WE09 0.489532  881         14
## 203942  UGA_173101_P020_WH03 UGA_173101_P020_WA07 0.490099  889         14
## 203944  UGA_173101_P020_WH03 UGA_173101_P016_WD06 0.484056  851         14
## 203975  UGA_173101_P020_WH03 UGA_173101_P017_WA02 0.492225  913         14
## 203988  UGA_173101_P020_WH03 UGA_173101_P015_WH07 0.489808  907         14
## 204014  UGA_173101_P020_WH03 UGA_173101_P020_WA05 0.493483  910         14
## 204076  UGA_173101_P020_WH03 UGA_173101_P018_WE02 0.491647  911         14
## 204136  UGA_173101_P020_WH03 UGA_173101_P016_WH06 0.484722  779         14
## 204138  UGA_173101_P020_WH03 UGA_173101_P016_WH08 0.484887  867         14
## 204160  UGA_173101_P020_WH03 UGA_173101_P017_WF04 0.489953  913         14
## 204186  UGA_173101_P020_WH03 UGA_173101_P017_WF08 0.475299  743         14
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## 204206  UGA_173101_P020_WH03 UGA_173101_P017_WH09 0.478754  772         14
## 204251  UGA_173101_P020_WH03 UGA_173101_P023_WG04 0.492908  914         14
## 204262  UGA_173101_P020_WH03 UGA_173101_P023_WB11 0.491546  905         14
## 204265  UGA_173101_P020_WH03 UGA_173101_P023_WB06 0.481692  897         14
## 204271  UGA_173101_P020_WH03 UGA_173101_P023_WB04 0.491647  908         14
## 204318  UGA_173101_P020_WH03 UGA_173101_P020_WG06 0.493499  910         14
## 204348  UGA_173101_P020_WH03 UGA_173101_P020_WH04 0.487762  916         14
## 204360  UGA_173101_P020_WH03 UGA_173101_P020_WG07 0.472122  608         14
## 204362  UGA_173101_P020_WH03 UGA_173101_P020_WH11 0.494090  918         14
## 204386  UGA_173101_P020_WH03 UGA_173101_P022_WG09 0.494090  914         14
## 204397  UGA_173101_P020_WH03 UGA_173101_P022_WA12 0.478426  865         14
## 204420  UGA_173101_P020_WH03 UGA_173101_P021_WC01 0.490964  895         14
## 204447  UGA_173101_P020_WH03 UGA_173101_P021_WH06 0.484375  895         14
## 232918            GTS_938-23 UGA_173101_P020_WA09 0.490741   94        250
## 232954            GTS_938-23 UGA_173101_P019_WC06 0.483871  119        250
## 232957            GTS_938-23 UGA_173101_P019_WC02 0.492188  119        250
## 232963            GTS_938-23 UGA_173101_P019_WD04 0.476562  122        250
## 232977            GTS_938-23 UGA_173101_P018_WF03 0.491667  110        250
## 232981            GTS_938-23 UGA_173101_P018_WE09 0.482143  105        250
## 233004            GTS_938-23 UGA_173101_P020_WA07 0.468750  104        250
## 233006            GTS_938-23 UGA_173101_P016_WD06 0.490385   99        250
## 233037            GTS_938-23 UGA_173101_P017_WA02 0.491935  117        250
## 233050            GTS_938-23 UGA_173101_P015_WH07 0.476562  113        250
## 233076            GTS_938-23 UGA_173101_P020_WA05 0.476562  114        250
## 233138            GTS_938-23 UGA_173101_P018_WE02 0.491935  115        250
## 233198            GTS_938-23 UGA_173101_P016_WH06 0.490385   90        250
## 233200            GTS_938-23 UGA_173101_P016_WH08 0.482759  102        250
## 233222            GTS_938-23 UGA_173101_P017_WF04 0.476562  117        250
## 233248            GTS_938-23 UGA_173101_P017_WF08 0.456522   94        250
## 233268            GTS_938-23 UGA_173101_P017_WH09 0.462963   96        250
## 233313            GTS_938-23 UGA_173101_P023_WG04 0.484375  115        250
## 233324            GTS_938-23 UGA_173101_P023_WB11 0.491935  106        250
## 233327            GTS_938-23 UGA_173101_P023_WB06 0.483333  107        250
## 233333            GTS_938-23 UGA_173101_P023_WB04 0.482759  108        250
## 233380            GTS_938-23 UGA_173101_P020_WG06 0.483871  113        250
## 233410            GTS_938-23 UGA_173101_P020_WH04 0.484375  117        250
## 233422            GTS_938-23 UGA_173101_P020_WG07 0.466667   57        250
## 233424            GTS_938-23 UGA_173101_P020_WH11 0.484375  118        250
## 233448            GTS_938-23 UGA_173101_P022_WG09 0.484375  114        250
## 233459            GTS_938-23 UGA_173101_P022_WA12 0.461538   94        250
## 233482            GTS_938-23 UGA_173101_P021_WC01 0.473214  100        250
## 233509            GTS_938-23 UGA_173101_P021_WH06 0.480769  102        250
## 233543            GTS_938-23 UGA_173101_P020_WH03 0.468750  118        250
## 235656            GTS_944-23 UGA_173101_P020_WA09 0.490741   93        250
## 235692            GTS_944-23 UGA_173101_P019_WC06 0.491935  117        250
## 235695            GTS_944-23 UGA_173101_P019_WC02 0.492188  118        250
## 235701            GTS_944-23 UGA_173101_P019_WD04 0.484375  120        250
## 235715            GTS_944-23 UGA_173101_P018_WF03 0.491667  109        250
## 235719            GTS_944-23 UGA_173101_P018_WE09 0.482143  104        250
## 235742            GTS_944-23 UGA_173101_P020_WA07 0.468750  103        250
## 235744            GTS_944-23 UGA_173101_P016_WD06 0.490385   98        250
## 235775            GTS_944-23 UGA_173101_P017_WA02 0.491935  116        250
## 235788            GTS_944-23 UGA_173101_P015_WH07 0.484375  111        250
## 235814            GTS_944-23 UGA_173101_P020_WA05 0.484375  112        250
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## 235876            GTS_944-23 UGA_173101_P018_WE02 0.491935  114        250
## 235936            GTS_944-23 UGA_173101_P016_WH06 0.490385   89        250
## 235938            GTS_944-23 UGA_173101_P016_WH08 0.482759  101        250
## 235960            GTS_944-23 UGA_173101_P017_WF04 0.476562  116        250
## 235986            GTS_944-23 UGA_173101_P017_WF08 0.456522   93        250
## 236006            GTS_944-23 UGA_173101_P017_WH09 0.462963   95        250
## 236051            GTS_944-23 UGA_173101_P023_WG04 0.492188  113        250
## 236062            GTS_944-23 UGA_173101_P023_WB11 0.491935  105        250
## 236065            GTS_944-23 UGA_173101_P023_WB06 0.483333  106        250
## 236071            GTS_944-23 UGA_173101_P023_WB04 0.482759  107        250
## 236118            GTS_944-23 UGA_173101_P020_WG06 0.491935  111        250
## 236148            GTS_944-23 UGA_173101_P020_WH04 0.492188  115        250
## 236160            GTS_944-23 UGA_173101_P020_WG07 0.466667   57        250
## 236162            GTS_944-23 UGA_173101_P020_WH11 0.492188  116        250
## 236186            GTS_944-23 UGA_173101_P022_WG09 0.492188  112        250
## 236197            GTS_944-23 UGA_173101_P022_WA12 0.471154   92        250
## 236220            GTS_944-23 UGA_173101_P021_WC01 0.482143   98        250
## 236247            GTS_944-23 UGA_173101_P021_WH06 0.480769  101        250
## 236281            GTS_944-23 UGA_173101_P020_WH03 0.476562  116        250
## 236325            GTS_944-23           GTS_938-23 0.500000  121        250
## 253131            GTS_631-23 UGA_173101_P020_WA09 0.451923   93        250
## 253167            GTS_631-23 UGA_173101_P019_WC06 0.450000  117        250
## 253170            GTS_631-23 UGA_173101_P019_WC02 0.459677  117        250
## 253176            GTS_631-23 UGA_173101_P019_WD04 0.443548  120        250
## 253190            GTS_631-23 UGA_173101_P018_WF03 0.456897  108        250
## 253194            GTS_631-23 UGA_173101_P018_WE09 0.444444  103        250
## 253217            GTS_631-23 UGA_173101_P020_WA07 0.423913  102        250
## 253219            GTS_631-23 UGA_173101_P016_WD06 0.450000   97        250
## 253250            GTS_631-23 UGA_173101_P017_WA02 0.458333  115        250
## 253263            GTS_631-23 UGA_173101_P015_WH07 0.443548  111        250
## 253289            GTS_631-23 UGA_173101_P020_WA05 0.443548  112        250
## 253351            GTS_631-23 UGA_173101_P018_WE02 0.458333  113        250
## 253411            GTS_631-23 UGA_173101_P016_WH06 0.450000   89        250
## 253413            GTS_631-23 UGA_173101_P016_WH08 0.446429  100        250
## 253435            GTS_631-23 UGA_173101_P017_WF04 0.443548  115        250
## 253461            GTS_631-23 UGA_173101_P017_WF08 0.409091   93        250
## 253481            GTS_631-23 UGA_173101_P017_WH09 0.423077   94        250
## 253526            GTS_631-23 UGA_173101_P023_WG04 0.451613  113        250
## 253537            GTS_631-23 UGA_173101_P023_WB11 0.458333  104        250
## 253540            GTS_631-23 UGA_173101_P023_WB06 0.448276  105        250
## 253546            GTS_631-23 UGA_173101_P023_WB04 0.446429  106        250
## 253593            GTS_631-23 UGA_173101_P020_WG06 0.450000  111        250
## 253623            GTS_631-23 UGA_173101_P020_WH04 0.451613  115        250
## 253635            GTS_631-23 UGA_173101_P020_WG07 0.400000   58        250
## 253637            GTS_631-23 UGA_173101_P020_WH11 0.451613  116        250
## 253661            GTS_631-23 UGA_173101_P022_WG09 0.451613  112        250
## 253672            GTS_631-23 UGA_173101_P022_WA12 0.420000   92        250
## 253695            GTS_631-23 UGA_173101_P021_WC01 0.435185   98        250
## 253722            GTS_631-23 UGA_173101_P021_WH06 0.440000  100        250
## 253756            GTS_631-23 UGA_173101_P020_WH03 0.435484  116        250
## 253800            GTS_631-23           GTS_938-23 0.500000  120        250
## 253804            GTS_631-23           GTS_944-23 0.500000  119        250
## 258136            GTS_781-23 UGA_173101_P020_WA09 0.453704   94        269
## 258172            GTS_781-23 UGA_173101_P019_WC06 0.460938  119        269
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## 258175             GTS_781-23 UGA_173101_P019_WC02 0.460938  119         269
## 258181             GTS_781-23 UGA_173101_P019_WD04 0.454545  122         269
## 258195             GTS_781-23 UGA_173101_P018_WF03 0.458333  110         269
## 258199             GTS_781-23 UGA_173101_P018_WE09 0.446429  105         269
## 258222             GTS_781-23 UGA_173101_P020_WA07 0.427083  104         269
## 258224             GTS_781-23 UGA_173101_P016_WD06 0.451923   99         269
## 258255             GTS_781-23 UGA_173101_P017_WA02 0.459677  117         269
## 258268             GTS_781-23 UGA_173101_P015_WH07 0.454545  113         269
## 258294             GTS_781-23 UGA_173101_P020_WA05 0.454545  114         269
## 258356             GTS_781-23 UGA_173101_P018_WE02 0.459677  115         269
## 258416             GTS_781-23 UGA_173101_P016_WH06 0.451923   90         269
## 258418             GTS_781-23 UGA_173101_P016_WH08 0.448276  102         269
## 258440             GTS_781-23 UGA_173101_P017_WF04 0.445312  117         269
## 258466             GTS_781-23 UGA_173101_P017_WF08 0.413043   94         269
## 258486             GTS_781-23 UGA_173101_P017_WH09 0.425926   96         269
## 258531             GTS_781-23 UGA_173101_P023_WG04 0.462121  115         269
## 258542             GTS_781-23 UGA_173101_P023_WB11 0.459677  106         269
## 258545             GTS_781-23 UGA_173101_P023_WB06 0.450000  107         269
## 258551             GTS_781-23 UGA_173101_P023_WB04 0.448276  108         269
## 258598             GTS_781-23 UGA_173101_P020_WG06 0.460938  113         269
## 258628             GTS_781-23 UGA_173101_P020_WH04 0.462121  117         269
## 258640             GTS_781-23 UGA_173101_P020_WG07 0.400000   58         269
## 258642             GTS_781-23 UGA_173101_P020_WH11 0.462121  118         269
## 258666             GTS_781-23 UGA_173101_P022_WG09 0.462121  114         269
## 258677             GTS_781-23 UGA_173101_P022_WA12 0.435185   94         269
## 258700             GTS_781-23 UGA_173101_P021_WC01 0.448276  100         269
## 258727             GTS_781-23 UGA_173101_P021_WH06 0.442308  102         269
## 258761             GTS_781-23 UGA_173101_P020_WH03 0.446970  118         269
## 258805             GTS_781-23           GTS_938-23 0.492188  122         269
## 258809             GTS_781-23           GTS_944-23 0.500000  121         269
## 258834             GTS_781-23           GTS_631-23 0.491935  121         269
## 264643            GTS_781-23R UGA_173101_P020_WA09 0.490741   93         269
## 264679            GTS_781-23R UGA_173101_P019_WC06 0.492188  118         269
## 264682            GTS_781-23R UGA_173101_P019_WC02 0.492188  118         269
## 264688            GTS_781-23R UGA_173101_P019_WD04 0.484848  121         269
## 264702            GTS_781-23R UGA_173101_P018_WF03 0.491667  109         269
## 264706            GTS_781-23R UGA_173101_P018_WE09 0.482143  104         269
## 264729            GTS_781-23R UGA_173101_P020_WA07 0.468750  103         269
## 264731            GTS_781-23R UGA_173101_P016_WD06 0.490385   98         269
## 264762            GTS_781-23R UGA_173101_P017_WA02 0.491935  116         269
## 264775            GTS_781-23R UGA_173101_P015_WH07 0.484848  112         269
## 264801            GTS_781-23R UGA_173101_P020_WA05 0.484848  113         269
## 264863            GTS_781-23R UGA_173101_P018_WE02 0.491935  114         269
## 264923            GTS_781-23R UGA_173101_P016_WH06 0.490385   89         269
## 264925            GTS_781-23R UGA_173101_P016_WH08 0.482759  101         269
## 264947            GTS_781-23R UGA_173101_P017_WF04 0.476562  116         269
## 264973            GTS_781-23R UGA_173101_P017_WF08 0.456522   93         269
## 264993            GTS_781-23R UGA_173101_P017_WH09 0.462963   95         269
## 265038            GTS_781-23R UGA_173101_P023_WG04 0.492424  114         269
## 265049            GTS_781-23R UGA_173101_P023_WB11 0.491935  105         269
## 265052            GTS_781-23R UGA_173101_P023_WB06 0.483333  106         269
## 265058            GTS_781-23R UGA_173101_P023_WB04 0.482759  107         269
## 265105            GTS_781-23R UGA_173101_P020_WG06 0.492188  112         269
## 265135            GTS_781-23R UGA_173101_P020_WH04 0.492424  116         269
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## 265147            GTS_781-23R UGA_173101_P020_WG07 0.466667   57         269
## 265149            GTS_781-23R UGA_173101_P020_WH11 0.492424  117         269
## 265173            GTS_781-23R UGA_173101_P022_WG09 0.492424  113         269
## 265184            GTS_781-23R UGA_173101_P022_WA12 0.472222   93         269
## 265207            GTS_781-23R UGA_173101_P021_WC01 0.482759   99         269
## 265234            GTS_781-23R UGA_173101_P021_WH06 0.480769  101         269
## 265268            GTS_781-23R UGA_173101_P020_WH03 0.477273  117         269
## 265312            GTS_781-23R           GTS_938-23 0.492188  122         269
## 265316            GTS_781-23R           GTS_944-23 0.500000  121         269
## 265341            GTS_781-23R           GTS_631-23 0.491935  120         269
## 265348            GTS_781-23R           GTS_781-23 0.500000  122         269
## 266830             GTS_638-23 UGA_173101_P020_WA09 0.480769   94         250
## 266866             GTS_638-23 UGA_173101_P019_WC06 0.483333  116         250
## 266869             GTS_638-23 UGA_173101_P019_WC02 0.483871  117         250
## 266875             GTS_638-23 UGA_173101_P019_WD04 0.475806  119         250
## 266889             GTS_638-23 UGA_173101_P018_WF03 0.482759  108         250
## 266893             GTS_638-23 UGA_173101_P018_WE09 0.472222  104         250
## 266916             GTS_638-23 UGA_173101_P020_WA07 0.458333  102         250
## 266918             GTS_638-23 UGA_173101_P016_WD06 0.480000   97         250
## 266949             GTS_638-23 UGA_173101_P017_WA02 0.483333  115         250
## 266962             GTS_638-23 UGA_173101_P015_WH07 0.475806  110         250
## 266988             GTS_638-23 UGA_173101_P020_WA05 0.475806  111         250
## 267050             GTS_638-23 UGA_173101_P018_WE02 0.483333  113         250
## 267110             GTS_638-23 UGA_173101_P016_WH06 0.480000   89         250
## 267112             GTS_638-23 UGA_173101_P016_WH08 0.473214  100         250
## 267134             GTS_638-23 UGA_173101_P017_WF04 0.467742  115         250
## 267160             GTS_638-23 UGA_173101_P017_WF08 0.445652   94         250
## 267180             GTS_638-23 UGA_173101_P017_WH09 0.451923   95         250
## 267225             GTS_638-23 UGA_173101_P023_WG04 0.483871  112         250
## 267236             GTS_638-23 UGA_173101_P023_WB11 0.483333  105         250
## 267239             GTS_638-23 UGA_173101_P023_WB06 0.474138  105         250
## 267245             GTS_638-23 UGA_173101_P023_WB04 0.473214  107         250
## 267292             GTS_638-23 UGA_173101_P020_WG06 0.483333  110         250
## 267322             GTS_638-23 UGA_173101_P020_WH04 0.483871  114         250
## 267334             GTS_638-23 UGA_173101_P020_WG07 0.466667   57         250
## 267336             GTS_638-23 UGA_173101_P020_WH11 0.483871  115         250
## 267360             GTS_638-23 UGA_173101_P022_WG09 0.483871  111         250
## 267371             GTS_638-23 UGA_173101_P022_WA12 0.460000   92         250
## 267394             GTS_638-23 UGA_173101_P021_WC01 0.472222   98         250
## 267421             GTS_638-23 UGA_173101_P021_WH06 0.470000  100         250
## 267455             GTS_638-23 UGA_173101_P020_WH03 0.467742  115         250
## 267499             GTS_638-23           GTS_938-23 0.491935  120         250
## 267503             GTS_638-23           GTS_944-23 0.491935  119         250
## 267528             GTS_638-23           GTS_631-23 0.500000  118         250
## 267535             GTS_638-23           GTS_781-23 0.491935  119         250
## 267544             GTS_638-23          GTS_781-23R 0.491935  119         250
## 271968             GTS_972-23 UGA_173101_P020_WA09 0.490741   93         269
## 272004             GTS_972-23 UGA_173101_P019_WC06 0.491935  117         269
## 272007             GTS_972-23 UGA_173101_P019_WC02 0.492188  118         269
## 272013             GTS_972-23 UGA_173101_P019_WD04 0.484375  120         269
## 272027             GTS_972-23 UGA_173101_P018_WF03 0.491667  109         269
## 272031             GTS_972-23 UGA_173101_P018_WE09 0.482143  104         269
## 272054             GTS_972-23 UGA_173101_P020_WA07 0.468750  103         269
## 272056             GTS_972-23 UGA_173101_P016_WD06 0.490385   98         269
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## 272087             GTS_972-23 UGA_173101_P017_WA02 0.491935  116        269
## 272100             GTS_972-23 UGA_173101_P015_WH07 0.484375  111        269
## 272126             GTS_972-23 UGA_173101_P020_WA05 0.484375  112        269
## 272188             GTS_972-23 UGA_173101_P018_WE02 0.491935  114        269
## 272248             GTS_972-23 UGA_173101_P016_WH06 0.490385   89        269
## 272250             GTS_972-23 UGA_173101_P016_WH08 0.482759  101        269
## 272272             GTS_972-23 UGA_173101_P017_WF04 0.476562  116        269
## 272298             GTS_972-23 UGA_173101_P017_WF08 0.456522   93        269
## 272318             GTS_972-23 UGA_173101_P017_WH09 0.462963   95        269
## 272363             GTS_972-23 UGA_173101_P023_WG04 0.492188  113        269
## 272374             GTS_972-23 UGA_173101_P023_WB11 0.491935  105        269
## 272377             GTS_972-23 UGA_173101_P023_WB06 0.483333  106        269
## 272383             GTS_972-23 UGA_173101_P023_WB04 0.482759  107        269
## 272430             GTS_972-23 UGA_173101_P020_WG06 0.491935  111        269
## 272460             GTS_972-23 UGA_173101_P020_WH04 0.492188  115        269
## 272472             GTS_972-23 UGA_173101_P020_WG07 0.466667   57        269
## 272474             GTS_972-23 UGA_173101_P020_WH11 0.492188  116        269
## 272498             GTS_972-23 UGA_173101_P022_WG09 0.492188  112        269
## 272509             GTS_972-23 UGA_173101_P022_WA12 0.471154   92        269
## 272532             GTS_972-23 UGA_173101_P021_WC01 0.482143   98        269
## 272559             GTS_972-23 UGA_173101_P021_WH06 0.480769  101        269
## 272593             GTS_972-23 UGA_173101_P020_WH03 0.476562  116        269
## 272637             GTS_972-23           GTS_938-23 0.500000  121        269
## 272641             GTS_972-23           GTS_944-23 0.500000  121        269
## 272666             GTS_972-23           GTS_631-23 0.500000  119        269
## 272673             GTS_972-23           GTS_781-23 0.500000  121        269
## 272682             GTS_972-23          GTS_781-23R 0.500000  121        269
## 272685             GTS_972-23           GTS_638-23 0.491935  119        269
## 288435             GTS_651-23 UGA_173101_P020_WA09 0.490741   94        250
## 288471             GTS_651-23 UGA_173101_P019_WC06 0.491935  118        250
## 288474             GTS_651-23 UGA_173101_P019_WC02 0.492188  119        250
## 288480             GTS_651-23 UGA_173101_P019_WD04 0.484375  121        250
## 288494             GTS_651-23 UGA_173101_P018_WF03 0.491667  110        250
## 288498             GTS_651-23 UGA_173101_P018_WE09 0.482143  105        250
## 288521             GTS_651-23 UGA_173101_P020_WA07 0.468750  104        250
## 288523             GTS_651-23 UGA_173101_P016_WD06 0.490385   99        250
## 288554             GTS_651-23 UGA_173101_P017_WA02 0.491935  117        250
## 288567             GTS_651-23 UGA_173101_P015_WH07 0.484375  112        250
## 288593             GTS_651-23 UGA_173101_P020_WA05 0.484375  113        250
## 288655             GTS_651-23 UGA_173101_P018_WE02 0.491935  115        250
## 288715             GTS_651-23 UGA_173101_P016_WH06 0.490385   90        250
## 288717             GTS_651-23 UGA_173101_P016_WH08 0.482759  102        250
## 288739             GTS_651-23 UGA_173101_P017_WF04 0.476562  117        250
## 288765             GTS_651-23 UGA_173101_P017_WF08 0.456522   94        250
## 288785             GTS_651-23 UGA_173101_P017_WH09 0.462963   96        250
## 288830             GTS_651-23 UGA_173101_P023_WG04 0.492188  114        250
## 288841             GTS_651-23 UGA_173101_P023_WB11 0.491935  106        250
## 288844             GTS_651-23 UGA_173101_P023_WB06 0.483333  107        250
## 288850             GTS_651-23 UGA_173101_P023_WB04 0.482759  108        250
## 288897             GTS_651-23 UGA_173101_P020_WG06 0.491935  112        250
## 288927             GTS_651-23 UGA_173101_P020_WH04 0.492188  116        250
## 288939             GTS_651-23 UGA_173101_P020_WG07 0.466667   57        250
## 288941             GTS_651-23 UGA_173101_P020_WH11 0.492188  117        250
## 288965             GTS_651-23 UGA_173101_P022_WG09 0.492188  113        250
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## 288976          GTS_651-23 UGA_173101_P022_WA12 0.471154   93        250
## 288999          GTS_651-23 UGA_173101_P021_WC01 0.482143   99        250
## 289026          GTS_651-23 UGA_173101_P021_WH06 0.480769  102        250
## 289060          GTS_651-23 UGA_173101_P020_WH03 0.476562  117        250
## 289104          GTS_651-23           GTS_938-23 0.500000  122        250
## 289108          GTS_651-23           GTS_944-23 0.500000  121        250
## 289133          GTS_651-23           GTS_631-23 0.500000  119        250
## 289140          GTS_651-23           GTS_781-23 0.500000  121        250
## 289149          GTS_651-23          GTS_781-23R 0.500000  121        250
## 289152          GTS_651-23           GTS_638-23 0.491935  120        250
## 289159          GTS_651-23           GTS_972-23 0.500000  121        250
## 289195          GTS_982-23 UGA_173101_P020_WA09 0.490741   94        250
## 289231          GTS_982-23 UGA_173101_P019_WC06 0.491935  118        250
## 289234          GTS_982-23 UGA_173101_P019_WC02 0.492188  119        250
## 289240          GTS_982-23 UGA_173101_P019_WD04 0.484375  121        250
## 289254          GTS_982-23 UGA_173101_P018_WF03 0.491667  110        250
## 289258          GTS_982-23 UGA_173101_P018_WE09 0.482143  105        250
## 289281          GTS_982-23 UGA_173101_P020_WA07 0.468750  104        250
## 289283          GTS_982-23 UGA_173101_P016_WD06 0.490385   99        250
## 289314          GTS_982-23 UGA_173101_P017_WA02 0.491935  117        250
## 289327          GTS_982-23 UGA_173101_P015_WH07 0.484375  112        250
## 289353          GTS_982-23 UGA_173101_P020_WA05 0.484375  113        250
## 289415          GTS_982-23 UGA_173101_P018_WE02 0.491935  115        250
## 289475          GTS_982-23 UGA_173101_P016_WH06 0.490385   90        250
## 289477          GTS_982-23 UGA_173101_P016_WH08 0.482759  102        250
## 289499          GTS_982-23 UGA_173101_P017_WF04 0.476562  117        250
## 289525          GTS_982-23 UGA_173101_P017_WF08 0.456522   94        250
## 289545          GTS_982-23 UGA_173101_P017_WH09 0.462963   96        250
## 289590          GTS_982-23 UGA_173101_P023_WG04 0.492188  114        250
## 289601          GTS_982-23 UGA_173101_P023_WB11 0.491935  106        250
## 289604          GTS_982-23 UGA_173101_P023_WB06 0.483333  107        250
## 289610          GTS_982-23 UGA_173101_P023_WB04 0.482759  108        250
## 289657          GTS_982-23 UGA_173101_P020_WG06 0.491935  112        250
## 289687          GTS_982-23 UGA_173101_P020_WH04 0.492188  116        250
## 289699          GTS_982-23 UGA_173101_P020_WG07 0.466667   57        250
## 289701          GTS_982-23 UGA_173101_P020_WH11 0.492188  117        250
## 289725          GTS_982-23 UGA_173101_P022_WG09 0.492188  113        250
## 289736          GTS_982-23 UGA_173101_P022_WA12 0.471154   93        250
## 289759          GTS_982-23 UGA_173101_P021_WC01 0.482143   99        250
## 289786          GTS_982-23 UGA_173101_P021_WH06 0.480769  102        250
## 289820          GTS_982-23 UGA_173101_P020_WH03 0.476562  117        250
## 289864          GTS_982-23           GTS_938-23 0.500000  122        250
## 289868          GTS_982-23           GTS_944-23 0.500000  121        250
## 289893          GTS_982-23           GTS_631-23 0.500000  119        250
## 289900          GTS_982-23           GTS_781-23 0.500000  121        250
## 289909          GTS_982-23          GTS_781-23R 0.500000  121        250
## 289912          GTS_982-23           GTS_638-23 0.491935  120        250
## 289919          GTS_982-23           GTS_972-23 0.500000  121        250
## 289941          GTS_982-23           GTS_651-23 0.500000  122        250
## 296080         GTS_651-23R UGA_173101_P020_WA09 0.490741   94        250
## 296116         GTS_651-23R UGA_173101_P019_WC06 0.483871  119        250
## 296119         GTS_651-23R UGA_173101_P019_WC02 0.492188  119        250
## 296125         GTS_651-23R UGA_173101_P019_WD04 0.476562  122        250
## 296139         GTS_651-23R UGA_173101_P018_WF03 0.491667  110        250
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## 296143            GTS_651-23R UGA_173101_P018_WE09 0.482143  105         250
## 296166            GTS_651-23R UGA_173101_P020_WA07 0.468750  104         250
## 296168            GTS_651-23R UGA_173101_P016_WD06 0.490385   99         250
## 296199            GTS_651-23R UGA_173101_P017_WA02 0.491935  117         250
## 296212            GTS_651-23R UGA_173101_P015_WH07 0.476562  113         250
## 296238            GTS_651-23R UGA_173101_P020_WA05 0.476562  114         250
## 296300            GTS_651-23R UGA_173101_P018_WE02 0.491935  115         250
## 296360            GTS_651-23R UGA_173101_P016_WH06 0.490385   90         250
## 296362            GTS_651-23R UGA_173101_P016_WH08 0.482759  102         250
## 296384            GTS_651-23R UGA_173101_P017_WF04 0.476562  117         250
## 296410            GTS_651-23R UGA_173101_P017_WF08 0.456522   94         250
## 296430            GTS_651-23R UGA_173101_P017_WH09 0.462963   96         250
## 296475            GTS_651-23R UGA_173101_P023_WG04 0.484375  115         250
## 296486            GTS_651-23R UGA_173101_P023_WB11 0.491935  106         250
## 296489            GTS_651-23R UGA_173101_P023_WB06 0.483333  107         250
## 296495            GTS_651-23R UGA_173101_P023_WB04 0.482759  108         250
## 296542            GTS_651-23R UGA_173101_P020_WG06 0.483871  113         250
## 296572            GTS_651-23R UGA_173101_P020_WH04 0.484375  117         250
## 296584            GTS_651-23R UGA_173101_P020_WG07 0.466667   57         250
## 296586            GTS_651-23R UGA_173101_P020_WH11 0.484375  118         250
## 296610            GTS_651-23R UGA_173101_P022_WG09 0.484375  114         250
## 296621            GTS_651-23R UGA_173101_P022_WA12 0.461538   94         250
## 296644            GTS_651-23R UGA_173101_P021_WC01 0.473214  100         250
## 296671            GTS_651-23R UGA_173101_P021_WH06 0.480769  102         250
## 296705            GTS_651-23R UGA_173101_P020_WH03 0.468750  118         250
## 296749            GTS_651-23R           GTS_938-23 0.500000  123         250
## 296753            GTS_651-23R           GTS_944-23 0.500000  121         250
## 296778            GTS_651-23R           GTS_631-23 0.500000  120         250
## 296785            GTS_651-23R           GTS_781-23 0.492188  122         250
## 296794            GTS_651-23R          GTS_781-23R 0.492188  122         250
## 296797            GTS_651-23R           GTS_638-23 0.491935  120         250
## 296804            GTS_651-23R           GTS_972-23 0.500000  121         250
## 296826            GTS_651-23R           GTS_651-23 0.500000  122         250
## 296827            GTS_651-23R           GTS_982-23 0.500000  122         250
## 300715             GTS_803-23 UGA_173101_P020_WA09 0.490741   94         269
## 300751             GTS_803-23 UGA_173101_P019_WC06 0.492188  119         269
## 300754             GTS_803-23 UGA_173101_P019_WC02 0.492188  119         269
## 300760             GTS_803-23 UGA_173101_P019_WD04 0.484848  122         269
## 300774             GTS_803-23 UGA_173101_P018_WF03 0.491667  110         269
## 300778             GTS_803-23 UGA_173101_P018_WE09 0.482143  105         269
## 300801             GTS_803-23 UGA_173101_P020_WA07 0.468750  104         269
## 300803             GTS_803-23 UGA_173101_P016_WD06 0.490385   99         269
## 300834             GTS_803-23 UGA_173101_P017_WA02 0.491935  117         269
## 300847             GTS_803-23 UGA_173101_P015_WH07 0.484848  113         269
## 300873             GTS_803-23 UGA_173101_P020_WA05 0.484848  114         269
## 300935             GTS_803-23 UGA_173101_P018_WE02 0.491935  115         269
## 300995             GTS_803-23 UGA_173101_P016_WH06 0.490385   90         269
## 300997             GTS_803-23 UGA_173101_P016_WH08 0.482759  102         269
## 301019             GTS_803-23 UGA_173101_P017_WF04 0.476562  117         269
## 301045             GTS_803-23 UGA_173101_P017_WF08 0.456522   94         269
## 301065             GTS_803-23 UGA_173101_P017_WH09 0.462963   96         269
## 301110             GTS_803-23 UGA_173101_P023_WG04 0.492424  115         269
## 301121             GTS_803-23 UGA_173101_P023_WB11 0.491935  106         269
## 301124             GTS_803-23 UGA_173101_P023_WB06 0.483333  107         269
```

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## 301130            GTS_803-23 UGA_173101_P023_WB04 0.482759  108        269
## 301177            GTS_803-23 UGA_173101_P020_WG06 0.492188  113        269
## 301207            GTS_803-23 UGA_173101_P020_WH04 0.492424  117        269
## 301219            GTS_803-23 UGA_173101_P020_WG07 0.466667   57        269
## 301221            GTS_803-23 UGA_173101_P020_WH11 0.492424  118        269
## 301245            GTS_803-23 UGA_173101_P022_WG09 0.492424  114        269
## 301256            GTS_803-23 UGA_173101_P022_WA12 0.472222   94        269
## 301279            GTS_803-23 UGA_173101_P021_WC01 0.482759  100        269
## 301306            GTS_803-23 UGA_173101_P021_WH06 0.480769  102        269
## 301340            GTS_803-23 UGA_173101_P020_WH03 0.477273  118        269
## 301384            GTS_803-23           GTS_938-23 0.492188  123        269
## 301388            GTS_803-23           GTS_944-23 0.500000  121        269
## 301413            GTS_803-23           GTS_631-23 0.491935  120        269
## 301420            GTS_803-23           GTS_781-23 0.500000  122        269
## 301429            GTS_803-23          GTS_781-23R 0.500000  122        269
## 301432            GTS_803-23           GTS_638-23 0.491935  120        269
## 301439            GTS_803-23           GTS_972-23 0.500000  121        269
## 301461            GTS_803-23           GTS_651-23 0.500000  122        269
## 301462            GTS_803-23           GTS_982-23 0.500000  122        269
## 301471            GTS_803-23          GTS_651-23R 0.492188  123        269
## 313251            GTS_809-23 UGA_173101_P020_WA09 0.490741   94        269
## 313287            GTS_809-23 UGA_173101_P019_WC06 0.492188  119        269
## 313290            GTS_809-23 UGA_173101_P019_WC02 0.492188  119        269
## 313296            GTS_809-23 UGA_173101_P019_WD04 0.484848  122        269
## 313310            GTS_809-23 UGA_173101_P018_WF03 0.491667  110        269
## 313314            GTS_809-23 UGA_173101_P018_WE09 0.482143  105        269
## 313337            GTS_809-23 UGA_173101_P020_WA07 0.468750  104        269
## 313339            GTS_809-23 UGA_173101_P016_WD06 0.490385   99        269
## 313370            GTS_809-23 UGA_173101_P017_WA02 0.491935  117        269
## 313383            GTS_809-23 UGA_173101_P015_WH07 0.484848  113        269
## 313409            GTS_809-23 UGA_173101_P020_WA05 0.484848  114        269
## 313471            GTS_809-23 UGA_173101_P018_WE02 0.491935  115        269
## 313531            GTS_809-23 UGA_173101_P016_WH06 0.490385   90        269
## 313533            GTS_809-23 UGA_173101_P016_WH08 0.482759  102        269
## 313555            GTS_809-23 UGA_173101_P017_WF04 0.476562  117        269
## 313581            GTS_809-23 UGA_173101_P017_WF08 0.456522   94        269
## 313601            GTS_809-23 UGA_173101_P017_WH09 0.462963   96        269
## 313646            GTS_809-23 UGA_173101_P023_WG04 0.492424  115        269
## 313657            GTS_809-23 UGA_173101_P023_WB11 0.491935  106        269
## 313660            GTS_809-23 UGA_173101_P023_WB06 0.483333  107        269
## 313666            GTS_809-23 UGA_173101_P023_WB04 0.482759  108        269
## 313713            GTS_809-23 UGA_173101_P020_WG06 0.492188  113        269
## 313743            GTS_809-23 UGA_173101_P020_WH04 0.492424  117        269
## 313755            GTS_809-23 UGA_173101_P020_WG07 0.466667   57        269
## 313757            GTS_809-23 UGA_173101_P020_WH11 0.492424  118        269
## 313781            GTS_809-23 UGA_173101_P022_WG09 0.492424  114        269
## 313792            GTS_809-23 UGA_173101_P022_WA12 0.472222   94        269
## 313815            GTS_809-23 UGA_173101_P021_WC01 0.482759  100        269
## 313842            GTS_809-23 UGA_173101_P021_WH06 0.480769  102        269
## 313876            GTS_809-23 UGA_173101_P020_WH03 0.477273  118        269
## 313920            GTS_809-23           GTS_938-23 0.492188  123        269
## 313924            GTS_809-23           GTS_944-23 0.500000  121        269
## 313949            GTS_809-23           GTS_631-23 0.491935  120        269
## 313956            GTS_809-23           GTS_781-23 0.500000  122        269
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## 313965            GTS_809-23          GTS_781-23R 0.500000  122       269
## 313968            GTS_809-23           GTS_638-23 0.491935  120       269
## 313975            GTS_809-23           GTS_972-23 0.500000  121       269
## 313997            GTS_809-23           GTS_651-23 0.500000  122       269
## 313998            GTS_809-23           GTS_982-23 0.500000  122       269
## 314007            GTS_809-23          GTS_651-23R 0.492188  123       269
## 314013            GTS_809-23           GTS_803-23 0.500000  123       269
## 327660           GTS_1009-23 UGA_173101_P020_WA09 0.490741   94       250
## 327696           GTS_1009-23 UGA_173101_P019_WC06 0.491935  118       250
## 327699           GTS_1009-23 UGA_173101_P019_WC02 0.492188  119       250
## 327705           GTS_1009-23 UGA_173101_P019_WD04 0.484375  121       250
## 327719           GTS_1009-23 UGA_173101_P018_WF03 0.491667  110       250
## 327723           GTS_1009-23 UGA_173101_P018_WE09 0.482143  105       250
## 327746           GTS_1009-23 UGA_173101_P020_WA07 0.468750  104       250
## 327748           GTS_1009-23 UGA_173101_P016_WD06 0.490385   99       250
## 327779           GTS_1009-23 UGA_173101_P017_WA02 0.491935  117       250
## 327792           GTS_1009-23 UGA_173101_P015_WH07 0.484375  112       250
## 327818           GTS_1009-23 UGA_173101_P020_WA05 0.484375  113       250
## 327880           GTS_1009-23 UGA_173101_P018_WE02 0.491935  115       250
## 327940           GTS_1009-23 UGA_173101_P016_WH06 0.490385   90       250
## 327942           GTS_1009-23 UGA_173101_P016_WH08 0.482759  102       250
## 327964           GTS_1009-23 UGA_173101_P017_WF04 0.476562  117       250
## 327990           GTS_1009-23 UGA_173101_P017_WF08 0.456522   94       250
## 328010           GTS_1009-23 UGA_173101_P017_WH09 0.462963   96       250
## 328055           GTS_1009-23 UGA_173101_P023_WG04 0.492188  114       250
## 328066           GTS_1009-23 UGA_173101_P023_WB11 0.491935  106       250
## 328069           GTS_1009-23 UGA_173101_P023_WB06 0.483333  107       250
## 328075           GTS_1009-23 UGA_173101_P023_WB04 0.482759  108       250
## 328122           GTS_1009-23 UGA_173101_P020_WG06 0.491935  112       250
## 328152           GTS_1009-23 UGA_173101_P020_WH04 0.492188  116       250
## 328164           GTS_1009-23 UGA_173101_P020_WG07 0.466667   57       250
## 328166           GTS_1009-23 UGA_173101_P020_WH11 0.492188  117       250
## 328190           GTS_1009-23 UGA_173101_P022_WG09 0.492188  113       250
## 328201           GTS_1009-23 UGA_173101_P022_WA12 0.471154   93       250
## 328224           GTS_1009-23 UGA_173101_P021_WC01 0.482143   99       250
## 328251           GTS_1009-23 UGA_173101_P021_WH06 0.480769  102       250
## 328285           GTS_1009-23 UGA_173101_P020_WH03 0.476562  117       250
## 328329           GTS_1009-23           GTS_938-23 0.500000  122       250
## 328333           GTS_1009-23           GTS_944-23 0.500000  121       250
## 328358           GTS_1009-23           GTS_631-23 0.500000  119       250
## 328365           GTS_1009-23           GTS_781-23 0.500000  121       250
## 328374           GTS_1009-23          GTS_781-23R 0.500000  121       250
## 328377           GTS_1009-23           GTS_638-23 0.491935  120       250
## 328384           GTS_1009-23           GTS_972-23 0.500000  121       250
## 328406           GTS_1009-23           GTS_651-23 0.500000  122       250
## 328407           GTS_1009-23           GTS_982-23 0.500000  122       250
## 328416           GTS_1009-23          GTS_651-23R 0.500000  122       250
## 328422           GTS_1009-23           GTS_803-23 0.500000  122       250
## 328438           GTS_1009-23           GTS_809-23 0.500000  122       250
## 331720           GTS_1013-23 UGA_173101_P020_WA09 0.490741   94       250
## 331756           GTS_1013-23 UGA_173101_P019_WC06 0.483871  119       250
## 331759           GTS_1013-23 UGA_173101_P019_WC02 0.492188  119       250
## 331765           GTS_1013-23 UGA_173101_P019_WD04 0.476562  122       250
## 331779           GTS_1013-23 UGA_173101_P018_WF03 0.491667  110       250
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## 331783            GTS_1013-23 UGA_173101_P018_WE09 0.482143  105        250
## 331806            GTS_1013-23 UGA_173101_P020_WA07 0.468750  104        250
## 331808            GTS_1013-23 UGA_173101_P016_WD06 0.490385   99        250
## 331839            GTS_1013-23 UGA_173101_P017_WA02 0.491935  117        250
## 331852            GTS_1013-23 UGA_173101_P015_WH07 0.476562  113        250
## 331878            GTS_1013-23 UGA_173101_P020_WA05 0.476562  114        250
## 331940            GTS_1013-23 UGA_173101_P018_WE02 0.491935  115        250
## 332000            GTS_1013-23 UGA_173101_P016_WH06 0.490385   90        250
## 332002            GTS_1013-23 UGA_173101_P016_WH08 0.482759  102        250
## 332024            GTS_1013-23 UGA_173101_P017_WF04 0.476562  117        250
## 332050            GTS_1013-23 UGA_173101_P017_WF08 0.456522   94        250
## 332070            GTS_1013-23 UGA_173101_P017_WH09 0.462963   96        250
## 332115            GTS_1013-23 UGA_173101_P023_WG04 0.484375  115        250
## 332126            GTS_1013-23 UGA_173101_P023_WB11 0.491935  106        250
## 332129            GTS_1013-23 UGA_173101_P023_WB06 0.483333  107        250
## 332135            GTS_1013-23 UGA_173101_P023_WB04 0.482759  108        250
## 332182            GTS_1013-23 UGA_173101_P020_WG06 0.483871  113        250
## 332212            GTS_1013-23 UGA_173101_P020_WH04 0.484375  117        250
## 332224            GTS_1013-23 UGA_173101_P020_WG07 0.466667   57        250
## 332226            GTS_1013-23 UGA_173101_P020_WH11 0.484375  118        250
## 332250            GTS_1013-23 UGA_173101_P022_WG09 0.484375  114        250
## 332261            GTS_1013-23 UGA_173101_P022_WA12 0.461538   94        250
## 332284            GTS_1013-23 UGA_173101_P021_WC01 0.473214  100        250
## 332311            GTS_1013-23 UGA_173101_P021_WH06 0.480769  102        250
## 332345            GTS_1013-23 UGA_173101_P020_WH03 0.468750  118        250
## 332389            GTS_1013-23           GTS_938-23 0.500000  123        250
## 332393            GTS_1013-23           GTS_944-23 0.500000  121        250
## 332418            GTS_1013-23           GTS_631-23 0.500000  120        250
## 332425            GTS_1013-23           GTS_781-23 0.492188  122        250
## 332434            GTS_1013-23          GTS_781-23R 0.492188  122        250
## 332437            GTS_1013-23           GTS_638-23 0.491935  120        250
## 332444            GTS_1013-23           GTS_972-23 0.500000  121        250
## 332466            GTS_1013-23           GTS_651-23 0.500000  122        250
## 332467            GTS_1013-23           GTS_982-23 0.500000  122        250
## 332476            GTS_1013-23          GTS_651-23R 0.500000  123        250
## 332482            GTS_1013-23           GTS_803-23 0.492188  123        250
## 332498            GTS_1013-23           GTS_809-23 0.492188  123        250
## 332516            GTS_1013-23          GTS_1009-23 0.500000  122        250
## 388536             GTS_871-23 UGA_173101_P020_WA09 0.490741   93        269
## 388572             GTS_871-23 UGA_173101_P019_WC06 0.492188  118        269
## 388575             GTS_871-23 UGA_173101_P019_WC02 0.492188  118        269
## 388581             GTS_871-23 UGA_173101_P019_WD04 0.484848  121        269
## 388595             GTS_871-23 UGA_173101_P018_WF03 0.491667  109        269
## 388599             GTS_871-23 UGA_173101_P018_WE09 0.482143  104        269
## 388622             GTS_871-23 UGA_173101_P020_WA07 0.468750  103        269
## 388624             GTS_871-23 UGA_173101_P016_WD06 0.490385   98        269
## 388655             GTS_871-23 UGA_173101_P017_WA02 0.491935  116        269
## 388668             GTS_871-23 UGA_173101_P015_WH07 0.484848  112        269
## 388694             GTS_871-23 UGA_173101_P020_WA05 0.484848  113        269
## 388756             GTS_871-23 UGA_173101_P018_WE02 0.491935  114        269
## 388816             GTS_871-23 UGA_173101_P016_WH06 0.490385   89        269
## 388818             GTS_871-23 UGA_173101_P016_WH08 0.482759  101        269
## 388840             GTS_871-23 UGA_173101_P017_WF04 0.476562  116        269
## 388866             GTS_871-23 UGA_173101_P017_WF08 0.456522   93        269
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## 388886             GTS_871-23 UGA_173101_P017_WH09 0.462963   95        269
## 388931             GTS_871-23 UGA_173101_P023_WG04 0.492424  114        269
## 388942             GTS_871-23 UGA_173101_P023_WB11 0.491935  105        269
## 388945             GTS_871-23 UGA_173101_P023_WB06 0.483333  106        269
## 388951             GTS_871-23 UGA_173101_P023_WB04 0.482759  107        269
## 388998             GTS_871-23 UGA_173101_P020_WG06 0.492188  112        269
## 389028             GTS_871-23 UGA_173101_P020_WH04 0.492424  116        269
## 389040             GTS_871-23 UGA_173101_P020_WG07 0.466667   57        269
## 389042             GTS_871-23 UGA_173101_P020_WH11 0.492424  117        269
## 389066             GTS_871-23 UGA_173101_P022_WG09 0.492424  113        269
## 389077             GTS_871-23 UGA_173101_P022_WA12 0.472222   93        269
## 389100             GTS_871-23 UGA_173101_P021_WC01 0.482759   99        269
## 389127             GTS_871-23 UGA_173101_P021_WH06 0.480769  101        269
## 389161             GTS_871-23 UGA_173101_P020_WH03 0.477273  117        269
## 389205             GTS_871-23           GTS_938-23 0.492188  122        269
## 389209             GTS_871-23           GTS_944-23 0.500000  121        269
## 389234             GTS_871-23           GTS_631-23 0.491935  120        269
## 389241             GTS_871-23           GTS_781-23 0.500000  122        269
## 389250             GTS_871-23          GTS_781-23R 0.500000  122        269
## 389253             GTS_871-23           GTS_638-23 0.491935  119        269
## 389260             GTS_871-23           GTS_972-23 0.500000  121        269
## 389282             GTS_871-23           GTS_651-23 0.500000  121        269
## 389283             GTS_871-23           GTS_982-23 0.500000  121        269
## 389292             GTS_871-23          GTS_651-23R 0.492188  122        269
## 389298             GTS_871-23           GTS_803-23 0.500000  122        269
## 389314             GTS_871-23           GTS_809-23 0.500000  122        269
## 389332             GTS_871-23          GTS_1009-23 0.500000  121        269
## 389337             GTS_871-23          GTS_1013-23 0.492188  122        269
## 392956             GTS_713-23 UGA_173101_P020_WA09 0.482143   93        250
## 392992             GTS_713-23 UGA_173101_P019_WC06 0.484375  117        250
## 392995             GTS_713-23 UGA_173101_P019_WC02 0.484848  118        250
## 393001             GTS_713-23 UGA_173101_P019_WD04 0.477273  120        250
## 393015             GTS_713-23 UGA_173101_P018_WF03 0.483871  109        250
## 393019             GTS_713-23 UGA_173101_P018_WE09 0.473214  104        250
## 393042             GTS_713-23 UGA_173101_P020_WA07 0.458333  103        250
## 393044             GTS_713-23 UGA_173101_P016_WD06 0.481481   98        250
## 393075             GTS_713-23 UGA_173101_P017_WA02 0.484375  116        250
## 393088             GTS_713-23 UGA_173101_P015_WH07 0.477273  112        250
## 393114             GTS_713-23 UGA_173101_P020_WA05 0.476562  112        250
## 393176             GTS_713-23 UGA_173101_P018_WE02 0.484375  114        250
## 393236             GTS_713-23 UGA_173101_P016_WH06 0.490385   90        250
## 393238             GTS_713-23 UGA_173101_P016_WH08 0.474138  102        250
## 393260             GTS_713-23 UGA_173101_P017_WF04 0.469697  117        250
## 393286             GTS_713-23 UGA_173101_P017_WF08 0.445652   93        250
## 393306             GTS_713-23 UGA_173101_P017_WH09 0.455357   96        250
## 393351             GTS_713-23 UGA_173101_P023_WG04 0.484848  113        250
## 393362             GTS_713-23 UGA_173101_P023_WB11 0.484375  105        250
## 393365             GTS_713-23 UGA_173101_P023_WB06 0.475000  106        250
## 393371             GTS_713-23 UGA_173101_P023_WB04 0.475000  107        250
## 393418             GTS_713-23 UGA_173101_P020_WG06 0.484375  111        250
## 393448             GTS_713-23 UGA_173101_P020_WH04 0.484848  115        250
## 393460             GTS_713-23 UGA_173101_P020_WG07 0.466667   57        250
## 393462             GTS_713-23 UGA_173101_P020_WH11 0.484848  116        250
## 393486             GTS_713-23 UGA_173101_P022_WG09 0.484848  112        250
```

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## 393497             GTS_713-23 UGA_173101_P022_WA12 0.471154   92         250
## 393520             GTS_713-23 UGA_173101_P021_WC01 0.474138   98         250
## 393547             GTS_713-23 UGA_173101_P021_WH06 0.471154  102         250
## 393581             GTS_713-23 UGA_173101_P020_WH03 0.469697  116         250
## 393625             GTS_713-23           GTS_938-23 0.492188  121         250
## 393629             GTS_713-23           GTS_944-23 0.492188  120         250
## 393654             GTS_713-23           GTS_631-23 0.491935  118         250
## 393661             GTS_713-23           GTS_781-23 0.492188  120         250
## 393670             GTS_713-23          GTS_781-23R 0.492188  120         250
## 393673             GTS_713-23           GTS_638-23 0.483871  119         250
## 393680             GTS_713-23           GTS_972-23 0.492188  120         250
## 393702             GTS_713-23           GTS_651-23 0.492188  121         250
## 393703             GTS_713-23           GTS_982-23 0.492188  121         250
## 393712             GTS_713-23          GTS_651-23R 0.492188  121         250
## 393718             GTS_713-23           GTS_803-23 0.492188  121         250
## 393734             GTS_713-23           GTS_809-23 0.492188  121         250
## 393752             GTS_713-23          GTS_1009-23 0.492188  121         250
## 393757             GTS_713-23          GTS_1013-23 0.492188  121         250
## 393824             GTS_713-23           GTS_871-23 0.492188  120         250
## 413610             GTS_732-23 UGA_173101_P020_WA09 0.490385   93         250
## 413646             GTS_732-23 UGA_173101_P019_WC06 0.491667  117         250
## 413649             GTS_732-23 UGA_173101_P019_WC02 0.491935  118         250
## 413655             GTS_732-23 UGA_173101_P019_WD04 0.483871  120         250
## 413669             GTS_732-23 UGA_173101_P018_WF03 0.491379  109         250
## 413673             GTS_732-23 UGA_173101_P018_WE09 0.481481  104         250
## 413696             GTS_732-23 UGA_173101_P020_WA07 0.467391  103         250
## 413698             GTS_732-23 UGA_173101_P016_WD06 0.490000   98         250
## 413729             GTS_732-23 UGA_173101_P017_WA02 0.491667  116         250
## 413742             GTS_732-23 UGA_173101_P015_WH07 0.483871  111         250
## 413768             GTS_732-23 UGA_173101_P020_WA05 0.483871  112         250
## 413830             GTS_732-23 UGA_173101_P018_WE02 0.491667  114         250
## 413890             GTS_732-23 UGA_173101_P016_WH06 0.490000   89         250
## 413892             GTS_732-23 UGA_173101_P016_WH08 0.482143  101         250
## 413914             GTS_732-23 UGA_173101_P017_WF04 0.475806  116         250
## 413940             GTS_732-23 UGA_173101_P017_WF08 0.454545   93         250
## 413960             GTS_732-23 UGA_173101_P017_WH09 0.461538   95         250
## 414005             GTS_732-23 UGA_173101_P023_WG04 0.491935  113         250
## 414016             GTS_732-23 UGA_173101_P023_WB11 0.491667  105         250
## 414019             GTS_732-23 UGA_173101_P023_WB06 0.482759  106         250
## 414025             GTS_732-23 UGA_173101_P023_WB04 0.482143  107         250
## 414072             GTS_732-23 UGA_173101_P020_WG06 0.491667  111         250
## 414102             GTS_732-23 UGA_173101_P020_WH04 0.491935  115         250
## 414114             GTS_732-23 UGA_173101_P020_WG07 0.466667   57         250
## 414116             GTS_732-23 UGA_173101_P020_WH11 0.491935  116         250
## 414140             GTS_732-23 UGA_173101_P022_WG09 0.491935  112         250
## 414151             GTS_732-23 UGA_173101_P022_WA12 0.470000   92         250
## 414174             GTS_732-23 UGA_173101_P021_WC01 0.481481   98         250
## 414201             GTS_732-23 UGA_173101_P021_WH06 0.480000  101         250
## 414235             GTS_732-23 UGA_173101_P020_WH03 0.475806  116         250
## 414279             GTS_732-23           GTS_938-23 0.500000  121         250
## 414283             GTS_732-23           GTS_944-23 0.500000  120         250
## 414308             GTS_732-23           GTS_631-23 0.500000  119         250
## 414315             GTS_732-23           GTS_781-23 0.500000  120         250
## 414324             GTS_732-23          GTS_781-23R 0.500000  120         250
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## 414327             GTS_732-23           GTS_638-23 0.500000  119        250
## 414334             GTS_732-23           GTS_972-23 0.500000  120        250
## 414356             GTS_732-23           GTS_651-23 0.500000  121        250
## 414357             GTS_732-23           GTS_982-23 0.500000  121        250
## 414366             GTS_732-23          GTS_651-23R 0.500000  121        250
## 414372             GTS_732-23           GTS_803-23 0.500000  121        250
## 414388             GTS_732-23           GTS_809-23 0.500000  121        250
## 414406             GTS_732-23          GTS_1009-23 0.500000  121        250
## 414411             GTS_732-23          GTS_1013-23 0.500000  121        250
## 414478             GTS_732-23           GTS_871-23 0.500000  120        250
## 414483             GTS_732-23           GTS_713-23 0.491935  120        250
## 446055             GTS_919-23 UGA_173101_P020_WA09 0.490741   94        269
## 446091             GTS_919-23 UGA_173101_P019_WC06 0.492188  119        269
## 446094             GTS_919-23 UGA_173101_P019_WC02 0.492188  119        269
## 446100             GTS_919-23 UGA_173101_P019_WD04 0.484848  122        269
## 446114             GTS_919-23 UGA_173101_P018_WF03 0.491667  110        269
## 446118             GTS_919-23 UGA_173101_P018_WE09 0.482143  105        269
## 446141             GTS_919-23 UGA_173101_P020_WA07 0.468750  104        269
## 446143             GTS_919-23 UGA_173101_P016_WD06 0.490385   99        269
## 446174             GTS_919-23 UGA_173101_P017_WA02 0.491935  117        269
## 446187             GTS_919-23 UGA_173101_P015_WH07 0.484848  113        269
## 446213             GTS_919-23 UGA_173101_P020_WA05 0.484848  114        269
## 446275             GTS_919-23 UGA_173101_P018_WE02 0.491935  115        269
## 446335             GTS_919-23 UGA_173101_P016_WH06 0.490385   90        269
## 446337             GTS_919-23 UGA_173101_P016_WH08 0.482759  102        269
## 446359             GTS_919-23 UGA_173101_P017_WF04 0.476562  117        269
## 446385             GTS_919-23 UGA_173101_P017_WF08 0.456522   94        269
## 446405             GTS_919-23 UGA_173101_P017_WH09 0.462963   96        269
## 446450             GTS_919-23 UGA_173101_P023_WG04 0.492424  115        269
## 446461             GTS_919-23 UGA_173101_P023_WB11 0.491935  106        269
## 446464             GTS_919-23 UGA_173101_P023_WB06 0.483333  107        269
## 446470             GTS_919-23 UGA_173101_P023_WB04 0.482759  108        269
## 446517             GTS_919-23 UGA_173101_P020_WG06 0.492188  113        269
## 446547             GTS_919-23 UGA_173101_P020_WH04 0.492424  117        269
## 446559             GTS_919-23 UGA_173101_P020_WG07 0.466667   57        269
## 446561             GTS_919-23 UGA_173101_P020_WH11 0.492424  118        269
## 446585             GTS_919-23 UGA_173101_P022_WG09 0.492424  114        269
## 446596             GTS_919-23 UGA_173101_P022_WA12 0.472222   94        269
## 446619             GTS_919-23 UGA_173101_P021_WC01 0.482759  100        269
## 446646             GTS_919-23 UGA_173101_P021_WH06 0.480769  102        269
## 446680             GTS_919-23 UGA_173101_P020_WH03 0.477273  118        269
## 446724             GTS_919-23           GTS_938-23 0.492188  123        269
## 446728             GTS_919-23           GTS_944-23 0.500000  121        269
## 446753             GTS_919-23           GTS_631-23 0.491935  120        269
## 446760             GTS_919-23           GTS_781-23 0.500000  122        269
## 446769             GTS_919-23          GTS_781-23R 0.500000  122        269
## 446772             GTS_919-23           GTS_638-23 0.491935  120        269
## 446779             GTS_919-23           GTS_972-23 0.500000  121        269
## 446801             GTS_919-23           GTS_651-23 0.500000  122        269
## 446802             GTS_919-23           GTS_982-23 0.500000  122        269
## 446811             GTS_919-23          GTS_651-23R 0.492188  123        269
## 446817             GTS_919-23           GTS_803-23 0.500000  123        269
## 446833             GTS_919-23           GTS_809-23 0.500000  123        269
## 446851             GTS_919-23          GTS_1009-23 0.500000  122        269
```

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## 446856            GTS_919-23          GTS_1013-23 0.492188  123        269
## 446923            GTS_919-23           GTS_871-23 0.500000  122        269
## 446928            GTS_919-23           GTS_713-23 0.492188  121        269
## 446951            GTS_919-23           GTS_732-23 0.500000  121        269
## 465145            GTS_671-23 UGA_173101_P020_WA09 0.490385   93        250
## 465181            GTS_671-23 UGA_173101_P019_WC06 0.491667  116        250
## 465184            GTS_671-23 UGA_173101_P019_WC02 0.491935  117        250
## 465190            GTS_671-23 UGA_173101_P019_WD04 0.483871  119        250
## 465204            GTS_671-23 UGA_173101_P018_WF03 0.491379  108        250
## 465208            GTS_671-23 UGA_173101_P018_WE09 0.481481  104        250
## 465231            GTS_671-23 UGA_173101_P020_WA07 0.467391  102        250
## 465233            GTS_671-23 UGA_173101_P016_WD06 0.490000   97        250
## 465264            GTS_671-23 UGA_173101_P017_WA02 0.491667  115        250
## 465277            GTS_671-23 UGA_173101_P015_WH07 0.483871  110        250
## 465303            GTS_671-23 UGA_173101_P020_WA05 0.483871  111        250
## 465365            GTS_671-23 UGA_173101_P018_WE02 0.491667  113        250
## 465425            GTS_671-23 UGA_173101_P016_WH06 0.490000   88        250
## 465427            GTS_671-23 UGA_173101_P016_WH08 0.482143  100        250
## 465449            GTS_671-23 UGA_173101_P017_WF04 0.475806  115        250
## 465475            GTS_671-23 UGA_173101_P017_WF08 0.454545   93        250
## 465495            GTS_671-23 UGA_173101_P017_WH09 0.461538   95        250
## 465540            GTS_671-23 UGA_173101_P023_WG04 0.491935  112        250
## 465551            GTS_671-23 UGA_173101_P023_WB11 0.491667  105        250
## 465554            GTS_671-23 UGA_173101_P023_WB06 0.482759  105        250
## 465560            GTS_671-23 UGA_173101_P023_WB04 0.482143  107        250
## 465607            GTS_671-23 UGA_173101_P020_WG06 0.491667  110        250
## 465637            GTS_671-23 UGA_173101_P020_WH04 0.491935  114        250
## 465649            GTS_671-23 UGA_173101_P020_WG07 0.466667   57        250
## 465651            GTS_671-23 UGA_173101_P020_WH11 0.491935  115        250
## 465675            GTS_671-23 UGA_173101_P022_WG09 0.491935  111        250
## 465686            GTS_671-23 UGA_173101_P022_WA12 0.470000   92        250
## 465709            GTS_671-23 UGA_173101_P021_WC01 0.481481   98        250
## 465736            GTS_671-23 UGA_173101_P021_WH06 0.480000  100        250
## 465770            GTS_671-23 UGA_173101_P020_WH03 0.475806  115        250
## 465814            GTS_671-23           GTS_938-23 0.500000  120        250
## 465818            GTS_671-23           GTS_944-23 0.500000  119        250
## 465843            GTS_671-23           GTS_631-23 0.500000  118        250
## 465850            GTS_671-23           GTS_781-23 0.500000  119        250
## 465859            GTS_671-23          GTS_781-23R 0.500000  120        250
## 465862            GTS_671-23           GTS_638-23 0.500000  119        250
## 465869            GTS_671-23           GTS_972-23 0.500000  119        250
## 465891            GTS_671-23           GTS_651-23 0.500000  120        250
## 465892            GTS_671-23           GTS_982-23 0.500000  120        250
## 465901            GTS_671-23          GTS_651-23R 0.500000  120        250
## 465907            GTS_671-23           GTS_803-23 0.500000  120        250
## 465923            GTS_671-23           GTS_809-23 0.500000  120        250
## 465941            GTS_671-23          GTS_1009-23 0.500000  120        250
## 465946            GTS_671-23          GTS_1013-23 0.500000  120        250
## 466013            GTS_671-23           GTS_871-23 0.500000  119        250
## 466018            GTS_671-23           GTS_713-23 0.491935  119        250
## 466041            GTS_671-23           GTS_732-23 0.500000  120        250
## 466076            GTS_671-23           GTS_919-23 0.500000  120        250
## 489570            GTS_706-23 UGA_173101_P020_WA09 0.490385   93        250
## 489606            GTS_706-23 UGA_173101_P019_WC06 0.491667  117        250
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## 489609             GTS_706-23 UGA_173101_P019_WC02 0.491935  118         250
## 489615             GTS_706-23 UGA_173101_P019_WD04 0.483871  120         250
## 489629             GTS_706-23 UGA_173101_P018_WF03 0.491379  109         250
## 489633             GTS_706-23 UGA_173101_P018_WE09 0.481481  104         250
## 489656             GTS_706-23 UGA_173101_P020_WA07 0.467391  103         250
## 489658             GTS_706-23 UGA_173101_P016_WD06 0.490000   98         250
## 489689             GTS_706-23 UGA_173101_P017_WA02 0.491667  116         250
## 489702             GTS_706-23 UGA_173101_P015_WH07 0.483871  111         250
## 489728             GTS_706-23 UGA_173101_P020_WA05 0.483871  112         250
## 489790             GTS_706-23 UGA_173101_P018_WE02 0.491667  114         250
## 489850             GTS_706-23 UGA_173101_P016_WH06 0.490000   89         250
## 489852             GTS_706-23 UGA_173101_P016_WH08 0.482143  101         250
## 489874             GTS_706-23 UGA_173101_P017_WF04 0.475806  116         250
## 489900             GTS_706-23 UGA_173101_P017_WF08 0.454545   93         250
## 489920             GTS_706-23 UGA_173101_P017_WH09 0.461538   95         250
## 489965             GTS_706-23 UGA_173101_P023_WG04 0.491935  113         250
## 489976             GTS_706-23 UGA_173101_P023_WB11 0.491667  105         250
## 489979             GTS_706-23 UGA_173101_P023_WB06 0.482759  106         250
## 489985             GTS_706-23 UGA_173101_P023_WB04 0.482143  107         250
## 490032             GTS_706-23 UGA_173101_P020_WG06 0.491667  111         250
## 490062             GTS_706-23 UGA_173101_P020_WH04 0.491935  115         250
## 490074             GTS_706-23 UGA_173101_P020_WG07 0.466667   57         250
## 490076             GTS_706-23 UGA_173101_P020_WH11 0.491935  116         250
## 490100             GTS_706-23 UGA_173101_P022_WG09 0.491935  112         250
## 490111             GTS_706-23 UGA_173101_P022_WA12 0.470000   92         250
## 490134             GTS_706-23 UGA_173101_P021_WC01 0.481481   98         250
## 490161             GTS_706-23 UGA_173101_P021_WH06 0.480000  101         250
## 490195             GTS_706-23 UGA_173101_P020_WH03 0.475806  116         250
## 490239             GTS_706-23           GTS_938-23 0.500000  121         250
## 490243             GTS_706-23           GTS_944-23 0.500000  120         250
## 490268             GTS_706-23           GTS_631-23 0.500000  119         250
## 490275             GTS_706-23           GTS_781-23 0.500000  120         250
## 490284             GTS_706-23          GTS_781-23R 0.500000  121         250
## 490287             GTS_706-23           GTS_638-23 0.500000  119         250
## 490294             GTS_706-23           GTS_972-23 0.500000  120         250
## 490316             GTS_706-23           GTS_651-23 0.500000  121         250
## 490317             GTS_706-23           GTS_982-23 0.500000  121         250
## 490326             GTS_706-23          GTS_651-23R 0.500000  121         250
## 490332             GTS_706-23           GTS_803-23 0.500000  121         250
## 490348             GTS_706-23           GTS_809-23 0.500000  121         250
## 490366             GTS_706-23          GTS_1009-23 0.500000  121         250
## 490371             GTS_706-23          GTS_1013-23 0.500000  121         250
## 490438             GTS_706-23           GTS_871-23 0.500000  120         250
## 490443             GTS_706-23           GTS_713-23 0.491935  120         250
## 490466             GTS_706-23           GTS_732-23 0.500000  121         250
## 490501             GTS_706-23           GTS_919-23 0.500000  121         250
## 490521             GTS_706-23           GTS_671-23 0.500000  121         250
## 511581             GTS_776-23 UGA_173101_P020_WA09 0.490741   94         250
## 511617             GTS_776-23 UGA_173101_P019_WC06 0.483871  119         250
## 511620             GTS_776-23 UGA_173101_P019_WC02 0.492188  119         250
## 511626             GTS_776-23 UGA_173101_P019_WD04 0.476562  122         250
## 511640             GTS_776-23 UGA_173101_P018_WF03 0.491667  110         250
## 511644             GTS_776-23 UGA_173101_P018_WE09 0.482143  105         250
## 511667             GTS_776-23 UGA_173101_P020_WA07 0.468750  104         250
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## 511669            GTS_776-23 UGA_173101_P016_WD06 0.490385   99        250
## 511700            GTS_776-23 UGA_173101_P017_WA02 0.491935  117        250
## 511713            GTS_776-23 UGA_173101_P015_WH07 0.476562  113        250
## 511739            GTS_776-23 UGA_173101_P020_WA05 0.476562  114        250
## 511801            GTS_776-23 UGA_173101_P018_WE02 0.491935  115        250
## 511861            GTS_776-23 UGA_173101_P016_WH06 0.490385   90        250
## 511863            GTS_776-23 UGA_173101_P016_WH08 0.482759  102        250
## 511885            GTS_776-23 UGA_173101_P017_WF04 0.476562  117        250
## 511911            GTS_776-23 UGA_173101_P017_WF08 0.456522   94        250
## 511931            GTS_776-23 UGA_173101_P017_WH09 0.462963   96        250
## 511976            GTS_776-23 UGA_173101_P023_WG04 0.484375  115        250
## 511987            GTS_776-23 UGA_173101_P023_WB11 0.491935  106        250
## 511990            GTS_776-23 UGA_173101_P023_WB06 0.483333  107        250
## 511996            GTS_776-23 UGA_173101_P023_WB04 0.482759  108        250
## 512043            GTS_776-23 UGA_173101_P020_WG06 0.483871  113        250
## 512073            GTS_776-23 UGA_173101_P020_WH04 0.484375  117        250
## 512085            GTS_776-23 UGA_173101_P020_WG07 0.466667   57        250
## 512087            GTS_776-23 UGA_173101_P020_WH11 0.484375  118        250
## 512111            GTS_776-23 UGA_173101_P022_WG09 0.484375  114        250
## 512122            GTS_776-23 UGA_173101_P022_WA12 0.461538   94        250
## 512145            GTS_776-23 UGA_173101_P021_WC01 0.473214  100        250
## 512172            GTS_776-23 UGA_173101_P021_WH06 0.480769  102        250
## 512206            GTS_776-23 UGA_173101_P020_WH03 0.468750  118        250
## 512250            GTS_776-23           GTS_938-23 0.500000  123        250
## 512254            GTS_776-23           GTS_944-23 0.500000  121        250
## 512279            GTS_776-23           GTS_631-23 0.500000  120        250
## 512286            GTS_776-23           GTS_781-23 0.492188  122        250
## 512295            GTS_776-23          GTS_781-23R 0.492424  123        250
## 512298            GTS_776-23           GTS_638-23 0.491935  120        250
## 512305            GTS_776-23           GTS_972-23 0.500000  121        250
## 512327            GTS_776-23           GTS_651-23 0.500000  122        250
## 512328            GTS_776-23           GTS_982-23 0.500000  122        250
## 512337            GTS_776-23          GTS_651-23R 0.500000  123        250
## 512343            GTS_776-23           GTS_803-23 0.492188  123        250
## 512359            GTS_776-23           GTS_809-23 0.492188  123        250
## 512377            GTS_776-23          GTS_1009-23 0.500000  122        250
## 512382            GTS_776-23          GTS_1013-23 0.500000  123        250
## 512449            GTS_776-23           GTS_871-23 0.492188  122        250
## 512454            GTS_776-23           GTS_713-23 0.492188  121        250
## 512477            GTS_776-23           GTS_732-23 0.500000  121        250
## 512512            GTS_776-23           GTS_919-23 0.492188  123        250
## 512532            GTS_776-23           GTS_671-23 0.500000  121        250
## 512557            GTS_776-23           GTS_706-23 0.500000  122        250
## 536145            GTS_847-23 UGA_173101_P020_WA09 0.453704   94        250
## 536181            GTS_847-23 UGA_173101_P019_WC06 0.451613  119        250
## 536184            GTS_847-23 UGA_173101_P019_WC02 0.460938  119        250
## 536190            GTS_847-23 UGA_173101_P019_WD04 0.445312  122        250
## 536204            GTS_847-23 UGA_173101_P018_WF03 0.458333  110        250
## 536208            GTS_847-23 UGA_173101_P018_WE09 0.446429  105        250
## 536231            GTS_847-23 UGA_173101_P020_WA07 0.427083  104        250
## 536233            GTS_847-23 UGA_173101_P016_WD06 0.451923   99        250
## 536264            GTS_847-23 UGA_173101_P017_WA02 0.459677  117        250
## 536277            GTS_847-23 UGA_173101_P015_WH07 0.445312  113        250
## 536303            GTS_847-23 UGA_173101_P020_WA05 0.445312  114        250
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## 536365             GTS_847-23 UGA_173101_P018_WE02 0.459677  115        250
## 536425             GTS_847-23 UGA_173101_P016_WH06 0.451923   90        250
## 536427             GTS_847-23 UGA_173101_P016_WH08 0.448276  102        250
## 536449             GTS_847-23 UGA_173101_P017_WF04 0.445312  117        250
## 536475             GTS_847-23 UGA_173101_P017_WF08 0.413043   94        250
## 536495             GTS_847-23 UGA_173101_P017_WH09 0.425926   96        250
## 536540             GTS_847-23 UGA_173101_P023_WG04 0.453125  115        250
## 536551             GTS_847-23 UGA_173101_P023_WB11 0.459677  106        250
## 536554             GTS_847-23 UGA_173101_P023_WB06 0.450000  107        250
## 536560             GTS_847-23 UGA_173101_P023_WB04 0.448276  108        250
## 536607             GTS_847-23 UGA_173101_P020_WG06 0.451613  113        250
## 536637             GTS_847-23 UGA_173101_P020_WH04 0.453125  117        250
## 536649             GTS_847-23 UGA_173101_P020_WG07 0.400000   58        250
## 536651             GTS_847-23 UGA_173101_P020_WH11 0.453125  118        250
## 536675             GTS_847-23 UGA_173101_P022_WG09 0.453125  114        250
## 536686             GTS_847-23 UGA_173101_P022_WA12 0.423077   94        250
## 536709             GTS_847-23 UGA_173101_P021_WC01 0.437500  100        250
## 536736             GTS_847-23 UGA_173101_P021_WH06 0.442308  102        250
## 536770             GTS_847-23 UGA_173101_P020_WH03 0.437500  118        250
## 536814             GTS_847-23           GTS_938-23 0.500000  122        250
## 536818             GTS_847-23           GTS_944-23 0.500000  121        250
## 536843             GTS_847-23           GTS_631-23 0.500000  121        250
## 536850             GTS_847-23           GTS_781-23 0.492188  123        250
## 536859             GTS_847-23          GTS_781-23R 0.492424  123        250
## 536862             GTS_847-23           GTS_638-23 0.491935  119        250
## 536869             GTS_847-23           GTS_972-23 0.500000  121        250
## 536891             GTS_847-23           GTS_651-23 0.500000  121        250
## 536892             GTS_847-23           GTS_982-23 0.500000  121        250
## 536901             GTS_847-23          GTS_651-23R 0.500000  122        250
## 536907             GTS_847-23           GTS_803-23 0.492188  122        250
## 536923             GTS_847-23           GTS_809-23 0.492188  122        250
## 536941             GTS_847-23          GTS_1009-23 0.500000  121        250
## 536946             GTS_847-23          GTS_1013-23 0.500000  122        250
## 537013             GTS_847-23           GTS_871-23 0.492188  122        250
## 537018             GTS_847-23           GTS_713-23 0.492188  120        250
## 537041             GTS_847-23           GTS_732-23 0.500000  120        250
## 537076             GTS_847-23           GTS_919-23 0.492188  122        250
## 537096             GTS_847-23           GTS_671-23 0.500000  120        250
## 537121             GTS_847-23           GTS_706-23 0.500000  121        250
## 537143             GTS_847-23           GTS_776-23 0.500000  123        250
## 561285             GTS_881-23 UGA_173101_P020_WA09 0.490385   92        250
## 561321             GTS_881-23 UGA_173101_P019_WC06 0.483333  117        250
## 561324             GTS_881-23 UGA_173101_P019_WC02 0.491935  117        250
## 561330             GTS_881-23 UGA_173101_P019_WD04 0.475806  120        250
## 561344             GTS_881-23 UGA_173101_P018_WF03 0.491379  108        250
## 561348             GTS_881-23 UGA_173101_P018_WE09 0.481481  103        250
## 561371             GTS_881-23 UGA_173101_P020_WA07 0.467391  102        250
## 561373             GTS_881-23 UGA_173101_P016_WD06 0.490000   97        250
## 561404             GTS_881-23 UGA_173101_P017_WA02 0.491667  115        250
## 561417             GTS_881-23 UGA_173101_P015_WH07 0.475806  111        250
## 561443             GTS_881-23 UGA_173101_P020_WA05 0.475806  112        250
## 561505             GTS_881-23 UGA_173101_P018_WE02 0.491667  113        250
## 561565             GTS_881-23 UGA_173101_P016_WH06 0.490000   88        250
## 561567             GTS_881-23 UGA_173101_P016_WH08 0.482143  100        250
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## 561589             GTS_881-23 UGA_173101_P017_WF04 0.475806  115        250
## 561615             GTS_881-23 UGA_173101_P017_WF08 0.454545   92        250
## 561635             GTS_881-23 UGA_173101_P017_WH09 0.461538   94        250
## 561680             GTS_881-23 UGA_173101_P023_WG04 0.483871  113        250
## 561691             GTS_881-23 UGA_173101_P023_WB11 0.491667  104        250
## 561694             GTS_881-23 UGA_173101_P023_WB06 0.482759  105        250
## 561700             GTS_881-23 UGA_173101_P023_WB04 0.482143  106        250
## 561747             GTS_881-23 UGA_173101_P020_WG06 0.483333  111        250
## 561777             GTS_881-23 UGA_173101_P020_WH04 0.483871  115        250
## 561789             GTS_881-23 UGA_173101_P020_WG07 0.466667   57        250
## 561791             GTS_881-23 UGA_173101_P020_WH11 0.483871  116        250
## 561815             GTS_881-23 UGA_173101_P022_WG09 0.483871  112        250
## 561826             GTS_881-23 UGA_173101_P022_WA12 0.460000   92        250
## 561849             GTS_881-23 UGA_173101_P021_WC01 0.472222   98        250
## 561876             GTS_881-23 UGA_173101_P021_WH06 0.480000  100        250
## 561910             GTS_881-23 UGA_173101_P020_WH03 0.467742  116        250
## 561954             GTS_881-23           GTS_938-23 0.500000  121        250
## 561958             GTS_881-23           GTS_944-23 0.500000  120        250
## 561983             GTS_881-23           GTS_631-23 0.500000  120        250
## 561990             GTS_881-23           GTS_781-23 0.491935  121        250
## 561999             GTS_881-23          GTS_781-23R 0.492188  122        250
## 562002             GTS_881-23           GTS_638-23 0.500000  118        250
## 562009             GTS_881-23           GTS_972-23 0.500000  120        250
## 562031             GTS_881-23           GTS_651-23 0.500000  120        250
## 562032             GTS_881-23           GTS_982-23 0.500000  120        250
## 562041             GTS_881-23          GTS_651-23R 0.500000  121        250
## 562047             GTS_881-23           GTS_803-23 0.491935  121        250
## 562063             GTS_881-23           GTS_809-23 0.491935  121        250
## 562081             GTS_881-23          GTS_1009-23 0.500000  120        250
## 562086             GTS_881-23          GTS_1013-23 0.500000  121        250
## 562153             GTS_881-23           GTS_871-23 0.491935  121        250
## 562158             GTS_881-23           GTS_713-23 0.491935  119        250
## 562181             GTS_881-23           GTS_732-23 0.500000  120        250
## 562216             GTS_881-23           GTS_919-23 0.491935  121        250
## 562236             GTS_881-23           GTS_671-23 0.500000  120        250
## 562261             GTS_881-23           GTS_706-23 0.500000  121        250
## 562283             GTS_881-23           GTS_776-23 0.500000  122        250
## 562307             GTS_881-23           GTS_847-23 0.500000  122        250
## 565531             GTS_890-23 UGA_173101_P020_WA09 0.453704   94        250
## 565567             GTS_890-23 UGA_173101_P019_WC06 0.451613  119        250
## 565570             GTS_890-23 UGA_173101_P019_WC02 0.460938  119        250
## 565576             GTS_890-23 UGA_173101_P019_WD04 0.445312  122        250
## 565590             GTS_890-23 UGA_173101_P018_WF03 0.458333  110        250
## 565594             GTS_890-23 UGA_173101_P018_WE09 0.446429  105        250
## 565617             GTS_890-23 UGA_173101_P020_WA07 0.427083  104        250
## 565619             GTS_890-23 UGA_173101_P016_WD06 0.451923   99        250
## 565650             GTS_890-23 UGA_173101_P017_WA02 0.459677  117        250
## 565663             GTS_890-23 UGA_173101_P015_WH07 0.445312  113        250
## 565689             GTS_890-23 UGA_173101_P020_WA05 0.445312  114        250
## 565751             GTS_890-23 UGA_173101_P018_WE02 0.459677  115        250
## 565811             GTS_890-23 UGA_173101_P016_WH06 0.451923   90        250
## 565813             GTS_890-23 UGA_173101_P016_WH08 0.448276  102        250
## 565835             GTS_890-23 UGA_173101_P017_WF04 0.445312  117        250
## 565861             GTS_890-23 UGA_173101_P017_WF08 0.413043   94        250
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## 565881             GTS_890-23 UGA_173101_P017_WH09 0.425926   96         250
## 565926             GTS_890-23 UGA_173101_P023_WG04 0.453125  115         250
## 565937             GTS_890-23 UGA_173101_P023_WB11 0.459677  106         250
## 565940             GTS_890-23 UGA_173101_P023_WB06 0.450000  107         250
## 565946             GTS_890-23 UGA_173101_P023_WB04 0.448276  108         250
## 565993             GTS_890-23 UGA_173101_P020_WG06 0.451613  113         250
## 566023             GTS_890-23 UGA_173101_P020_WH04 0.453125  117         250
## 566035             GTS_890-23 UGA_173101_P020_WG07 0.400000   58         250
## 566037             GTS_890-23 UGA_173101_P020_WH11 0.453125  118         250
## 566061             GTS_890-23 UGA_173101_P022_WG09 0.453125  114         250
## 566072             GTS_890-23 UGA_173101_P022_WA12 0.423077   94         250
## 566095             GTS_890-23 UGA_173101_P021_WC01 0.437500  100         250
## 566122             GTS_890-23 UGA_173101_P021_WH06 0.442308  102         250
## 566156             GTS_890-23 UGA_173101_P020_WH03 0.437500  118         250
## 566200             GTS_890-23           GTS_938-23 0.500000  122         250
## 566204             GTS_890-23           GTS_944-23 0.500000  121         250
## 566229             GTS_890-23           GTS_631-23 0.500000  121         250
## 566236             GTS_890-23           GTS_781-23 0.492188  123         250
## 566245             GTS_890-23          GTS_781-23R 0.492424  123         250
## 566248             GTS_890-23           GTS_638-23 0.491935  119         250
## 566255             GTS_890-23           GTS_972-23 0.500000  121         250
## 566277             GTS_890-23           GTS_651-23 0.500000  121         250
## 566278             GTS_890-23           GTS_982-23 0.500000  121         250
## 566287             GTS_890-23          GTS_651-23R 0.500000  122         250
## 566293             GTS_890-23           GTS_803-23 0.492188  122         250
## 566309             GTS_890-23           GTS_809-23 0.492188  122         250
## 566327             GTS_890-23          GTS_1009-23 0.500000  121         250
## 566332             GTS_890-23          GTS_1013-23 0.500000  122         250
## 566399             GTS_890-23           GTS_871-23 0.492188  122         250
## 566404             GTS_890-23           GTS_713-23 0.492188  120         250
## 566427             GTS_890-23           GTS_732-23 0.500000  120         250
## 566462             GTS_890-23           GTS_919-23 0.492188  122         250
## 566482             GTS_890-23           GTS_671-23 0.500000  120         250
## 566507             GTS_890-23           GTS_706-23 0.500000  121         250
## 566529             GTS_890-23           GTS_776-23 0.500000  123         250
## 566553             GTS_890-23           GTS_847-23 0.500000  124         250
## 566577             GTS_890-23           GTS_881-23 0.500000  122         250
## 613293            GTS_1002-23 UGA_173101_P020_WA09 0.490741   93         250
## 613329            GTS_1002-23 UGA_173101_P019_WC06 0.483871  118         250
## 613332            GTS_1002-23 UGA_173101_P019_WC02 0.492188  118         250
## 613338            GTS_1002-23 UGA_173101_P019_WD04 0.476562  121         250
## 613352            GTS_1002-23 UGA_173101_P018_WF03 0.491667  109         250
## 613356            GTS_1002-23 UGA_173101_P018_WE09 0.482143  104         250
## 613379            GTS_1002-23 UGA_173101_P020_WA07 0.468750  103         250
## 613381            GTS_1002-23 UGA_173101_P016_WD06 0.490385   98         250
## 613412            GTS_1002-23 UGA_173101_P017_WA02 0.491935  116         250
## 613425            GTS_1002-23 UGA_173101_P015_WH07 0.476562  112         250
## 613451            GTS_1002-23 UGA_173101_P020_WA05 0.476562  113         250
## 613513            GTS_1002-23 UGA_173101_P018_WE02 0.491935  114         250
## 613573            GTS_1002-23 UGA_173101_P016_WH06 0.490385   89         250
## 613575            GTS_1002-23 UGA_173101_P016_WH08 0.482759  101         250
## 613597            GTS_1002-23 UGA_173101_P017_WF04 0.476562  116         250
## 613623            GTS_1002-23 UGA_173101_P017_WF08 0.456522   93         250
## 613643            GTS_1002-23 UGA_173101_P017_WH09 0.462963   95         250
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## 613688           GTS_1002-23 UGA_173101_P023_WG04 0.484375  114        250
## 613699           GTS_1002-23 UGA_173101_P023_WB11 0.491935  105        250
## 613702           GTS_1002-23 UGA_173101_P023_WB06 0.483333  106        250
## 613708           GTS_1002-23 UGA_173101_P023_WB04 0.482759  107        250
## 613755           GTS_1002-23 UGA_173101_P020_WG06 0.483871  112        250
## 613785           GTS_1002-23 UGA_173101_P020_WH04 0.484375  116        250
## 613797           GTS_1002-23 UGA_173101_P020_WG07 0.466667   57        250
## 613799           GTS_1002-23 UGA_173101_P020_WH11 0.484375  117        250
## 613823           GTS_1002-23 UGA_173101_P022_WG09 0.484375  113        250
## 613834           GTS_1002-23 UGA_173101_P022_WA12 0.461538   93        250
## 613857           GTS_1002-23 UGA_173101_P021_WC01 0.473214   99        250
## 613884           GTS_1002-23 UGA_173101_P021_WH06 0.480769  101        250
## 613918           GTS_1002-23 UGA_173101_P020_WH03 0.468750  117        250
## 613962           GTS_1002-23           GTS_938-23 0.500000  122        250
## 613966           GTS_1002-23           GTS_944-23 0.500000  121        250
## 613991           GTS_1002-23           GTS_631-23 0.500000  120        250
## 613998           GTS_1002-23           GTS_781-23 0.492188  122        250
## 614007           GTS_1002-23          GTS_781-23R 0.492424  123        250
## 614010           GTS_1002-23           GTS_638-23 0.491935  119        250
## 614017           GTS_1002-23           GTS_972-23 0.500000  121        250
## 614039           GTS_1002-23           GTS_651-23 0.500000  121        250
## 614040           GTS_1002-23           GTS_982-23 0.500000  121        250
## 614049           GTS_1002-23          GTS_651-23R 0.500000  122        250
## 614055           GTS_1002-23           GTS_803-23 0.492188  122        250
## 614071           GTS_1002-23           GTS_809-23 0.492188  122        250
## 614089           GTS_1002-23          GTS_1009-23 0.500000  121        250
## 614094           GTS_1002-23          GTS_1013-23 0.500000  122        250
## 614161           GTS_1002-23           GTS_871-23 0.492188  122        250
## 614166           GTS_1002-23           GTS_713-23 0.492188  120        250
## 614189           GTS_1002-23           GTS_732-23 0.500000  120        250
## 614224           GTS_1002-23           GTS_919-23 0.492188  122        250
## 614244           GTS_1002-23           GTS_671-23 0.500000  120        250
## 614269           GTS_1002-23           GTS_706-23 0.500000  121        250
## 614291           GTS_1002-23           GTS_776-23 0.500000  123        250
## 614315           GTS_1002-23           GTS_847-23 0.500000  123        250
## 614339           GTS_1002-23           GTS_881-23 0.500000  122        250
## 614343           GTS_1002-23           GTS_890-23 0.500000  123        250
## 1103370           GTS_399-23 UGA_173101_P020_WA09 0.490741   93        269
## 1103406           GTS_399-23 UGA_173101_P019_WC06 0.492188  119        269
## 1103409           GTS_399-23 UGA_173101_P019_WC02 0.492188  119        269
## 1103415           GTS_399-23 UGA_173101_P019_WD04 0.484848  122        269
## 1103429           GTS_399-23 UGA_173101_P018_WF03 0.491667  110        269
## 1103433           GTS_399-23 UGA_173101_P018_WE09 0.482143  104        269
## 1103456           GTS_399-23 UGA_173101_P020_WA07 0.468750  103        269
## 1103458           GTS_399-23 UGA_173101_P016_WD06 0.490385   98        269
## 1103489           GTS_399-23 UGA_173101_P017_WA02 0.491935  116        269
## 1103502           GTS_399-23 UGA_173101_P015_WH07 0.484848  112        269
## 1103528           GTS_399-23 UGA_173101_P020_WA05 0.484848  113        269
## 1103590           GTS_399-23 UGA_173101_P018_WE02 0.491935  115        269
## 1103650           GTS_399-23 UGA_173101_P016_WH06 0.490385   89        269
## 1103652           GTS_399-23 UGA_173101_P016_WH08 0.482759  101        269
## 1103674           GTS_399-23 UGA_173101_P017_WF04 0.476562  117        269
## 1103700           GTS_399-23 UGA_173101_P017_WF08 0.456522   93        269
## 1103720           GTS_399-23 UGA_173101_P017_WH09 0.462963   95        269
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## 1103765           GTS_399-23 UGA_173101_P023_WG04 0.492424  114        269
## 1103776           GTS_399-23 UGA_173101_P023_WB11 0.491935  106        269
## 1103779           GTS_399-23 UGA_173101_P023_WB06 0.483333  107        269
## 1103785           GTS_399-23 UGA_173101_P023_WB04 0.482759  108        269
## 1103832           GTS_399-23 UGA_173101_P020_WG06 0.492188  112        269
## 1103862           GTS_399-23 UGA_173101_P020_WH04 0.492424  117        269
## 1103874           GTS_399-23 UGA_173101_P020_WG07 0.466667   57        269
## 1103876           GTS_399-23 UGA_173101_P020_WH11 0.492424  118        269
## 1103900           GTS_399-23 UGA_173101_P022_WG09 0.492424  114        269
## 1103911           GTS_399-23 UGA_173101_P022_WA12 0.472222   93        269
## 1103934           GTS_399-23 UGA_173101_P021_WC01 0.482759   99        269
## 1103961           GTS_399-23 UGA_173101_P021_WH06 0.480769  101        269
## 1103995           GTS_399-23 UGA_173101_P020_WH03 0.477273  118        269
## 1104039           GTS_399-23           GTS_938-23 0.492188  122        269
## 1104043           GTS_399-23           GTS_944-23 0.500000  121        269
## 1104068           GTS_399-23           GTS_631-23 0.491935  120        269
## 1104075           GTS_399-23           GTS_781-23 0.500000  122        269
## 1104084           GTS_399-23          GTS_781-23R 0.500000  123        269
## 1104087           GTS_399-23           GTS_638-23 0.491935  119        269
## 1104094           GTS_399-23           GTS_972-23 0.500000  121        269
## 1104116           GTS_399-23           GTS_651-23 0.500000  121        269
## 1104117           GTS_399-23           GTS_982-23 0.500000  121        269
## 1104126           GTS_399-23          GTS_651-23R 0.492188  122        269
## 1104132           GTS_399-23           GTS_803-23 0.500000  122        269
## 1104148           GTS_399-23           GTS_809-23 0.500000  122        269
## 1104166           GTS_399-23          GTS_1009-23 0.500000  121        269
## 1104171           GTS_399-23          GTS_1013-23 0.492188  122        269
## 1104238           GTS_399-23           GTS_871-23 0.500000  122        269
## 1104243           GTS_399-23           GTS_713-23 0.492188  120        269
## 1104266           GTS_399-23           GTS_732-23 0.500000  120        269
## 1104301           GTS_399-23           GTS_919-23 0.500000  122        269
## 1104321           GTS_399-23           GTS_671-23 0.500000  120        269
## 1104346           GTS_399-23           GTS_706-23 0.500000  121        269
## 1104368           GTS_399-23           GTS_776-23 0.492424  123        269
## 1104392           GTS_399-23           GTS_847-23 0.492424  123        269
## 1104416           GTS_399-23           GTS_881-23 0.492188  122        269
## 1104420           GTS_399-23           GTS_890-23 0.492424  123        269
## 1104464           GTS_399-23          GTS_1002-23 0.492424  123        269
## 1113793           GTS_104-23 UGA_173101_P020_WA09 0.487500   83        269
## 1113829           GTS_104-23 UGA_173101_P019_WC06 0.489583  107        269
## 1113832           GTS_104-23 UGA_173101_P019_WC02 0.489130  106        269
## 1113838           GTS_104-23 UGA_173101_P019_WD04 0.479167  109        269
## 1113852           GTS_104-23 UGA_173101_P018_WF03 0.488636   98        269
## 1113856           GTS_104-23 UGA_173101_P018_WE09 0.487500   92        269
## 1113879           GTS_104-23 UGA_173101_P020_WA07 0.472222   92        269
## 1113881           GTS_104-23 UGA_173101_P016_WD06 0.487500   88        269
## 1113912           GTS_104-23 UGA_173101_P017_WA02 0.489130  104        269
## 1113925           GTS_104-23 UGA_173101_P015_WH07 0.479167  100        269
## 1113951           GTS_104-23 UGA_173101_P020_WA05 0.479167  100        269
## 1114013           GTS_104-23 UGA_173101_P018_WE02 0.489130  103        269
## 1114073           GTS_104-23 UGA_173101_P016_WH06 0.486111   80        269
## 1114075           GTS_104-23 UGA_173101_P016_WH08 0.476190   90        269
## 1114097           GTS_104-23 UGA_173101_P017_WF04 0.467391  105        269
## 1114123           GTS_104-23 UGA_173101_P017_WF08 0.441176   84        269
```

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## 1114143           GTS_104-23 UGA_173101_P017_WH09 0.450000   85         269
## 1114188           GTS_104-23 UGA_173101_P023_WG04 0.489583  101         269
## 1114199           GTS_104-23 UGA_173101_P023_WB11 0.488636   93         269
## 1114202           GTS_104-23 UGA_173101_P023_WB06 0.488095   94         269
## 1114208           GTS_104-23 UGA_173101_P023_WB04 0.475000   95         269
## 1114255           GTS_104-23 UGA_173101_P020_WG06 0.489130   99         269
## 1114285           GTS_104-23 UGA_173101_P020_WH04 0.489583  104         269
## 1114297           GTS_104-23 UGA_173101_P020_WG07 0.450000   50         269
## 1114299           GTS_104-23 UGA_173101_P020_WH11 0.489583  105         269
## 1114323           GTS_104-23 UGA_173101_P022_WG09 0.489583  101         269
## 1114334           GTS_104-23 UGA_173101_P022_WA12 0.460526   81         269
## 1114357           GTS_104-23 UGA_173101_P021_WC01 0.476190   87         269
## 1114384           GTS_104-23 UGA_173101_P021_WH06 0.472222   90         269
## 1114418           GTS_104-23 UGA_173101_P020_WH03 0.468750  105         269
## 1114462           GTS_104-23           GTS_938-23 0.489130  108         269
## 1114466           GTS_104-23           GTS_944-23 0.500000  107         269
## 1114491           GTS_104-23           GTS_631-23 0.488636  107         269
## 1114498           GTS_104-23           GTS_781-23 0.500000  108         269
## 1114507           GTS_104-23          GTS_781-23R 0.500000  108         269
## 1114510           GTS_104-23           GTS_638-23 0.488636  106         269
## 1114517           GTS_104-23           GTS_972-23 0.500000  107         269
## 1114539           GTS_104-23           GTS_651-23 0.500000  107         269
## 1114540           GTS_104-23           GTS_982-23 0.500000  107         269
## 1114549           GTS_104-23          GTS_651-23R 0.489130  108         269
## 1114555           GTS_104-23           GTS_803-23 0.500000  108         269
## 1114571           GTS_104-23           GTS_809-23 0.500000  108         269
## 1114589           GTS_104-23          GTS_1009-23 0.500000  107         269
## 1114594           GTS_104-23          GTS_1013-23 0.489130  108         269
## 1114661           GTS_104-23           GTS_871-23 0.500000  108         269
## 1114666           GTS_104-23           GTS_713-23 0.489130  107         269
## 1114689           GTS_104-23           GTS_732-23 0.500000  106         269
## 1114724           GTS_104-23           GTS_919-23 0.500000  108         269
## 1114744           GTS_104-23           GTS_671-23 0.500000  105         269
## 1114769           GTS_104-23           GTS_706-23 0.500000  106         269
## 1114791           GTS_104-23           GTS_776-23 0.489130  108         269
## 1114815           GTS_104-23           GTS_847-23 0.489130  108         269
## 1114839           GTS_104-23           GTS_881-23 0.488636  107         269
## 1114843           GTS_104-23           GTS_890-23 0.489130  108         269
## 1114887           GTS_104-23          GTS_1002-23 0.489130  108         269
## 1115265           GTS_104-23           GTS_399-23 0.500000  109         269
## 1142331           GTS_139-23 UGA_173101_P020_WA09 0.490741   93         269
## 1142367           GTS_139-23 UGA_173101_P019_WC06 0.492188  119         269
## 1142370           GTS_139-23 UGA_173101_P019_WC02 0.492188  119         269
## 1142376           GTS_139-23 UGA_173101_P019_WD04 0.484848  122         269
## 1142390           GTS_139-23 UGA_173101_P018_WF03 0.491667  110         269
## 1142394           GTS_139-23 UGA_173101_P018_WE09 0.482143  104         269
## 1142417           GTS_139-23 UGA_173101_P020_WA07 0.468750  103         269
## 1142419           GTS_139-23 UGA_173101_P016_WD06 0.490385   98         269
## 1142450           GTS_139-23 UGA_173101_P017_WA02 0.491935  116         269
## 1142463           GTS_139-23 UGA_173101_P015_WH07 0.484848  112         269
## 1142489           GTS_139-23 UGA_173101_P020_WA05 0.484848  113         269
## 1142551           GTS_139-23 UGA_173101_P018_WE02 0.491935  115         269
## 1142611           GTS_139-23 UGA_173101_P016_WH06 0.490385   89         269
## 1142613           GTS_139-23 UGA_173101_P016_WH08 0.482759  101         269
```

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## 1142635           GTS_139-23 UGA_173101_P017_WF04 0.476562  117        269
## 1142661           GTS_139-23 UGA_173101_P017_WF08 0.456522   93        269
## 1142681           GTS_139-23 UGA_173101_P017_WH09 0.462963   95        269
## 1142726           GTS_139-23 UGA_173101_P023_WG04 0.492424  114        269
## 1142737           GTS_139-23 UGA_173101_P023_WB11 0.491935  106        269
## 1142740           GTS_139-23 UGA_173101_P023_WB06 0.483333  107        269
## 1142746           GTS_139-23 UGA_173101_P023_WB04 0.482759  108        269
## 1142793           GTS_139-23 UGA_173101_P020_WG06 0.492188  112        269
## 1142823           GTS_139-23 UGA_173101_P020_WH04 0.492424  117        269
## 1142835           GTS_139-23 UGA_173101_P020_WG07 0.466667   57        269
## 1142837           GTS_139-23 UGA_173101_P020_WH11 0.492424  118        269
## 1142861           GTS_139-23 UGA_173101_P022_WG09 0.492424  114        269
## 1142872           GTS_139-23 UGA_173101_P022_WA12 0.472222   93        269
## 1142895           GTS_139-23 UGA_173101_P021_WC01 0.482759   99        269
## 1142922           GTS_139-23 UGA_173101_P021_WH06 0.480769  101        269
## 1142956           GTS_139-23 UGA_173101_P020_WH03 0.477273  118        269
## 1143000           GTS_139-23           GTS_938-23 0.492188  122        269
## 1143004           GTS_139-23           GTS_944-23 0.500000  121        269
## 1143029           GTS_139-23           GTS_631-23 0.491935  120        269
## 1143036           GTS_139-23           GTS_781-23 0.500000  122        269
## 1143045           GTS_139-23          GTS_781-23R 0.500000  123        269
## 1143048           GTS_139-23           GTS_638-23 0.491935  119        269
## 1143055           GTS_139-23           GTS_972-23 0.500000  121        269
## 1143077           GTS_139-23           GTS_651-23 0.500000  121        269
## 1143078           GTS_139-23           GTS_982-23 0.500000  121        269
## 1143087           GTS_139-23          GTS_651-23R 0.492188  122        269
## 1143093           GTS_139-23           GTS_803-23 0.500000  122        269
## 1143109           GTS_139-23           GTS_809-23 0.500000  122        269
## 1143127           GTS_139-23          GTS_1009-23 0.500000  121        269
## 1143132           GTS_139-23          GTS_1013-23 0.492188  122        269
## 1143199           GTS_139-23           GTS_871-23 0.500000  122        269
## 1143204           GTS_139-23           GTS_713-23 0.492188  120        269
## 1143227           GTS_139-23           GTS_732-23 0.500000  120        269
## 1143262           GTS_139-23           GTS_919-23 0.500000  122        269
## 1143282           GTS_139-23           GTS_671-23 0.500000  120        269
## 1143307           GTS_139-23           GTS_706-23 0.500000  121        269
## 1143329           GTS_139-23           GTS_776-23 0.492424  123        269
## 1143353           GTS_139-23           GTS_847-23 0.492424  123        269
## 1143377           GTS_139-23           GTS_881-23 0.492188  122        269
## 1143381           GTS_139-23           GTS_890-23 0.492424  123        269
## 1143425           GTS_139-23          GTS_1002-23 0.492424  123        269
## 1143803           GTS_139-23           GTS_399-23 0.500000  124        269
## 1143810           GTS_139-23           GTS_104-23 0.500000  109        269
## 1152936           GTS_150-23 UGA_173101_P020_WA09 0.488095   86        269
## 1152972           GTS_150-23 UGA_173101_P019_WC06 0.490385  112        269
## 1152975           GTS_150-23 UGA_173101_P019_WC02 0.490000  111        269
## 1152981           GTS_150-23 UGA_173101_P019_WD04 0.480769  114        269
## 1152995           GTS_150-23 UGA_173101_P018_WF03 0.489583  103        269
## 1152999           GTS_150-23 UGA_173101_P018_WE09 0.488636   97        269
## 1153022           GTS_150-23 UGA_173101_P020_WA07 0.473684   96        269
## 1153024           GTS_150-23 UGA_173101_P016_WD06 0.488636   93        269
## 1153055           GTS_150-23 UGA_173101_P017_WA02 0.490000  109        269
## 1153068           GTS_150-23 UGA_173101_P015_WH07 0.480769  105        269
## 1153094           GTS_150-23 UGA_173101_P020_WA05 0.480769  105        269
```

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## 1153156            GTS_150-23 UGA_173101_P018_WE02 0.490000  108        269
## 1153216            GTS_150-23 UGA_173101_P016_WH06 0.487500   83        269
## 1153218            GTS_150-23 UGA_173101_P016_WH08 0.478261   95        269
## 1153240            GTS_150-23 UGA_173101_P017_WF04 0.470000  110        269
## 1153266            GTS_150-23 UGA_173101_P017_WF08 0.460526   87        269
## 1153286            GTS_150-23 UGA_173101_P017_WH09 0.454545   90        269
## 1153331            GTS_150-23 UGA_173101_P023_WG04 0.490385  106        269
## 1153342            GTS_150-23 UGA_173101_P023_WB11 0.489583   98        269
## 1153345            GTS_150-23 UGA_173101_P023_WB06 0.478261   99        269
## 1153351            GTS_150-23 UGA_173101_P023_WB04 0.477273  100        269
## 1153398            GTS_150-23 UGA_173101_P020_WG06 0.490000  104        269
## 1153428            GTS_150-23 UGA_173101_P020_WH04 0.490385  109        269
## 1153440            GTS_150-23 UGA_173101_P020_WG07 0.458333   54        269
## 1153442            GTS_150-23 UGA_173101_P020_WH11 0.490385  110        269
## 1153466            GTS_150-23 UGA_173101_P022_WG09 0.490385  106        269
## 1153477            GTS_150-23 UGA_173101_P022_WA12 0.464286   86        269
## 1153500            GTS_150-23 UGA_173101_P021_WC01 0.478261   92        269
## 1153527            GTS_150-23 UGA_173101_P021_WH06 0.475000   95        269
## 1153561            GTS_150-23 UGA_173101_P020_WH03 0.471154  110        269
## 1153605            GTS_150-23           GTS_938-23 0.490000  113        269
## 1153609            GTS_150-23           GTS_944-23 0.500000  112        269
## 1153634            GTS_150-23           GTS_631-23 0.489583  111        269
## 1153641            GTS_150-23           GTS_781-23 0.500000  113        269
## 1153650            GTS_150-23          GTS_781-23R 0.500000  113        269
## 1153653            GTS_150-23           GTS_638-23 0.489583  110        269
## 1153660            GTS_150-23           GTS_972-23 0.500000  112        269
## 1153682            GTS_150-23           GTS_651-23 0.500000  112        269
## 1153683            GTS_150-23           GTS_982-23 0.500000  112        269
## 1153692            GTS_150-23          GTS_651-23R 0.490000  113        269
## 1153698            GTS_150-23           GTS_803-23 0.500000  113        269
## 1153714            GTS_150-23           GTS_809-23 0.500000  113        269
## 1153732            GTS_150-23          GTS_1009-23 0.500000  112        269
## 1153737            GTS_150-23          GTS_1013-23 0.490000  113        269
## 1153804            GTS_150-23           GTS_871-23 0.500000  113        269
## 1153809            GTS_150-23           GTS_713-23 0.490000  112        269
## 1153832            GTS_150-23           GTS_732-23 0.500000  111        269
## 1153867            GTS_150-23           GTS_919-23 0.500000  113        269
## 1153887            GTS_150-23           GTS_671-23 0.500000  110        269
## 1153912            GTS_150-23           GTS_706-23 0.500000  111        269
## 1153934            GTS_150-23           GTS_776-23 0.490000  113        269
## 1153958            GTS_150-23           GTS_847-23 0.490000  113        269
## 1153982            GTS_150-23           GTS_881-23 0.489583  112        269
## 1153986            GTS_150-23           GTS_890-23 0.490000  113        269
## 1154030            GTS_150-23          GTS_1002-23 0.490000  113        269
## 1154408            GTS_150-23           GTS_399-23 0.500000  114        269
## 1154415            GTS_150-23           GTS_104-23 0.500000  107        269
## 1154434            GTS_150-23           GTS_139-23 0.500000  114        269
## 1180431            GTS_181-23 UGA_173101_P020_WA09 0.490741   93        269
## 1180467            GTS_181-23 UGA_173101_P019_WC06 0.492188  119        269
## 1180470            GTS_181-23 UGA_173101_P019_WC02 0.492188  119        269
## 1180476            GTS_181-23 UGA_173101_P019_WD04 0.484848  122        269
## 1180490            GTS_181-23 UGA_173101_P018_WF03 0.491667  110        269
## 1180494            GTS_181-23 UGA_173101_P018_WE09 0.482143  104        269
## 1180517            GTS_181-23 UGA_173101_P020_WA07 0.468750  103        269
```

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## 1180519          GTS_181-23 UGA_173101_P016_WD06 0.490385   98        269
## 1180550          GTS_181-23 UGA_173101_P017_WA02 0.491935  116        269
## 1180563          GTS_181-23 UGA_173101_P015_WH07 0.484848  112        269
## 1180589          GTS_181-23 UGA_173101_P020_WA05 0.484848  113        269
## 1180651          GTS_181-23 UGA_173101_P018_WE02 0.491935  115        269
## 1180711          GTS_181-23 UGA_173101_P016_WH06 0.490385   89        269
## 1180713          GTS_181-23 UGA_173101_P016_WH08 0.482759  101        269
## 1180735          GTS_181-23 UGA_173101_P017_WF04 0.476562  117        269
## 1180761          GTS_181-23 UGA_173101_P017_WF08 0.456522   93        269
## 1180781          GTS_181-23 UGA_173101_P017_WH09 0.462963   95        269
## 1180826          GTS_181-23 UGA_173101_P023_WG04 0.492424  114        269
## 1180837          GTS_181-23 UGA_173101_P023_WB11 0.491935  106        269
## 1180840          GTS_181-23 UGA_173101_P023_WB06 0.483333  107        269
## 1180846          GTS_181-23 UGA_173101_P023_WB04 0.482759  108        269
## 1180893          GTS_181-23 UGA_173101_P020_WG06 0.492188  112        269
## 1180923          GTS_181-23 UGA_173101_P020_WH04 0.492424  117        269
## 1180935          GTS_181-23 UGA_173101_P020_WG07 0.466667   57        269
## 1180937          GTS_181-23 UGA_173101_P020_WH11 0.492424  118        269
## 1180961          GTS_181-23 UGA_173101_P022_WG09 0.492424  114        269
## 1180972          GTS_181-23 UGA_173101_P022_WA12 0.472222   93        269
## 1180995          GTS_181-23 UGA_173101_P021_WC01 0.482759   99        269
## 1181022          GTS_181-23 UGA_173101_P021_WH06 0.480769  101        269
## 1181056          GTS_181-23 UGA_173101_P020_WH03 0.477273  118        269
## 1181100          GTS_181-23           GTS_938-23 0.492188  122        269
## 1181104          GTS_181-23           GTS_944-23 0.500000  121        269
## 1181129          GTS_181-23           GTS_631-23 0.491935  120        269
## 1181136          GTS_181-23           GTS_781-23 0.500000  122        269
## 1181145          GTS_181-23          GTS_781-23R 0.500000  123        269
## 1181148          GTS_181-23           GTS_638-23 0.491935  119        269
## 1181155          GTS_181-23           GTS_972-23 0.500000  121        269
## 1181177          GTS_181-23           GTS_651-23 0.500000  121        269
## 1181178          GTS_181-23           GTS_982-23 0.500000  121        269
## 1181187          GTS_181-23          GTS_651-23R 0.492188  122        269
## 1181193          GTS_181-23           GTS_803-23 0.500000  122        269
## 1181209          GTS_181-23           GTS_809-23 0.500000  122        269
## 1181227          GTS_181-23          GTS_1009-23 0.500000  121        269
## 1181232          GTS_181-23          GTS_1013-23 0.492188  122        269
## 1181299          GTS_181-23           GTS_871-23 0.500000  122        269
## 1181304          GTS_181-23           GTS_713-23 0.492188  120        269
## 1181327          GTS_181-23           GTS_732-23 0.500000  120        269
## 1181362          GTS_181-23           GTS_919-23 0.500000  122        269
## 1181382          GTS_181-23           GTS_671-23 0.500000  120        269
## 1181407          GTS_181-23           GTS_706-23 0.500000  121        269
## 1181429          GTS_181-23           GTS_776-23 0.492424  123        269
## 1181453          GTS_181-23           GTS_847-23 0.492424  123        269
## 1181477          GTS_181-23           GTS_881-23 0.492188  122        269
## 1181481          GTS_181-23           GTS_890-23 0.492424  123        269
## 1181525          GTS_181-23          GTS_1002-23 0.492424  123        269
## 1181903          GTS_181-23           GTS_399-23 0.500000  124        269
## 1181910          GTS_181-23           GTS_104-23 0.500000  109        269
## 1181929          GTS_181-23           GTS_139-23 0.500000  124        269
## 1181936          GTS_181-23           GTS_150-23 0.500000  114        269
## 1197393          GTS_187-23 UGA_173101_P020_WA09 0.490000   91        269
## 1197429          GTS_187-23 UGA_173101_P019_WC06 0.491379  116        269
```

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## 1197432           GTS_187-23 UGA_173101_P019_WC02 0.491379  116        269
## 1197438           GTS_187-23 UGA_173101_P019_WD04 0.483333  119        269
## 1197452           GTS_187-23 UGA_173101_P018_WF03 0.490741  107        269
## 1197456           GTS_187-23 UGA_173101_P018_WE09 0.490385  102        269
## 1197479           GTS_187-23 UGA_173101_P020_WA07 0.467391  102        269
## 1197481           GTS_187-23 UGA_173101_P016_WD06 0.489583   96        269
## 1197512           GTS_187-23 UGA_173101_P017_WA02 0.491071  113        269
## 1197525           GTS_187-23 UGA_173101_P015_WH07 0.483333  109        269
## 1197551           GTS_187-23 UGA_173101_P020_WA05 0.483333  110        269
## 1197613           GTS_187-23 UGA_173101_P018_WE02 0.491071  112        269
## 1197673           GTS_187-23 UGA_173101_P016_WH06 0.489583   87        269
## 1197675           GTS_187-23 UGA_173101_P016_WH08 0.490385   98        269
## 1197697           GTS_187-23 UGA_173101_P017_WF04 0.474138  114        269
## 1197723           GTS_187-23 UGA_173101_P017_WF08 0.452381   91        269
## 1197743           GTS_187-23 UGA_173101_P017_WH09 0.468750   92        269
## 1197788           GTS_187-23 UGA_173101_P023_WG04 0.491667  111        269
## 1197799           GTS_187-23 UGA_173101_P023_WB11 0.491379  104        269
## 1197802           GTS_187-23 UGA_173101_P023_WB06 0.481481  104        269
## 1197808           GTS_187-23 UGA_173101_P023_WB04 0.481481  106        269
## 1197855           GTS_187-23 UGA_173101_P020_WG06 0.491379  109        269
## 1197885           GTS_187-23 UGA_173101_P020_WH04 0.491667  114        269
## 1197897           GTS_187-23 UGA_173101_P020_WG07 0.482143   55        269
## 1197899           GTS_187-23 UGA_173101_P020_WH11 0.491667  115        269
## 1197923           GTS_187-23 UGA_173101_P022_WG09 0.491667  111        269
## 1197934           GTS_187-23 UGA_173101_P022_WA12 0.470000   91        269
## 1197957           GTS_187-23 UGA_173101_P021_WC01 0.481481   97        269
## 1197984           GTS_187-23 UGA_173101_P021_WH06 0.489583   99        269
## 1198018           GTS_187-23 UGA_173101_P020_WH03 0.475000  115        269
## 1198062           GTS_187-23           GTS_938-23 0.491379  119        269
## 1198066           GTS_187-23           GTS_944-23 0.500000  118        269
## 1198091           GTS_187-23           GTS_631-23 0.491071  117        269
## 1198098           GTS_187-23           GTS_781-23 0.500000  119        269
## 1198107           GTS_187-23          GTS_781-23R 0.500000  119        269
## 1198110           GTS_187-23           GTS_638-23 0.491071  116        269
## 1198117           GTS_187-23           GTS_972-23 0.500000  118        269
## 1198139           GTS_187-23           GTS_651-23 0.500000  118        269
## 1198140           GTS_187-23           GTS_982-23 0.500000  118        269
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## 1198155           GTS_187-23           GTS_803-23 0.500000  119        269
## 1198171           GTS_187-23           GTS_809-23 0.500000  119        269
## 1198189           GTS_187-23          GTS_1009-23 0.500000  118        269
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## 1198289           GTS_187-23           GTS_732-23 0.500000  117        269
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## 1198865           GTS_187-23           GTS_399-23 0.500000  120        269
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## 565940            14           TRUE       FALSE
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```

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## 1103652        14        TRUE      FALSE
## 1103674        14        TRUE      FALSE
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## 1103765        14        TRUE      FALSE
## 1103776        14        TRUE      FALSE
## 1103779        14        TRUE      FALSE
## 1103785        14        TRUE      FALSE
## 1103832        14        TRUE      FALSE
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## 1198916         269           TRUE         TRUE
```

```r
#Clone 33 - 62 M, 1 F, 5 undet, no direct captures
table(smd$sexGT[smd$finalClone == 33])
```

```
##
##     F     M undet
##     1    62     5
```

```r
smd[which(smd$finalClone == 33),]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 132                     UGA_173101_P016_WC12  452-23   TRUE  FALSE    7 2023
## 152                     UGA_173101_P016_WE08  194-24   TRUE  FALSE    2 2024
## 163                     UGA_173101_P016_WF07  496-24   TRUE  FALSE    3 2024
## 189                     UGA_173101_P016_WH09  480-23   TRUE  FALSE    6 2023
## 218                     UGA_173101_P017_WC02  216-23   TRUE  FALSE    7 2023
## 228                     UGA_173101_P017_WC12  101-24   TRUE  FALSE    3 2024
## 237                     UGA_173101_P017_WD09  318-23   TRUE  FALSE    7 2023
## 304                     UGA_173101_P018_WB04  186-24   TRUE  FALSE    3 2024
## 306                     UGA_173101_P018_WB06  490-23   TRUE  FALSE    3 2023
## 323                     UGA_173101_P018_WC11  358-24   TRUE   TRUE    3 2024
## 334                     UGA_173101_P018_WD10  334-23   TRUE  FALSE    6 2023
## 367                     UGA_173101_P019_WA01   25-23   TRUE  FALSE    6 2023
## 368                     UGA_173101_P019_WA02   32-23   TRUE  FALSE    7 2023
## 370                     UGA_173101_P019_WA04   48-23   TRUE  FALSE    6 2023
## 419                     UGA_173101_P019_WE05   48-24   TRUE  FALSE    3 2024
## 441                     UGA_173101_P019_WG03  358-24   TRUE   TRUE    3 2024
## 553                     UGA_173101_P020_WH07   33-23   TRUE  FALSE    7 2023
## 619                     UGA_173101_P021_WF01  274-23   TRUE  FALSE    7 2023
## 663                     UGA_173101_P022_WA09  272-23   TRUE  FALSE    6 2023
## 692                     UGA_173101_P022_WD02   63-24   TRUE  FALSE    3 2024
## 754                     UGA_173101_P023_WA04  189-23   TRUE  FALSE    3 2023
## 757                     UGA_173101_P023_WA07  307-23   TRUE  FALSE    7 2023
## 774                     UGA_173101_P023_WB12  374-23   TRUE  FALSE    6 2023
## 815                     UGA_173101_P023_WF05  333-24   TRUE  FALSE    3 2024
## 873    GTseek_024_384i5_046_P098.003_949-23  949-23   TRUE  FALSE    6 2023
## 936    GTseek_024_384i5_109_P098.001_646-23  646-23   TRUE  FALSE    7 2023
## 951    GTseek_024_384i5_124_P098.004_616-24  616-24   TRUE  FALSE    2 2024
## 962    GTseek_024_384i5_135_P098.001_655-23  655-23   TRUE  FALSE    6 2023
## 963    GTseek_024_384i5_136_P098.003_983-23  983-23   TRUE  FALSE    3 2023
## 974    GTseek_024_384i5_148_P098.004_630-24  630-24   TRUE  FALSE    3 2024
## 993    GTseek_024_384i5_167_P098.001_665-23  665-23   TRUE  FALSE    7 2023
## 998    GTseek_024_384i5_172_P098.003_997-23  997-23   TRUE  FALSE    7 2023
## 1006   GTseek_024_384i5_180_P098.004_644-24  644-24   TRUE  FALSE    3 2024
## 1013   GTseek_024_384i5_187_P098.002_818-23  818-23   TRUE  FALSE    6 2023
## 1029   GTseek_024_384i5_203_P098.001_677-23  677-23   TRUE  FALSE    3 2023
## 1030  GTseek_024_384i5_204_P098.003_1012-23 1012-23   TRUE  FALSE    7 2023
## 1038   GTseek_024_384i5_212_P098.004_657-24  657-24   TRUE  FALSE    3 2024
## 1061   GTseek_024_384i5_235_P098.001_691-23  691-23   TRUE  FALSE    6 2023
## 1151   GTseek_024_384i5_325_P098.001_725-23  725-23   TRUE  FALSE    3 2023
## 1161   GTseek_024_384i5_335_P098.001_736-23  736-23   TRUE  FALSE    6 2023
## 1294  GTseek_025_384i5_084_P098.008_1202-24 1202-24   TRUE  FALSE    3 2024
## 1314   GTseek_025_384i5_104_P098.007_850-23  850-23   TRUE  FALSE    6 2023
## 1353   GTseek_025_384i5_143_P098.005_825-24  825-24   TRUE  FALSE    3 2024
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## 1374   GTseek_025_384i5_164_P098.007_935-23  935-23   TRUE   TRUE    7 2023
## 1386  GTseek_025_384i5_176_P098.007_935-23R 935-23R   TRUE   TRUE    7 2023
## 1411   GTseek_025_384i5_201_P098.005_855-24  855-24   TRUE  FALSE    3 2024
## 1426  GTseek_025_384i5_216_P098.008_1344-24 1344-24   TRUE  FALSE    3 2024
## 1473   GTseek_025_384i5_263_P098.005_886-24  886-24   TRUE  FALSE    3 2024
## 1490  GTseek_025_384i5_280_P098.008_1403-24 1403-24   TRUE  FALSE    3 2024
## 1513   GTseek_025_384i5_303_P098.005_931-24  931-24   TRUE  FALSE    3 2024
## 1518  GTseek_025_384i5_308_P098.008_1428-24 1428-24   TRUE  FALSE    2 2024
## 1527  GTseek_025_384i5_317_P098.006_3012-24 3012-24   TRUE  FALSE    6 2024
## 1548  GTseek_025_384i5_338_P098.008_1451-24 1451-24   TRUE  FALSE    3 2024
## 1563   GTseek_025_384i5_353_P098.005_955-24  955-24   TRUE  FALSE    2 2024
## 1595  GTseek_026_384i5_001_P098.009_1516-24 1516-24   TRUE  FALSE    3 2024
## 1641  GTseek_026_384i5_061_P098.010_2777-24 2777-24   TRUE  FALSE    3 2024
## 1677  GTseek_026_384i5_107_P098.009_1633-24 1633-24   TRUE  FALSE    3 2024
## 1691  GTseek_026_384i5_129_P098.009_1672-24 1672-24   TRUE  FALSE    3 2024
## 1717  GTseek_026_384i5_167_P098.009_2376-24 2376-24   TRUE  FALSE    6 2024
## 1747  GTseek_026_384i5_227_P098.009_2669-24 2669-24   TRUE  FALSE    6 2024
## 1756  GTseek_026_384i5_245_P098.010_2857-24 2857-24   TRUE  FALSE    3 2024
## 1782  GTseek_026_384i5_297_P098.009_2697-24 2697-24   TRUE  FALSE    2 2024
## 1805  GTseek_026_384i5_343_P098.010_2906-24 2906-24   TRUE  FALSE    6 2024
## 1903  GTseek_028_384i5_078_P098.UGA3_151-24  151-24   TRUE   TRUE    3 2024
## 1920  GTseek_028_384i5_095_P098.UGA2_426-23  426-23   TRUE  FALSE    6 2023
## 1937 GTseek_028_384i5_112_P098.UGA3_151-24R 151-24R   TRUE   TRUE    3 2024
## 1964  GTseek_028_384i5_139_P098.UGA1_146-23  146-23   TRUE  FALSE    3 2023
## 2185  GTseek_028_384i5_360_P098.UGA3_311-24  311-24   TRUE  FALSE    3 2024
##        snare       utmSnare
## 132  HS 009 259209x3585507
## 152  HS 014 261409x3586557
## 163  HS 017 260417x3588089
## 189  HS 009 259209x3585507
## 218  HS 009 259209x3585507
## 228  HS 012 258412x3586648
## 237  HS 002 255814x3587061
## 304  HS 011 257416x3586308
## 306  HS 012 258412x3586648
## 323  HS 017 260417x3588089
## 334  HS 012 258412x3586648
## 367  HS 009 259209x3585507
## 368  HS 009 259209x3585507
## 370  HS 012 258412x3586648
## 419  HS 017 260417x3588089
## 441  HS 017 260417x3588089
## 553  HS 012 258412x3586648
## 619  HS 007 255057x3589730
## 663  HS 009 259209x3585507
## 692  HS 011 257416x3586308
## 754  HS 012 258412x3586648
## 757  HS 009 259209x3585507
## 774  HS 009 259209x3585507
## 815  HS 017 260417x3588089
## 873  HS 009 259209x3585507
## 936  HS 009 259209x3585507
## 951  HS 014 261409x3586557
## 962  HS 012 258412x3586648
```

```
## 963  HS 012 258412x3586648
## 974  HS 012 258412x3586648
## 993  HS 002 255814x3587061
## 998  HS 009 259209x3585507
## 1006 HS 012 258412x3586648
## 1013 HS 009 259209x3585507
## 1029 HS 012 258412x3586648
## 1030 HS 002 255814x3587061
## 1038 HS 012 258412x3586648
## 1061 HS 012 258412x3586648
## 1151 HS 012 258412x3586648
## 1161 HS 012 258412x3586648
## 1294 HS 012 258412x3586648
## 1314 HS 012 258412x3586648
## 1353 HS 017 260417x3588089
## 1374 HS 002 255814x3587061
## 1386 HS 002 255814x3587061
## 1411 HS 011 257416x3586308
## 1426 HS 011 257416x3586308
## 1473 HS 017 260417x3588089
## 1490 HS 012 258412x3586648
## 1513 HS 011 257416x3586308
## 1518 HS 014 261409x3586557
## 1527 HS 009 259092x3585434
## 1548 HS 012 258412x3586648
## 1563 HS 014 261409x3586557
## 1595 HS 017 260417x3588089
## 1641 HS 011 257416x3586308
## 1677 HS 011 257416x3586308
## 1691 HS 011 257416x3586308
## 1717 HS 009 259092x3585434
## 1747 HS 009 259092x3585434
## 1756 HS 011 257416x3586308
## 1782 HS 014 261409x3586557
## 1805 HS 009 259092x3585434
## 1903 HS 012 258412x3586648
## 1920 HS 012 258412x3586648
## 1937 HS 012 258412x3586648
## 1964 HS 012 258412x3586648
## 2185 HS 012 258412x3586648
##                                                                            fqfile
## 132  RAPiD-Genomics_F534_UGA_173101_P016_WC12_i5-504_i7-132_S14870_L003_R1_001.fastq
## 152  RAPiD-Genomics_F534_UGA_173101_P016_WE08_i5-504_i7-152_S14889_L003_R1_001.fastq
## 163  RAPiD-Genomics_F534_UGA_173101_P016_WF07_i5-504_i7-163_S14900_L003_R1_001.fastq
## 189  RAPiD-Genomics_F534_UGA_173101_P016_WH09_i5-504_i7-189_S14926_L003_R1_001.fastq
## 218  RAPiD-Genomics_F534_UGA_173101_P017_WC02_i5-512_i7-122_S14955_L003_R1_001.fastq
## 228  RAPiD-Genomics_F534_UGA_173101_P017_WC12_i5-512_i7-132_S14965_L003_R1_001.fastq
## 237  RAPiD-Genomics_F534_UGA_173101_P017_WD09_i5-512_i7-141_S14974_L003_R1_001.fastq
## 304  RAPiD-Genomics_F534_UGA_173101_P018_WB04_i5-537_i7-112_S15041_L003_R1_001.fastq
## 306  RAPiD-Genomics_F534_UGA_173101_P018_WB06_i5-537_i7-114_S15043_L003_R1_001.fastq
## 323  RAPiD-Genomics_F534_UGA_173101_P018_WC11_i5-537_i7-131_S15060_L003_R1_001.fastq
## 334  RAPiD-Genomics_F534_UGA_173101_P018_WD10_i5-537_i7-142_S15071_L003_R1_001.fastq
## 367  RAPiD-Genomics_F534_UGA_173101_P019_WA01_i5-1102_i7-59_S15106_L003_R1_001.fastq
## 368  RAPiD-Genomics_F534_UGA_173101_P019_WA02_i5-1102_i7-27_S15107_L003_R1_001.fastq
```

```
## 370   RAPiD-Genomics_F534_UGA_173101_P019_WA04_i5-1102_i7-7_S15109_L003_R1_001.fastq
## 419  RAPiD-Genomics_F534_UGA_173101_P019_WE05_i5-1102_i7-94_S15158_L003_R1_001.fastq
## 441  RAPiD-Genomics_F534_UGA_173101_P019_WG03_i5-1102_i7-22_S15277_L003_R1_001.fastq
## 553     RAPiD-Genomics_F566_UGA_173101_P020_WH07_i5-505_i7-1_S3981_L001_R1_001.fastq
## 619      RAPiD-Genomics_F566_UGA_173101_P021_WF01_i5-501_i7-28_S61_L003_R1_001.fastq
## 663   RAPiD-Genomics_F566_UGA_173101_P022_WA09_i5-1098_i7-36_S7591_L003_R1_001.fastq
## 692   RAPiD-Genomics_F566_UGA_173101_P022_WD02_i5-1098_i7-72_S7620_L003_R1_001.fastq
## 754    RAPiD-Genomics_F566_UGA_173101_P023_WA04_i5-1099_i7-7_S7682_L003_R1_001.fastq
## 757   RAPiD-Genomics_F566_UGA_173101_P023_WA07_i5-1099_i7-77_S7685_L003_R1_001.fastq
## 774   RAPiD-Genomics_F566_UGA_173101_P023_WB12_i5-1099_i7-75_S7702_L003_R1_001.fastq
## 815   RAPiD-Genomics_F566_UGA_173101_P023_WF05_i5-1099_i7-92_S7743_L003_R1_001.fastq
## 873                                                                              <NA>
## 936                                                                              <NA>
## 951                                                                              <NA>
## 962                                                                              <NA>
## 963                                                                              <NA>
## 974                                                                              <NA>
## 993                                                                              <NA>
## 998                                                                              <NA>
## 1006                                                                             <NA>
## 1013                                                                             <NA>
## 1029                                                                             <NA>
## 1030                                                                             <NA>
## 1038                                                                             <NA>
## 1061                                                                             <NA>
## 1151                                                                             <NA>
## 1161                                                                             <NA>
## 1294                                                                             <NA>
## 1314                                                                             <NA>
## 1353                                                                             <NA>
## 1374                                                                             <NA>
## 1386                                                                             <NA>
## 1411                                                                             <NA>
## 1426                                                                             <NA>
## 1473                                                                             <NA>
## 1490                                                                             <NA>
## 1513                                                                             <NA>
## 1518                                                                             <NA>
## 1527                                                                             <NA>
## 1548                                                                             <NA>
## 1563                                                                             <NA>
## 1595                                                                             <NA>
## 1641                                                                             <NA>
## 1677                                                                             <NA>
## 1691                                                                             <NA>
## 1717                                                                             <NA>
## 1747                                                                             <NA>
## 1756                                                                             <NA>
## 1782                                                                             <NA>
## 1805                                                                             <NA>
## 1903                                                                             <NA>
## 1920                                                                             <NA>
## 1937                                                                             <NA>
## 1964                                                                             <NA>
```

```
## 2185                                                                              <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 132           Hair Snare    CGA    902 UGA_173101_P016_WC12 FALSE       LGC
## 152           Hair Snare    CGA    917 UGA_173101_P016_WE08 FALSE       LGC
## 163           Hair Snare    CGA    873 UGA_173101_P016_WF07 FALSE       LGC
## 189           Hair Snare    CGA    911 UGA_173101_P016_WH09 FALSE       LGC
## 218           Hair Snare    CGA    916 UGA_173101_P017_WC02 FALSE       LGC
## 228           Hair Snare    CGA    917 UGA_173101_P017_WC12 FALSE       LGC
## 237           Hair Snare    CGA    906 UGA_173101_P017_WD09 FALSE       LGC
## 304           Hair Snare    CGA    917 UGA_173101_P018_WB04 FALSE       LGC
## 306           Hair Snare    CGA    475 UGA_173101_P018_WB06 FALSE       LGC
## 323           Hair Snare    CGA    922 UGA_173101_P018_WC11 FALSE       LGC
## 334           Hair Snare    CGA    887 UGA_173101_P018_WD10 FALSE       LGC
## 367           Hair Snare    CGA    921 UGA_173101_P019_WA01 FALSE       LGC
## 368           Hair Snare    CGA    919 UGA_173101_P019_WA02 FALSE       LGC
## 370           Hair Snare    CGA    761 UGA_173101_P019_WA04 FALSE       LGC
## 419           Hair Snare    CGA    906 UGA_173101_P019_WE05 FALSE       LGC
## 441           Hair Snare    CGA    922 UGA_173101_P019_WG03 FALSE       LGC
## 553           Hair Snare    CGA    654 UGA_173101_P020_WH07 FALSE       LGC
## 619           Hair Snare    CGA    896 UGA_173101_P021_WF01 FALSE       LGC
## 663           Hair Snare    CGA    899 UGA_173101_P022_WA09 FALSE       LGC
## 692           Hair Snare    CGA    468 UGA_173101_P022_WD02 FALSE       LGC
## 754           Hair Snare    CGA    879 UGA_173101_P023_WA04 FALSE       LGC
## 757           Hair Snare    CGA    903 UGA_173101_P023_WA07 FALSE       LGC
## 774           Hair Snare    CGA    911 UGA_173101_P023_WB12 FALSE       LGC
## 815           Hair Snare    CGA    861 UGA_173101_P023_WF05 FALSE       LGC
## 873      <NA> Hair Sample    CGA    124           GTS_949-23 FALSE    GTSeek
## 936      <NA> Hair Sample    CGA    120           GTS_646-23 FALSE    GTSeek
## 951      <NA> Hair Sample    CGA    122           GTS_616-24 FALSE    GTSeek
## 962      <NA> Hair Sample    CGA    116           GTS_655-23 FALSE    GTSeek
## 963      <NA> Hair Sample    CGA    121           GTS_983-23 FALSE    GTSeek
## 974      <NA> Hair Sample    CGA    119           GTS_630-24 FALSE    GTSeek
## 993      <NA> Hair Sample    CGA    122           GTS_665-23 FALSE    GTSeek
## 998      <NA> Hair Sample    CGA    123           GTS_997-23 FALSE    GTSeek
## 1006     <NA> Hair Sample    CGA    119           GTS_644-24 FALSE    GTSeek
## 1013     <NA> Hair Sample    CGA    123           GTS_818-23 FALSE    GTSeek
## 1029     <NA> Hair Sample    CGA    119           GTS_677-23 FALSE    GTSeek
## 1030     <NA> Hair Sample    CGA    121          GTS_1012-23 FALSE    GTSeek
## 1038     <NA> Hair Sample    CGA    120           GTS_657-24 FALSE    GTSeek
## 1061     <NA> Hair Sample    CGA    119           GTS_691-23 FALSE    GTSeek
## 1151     <NA> Hair Sample    CGA    119           GTS_725-23 FALSE    GTSeek
## 1161     <NA> Hair Sample    CGA    120           GTS_736-23 FALSE    GTSeek
## 1294     <NA> Hair Sample    CGA    122          GTS_1202-24 FALSE    GTSeek
## 1314     <NA> Hair Sample    CGA    124           GTS_850-23 FALSE    GTSeek
## 1353     <NA> Hair Sample    CGA    120           GTS_825-24 FALSE    GTSeek
## 1374     <NA> Hair Sample    CGA    123           GTS_935-23 FALSE    GTSeek
## 1386     <NA> Hair Sample    CGA    123          GTS_935-23R FALSE    GTSeek
## 1411     <NA> Hair Sample    CGA    123           GTS_855-24 FALSE    GTSeek
## 1426     <NA> Hair Sample    CGA    124          GTS_1344-24 FALSE    GTSeek
## 1473     <NA> Hair Sample    CGA    124           GTS_886-24 FALSE    GTSeek
## 1490     <NA> Hair Sample    CGA    122          GTS_1403-24 FALSE    GTSeek
## 1513     <NA> Hair Sample    CGA    119           GTS_931-24 FALSE    GTSeek
## 1518     <NA> Hair Sample    CGA    123          GTS_1428-24 FALSE    GTSeek
## 1527     <NA> Hair Sample    CGA    122          GTS_3012-24 FALSE    GTSeek
```

```
## 1548     <NA> Hair Sample    CGA     123          GTS_1451-24 FALSE    GTSeek
## 1563     <NA> Hair Sample    CGA     122           GTS_955-24 FALSE    GTSeek
## 1595     <NA> Hair Sample    CGA     120          GTS_1516-24 FALSE    GTSeek
## 1641     <NA> Hair Sample    CGA     121          GTS_2777-24 FALSE    GTSeek
## 1677     <NA> Hair Sample    CGA     124          GTS_1633-24 FALSE    GTSeek
## 1691     <NA> Hair Sample    CGA     121          GTS_1672-24 FALSE    GTSeek
## 1717     <NA> Hair Sample    CGA     123          GTS_2376-24 FALSE    GTSeek
## 1747     <NA> Hair Sample    CGA     123          GTS_2669-24 FALSE    GTSeek
## 1756     <NA> Hair Sample    CGA     124          GTS_2857-24 FALSE    GTSeek
## 1782     <NA> Hair Sample    CGA     122          GTS_2697-24 FALSE    GTSeek
## 1805     <NA> Hair Sample    CGA     124          GTS_2906-24 FALSE    GTSeek
## 1903     <NA> Hair Sample    CGA     122           GTS_151-24 FALSE    GTSeek
## 1920     <NA> Hair Sample    CGA     119           GTS_426-23 FALSE    GTSeek
## 1937     <NA> Hair Sample    CGA     118          GTS_151-24R FALSE    GTSeek
## 1964     <NA> Hair Sample    CGA     122           GTS_146-23 FALSE    GTSeek
## 2185     <NA> Hair Sample    CGA     121           GTS_311-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 132      M       33         33
## 152      M       33         33
## 163      M       33         33
## 189      M       33         33
## 218      M       33         33
## 228      M       33         33
## 237      M       33         33
## 304      M       33         33
## 306      M       33         33
## 323      M       33         33
## 334      M       33         33
## 367      M       33         33
## 368      M       33         33
## 370      M       33         33
## 419      M       33         33
## 441      M       33         33
## 553      M       33         33
## 619      M       33         33
## 663      M       33         33
## 692      M       33         33
## 754      M       33         33
## 757      M       33         33
## 774      M       33         33
## 815      M       33         33
## 873      M      255         33
## 936  undet      255         33
## 951      M      276         33
## 962      F      255         33
## 963      M      255         33
## 974      M      255         33
## 993  undet      255         33
## 998      M      255         33
## 1006     M      276         33
## 1013     M      255         33
## 1029     M      255         33
## 1030     M       33         33
## 1038     M      276         33
```

```
## 1061     M      289         33
## 1151 undet       33         33
## 1161 undet      276         33
## 1294     M      289         33
## 1314     M      301         33
## 1353     M      301         33
## 1374     M      255         33
## 1386     M      289         33
## 1411     M      255         33
## 1426     M      255         33
## 1473     M      255         33
## 1490     M      255         33
## 1513     M      255         33
## 1518     M      255         33
## 1527     M      255         33
## 1548     M      255         33
## 1563     M      255         33
## 1595     M      276         33
## 1641     M      255         33
## 1677     M      289         33
## 1691     M      255         33
## 1717     M      255         33
## 1747     M      255         33
## 1756     M      255         33
## 1782     M      255         33
## 1805     M      255         33
## 1903     M      255         33
## 1920     M      255         33
## 1937     M       33         33
## 1964     M       33         33
## 2185 undet       33         33
```

```
unique(smd$COLclone[which(smd$finalClone == 33)])
```

```
## [1]  33 255 276 289 301
```

```
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 33)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 33)],]
```

```
##                          id1                  id2  kinship nsnp colclone1
## 2056    UGA_173101_P019_WA04 UGA_173101_P019_WE05 0.480702  753        33
## 5500    UGA_173101_P016_WC12 UGA_173101_P019_WE05 0.496408  892        33
## 5525    UGA_173101_P016_WC12 UGA_173101_P019_WA04 0.481707  755        33
## 5605    UGA_173101_P017_WC02 UGA_173101_P019_WE05 0.496429  904        33
## 5630    UGA_173101_P017_WC02 UGA_173101_P019_WA04 0.479965  760        33
## 5671    UGA_173101_P017_WC02 UGA_173101_P016_WC12 0.495751  901        33
## 6368    UGA_173101_P018_WB04 UGA_173101_P019_WE05 0.497126  904        33
## 6393    UGA_173101_P018_WB04 UGA_173101_P019_WA04 0.480769  760        33
## 6434    UGA_173101_P018_WB04 UGA_173101_P016_WC12 0.496439  900        33
## 6435    UGA_173101_P018_WB04 UGA_173101_P017_WC02 0.496449  914        33
## 6710    UGA_173101_P016_WF07 UGA_173101_P019_WE05 0.489458  864        33
## 6735    UGA_173101_P016_WF07 UGA_173101_P019_WA04 0.472122  742        33
## 6776    UGA_173101_P016_WF07 UGA_173101_P016_WC12 0.487237  864        33
```

```
## 6777    UGA_173101_P016_WF07 UGA_173101_P017_WC02 0.487313  871        33
## 6784    UGA_173101_P016_WF07 UGA_173101_P018_WB04 0.490211  871        33
## 12760   UGA_173101_P019_WG03 UGA_173101_P019_WE05 0.498571  906        33
## 12785   UGA_173101_P019_WG03 UGA_173101_P019_WA04 0.482578  761        33
## 12826   UGA_173101_P019_WG03 UGA_173101_P016_WC12 0.497875  902        33
## 12827   UGA_173101_P019_WG03 UGA_173101_P017_WC02 0.497887  916        33
## 12834   UGA_173101_P019_WG03 UGA_173101_P018_WB04 0.498580  917        33
## 12837   UGA_173101_P019_WG03 UGA_173101_P016_WF07 0.489552  873        33
## 13570   UGA_173101_P016_WE08 UGA_173101_P019_WE05 0.498571  904        33
## 13595   UGA_173101_P016_WE08 UGA_173101_P019_WA04 0.482578  761        33
## 13636   UGA_173101_P016_WE08 UGA_173101_P016_WC12 0.497875  901        33
## 13637   UGA_173101_P016_WE08 UGA_173101_P017_WC02 0.497887  915        33
## 13644   UGA_173101_P016_WE08 UGA_173101_P018_WB04 0.498580  915        33
## 13647   UGA_173101_P016_WE08 UGA_173101_P016_WF07 0.489552  873        33
## 13691   UGA_173101_P016_WE08 UGA_173101_P019_WG03 0.500000  917        33
## 18761   UGA_173101_P017_WC12 UGA_173101_P019_WE05 0.497143  905        33
## 18786   UGA_173101_P017_WC12 UGA_173101_P019_WA04 0.482578  761        33
## 18827   UGA_173101_P017_WC12 UGA_173101_P016_WC12 0.496459  902        33
## 18828   UGA_173101_P017_WC12 UGA_173101_P017_WC02 0.496479  915        33
## 18835   UGA_173101_P017_WC12 UGA_173101_P018_WB04 0.497159  915        33
## 18838   UGA_173101_P017_WC12 UGA_173101_P016_WF07 0.488806  873        33
## 18882   UGA_173101_P017_WC12 UGA_173101_P019_WG03 0.498592  917        33
## 18887   UGA_173101_P017_WC12 UGA_173101_P016_WE08 0.498592  916        33
## 22831   UGA_173101_P019_WA02 UGA_173101_P019_WE05 0.497851  905        33
## 22856   UGA_173101_P019_WA02 UGA_173101_P019_WA04 0.483449  761        33
## 22897   UGA_173101_P019_WA02 UGA_173101_P016_WC12 0.498584  902        33
## 22898   UGA_173101_P019_WA02 UGA_173101_P017_WC02 0.497175  916        33
## 22905   UGA_173101_P019_WA02 UGA_173101_P018_WB04 0.497869  916        33
## 22908   UGA_173101_P019_WA02 UGA_173101_P016_WF07 0.488772  873        33
## 22952   UGA_173101_P019_WA02 UGA_173101_P019_WG03 0.499294  918        33
## 22957   UGA_173101_P019_WA02 UGA_173101_P016_WE08 0.499294  917        33
## 22986   UGA_173101_P019_WA02 UGA_173101_P017_WC12 0.497881  917        33
## 28720   UGA_173101_P018_WD10 UGA_173101_P019_WE05 0.492733  877        33
## 28745   UGA_173101_P018_WD10 UGA_173101_P019_WA04 0.477837  741        33
## 28786   UGA_173101_P018_WD10 UGA_173101_P016_WC12 0.491329  875        33
## 28787   UGA_173101_P018_WD10 UGA_173101_P017_WC02 0.492120  885        33
## 28794   UGA_173101_P018_WD10 UGA_173101_P018_WB04 0.492029  885        33
## 28797   UGA_173101_P018_WD10 UGA_173101_P016_WF07 0.482628  849        33
## 28841   UGA_173101_P018_WD10 UGA_173101_P019_WG03 0.493534  887        33
## 28846   UGA_173101_P018_WD10 UGA_173101_P016_WE08 0.493534  886        33
## 28875   UGA_173101_P018_WD10 UGA_173101_P017_WC12 0.492795  886        33
## 28895   UGA_173101_P018_WD10 UGA_173101_P019_WA02 0.492795  886        33
## 43996   UGA_173101_P016_WH09 UGA_173101_P019_WE05 0.494286  899        33
## 44021   UGA_173101_P016_WH09 UGA_173101_P019_WA04 0.479965  758        33
## 44062   UGA_173101_P016_WH09 UGA_173101_P016_WC12 0.495028  897        33
## 44063   UGA_173101_P016_WH09 UGA_173101_P017_WC02 0.493662  908        33
## 44070   UGA_173101_P016_WH09 UGA_173101_P018_WB04 0.494302  907        33
## 44073   UGA_173101_P016_WH09 UGA_173101_P016_WF07 0.485075  868        33
## 44117   UGA_173101_P016_WH09 UGA_173101_P019_WG03 0.495763  910        33
## 44122   UGA_173101_P016_WH09 UGA_173101_P016_WE08 0.495763  908        33
## 44151   UGA_173101_P016_WH09 UGA_173101_P017_WC12 0.494350  909        33
## 44171   UGA_173101_P016_WH09 UGA_173101_P019_WA02 0.496459  910        33
## 44197   UGA_173101_P016_WH09 UGA_173101_P018_WD10 0.489971  881        33
## 54325   UGA_173101_P017_WD09 UGA_173101_P019_WE05 0.495702  896        33
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## 54350   UGA_173101_P017_WD09 UGA_173101_P019_WA04 0.480769  756         33
## 54391   UGA_173101_P017_WD09 UGA_173101_P016_WC12 0.495014  894         33
## 54392   UGA_173101_P017_WD09 UGA_173101_P017_WC02 0.495042  905         33
## 54399   UGA_173101_P017_WD09 UGA_173101_P018_WB04 0.495714  904         33
## 54402   UGA_173101_P017_WD09 UGA_173101_P016_WF07 0.486567  869         33
## 54446   UGA_173101_P017_WD09 UGA_173101_P019_WG03 0.497167  906         33
## 54451   UGA_173101_P017_WD09 UGA_173101_P016_WE08 0.497167  906         33
## 54480   UGA_173101_P017_WD09 UGA_173101_P017_WC12 0.495751  906         33
## 54500   UGA_173101_P017_WD09 UGA_173101_P019_WA02 0.496449  906         33
## 54526   UGA_173101_P017_WD09 UGA_173101_P018_WD10 0.490634  881         33
## 54583   UGA_173101_P017_WD09 UGA_173101_P016_WH09 0.492918  901         33
## 60766   UGA_173101_P019_WA01 UGA_173101_P019_WE05 0.498571  906         33
## 60791   UGA_173101_P019_WA01 UGA_173101_P019_WA04 0.482578  761         33
## 60832   UGA_173101_P019_WA01 UGA_173101_P016_WC12 0.497875  902         33
## 60833   UGA_173101_P019_WA01 UGA_173101_P017_WC02 0.497887  916         33
## 60840   UGA_173101_P019_WA01 UGA_173101_P018_WB04 0.498580  917         33
## 60843   UGA_173101_P019_WA01 UGA_173101_P016_WF07 0.489552  873         33
## 60887   UGA_173101_P019_WA01 UGA_173101_P019_WG03 0.500000  921         33
## 60892   UGA_173101_P019_WA01 UGA_173101_P016_WE08 0.500000  917         33
## 60921   UGA_173101_P019_WA01 UGA_173101_P017_WC12 0.498592  917         33
## 60941   UGA_173101_P019_WA01 UGA_173101_P019_WA02 0.499294  918         33
## 60967   UGA_173101_P019_WA01 UGA_173101_P018_WD10 0.493534  887         33
## 61024   UGA_173101_P019_WA01 UGA_173101_P016_WH09 0.495763  909         33
## 61057   UGA_173101_P019_WA01 UGA_173101_P017_WD09 0.497167  906         33
## 70540   UGA_173101_P018_WB06 UGA_173101_P019_WE05 0.462707  473         33
## 70565   UGA_173101_P018_WB06 UGA_173101_P019_WA04 0.446774  414         33
## 70606   UGA_173101_P018_WB06 UGA_173101_P016_WC12 0.459722  472         33
## 70607   UGA_173101_P018_WB06 UGA_173101_P017_WC02 0.459945  475         33
## 70614   UGA_173101_P018_WB06 UGA_173101_P018_WB04 0.459722  473         33
## 70617   UGA_173101_P018_WB06 UGA_173101_P016_WF07 0.448276  464         33
## 70661   UGA_173101_P018_WB06 UGA_173101_P019_WG03 0.459945  475         33
## 70666   UGA_173101_P018_WB06 UGA_173101_P016_WE08 0.459945  475         33
## 70695   UGA_173101_P018_WB06 UGA_173101_P017_WC12 0.459945  475         33
## 70715   UGA_173101_P018_WB06 UGA_173101_P019_WA02 0.459945  475         33
## 70741   UGA_173101_P018_WB06 UGA_173101_P018_WD10 0.450843  465         33
## 70798   UGA_173101_P018_WB06 UGA_173101_P016_WH09 0.454420  474         33
## 70831   UGA_173101_P018_WB06 UGA_173101_P017_WD09 0.458333  473         33
## 70850   UGA_173101_P018_WB06 UGA_173101_P019_WA01 0.459945  475         33
## 79840   UGA_173101_P018_WC11 UGA_173101_P019_WE05 0.498571  906         33
## 79865   UGA_173101_P018_WC11 UGA_173101_P019_WA04 0.482578  761         33
## 79906   UGA_173101_P018_WC11 UGA_173101_P016_WC12 0.497875  902         33
## 79907   UGA_173101_P018_WC11 UGA_173101_P017_WC02 0.497887  916         33
## 79914   UGA_173101_P018_WC11 UGA_173101_P018_WB04 0.498580  917         33
## 79917   UGA_173101_P018_WC11 UGA_173101_P016_WF07 0.489552  873         33
## 79961   UGA_173101_P018_WC11 UGA_173101_P019_WG03 0.500000  922         33
## 79966   UGA_173101_P018_WC11 UGA_173101_P016_WE08 0.500000  917         33
## 79995   UGA_173101_P018_WC11 UGA_173101_P017_WC12 0.498592  917         33
## 80015   UGA_173101_P018_WC11 UGA_173101_P019_WA02 0.499294  918         33
## 80041   UGA_173101_P018_WC11 UGA_173101_P018_WD10 0.493534  887         33
## 80098   UGA_173101_P018_WC11 UGA_173101_P016_WH09 0.495763  910         33
## 80131   UGA_173101_P018_WC11 UGA_173101_P017_WD09 0.497167  906         33
## 80150   UGA_173101_P018_WC11 UGA_173101_P019_WA01 0.500000  921         33
## 80177   UGA_173101_P018_WC11 UGA_173101_P018_WB06 0.459945  475         33
## 84706   UGA_173101_P023_WF05 UGA_173101_P019_WE05 0.481688  853         33
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## 84731   UGA_173101_P023_WF05 UGA_173101_P019_WA04 0.465993  727         33
## 84772   UGA_173101_P023_WF05 UGA_173101_P016_WC12 0.481746  854         33
## 84773   UGA_173101_P023_WF05 UGA_173101_P017_WC02 0.481013  860         33
## 84780   UGA_173101_P023_WF05 UGA_173101_P018_WB04 0.481013  860         33
## 84783   UGA_173101_P023_WF05 UGA_173101_P016_WF07 0.473333  829         33
## 84827   UGA_173101_P023_WF05 UGA_173101_P019_WG03 0.482595  861         33
## 84832   UGA_173101_P023_WF05 UGA_173101_P016_WE08 0.482595  861         33
## 84861   UGA_173101_P023_WF05 UGA_173101_P017_WC12 0.481804  861         33
## 84881   UGA_173101_P023_WF05 UGA_173101_P019_WA02 0.481804  861         33
## 84907   UGA_173101_P023_WF05 UGA_173101_P018_WD10 0.475080  838         33
## 84964   UGA_173101_P023_WF05 UGA_173101_P016_WH09 0.477848  858         33
## 84997   UGA_173101_P023_WF05 UGA_173101_P017_WD09 0.481746  857         33
## 85016   UGA_173101_P023_WF05 UGA_173101_P019_WA01 0.482595  861         33
## 85043   UGA_173101_P023_WF05 UGA_173101_P018_WB06 0.438650  461         33
## 85067   UGA_173101_P023_WF05 UGA_173101_P018_WC11 0.482595  861         33
## 87611   UGA_173101_P023_WB12 UGA_173101_P019_WE05 0.495702  899         33
## 87636   UGA_173101_P023_WB12 UGA_173101_P019_WA04 0.480769  756         33
## 87677   UGA_173101_P023_WB12 UGA_173101_P016_WC12 0.495014  896         33
## 87678   UGA_173101_P023_WB12 UGA_173101_P017_WC02 0.495056  908         33
## 87685   UGA_173101_P023_WB12 UGA_173101_P018_WB04 0.497151  909         33
## 87688   UGA_173101_P023_WB12 UGA_173101_P016_WF07 0.490269  868         33
## 87732   UGA_173101_P023_WB12 UGA_173101_P019_WG03 0.497167  911         33
## 87737   UGA_173101_P023_WB12 UGA_173101_P016_WE08 0.497875  909         33
## 87766   UGA_173101_P023_WB12 UGA_173101_P017_WC12 0.496469  910         33
## 87786   UGA_173101_P023_WB12 UGA_173101_P019_WA02 0.496449  910         33
## 87812   UGA_173101_P023_WB12 UGA_173101_P018_WD10 0.491354  881         33
## 87869   UGA_173101_P023_WB12 UGA_173101_P016_WH09 0.492938  903         33
## 87902   UGA_173101_P023_WB12 UGA_173101_P017_WD09 0.495014  900         33
## 87921   UGA_173101_P023_WB12 UGA_173101_P019_WA01 0.497167  911         33
## 87948   UGA_173101_P023_WB12 UGA_173101_P018_WB06 0.459945  474         33
## 87972   UGA_173101_P023_WB12 UGA_173101_P018_WC11 0.497167  911         33
## 87984   UGA_173101_P023_WB12 UGA_173101_P023_WF05 0.481013  860         33
## 90991   UGA_173101_P023_WA04 UGA_173101_P019_WE05 0.491254  872         33
## 91016   UGA_173101_P023_WA04 UGA_173101_P019_WA04 0.475352  740         33
## 91057   UGA_173101_P023_WA04 UGA_173101_P016_WC12 0.490552  871         33
## 91058   UGA_173101_P023_WA04 UGA_173101_P017_WC02 0.490580  878         33
## 91065   UGA_173101_P023_WA04 UGA_173101_P018_WB04 0.491304  879         33
## 91068   UGA_173101_P023_WA04 UGA_173101_P016_WF07 0.484940  845         33
## 91112   UGA_173101_P023_WA04 UGA_173101_P019_WG03 0.492754  879         33
## 91117   UGA_173101_P023_WA04 UGA_173101_P016_WE08 0.492754  879         33
## 91146   UGA_173101_P023_WA04 UGA_173101_P017_WC12 0.492029  879         33
## 91166   UGA_173101_P023_WA04 UGA_173101_P019_WA02 0.492029  879         33
## 91192   UGA_173101_P023_WA04 UGA_173101_P018_WD10 0.485988  857         33
## 91249   UGA_173101_P023_WA04 UGA_173101_P016_WH09 0.489130  875         33
## 91282   UGA_173101_P023_WA04 UGA_173101_P017_WD09 0.489855  876         33
## 91301   UGA_173101_P023_WA04 UGA_173101_P019_WA01 0.492754  879         33
## 91328   UGA_173101_P023_WA04 UGA_173101_P018_WB06 0.458101  465         33
## 91352   UGA_173101_P023_WA04 UGA_173101_P018_WC11 0.492754  879         33
## 91364   UGA_173101_P023_WA04 UGA_173101_P023_WF05 0.474359  843         33
## 91371   UGA_173101_P023_WA04 UGA_173101_P023_WB12 0.492029  879         33
## 111668  UGA_173101_P023_WA07 UGA_173101_P019_WE05 0.493497  890         33
## 111693  UGA_173101_P023_WA07 UGA_173101_P019_WA04 0.476316  753         33
## 111734  UGA_173101_P023_WA07 UGA_173101_P016_WC12 0.492857  888         33
## 111735  UGA_173101_P023_WA07 UGA_173101_P017_WC02 0.494302  902         33
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## 111742  UGA_173101_P023_WA07 UGA_173101_P018_WB04 0.493553  901         33
## 111745  UGA_173101_P023_WA07 UGA_173101_P016_WF07 0.485693  860         33
## 111789  UGA_173101_P023_WA07 UGA_173101_P019_WG03 0.495014  902         33
## 111794  UGA_173101_P023_WA07 UGA_173101_P016_WE08 0.495014  901         33
## 111823  UGA_173101_P023_WA07 UGA_173101_P017_WC12 0.493590  901         33
## 111843  UGA_173101_P023_WA07 UGA_173101_P019_WA02 0.494302  902         33
## 111869  UGA_173101_P023_WA07 UGA_173101_P018_WD10 0.489130  873         33
## 111926  UGA_173101_P023_WA07 UGA_173101_P016_WH09 0.490714  894         33
## 111959  UGA_173101_P023_WA07 UGA_173101_P017_WD09 0.492143  894         33
## 111978  UGA_173101_P023_WA07 UGA_173101_P019_WA01 0.495014  902         33
## 112005  UGA_173101_P023_WA07 UGA_173101_P018_WB06 0.453911  470         33
## 112029  UGA_173101_P023_WA07 UGA_173101_P018_WC11 0.495014  902         33
## 112041  UGA_173101_P023_WA07 UGA_173101_P023_WF05 0.477707  854         33
## 112048  UGA_173101_P023_WA07 UGA_173101_P023_WB12 0.494269  898         33
## 112056  UGA_173101_P023_WA07 UGA_173101_P023_WA04 0.489067  872         33
## 120826  UGA_173101_P022_WD02 UGA_173101_P019_WE05 0.463235  467         33
## 120851  UGA_173101_P022_WD02 UGA_173101_P019_WA04 0.453642  402         33
## 120892  UGA_173101_P022_WD02 UGA_173101_P016_WC12 0.464912  466         33
## 120893  UGA_173101_P022_WD02 UGA_173101_P017_WC02 0.464912  468         33
## 120900  UGA_173101_P022_WD02 UGA_173101_P018_WB04 0.467836  468         33
## 120903  UGA_173101_P022_WD02 UGA_173101_P016_WF07 0.459091  452         33
## 120947  UGA_173101_P022_WD02 UGA_173101_P019_WG03 0.466374  468         33
## 120952  UGA_173101_P022_WD02 UGA_173101_P016_WE08 0.466374  468         33
## 120981  UGA_173101_P022_WD02 UGA_173101_P017_WC12 0.464912  468         33
## 121001  UGA_173101_P022_WD02 UGA_173101_P019_WA02 0.464912  468         33
## 121027  UGA_173101_P022_WD02 UGA_173101_P018_WD10 0.456845  457         33
## 121084  UGA_173101_P022_WD02 UGA_173101_P016_WH09 0.461988  468         33
## 121117  UGA_173101_P022_WD02 UGA_173101_P017_WD09 0.461988  467         33
## 121136  UGA_173101_P022_WD02 UGA_173101_P019_WA01 0.466374  468         33
## 121163  UGA_173101_P022_WD02 UGA_173101_P018_WB06 0.443878  265         33
## 121187  UGA_173101_P022_WD02 UGA_173101_P018_WC11 0.466374  468         33
## 121199  UGA_173101_P022_WD02 UGA_173101_P023_WF05 0.451807  459         33
## 121206  UGA_173101_P022_WD02 UGA_173101_P023_WB12 0.467836  468         33
## 121214  UGA_173101_P022_WD02 UGA_173101_P023_WA04 0.458824  461         33
## 121260  UGA_173101_P022_WD02 UGA_173101_P023_WA07 0.464912  468         33
## 160501  UGA_173101_P020_WH07 UGA_173101_P019_WE05 0.433014  651         33
## 160526  UGA_173101_P020_WH07 UGA_173101_P019_WA04 0.398305  555         33
## 160567  UGA_173101_P020_WH07 UGA_173101_P016_WC12 0.433649  646         33
## 160568  UGA_173101_P020_WH07 UGA_173101_P017_WC02 0.431280  654         33
## 160575  UGA_173101_P020_WH07 UGA_173101_P018_WB04 0.434834  653         33
## 160578  UGA_173101_P020_WH07 UGA_173101_P016_WF07 0.424623  629         33
## 160622  UGA_173101_P020_WH07 UGA_173101_P019_WG03 0.434834  654         33
## 160627  UGA_173101_P020_WH07 UGA_173101_P016_WE08 0.434834  654         33
## 160656  UGA_173101_P020_WH07 UGA_173101_P017_WC12 0.434834  654         33
## 160676  UGA_173101_P020_WH07 UGA_173101_P019_WA02 0.433649  654         33
## 160702  UGA_173101_P020_WH07 UGA_173101_P018_WD10 0.427885  641         33
## 160759  UGA_173101_P020_WH07 UGA_173101_P016_WH09 0.430952  649         33
## 160792  UGA_173101_P020_WH07 UGA_173101_P017_WD09 0.431818  649         33
## 160811  UGA_173101_P020_WH07 UGA_173101_P019_WA01 0.434834  654         33
## 160838  UGA_173101_P020_WH07 UGA_173101_P018_WB06 0.408482  351         33
## 160862  UGA_173101_P020_WH07 UGA_173101_P018_WC11 0.434834  654         33
## 160874  UGA_173101_P020_WH07 UGA_173101_P023_WF05 0.419598  626         33
## 160881  UGA_173101_P020_WH07 UGA_173101_P023_WB12 0.437204  653         33
## 160889  UGA_173101_P020_WH07 UGA_173101_P023_WA04 0.425971  639         33
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## 160935  UGA_173101_P020_WH07 UGA_173101_P023_WA07 0.430288  649        33
## 160954  UGA_173101_P020_WH07 UGA_173101_P022_WD02 0.403689  368        33
## 188845  UGA_173101_P021_WF01 UGA_173101_P019_WE05 0.483434  886        33
## 188870  UGA_173101_P021_WF01 UGA_173101_P019_WA04 0.467199  748        33
## 188911  UGA_173101_P021_WF01 UGA_173101_P016_WC12 0.482836  886        33
## 188912  UGA_173101_P021_WF01 UGA_173101_P017_WC02 0.482887  894        33
## 188919  UGA_173101_P021_WF01 UGA_173101_P018_WB04 0.483631  895        33
## 188922  UGA_173101_P021_WF01 UGA_173101_P016_WF07 0.474843  855        33
## 188966  UGA_173101_P021_WF01 UGA_173101_P019_WG03 0.485119  896        33
## 188971  UGA_173101_P021_WF01 UGA_173101_P016_WE08 0.485119  894        33
## 189000  UGA_173101_P021_WF01 UGA_173101_P017_WC12 0.484375  895        33
## 189020  UGA_173101_P021_WF01 UGA_173101_P019_WA02 0.484375  895        33
## 189046  UGA_173101_P021_WF01 UGA_173101_P018_WD10 0.479483  870        33
## 189103  UGA_173101_P021_WF01 UGA_173101_P016_WH09 0.480597  891        33
## 189136  UGA_173101_P021_WF01 UGA_173101_P017_WD09 0.482036  887        33
## 189155  UGA_173101_P021_WF01 UGA_173101_P019_WA01 0.485119  896        33
## 189182  UGA_173101_P021_WF01 UGA_173101_P018_WB06 0.450867  469        33
## 189206  UGA_173101_P021_WF01 UGA_173101_P018_WC11 0.485119  896        33
## 189218  UGA_173101_P021_WF01 UGA_173101_P023_WF05 0.467949  853        33
## 189225  UGA_173101_P021_WF01 UGA_173101_P023_WB12 0.483631  895        33
## 189233  UGA_173101_P021_WF01 UGA_173101_P023_WA04 0.480303  869        33
## 189279  UGA_173101_P021_WF01 UGA_173101_P023_WA07 0.481982  885        33
## 189298  UGA_173101_P021_WF01 UGA_173101_P022_WD02 0.445266  464        33
## 189373  UGA_173101_P021_WF01 UGA_173101_P020_WH07 0.419471  649        33
## 213571  UGA_173101_P022_WA09 UGA_173101_P019_WE05 0.495677  889        33
## 213596  UGA_173101_P022_WA09 UGA_173101_P019_WA04 0.478947  752        33
## 213637  UGA_173101_P022_WA09 UGA_173101_P016_WC12 0.494971  887        33
## 213638  UGA_173101_P022_WA09 UGA_173101_P017_WC02 0.496429  897        33
## 213645  UGA_173101_P022_WA09 UGA_173101_P018_WB04 0.497126  897        33
## 213648  UGA_173101_P022_WA09 UGA_173101_P016_WF07 0.488739  860        33
## 213692  UGA_173101_P022_WA09 UGA_173101_P019_WG03 0.497143  899        33
## 213697  UGA_173101_P022_WA09 UGA_173101_P016_WE08 0.497143  897        33
## 213726  UGA_173101_P022_WA09 UGA_173101_P017_WC12 0.496429  898        33
## 213746  UGA_173101_P022_WA09 UGA_173101_P019_WA02 0.496418  898        33
## 213772  UGA_173101_P022_WA09 UGA_173101_P018_WD10 0.491304  871        33
## 213829  UGA_173101_P022_WA09 UGA_173101_P016_WH09 0.492857  894        33
## 213862  UGA_173101_P022_WA09 UGA_173101_P017_WD09 0.494986  892        33
## 213881  UGA_173101_P022_WA09 UGA_173101_P019_WA01 0.497143  899        33
## 213908  UGA_173101_P022_WA09 UGA_173101_P018_WB06 0.459722  470        33
## 213932  UGA_173101_P022_WA09 UGA_173101_P018_WC11 0.497143  899        33
## 213944  UGA_173101_P022_WA09 UGA_173101_P023_WF05 0.480159  857        33
## 213951  UGA_173101_P022_WA09 UGA_173101_P023_WB12 0.497143  899        33
## 213959  UGA_173101_P022_WA09 UGA_173101_P023_WA04 0.491304  873        33
## 214005  UGA_173101_P022_WA09 UGA_173101_P023_WA07 0.494957  890        33
## 214024  UGA_173101_P022_WA09 UGA_173101_P022_WD02 0.467836  468        33
## 214099  UGA_173101_P022_WA09 UGA_173101_P020_WH07 0.433894  649        33
## 214147  UGA_173101_P022_WA09 UGA_173101_P021_WF01 0.483483  887        33
## 237745            GTS_949-23 UGA_173101_P019_WE05 0.490741  115       255
## 237770            GTS_949-23 UGA_173101_P019_WA04 0.489130   92       255
## 237811            GTS_949-23 UGA_173101_P016_WC12 0.491379  110       255
## 237812            GTS_949-23 UGA_173101_P017_WC02 0.483333  117       255
## 237819            GTS_949-23 UGA_173101_P018_WB04 0.491071  118       255
## 237822            GTS_949-23 UGA_173101_P016_WF07 0.462963  103       255
## 237866            GTS_949-23 UGA_173101_P019_WG03 0.491379  122       255
```

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## 237871            GTS_949-23 UGA_173101_P016_WE08 0.491379  117        255
## 237900            GTS_949-23 UGA_173101_P017_WC12 0.482143  117        255
## 237920            GTS_949-23 UGA_173101_P019_WA02 0.491379  119        255
## 237946            GTS_949-23 UGA_173101_P018_WD10 0.491071  111        255
## 238003            GTS_949-23 UGA_173101_P016_WH09 0.491071  115        255
## 238036            GTS_949-23 UGA_173101_P017_WD09 0.490741  109        255
## 238055            GTS_949-23 UGA_173101_P019_WA01 0.491379  121        255
## 238082            GTS_949-23 UGA_173101_P018_WB06 0.468750   56        255
## 238106            GTS_949-23 UGA_173101_P018_WC11 0.491379  122        255
## 238118            GTS_949-23 UGA_173101_P023_WF05 0.480769   94        255
## 238125            GTS_949-23 UGA_173101_P023_WB12 0.473214  111        255
## 238133            GTS_949-23 UGA_173101_P023_WA04 0.489130   93        255
## 238179            GTS_949-23 UGA_173101_P023_WA07 0.481481  108        255
## 238198            GTS_949-23 UGA_173101_P022_WD02 0.450000   52        255
## 238273            GTS_949-23 UGA_173101_P020_WH07 0.416667   78        255
## 238321            GTS_949-23 UGA_173101_P021_WF01 0.481481  104        255
## 238360            GTS_949-23 UGA_173101_P022_WA09 0.490000  102        255
## 272731            GTS_646-23 UGA_173101_P019_WE05 0.500000  112        255
## 272756            GTS_646-23 UGA_173101_P019_WA04 0.500000   90        255
## 272797            GTS_646-23 UGA_173101_P016_WC12 0.500000  107        255
## 272798            GTS_646-23 UGA_173101_P017_WC02 0.490741  114        255
## 272805            GTS_646-23 UGA_173101_P018_WB04 0.500000  115        255
## 272808            GTS_646-23 UGA_173101_P016_WF07 0.468750  100        255
## 272852            GTS_646-23 UGA_173101_P019_WG03 0.500000  119        255
## 272857            GTS_646-23 UGA_173101_P016_WE08 0.500000  114        255
## 272886            GTS_646-23 UGA_173101_P017_WC12 0.490385  114        255
## 272906            GTS_646-23 UGA_173101_P019_WA02 0.500000  116        255
## 272932            GTS_646-23 UGA_173101_P018_WD10 0.500000  108        255
## 272989            GTS_646-23 UGA_173101_P016_WH09 0.500000  112        255
## 273022            GTS_646-23 UGA_173101_P017_WD09 0.500000  106        255
## 273041            GTS_646-23 UGA_173101_P019_WA01 0.500000  118        255
## 273068            GTS_646-23 UGA_173101_P018_WB06 0.484375   55        255
## 273092            GTS_646-23 UGA_173101_P018_WC11 0.500000  119        255
## 273104            GTS_646-23 UGA_173101_P023_WF05 0.489583   92        255
## 273111            GTS_646-23 UGA_173101_P023_WB12 0.480000  108        255
## 273119            GTS_646-23 UGA_173101_P023_WA04 0.500000   90        255
## 273165            GTS_646-23 UGA_173101_P023_WA07 0.490000  105        255
## 273184            GTS_646-23 UGA_173101_P022_WD02 0.475000   51        255
## 273259            GTS_646-23 UGA_173101_P020_WH07 0.425000   76        255
## 273307            GTS_646-23 UGA_173101_P021_WF01 0.489583  101        255
## 273346            GTS_646-23 UGA_173101_P022_WA09 0.500000   99        255
## 273382            GTS_646-23           GTS_949-23 0.500000  120        255
## 283168            GTS_616-24 UGA_173101_P019_WE05 0.490385  114        276
## 283193            GTS_616-24 UGA_173101_P019_WA04 0.488636   91        276
## 283234            GTS_616-24 UGA_173101_P016_WC12 0.491071  109        276
## 283235            GTS_616-24 UGA_173101_P017_WC02 0.482759  116        276
## 283242            GTS_616-24 UGA_173101_P018_WB04 0.490741  117        276
## 283245            GTS_616-24 UGA_173101_P016_WF07 0.461538  102        276
## 283289            GTS_616-24 UGA_173101_P019_WG03 0.491071  120        276
## 283294            GTS_616-24 UGA_173101_P016_WE08 0.491071  116        276
## 283323            GTS_616-24 UGA_173101_P017_WC12 0.481481  116        276
## 283343            GTS_616-24 UGA_173101_P019_WA02 0.491071  118        276
## 283369            GTS_616-24 UGA_173101_P018_WD10 0.490741  109        276
## 283426            GTS_616-24 UGA_173101_P016_WH09 0.490741  114        276
```

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## 283459             GTS_616-24 UGA_173101_P017_WD09 0.490385  108        276
## 283478             GTS_616-24 UGA_173101_P019_WA01 0.491071  119        276
## 283505             GTS_616-24 UGA_173101_P018_WB06 0.468750   56        276
## 283529             GTS_616-24 UGA_173101_P018_WC11 0.491071  120        276
## 283541             GTS_616-24 UGA_173101_P023_WF05 0.480000   93        276
## 283548             GTS_616-24 UGA_173101_P023_WB12 0.472222  109        276
## 283556             GTS_616-24 UGA_173101_P023_WA04 0.488636   92        276
## 283602             GTS_616-24 UGA_173101_P023_WA07 0.480769  107        276
## 283621             GTS_616-24 UGA_173101_P022_WD02 0.450000   52        276
## 283696             GTS_616-24 UGA_173101_P020_WH07 0.416667   78        276
## 283744             GTS_616-24 UGA_173101_P021_WF01 0.480769  102        276
## 283783             GTS_616-24 UGA_173101_P022_WA09 0.489583  100        276
## 283819             GTS_616-24           GTS_949-23 0.491379  122        276
## 283868             GTS_616-24           GTS_646-23 0.500000  119        276
## 289981             GTS_655-23 UGA_173101_P019_WE05 0.489130  109        255
## 290006             GTS_655-23 UGA_173101_P019_WA04 0.488095   89        255
## 290047             GTS_655-23 UGA_173101_P016_WC12 0.490000  104        255
## 290048             GTS_655-23 UGA_173101_P017_WC02 0.480769  111        255
## 290055             GTS_655-23 UGA_173101_P018_WB04 0.489583  112        255
## 290058             GTS_655-23 UGA_173101_P016_WF07 0.458333   98        255
## 290102             GTS_655-23 UGA_173101_P019_WG03 0.490000  115        255
## 290107             GTS_655-23 UGA_173101_P016_WE08 0.490000  111        255
## 290136             GTS_655-23 UGA_173101_P017_WC12 0.490000  111        255
## 290156             GTS_655-23 UGA_173101_P019_WA02 0.490000  113        255
## 290182             GTS_655-23 UGA_173101_P018_WD10 0.489583  105        255
## 290239             GTS_655-23 UGA_173101_P016_WH09 0.489583  109        255
## 290272             GTS_655-23 UGA_173101_P017_WD09 0.489130  104        255
## 290291             GTS_655-23 UGA_173101_P019_WA01 0.490000  114        255
## 290318             GTS_655-23 UGA_173101_P018_WB06 0.466667   54        255
## 290342             GTS_655-23 UGA_173101_P018_WC11 0.490000  115        255
## 290354             GTS_655-23 UGA_173101_P023_WF05 0.479167   92        255
## 290361             GTS_655-23 UGA_173101_P023_WB12 0.470000  105        255
## 290369             GTS_655-23 UGA_173101_P023_WA04 0.487500   90        255
## 290415             GTS_655-23 UGA_173101_P023_WA07 0.478261  102        255
## 290434             GTS_655-23 UGA_173101_P022_WD02 0.444444   50        255
## 290509             GTS_655-23 UGA_173101_P020_WH07 0.416667   75        255
## 290557             GTS_655-23 UGA_173101_P021_WF01 0.479167   98        255
## 290596             GTS_655-23 UGA_173101_P022_WA09 0.488636   96        255
## 290632             GTS_655-23           GTS_949-23 0.500000  116        255
## 290681             GTS_655-23           GTS_646-23 0.500000  115        255
## 290695             GTS_655-23           GTS_616-24 0.500000  116        255
## 290743             GTS_983-23 UGA_173101_P019_WE05 0.500000  113        255
## 290768             GTS_983-23 UGA_173101_P019_WA04 0.500000   90        255
## 290809             GTS_983-23 UGA_173101_P016_WC12 0.500000  108        255
## 290810             GTS_983-23 UGA_173101_P017_WC02 0.491071  115        255
## 290817             GTS_983-23 UGA_173101_P018_WB04 0.500000  116        255
## 290820             GTS_983-23 UGA_173101_P016_WF07 0.470000  101        255
## 290864             GTS_983-23 UGA_173101_P019_WG03 0.500000  120        255
## 290869             GTS_983-23 UGA_173101_P016_WE08 0.500000  115        255
## 290898             GTS_983-23 UGA_173101_P017_WC12 0.490741  115        255
## 290918             GTS_983-23 UGA_173101_P019_WA02 0.500000  117        255
## 290944             GTS_983-23 UGA_173101_P018_WD10 0.500000  109        255
## 291001             GTS_983-23 UGA_173101_P016_WH09 0.500000  113        255
## 291034             GTS_983-23 UGA_173101_P017_WD09 0.500000  107        255
```

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## 291053           GTS_983-23 UGA_173101_P019_WA01 0.500000  119        255
## 291080           GTS_983-23 UGA_173101_P018_WB06 0.483333   54        255
## 291104           GTS_983-23 UGA_173101_P018_WC11 0.500000  120        255
## 291116           GTS_983-23 UGA_173101_P023_WF05 0.489583   92        255
## 291123           GTS_983-23 UGA_173101_P023_WB12 0.480769  109        255
## 291131           GTS_983-23 UGA_173101_P023_WA04 0.500000   91        255
## 291177           GTS_983-23 UGA_173101_P023_WA07 0.490385  106        255
## 291196           GTS_983-23 UGA_173101_P022_WD02 0.472222   50        255
## 291271           GTS_983-23 UGA_173101_P020_WH07 0.425000   76        255
## 291319           GTS_983-23 UGA_173101_P021_WF01 0.490000  102        255
## 291358           GTS_983-23 UGA_173101_P022_WA09 0.500000  100        255
## 291394           GTS_983-23           GTS_949-23 0.500000  121        255
## 291443           GTS_983-23           GTS_646-23 0.500000  119        255
## 291457           GTS_983-23           GTS_616-24 0.500000  119        255
## 291466           GTS_983-23           GTS_655-23 0.500000  114        255
## 299191           GTS_630-24 UGA_173101_P019_WE05 0.500000  112        255
## 299216           GTS_630-24 UGA_173101_P019_WA04 0.500000   89        255
## 299257           GTS_630-24 UGA_173101_P016_WC12 0.500000  107        255
## 299258           GTS_630-24 UGA_173101_P017_WC02 0.490741  114        255
## 299265           GTS_630-24 UGA_173101_P018_WB04 0.500000  115        255
## 299268           GTS_630-24 UGA_173101_P016_WF07 0.468750  100        255
## 299312           GTS_630-24 UGA_173101_P019_WG03 0.500000  118        255
## 299317           GTS_630-24 UGA_173101_P016_WE08 0.500000  114        255
## 299346           GTS_630-24 UGA_173101_P017_WC12 0.490385  114        255
## 299366           GTS_630-24 UGA_173101_P019_WA02 0.500000  116        255
## 299392           GTS_630-24 UGA_173101_P018_WD10 0.500000  107        255
## 299449           GTS_630-24 UGA_173101_P016_WH09 0.500000  112        255
## 299482           GTS_630-24 UGA_173101_P017_WD09 0.500000  106        255
## 299501           GTS_630-24 UGA_173101_P019_WA01 0.500000  117        255
## 299528           GTS_630-24 UGA_173101_P018_WB06 0.483333   54        255
## 299552           GTS_630-24 UGA_173101_P018_WC11 0.500000  118        255
## 299564           GTS_630-24 UGA_173101_P023_WF05 0.489130   91        255
## 299571           GTS_630-24 UGA_173101_P023_WB12 0.480000  107        255
## 299579           GTS_630-24 UGA_173101_P023_WA04 0.500000   90        255
## 299625           GTS_630-24 UGA_173101_P023_WA07 0.500000  105        255
## 299644           GTS_630-24 UGA_173101_P022_WD02 0.475000   51        255
## 299719           GTS_630-24 UGA_173101_P020_WH07 0.428571   77        255
## 299767           GTS_630-24 UGA_173101_P021_WF01 0.490000  101        255
## 299806           GTS_630-24 UGA_173101_P022_WA09 0.500000   98        255
## 299842           GTS_630-24           GTS_949-23 0.500000  119        255
## 299891           GTS_630-24           GTS_646-23 0.500000  118        255
## 299905           GTS_630-24           GTS_616-24 0.500000  119        255
## 299914           GTS_630-24           GTS_655-23 0.500000  114        255
## 299915           GTS_630-24           GTS_983-23 0.500000  118        255
## 306976           GTS_665-23 UGA_173101_P019_WE05 0.490741  115        255
## 307001           GTS_665-23 UGA_173101_P019_WA04 0.489130   92        255
## 307042           GTS_665-23 UGA_173101_P016_WC12 0.491379  110        255
## 307043           GTS_665-23 UGA_173101_P017_WC02 0.483333  117        255
## 307050           GTS_665-23 UGA_173101_P018_WB04 0.491071  118        255
## 307053           GTS_665-23 UGA_173101_P016_WF07 0.462963  103        255
## 307097           GTS_665-23 UGA_173101_P019_WG03 0.491379  121        255
## 307102           GTS_665-23 UGA_173101_P016_WE08 0.491379  117        255
## 307131           GTS_665-23 UGA_173101_P017_WC12 0.482143  117        255
## 307151           GTS_665-23 UGA_173101_P019_WA02 0.491379  119        255
```

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## 307177            GTS_665-23 UGA_173101_P018_WD10 0.491071  110        255
## 307234            GTS_665-23 UGA_173101_P016_WH09 0.491071  115        255
## 307267            GTS_665-23 UGA_173101_P017_WD09 0.490741  109        255
## 307286            GTS_665-23 UGA_173101_P019_WA01 0.491379  120        255
## 307313            GTS_665-23 UGA_173101_P018_WB06 0.468750   56        255
## 307337            GTS_665-23 UGA_173101_P018_WC11 0.491379  121        255
## 307349            GTS_665-23 UGA_173101_P023_WF05 0.480769   94        255
## 307356            GTS_665-23 UGA_173101_P023_WB12 0.473214  110        255
## 307364            GTS_665-23 UGA_173101_P023_WA04 0.489130   93        255
## 307410            GTS_665-23 UGA_173101_P023_WA07 0.481481  108        255
## 307429            GTS_665-23 UGA_173101_P022_WD02 0.450000   52        255
## 307504            GTS_665-23 UGA_173101_P020_WH07 0.416667   78        255
## 307552            GTS_665-23 UGA_173101_P021_WF01 0.481481  103        255
## 307591            GTS_665-23 UGA_173101_P022_WA09 0.490000  101        255
## 307627            GTS_665-23           GTS_949-23 0.500000  122        255
## 307676            GTS_665-23           GTS_646-23 0.500000  119        255
## 307690            GTS_665-23           GTS_616-24 0.500000  121        255
## 307699            GTS_665-23           GTS_655-23 0.500000  116        255
## 307700            GTS_665-23           GTS_983-23 0.500000  120        255
## 307711            GTS_665-23           GTS_630-24 0.500000  119        255
## 309331            GTS_997-23 UGA_173101_P019_WE05 0.490741  115        255
## 309356            GTS_997-23 UGA_173101_P019_WA04 0.489130   92        255
## 309397            GTS_997-23 UGA_173101_P016_WC12 0.491379  110        255
## 309398            GTS_997-23 UGA_173101_P017_WC02 0.483333  117        255
## 309405            GTS_997-23 UGA_173101_P018_WB04 0.491071  118        255
## 309408            GTS_997-23 UGA_173101_P016_WF07 0.462963  103        255
## 309452            GTS_997-23 UGA_173101_P019_WG03 0.491379  122        255
## 309457            GTS_997-23 UGA_173101_P016_WE08 0.491379  117        255
## 309486            GTS_997-23 UGA_173101_P017_WC12 0.482143  117        255
## 309506            GTS_997-23 UGA_173101_P019_WA02 0.491379  119        255
## 309532            GTS_997-23 UGA_173101_P018_WD10 0.491071  111        255
## 309589            GTS_997-23 UGA_173101_P016_WH09 0.491071  115        255
## 309622            GTS_997-23 UGA_173101_P017_WD09 0.490741  109        255
## 309641            GTS_997-23 UGA_173101_P019_WA01 0.491379  121        255
## 309668            GTS_997-23 UGA_173101_P018_WB06 0.468750   56        255
## 309692            GTS_997-23 UGA_173101_P018_WC11 0.491379  122        255
## 309704            GTS_997-23 UGA_173101_P023_WF05 0.480769   94        255
## 309711            GTS_997-23 UGA_173101_P023_WB12 0.473214  111        255
## 309719            GTS_997-23 UGA_173101_P023_WA04 0.489130   93        255
## 309765            GTS_997-23 UGA_173101_P023_WA07 0.481481  108        255
## 309784            GTS_997-23 UGA_173101_P022_WD02 0.450000   52        255
## 309859            GTS_997-23 UGA_173101_P020_WH07 0.416667   78        255
## 309907            GTS_997-23 UGA_173101_P021_WF01 0.481481  104        255
## 309946            GTS_997-23 UGA_173101_P022_WA09 0.490000  102        255
## 309982            GTS_997-23           GTS_949-23 0.500000  123        255
## 310031            GTS_997-23           GTS_646-23 0.500000  120        255
## 310045            GTS_997-23           GTS_616-24 0.500000  121        255
## 310054            GTS_997-23           GTS_655-23 0.500000  116        255
## 310055            GTS_997-23           GTS_983-23 0.500000  121        255
## 310066            GTS_997-23           GTS_630-24 0.500000  119        255
## 310076            GTS_997-23           GTS_665-23 0.500000  122        255
## 314068            GTS_644-24 UGA_173101_P019_WE05 0.500000  111        276
## 314093            GTS_644-24 UGA_173101_P019_WA04 0.500000   89        276
## 314134            GTS_644-24 UGA_173101_P016_WC12 0.500000  106        276
```

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## 314135             GTS_644-24 UGA_173101_P017_WC02 0.490741  113        276
## 314142             GTS_644-24 UGA_173101_P018_WB04 0.500000  114        276
## 314145             GTS_644-24 UGA_173101_P016_WF07 0.468750   99        276
## 314189             GTS_644-24 UGA_173101_P019_WG03 0.500000  117        276
## 314194             GTS_644-24 UGA_173101_P016_WE08 0.500000  113        276
## 314223             GTS_644-24 UGA_173101_P017_WC12 0.490385  113        276
## 314243             GTS_644-24 UGA_173101_P019_WA02 0.500000  115        276
## 314269             GTS_644-24 UGA_173101_P018_WD10 0.500000  106        276
## 314326             GTS_644-24 UGA_173101_P016_WH09 0.500000  111        276
## 314359             GTS_644-24 UGA_173101_P017_WD09 0.500000  105        276
## 314378             GTS_644-24 UGA_173101_P019_WA01 0.500000  116        276
## 314405             GTS_644-24 UGA_173101_P018_WB06 0.483333   54        276
## 314429             GTS_644-24 UGA_173101_P018_WC11 0.500000  117        276
## 314441             GTS_644-24 UGA_173101_P023_WF05 0.489130   91        276
## 314448             GTS_644-24 UGA_173101_P023_WB12 0.480000  106        276
## 314456             GTS_644-24 UGA_173101_P023_WA04 0.500000   90        276
## 314502             GTS_644-24 UGA_173101_P023_WA07 0.500000  104        276
## 314521             GTS_644-24 UGA_173101_P022_WD02 0.475000   51        276
## 314596             GTS_644-24 UGA_173101_P020_WH07 0.428571   77        276
## 314644             GTS_644-24 UGA_173101_P021_WF01 0.490000  101        276
## 314683             GTS_644-24 UGA_173101_P022_WA09 0.500000   97        276
## 314719             GTS_644-24           GTS_949-23 0.490741  119        276
## 314768             GTS_644-24           GTS_646-23 0.500000  117        276
## 314782             GTS_644-24           GTS_616-24 0.500000  119        276
## 314791             GTS_644-24           GTS_655-23 0.500000  113        276
## 314792             GTS_644-24           GTS_983-23 0.500000  117        276
## 314803             GTS_644-24           GTS_630-24 0.500000  118        276
## 314813             GTS_644-24           GTS_665-23 0.500000  118        276
## 314816             GTS_644-24           GTS_997-23 0.500000  118        276
## 318841             GTS_818-23 UGA_173101_P019_WE05 0.490741  115        255
## 318866             GTS_818-23 UGA_173101_P019_WA04 0.489130   92        255
## 318907             GTS_818-23 UGA_173101_P016_WC12 0.491379  110        255
## 318908             GTS_818-23 UGA_173101_P017_WC02 0.483333  117        255
## 318915             GTS_818-23 UGA_173101_P018_WB04 0.491071  118        255
## 318918             GTS_818-23 UGA_173101_P016_WF07 0.462963  103        255
## 318962             GTS_818-23 UGA_173101_P019_WG03 0.491379  122        255
## 318967             GTS_818-23 UGA_173101_P016_WE08 0.491379  117        255
## 318996             GTS_818-23 UGA_173101_P017_WC12 0.482143  117        255
## 319016             GTS_818-23 UGA_173101_P019_WA02 0.491379  119        255
## 319042             GTS_818-23 UGA_173101_P018_WD10 0.491071  111        255
## 319099             GTS_818-23 UGA_173101_P016_WH09 0.491071  115        255
## 319132             GTS_818-23 UGA_173101_P017_WD09 0.490741  109        255
## 319151             GTS_818-23 UGA_173101_P019_WA01 0.491379  121        255
## 319178             GTS_818-23 UGA_173101_P018_WB06 0.468750   56        255
## 319202             GTS_818-23 UGA_173101_P018_WC11 0.491379  122        255
## 319214             GTS_818-23 UGA_173101_P023_WF05 0.480769   94        255
## 319221             GTS_818-23 UGA_173101_P023_WB12 0.473214  111        255
## 319229             GTS_818-23 UGA_173101_P023_WA04 0.489130   93        255
## 319275             GTS_818-23 UGA_173101_P023_WA07 0.481481  108        255
## 319294             GTS_818-23 UGA_173101_P022_WD02 0.450000   52        255
## 319369             GTS_818-23 UGA_173101_P020_WH07 0.416667   78        255
## 319417             GTS_818-23 UGA_173101_P021_WF01 0.481481  104        255
## 319456             GTS_818-23 UGA_173101_P022_WA09 0.490000  102        255
## 319492             GTS_818-23           GTS_949-23 0.500000  123        255
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## 319541             GTS_818-23           GTS_646-23 0.500000  120        255
## 319555             GTS_818-23           GTS_616-24 0.500000  121        255
## 319564             GTS_818-23           GTS_655-23 0.500000  116        255
## 319565             GTS_818-23           GTS_983-23 0.500000  121        255
## 319576             GTS_818-23           GTS_630-24 0.500000  119        255
## 319586             GTS_818-23           GTS_665-23 0.500000  122        255
## 319589             GTS_818-23           GTS_997-23 0.500000  123        255
## 319595             GTS_818-23           GTS_644-24 0.500000  118        255
## 329306             GTS_677-23 UGA_173101_P019_WE05 0.489583  112        255
## 329331             GTS_677-23 UGA_173101_P019_WA04 0.487500   89        255
## 329372             GTS_677-23 UGA_173101_P016_WC12 0.490385  107        255
## 329373             GTS_677-23 UGA_173101_P017_WC02 0.481481  114        255
## 329380             GTS_677-23 UGA_173101_P018_WB04 0.490000  115        255
## 329383             GTS_677-23 UGA_173101_P016_WF07 0.458333  100        255
## 329427             GTS_677-23 UGA_173101_P019_WG03 0.490385  118        255
## 329432             GTS_677-23 UGA_173101_P016_WE08 0.490385  114        255
## 329461             GTS_677-23 UGA_173101_P017_WC12 0.480000  114        255
## 329481             GTS_677-23 UGA_173101_P019_WA02 0.490385  116        255
## 329507             GTS_677-23 UGA_173101_P018_WD10 0.490000  107        255
## 329564             GTS_677-23 UGA_173101_P016_WH09 0.490000  112        255
## 329597             GTS_677-23 UGA_173101_P017_WD09 0.489583  106        255
## 329616             GTS_677-23 UGA_173101_P019_WA01 0.490385  117        255
## 329643             GTS_677-23 UGA_173101_P018_WB06 0.464286   54        255
## 329667             GTS_677-23 UGA_173101_P018_WC11 0.490385  118        255
## 329679             GTS_677-23 UGA_173101_P023_WF05 0.478261   91        255
## 329686             GTS_677-23 UGA_173101_P023_WB12 0.470000  107        255
## 329694             GTS_677-23 UGA_173101_P023_WA04 0.487500   90        255
## 329740             GTS_677-23 UGA_173101_P023_WA07 0.490000  105        255
## 329759             GTS_677-23 UGA_173101_P022_WD02 0.444444   51        255
## 329834             GTS_677-23 UGA_173101_P020_WH07 0.412500   77        255
## 329882             GTS_677-23 UGA_173101_P021_WF01 0.480000  101        255
## 329921             GTS_677-23 UGA_173101_P022_WA09 0.488636   98        255
## 329957             GTS_677-23           GTS_949-23 0.500000  119        255
## 330006             GTS_677-23           GTS_646-23 0.500000  117        255
## 330020             GTS_677-23           GTS_616-24 0.500000  119        255
## 330029             GTS_677-23           GTS_655-23 0.500000  114        255
## 330030             GTS_677-23           GTS_983-23 0.500000  118        255
## 330041             GTS_677-23           GTS_630-24 0.500000  118        255
## 330051             GTS_677-23           GTS_665-23 0.500000  119        255
## 330054             GTS_677-23           GTS_997-23 0.500000  119        255
## 330060             GTS_677-23           GTS_644-24 0.500000  117        255
## 330066             GTS_677-23           GTS_818-23 0.500000  119        255
## 330118            GTS_1012-23 UGA_173101_P019_WE05 0.500000  113         33
## 330143            GTS_1012-23 UGA_173101_P019_WA04 0.500000   90         33
## 330184            GTS_1012-23 UGA_173101_P016_WC12 0.500000  108         33
## 330185            GTS_1012-23 UGA_173101_P017_WC02 0.491071  115         33
## 330192            GTS_1012-23 UGA_173101_P018_WB04 0.500000  116         33
## 330195            GTS_1012-23 UGA_173101_P016_WF07 0.470000  101         33
## 330239            GTS_1012-23 UGA_173101_P019_WG03 0.500000  120         33
## 330244            GTS_1012-23 UGA_173101_P016_WE08 0.500000  115         33
## 330273            GTS_1012-23 UGA_173101_P017_WC12 0.490741  115         33
## 330293            GTS_1012-23 UGA_173101_P019_WA02 0.500000  117         33
## 330319            GTS_1012-23 UGA_173101_P018_WD10 0.500000  109         33
## 330376            GTS_1012-23 UGA_173101_P016_WH09 0.500000  113         33
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## 330409           GTS_1012-23 UGA_173101_P017_WD09 0.500000  107         33
## 330428           GTS_1012-23 UGA_173101_P019_WA01 0.500000  119         33
## 330455           GTS_1012-23 UGA_173101_P018_WB06 0.483333   54         33
## 330479           GTS_1012-23 UGA_173101_P018_WC11 0.500000  120         33
## 330491           GTS_1012-23 UGA_173101_P023_WF05 0.489583   92         33
## 330498           GTS_1012-23 UGA_173101_P023_WB12 0.480769  109         33
## 330506           GTS_1012-23 UGA_173101_P023_WA04 0.500000   91         33
## 330552           GTS_1012-23 UGA_173101_P023_WA07 0.490385  106         33
## 330571           GTS_1012-23 UGA_173101_P022_WD02 0.472222   50         33
## 330646           GTS_1012-23 UGA_173101_P020_WH07 0.425000   76         33
## 330694           GTS_1012-23 UGA_173101_P021_WF01 0.490000  102         33
## 330733           GTS_1012-23 UGA_173101_P022_WA09 0.500000  100         33
## 330769           GTS_1012-23           GTS_949-23 0.500000  121         33
## 330818           GTS_1012-23           GTS_646-23 0.500000  119         33
## 330832           GTS_1012-23           GTS_616-24 0.500000  119         33
## 330841           GTS_1012-23           GTS_655-23 0.500000  114         33
## 330842           GTS_1012-23           GTS_983-23 0.500000  121         33
## 330853           GTS_1012-23           GTS_630-24 0.500000  118         33
## 330863           GTS_1012-23           GTS_665-23 0.500000  120         33
## 330866           GTS_1012-23           GTS_997-23 0.500000  121         33
## 330872           GTS_1012-23           GTS_644-24 0.500000  117         33
## 330878           GTS_1012-23           GTS_818-23 0.500000  121         33
## 330891           GTS_1012-23           GTS_677-23 0.500000  118         33
## 335011            GTS_657-24 UGA_173101_P019_WE05 0.490385  113        276
## 335036            GTS_657-24 UGA_173101_P019_WA04 0.488636   91        276
## 335077            GTS_657-24 UGA_173101_P016_WC12 0.491071  108        276
## 335078            GTS_657-24 UGA_173101_P017_WC02 0.482143  115        276
## 335085            GTS_657-24 UGA_173101_P018_WB04 0.490741  116        276
## 335088            GTS_657-24 UGA_173101_P016_WF07 0.460000  101        276
## 335132            GTS_657-24 UGA_173101_P019_WG03 0.491071  119        276
## 335137            GTS_657-24 UGA_173101_P016_WE08 0.491071  115        276
## 335166            GTS_657-24 UGA_173101_P017_WC12 0.482143  115        276
## 335186            GTS_657-24 UGA_173101_P019_WA02 0.491071  117        276
## 335212            GTS_657-24 UGA_173101_P018_WD10 0.490741  108        276
## 335269            GTS_657-24 UGA_173101_P016_WH09 0.490741  113        276
## 335302            GTS_657-24 UGA_173101_P017_WD09 0.490385  107        276
## 335321            GTS_657-24 UGA_173101_P019_WA01 0.491071  118        276
## 335348            GTS_657-24 UGA_173101_P018_WB06 0.484375   55        276
## 335372            GTS_657-24 UGA_173101_P018_WC11 0.491071  119        276
## 335384            GTS_657-24 UGA_173101_P023_WF05 0.479167   93        276
## 335391            GTS_657-24 UGA_173101_P023_WB12 0.471154  108        276
## 335399            GTS_657-24 UGA_173101_P023_WA04 0.488636   92        276
## 335445            GTS_657-24 UGA_173101_P023_WA07 0.481481  106        276
## 335464            GTS_657-24 UGA_173101_P022_WD02 0.475000   51        276
## 335539            GTS_657-24 UGA_173101_P020_WH07 0.428571   77        276
## 335587            GTS_657-24 UGA_173101_P021_WF01 0.480000  102        276
## 335626            GTS_657-24 UGA_173101_P022_WA09 0.489583   99        276
## 335662            GTS_657-24           GTS_949-23 0.491071  120        276
## 335711            GTS_657-24           GTS_646-23 0.500000  118        276
## 335725            GTS_657-24           GTS_616-24 0.500000  119        276
## 335734            GTS_657-24           GTS_655-23 0.500000  114        276
## 335735            GTS_657-24           GTS_983-23 0.490741  119        276
## 335746            GTS_657-24           GTS_630-24 0.500000  118        276
## 335756            GTS_657-24           GTS_665-23 0.491071  120        276
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## 335759           GTS_657-24          GTS_997-23 0.491071  120        276
## 335765           GTS_657-24          GTS_644-24 0.500000  118        276
## 335771           GTS_657-24          GTS_818-23 0.491071  120        276
## 335784           GTS_657-24          GTS_677-23 0.500000  117        276
## 335785           GTS_657-24         GTS_1012-23 0.490741  119        276
## 350743           GTS_691-23 UGA_173101_P019_WE05 0.458333  112        289
## 350768           GTS_691-23 UGA_173101_P019_WA04 0.450000   89        289
## 350809           GTS_691-23 UGA_173101_P016_WC12 0.461538  107        289
## 350810           GTS_691-23 UGA_173101_P017_WC02 0.451923  114        289
## 350817           GTS_691-23 UGA_173101_P018_WB04 0.460000  115        289
## 350820           GTS_691-23 UGA_173101_P016_WF07 0.423913  100        289
## 350864           GTS_691-23 UGA_173101_P019_WG03 0.461538  118        289
## 350869           GTS_691-23 UGA_173101_P016_WE08 0.461538  114        289
## 350898           GTS_691-23 UGA_173101_P017_WC12 0.450000  114        289
## 350918           GTS_691-23 UGA_173101_P019_WA02 0.461538  116        289
## 350944           GTS_691-23 UGA_173101_P018_WD10 0.460000  107        289
## 351001           GTS_691-23 UGA_173101_P016_WH09 0.460000  112        289
## 351034           GTS_691-23 UGA_173101_P017_WD09 0.458333  106        289
## 351053           GTS_691-23 UGA_173101_P019_WA01 0.461538  117        289
## 351080           GTS_691-23 UGA_173101_P018_WB06 0.403846   53        289
## 351104           GTS_691-23 UGA_173101_P018_WC11 0.461538  118        289
## 351116           GTS_691-23 UGA_173101_P023_WF05 0.443182   91        289
## 351123           GTS_691-23 UGA_173101_P023_WB12 0.437500  107        289
## 351131           GTS_691-23 UGA_173101_P023_WA04 0.450000   90        289
## 351177           GTS_691-23 UGA_173101_P023_WA07 0.460000  105        289
## 351196           GTS_691-23 UGA_173101_P022_WD02 0.361111   51        289
## 351271           GTS_691-23 UGA_173101_P020_WH07 0.368421   76        289
## 351319           GTS_691-23 UGA_173101_P021_WF01 0.447917  101        289
## 351358           GTS_691-23 UGA_173101_P022_WA09 0.454545   98        289
## 351394           GTS_691-23          GTS_949-23 0.490385  119        289
## 351443           GTS_691-23          GTS_646-23 0.500000  116        289
## 351457           GTS_691-23          GTS_616-24 0.490000  118        289
## 351466           GTS_691-23          GTS_655-23 0.488636  113        289
## 351467           GTS_691-23          GTS_983-23 0.500000  118        289
## 351478           GTS_691-23          GTS_630-24 0.500000  117        289
## 351488           GTS_691-23          GTS_665-23 0.490385  119        289
## 351491           GTS_691-23          GTS_997-23 0.490385  119        289
## 351497           GTS_691-23          GTS_644-24 0.500000  116        289
## 351503           GTS_691-23          GTS_818-23 0.490385  119        289
## 351516           GTS_691-23          GTS_677-23 0.490000  118        289
## 351517           GTS_691-23         GTS_1012-23 0.500000  118        289
## 351523           GTS_691-23          GTS_657-24 0.490000  117        289
## 410005           GTS_725-23 UGA_173101_P019_WE05 0.500000  112         33
## 410030           GTS_725-23 UGA_173101_P019_WA04 0.500000   89         33
## 410071           GTS_725-23 UGA_173101_P016_WC12 0.500000  107         33
## 410072           GTS_725-23 UGA_173101_P017_WC02 0.490741  114         33
## 410079           GTS_725-23 UGA_173101_P018_WB04 0.500000  115         33
## 410082           GTS_725-23 UGA_173101_P016_WF07 0.468750  100         33
## 410126           GTS_725-23 UGA_173101_P019_WG03 0.500000  118         33
## 410131           GTS_725-23 UGA_173101_P016_WE08 0.500000  114         33
## 410160           GTS_725-23 UGA_173101_P017_WC12 0.490385  114         33
## 410180           GTS_725-23 UGA_173101_P019_WA02 0.500000  116         33
## 410206           GTS_725-23 UGA_173101_P018_WD10 0.500000  107         33
## 410263           GTS_725-23 UGA_173101_P016_WH09 0.500000  112         33
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## 410296             GTS_725-23 UGA_173101_P017_WD09 0.500000  106         33
## 410315             GTS_725-23 UGA_173101_P019_WA01 0.500000  117         33
## 410342             GTS_725-23 UGA_173101_P018_WB06 0.483333   54         33
## 410366             GTS_725-23 UGA_173101_P018_WC11 0.500000  118         33
## 410378             GTS_725-23 UGA_173101_P023_WF05 0.489130   91         33
## 410385             GTS_725-23 UGA_173101_P023_WB12 0.480000  107         33
## 410393             GTS_725-23 UGA_173101_P023_WA04 0.500000   90         33
## 410439             GTS_725-23 UGA_173101_P023_WA07 0.500000  105         33
## 410458             GTS_725-23 UGA_173101_P022_WD02 0.475000   51         33
## 410533             GTS_725-23 UGA_173101_P020_WH07 0.428571   77         33
## 410581             GTS_725-23 UGA_173101_P021_WF01 0.490000  101         33
## 410620             GTS_725-23 UGA_173101_P022_WA09 0.500000   98         33
## 410656             GTS_725-23           GTS_949-23 0.500000  119         33
## 410705             GTS_725-23           GTS_646-23 0.500000  118         33
## 410719             GTS_725-23           GTS_616-24 0.500000  119         33
## 410728             GTS_725-23           GTS_655-23 0.500000  114         33
## 410729             GTS_725-23           GTS_983-23 0.500000  118         33
## 410740             GTS_725-23           GTS_630-24 0.500000  119         33
## 410750             GTS_725-23           GTS_665-23 0.500000  119         33
## 410753             GTS_725-23           GTS_997-23 0.500000  119         33
## 410759             GTS_725-23           GTS_644-24 0.500000  118         33
## 410765             GTS_725-23           GTS_818-23 0.500000  119         33
## 410778             GTS_725-23           GTS_677-23 0.500000  118         33
## 410779             GTS_725-23          GTS_1012-23 0.500000  118         33
## 410785             GTS_725-23           GTS_657-24 0.500000  118         33
## 410804             GTS_725-23           GTS_691-23 0.500000  117         33
## 418195             GTS_736-23 UGA_173101_P019_WE05 0.470000  113        276
## 418220             GTS_736-23 UGA_173101_P019_WA04 0.464286   91        276
## 418261             GTS_736-23 UGA_173101_P016_WC12 0.472222  108        276
## 418262             GTS_736-23 UGA_173101_P017_WC02 0.462963  115        276
## 418269             GTS_736-23 UGA_173101_P018_WB04 0.471154  116        276
## 418272             GTS_736-23 UGA_173101_P016_WF07 0.437500  101        276
## 418316             GTS_736-23 UGA_173101_P019_WG03 0.472222  119        276
## 418321             GTS_736-23 UGA_173101_P016_WE08 0.472222  115        276
## 418350             GTS_736-23 UGA_173101_P017_WC12 0.462963  115        276
## 418370             GTS_736-23 UGA_173101_P019_WA02 0.472222  117        276
## 418396             GTS_736-23 UGA_173101_P018_WD10 0.471154  108        276
## 418453             GTS_736-23 UGA_173101_P016_WH09 0.471154  113        276
## 418486             GTS_736-23 UGA_173101_P017_WD09 0.470000  107        276
## 418505             GTS_736-23 UGA_173101_P019_WA01 0.472222  118        276
## 418532             GTS_736-23 UGA_173101_P018_WB06 0.450000   55        276
## 418556             GTS_736-23 UGA_173101_P018_WC11 0.472222  119        276
## 418568             GTS_736-23 UGA_173101_P023_WF05 0.456522   93        276
## 418575             GTS_736-23 UGA_173101_P023_WB12 0.450000  108        276
## 418583             GTS_736-23 UGA_173101_P023_WA04 0.464286   92        276
## 418629             GTS_736-23 UGA_173101_P023_WA07 0.442308  106        276
## 418648             GTS_736-23 UGA_173101_P022_WD02 0.444444   51        276
## 418723             GTS_736-23 UGA_173101_P020_WH07 0.412500   77        276
## 418771             GTS_736-23 UGA_173101_P021_WF01 0.470000  102        276
## 418810             GTS_736-23 UGA_173101_P022_WA09 0.467391   99        276
## 418846             GTS_736-23           GTS_949-23 0.481481  120        276
## 418895             GTS_736-23           GTS_646-23 0.490000  117        276
## 418909             GTS_736-23           GTS_616-24 0.490741  119        276
## 418918             GTS_736-23           GTS_655-23 0.489583  114        276
```

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## 418919         GTS_736-23          GTS_983-23 0.480769  119        276
## 418930         GTS_736-23          GTS_630-24 0.500000  117        276
## 418940         GTS_736-23          GTS_665-23 0.481481  120        276
## 418943         GTS_736-23          GTS_997-23 0.481481  120        276
## 418949         GTS_736-23          GTS_644-24 0.500000  117        276
## 418955         GTS_736-23          GTS_818-23 0.481481  120        276
## 418968         GTS_736-23          GTS_677-23 0.500000  118        276
## 418969         GTS_736-23         GTS_1012-23 0.480769  119        276
## 418975         GTS_736-23          GTS_657-24 0.490385  119        276
## 418994         GTS_736-23          GTS_691-23 0.480769  118        276
## 419062         GTS_736-23          GTS_725-23 0.500000  117        276
## 518711        GTS_1202-24 UGA_173101_P019_WE05 0.451923  113        289
## 518736        GTS_1202-24 UGA_173101_P019_WA04 0.443182   92        289
## 518777        GTS_1202-24 UGA_173101_P016_WC12 0.455357  109        289
## 518778        GTS_1202-24 UGA_173101_P017_WC02 0.446429  116        289
## 518785        GTS_1202-24 UGA_173101_P018_WB04 0.453704  117        289
## 518788        GTS_1202-24 UGA_173101_P016_WF07 0.420000  102        289
## 518832        GTS_1202-24 UGA_173101_P019_WG03 0.455357  120        289
## 518837        GTS_1202-24 UGA_173101_P016_WE08 0.455357  116        289
## 518866        GTS_1202-24 UGA_173101_P017_WC12 0.446429  116        289
## 518886        GTS_1202-24 UGA_173101_P019_WA02 0.455357  118        289
## 518912        GTS_1202-24 UGA_173101_P018_WD10 0.453704  110        289
## 518969        GTS_1202-24 UGA_173101_P016_WH09 0.453704  114        289
## 519002        GTS_1202-24 UGA_173101_P017_WD09 0.451923  108        289
## 519021        GTS_1202-24 UGA_173101_P019_WA01 0.455357  119        289
## 519048        GTS_1202-24 UGA_173101_P018_WB06 0.406250   56        289
## 519072        GTS_1202-24 UGA_173101_P018_WC11 0.455357  120        289
## 519084        GTS_1202-24 UGA_173101_P023_WF05 0.437500   93        289
## 519091        GTS_1202-24 UGA_173101_P023_WB12 0.432692  110        289
## 519099        GTS_1202-24 UGA_173101_P023_WA04 0.443182   92        289
## 519145        GTS_1202-24 UGA_173101_P023_WA07 0.444444  107        289
## 519164        GTS_1202-24 UGA_173101_P022_WD02 0.350000   51        289
## 519239        GTS_1202-24 UGA_173101_P020_WH07 0.362500   77        289
## 519287        GTS_1202-24 UGA_173101_P021_WF01 0.440000  103        289
## 519326        GTS_1202-24 UGA_173101_P022_WA09 0.447917  101        289
## 519362        GTS_1202-24          GTS_949-23 0.482759  122        289
## 519411        GTS_1202-24          GTS_646-23 0.490385  118        289
## 519425        GTS_1202-24          GTS_616-24 0.473214  120        289
## 519434        GTS_1202-24          GTS_655-23 0.479167  114        289
## 519435        GTS_1202-24          GTS_983-23 0.500000  119        289
## 519446        GTS_1202-24          GTS_630-24 0.490385  117        289
## 519456        GTS_1202-24          GTS_665-23 0.482143  120        289
## 519459        GTS_1202-24          GTS_997-23 0.482143  121        289
## 519465        GTS_1202-24          GTS_644-24 0.481481  117        289
## 519471        GTS_1202-24          GTS_818-23 0.482143  121        289
## 519484        GTS_1202-24          GTS_677-23 0.490385  117        289
## 519485        GTS_1202-24         GTS_1012-23 0.500000  119        289
## 519491        GTS_1202-24          GTS_657-24 0.482143  118        289
## 519510        GTS_1202-24          GTS_691-23 0.500000  117        289
## 519578        GTS_1202-24          GTS_725-23 0.490385  117        289
## 519587        GTS_1202-24          GTS_736-23 0.472222  118        289
## 537206         GTS_850-23 UGA_173101_P019_WE05 0.481481  115        301
## 537231         GTS_850-23 UGA_173101_P019_WA04 0.478261   92        301
## 537272         GTS_850-23 UGA_173101_P016_WC12 0.482759  110        301
```

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## 537273            GTS_850-23 UGA_173101_P017_WC02 0.474138  117        301
## 537280            GTS_850-23 UGA_173101_P018_WB04 0.482143  118        301
## 537283            GTS_850-23 UGA_173101_P016_WF07 0.451923  103        301
## 537327            GTS_850-23 UGA_173101_P019_WG03 0.482759  122        301
## 537332            GTS_850-23 UGA_173101_P016_WE08 0.482759  117        301
## 537361            GTS_850-23 UGA_173101_P017_WC12 0.473214  117        301
## 537381            GTS_850-23 UGA_173101_P019_WA02 0.482759  119        301
## 537407            GTS_850-23 UGA_173101_P018_WD10 0.482143  111        301
## 537464            GTS_850-23 UGA_173101_P016_WH09 0.482143  115        301
## 537497            GTS_850-23 UGA_173101_P017_WD09 0.481481  109        301
## 537516            GTS_850-23 UGA_173101_P019_WA01 0.482759  121        301
## 537543            GTS_850-23 UGA_173101_P018_WB06 0.453125   56        301
## 537567            GTS_850-23 UGA_173101_P018_WC11 0.482759  122        301
## 537579            GTS_850-23 UGA_173101_P023_WF05 0.470000   94        301
## 537586            GTS_850-23 UGA_173101_P023_WB12 0.462963  111        301
## 537594            GTS_850-23 UGA_173101_P023_WA04 0.478261   93        301
## 537640            GTS_850-23 UGA_173101_P023_WA07 0.472222  108        301
## 537659            GTS_850-23 UGA_173101_P022_WD02 0.425000   52        301
## 537734            GTS_850-23 UGA_173101_P020_WH07 0.404762   78        301
## 537782            GTS_850-23 UGA_173101_P021_WF01 0.471154  104        301
## 537821            GTS_850-23 UGA_173101_P022_WA09 0.480000  102        301
## 537857            GTS_850-23           GTS_949-23 0.491667  124        301
## 537906            GTS_850-23           GTS_646-23 0.490385  120        301
## 537920            GTS_850-23           GTS_616-24 0.482759  122        301
## 537929            GTS_850-23           GTS_655-23 0.490000  116        301
## 537930            GTS_850-23           GTS_983-23 0.500000  121        301
## 537941            GTS_850-23           GTS_630-24 0.490385  119        301
## 537951            GTS_850-23           GTS_665-23 0.491379  122        301
## 537954            GTS_850-23           GTS_997-23 0.491379  123        301
## 537960            GTS_850-23           GTS_644-24 0.481481  119        301
## 537966            GTS_850-23           GTS_818-23 0.491379  123        301
## 537979            GTS_850-23           GTS_677-23 0.500000  119        301
## 537980            GTS_850-23          GTS_1012-23 0.500000  121        301
## 537986            GTS_850-23           GTS_657-24 0.482143  120        301
## 538005            GTS_850-23           GTS_691-23 0.490385  119        301
## 538073            GTS_850-23           GTS_725-23 0.490385  119        301
## 538082            GTS_850-23           GTS_736-23 0.481481  120        301
## 538186            GTS_850-23          GTS_1202-24 0.491379  122        301
## 573025            GTS_825-24 UGA_173101_P019_WE05 0.480000  112        301
## 573050            GTS_825-24 UGA_173101_P019_WA04 0.476190   90        301
## 573091            GTS_825-24 UGA_173101_P016_WC12 0.481481  107        301
## 573092            GTS_825-24 UGA_173101_P017_WC02 0.472222  114        301
## 573099            GTS_825-24 UGA_173101_P018_WB04 0.480769  115        301
## 573102            GTS_825-24 UGA_173101_P016_WF07 0.447917  100        301
## 573146            GTS_825-24 UGA_173101_P019_WG03 0.481481  118        301
## 573151            GTS_825-24 UGA_173101_P016_WE08 0.481481  114        301
## 573180            GTS_825-24 UGA_173101_P017_WC12 0.471154  114        301
## 573200            GTS_825-24 UGA_173101_P019_WA02 0.481481  116        301
## 573226            GTS_825-24 UGA_173101_P018_WD10 0.480769  107        301
## 573283            GTS_825-24 UGA_173101_P016_WH09 0.480769  112        301
## 573316            GTS_825-24 UGA_173101_P017_WD09 0.480000  106        301
## 573335            GTS_825-24 UGA_173101_P019_WA01 0.481481  117        301
## 573362            GTS_825-24 UGA_173101_P018_WB06 0.450000   55        301
## 573386            GTS_825-24 UGA_173101_P018_WC11 0.481481  118        301
```

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## 573398            GTS_825-24 UGA_173101_P023_WF05 0.467391   92         301
## 573405            GTS_825-24 UGA_173101_P023_WB12 0.460000  107         301
## 573413            GTS_825-24 UGA_173101_P023_WA04 0.476190   91         301
## 573459            GTS_825-24 UGA_173101_P023_WA07 0.480769  105         301
## 573478            GTS_825-24 UGA_173101_P022_WD02 0.425000   52         301
## 573553            GTS_825-24 UGA_173101_P020_WH07 0.404762   78         301
## 573601            GTS_825-24 UGA_173101_P021_WF01 0.470000  102         301
## 573640            GTS_825-24 UGA_173101_P022_WA09 0.478261   98         301
## 573676            GTS_825-24           GTS_949-23 0.491071  120         301
## 573725            GTS_825-24           GTS_646-23 0.490000  117         301
## 573739            GTS_825-24           GTS_616-24 0.482143  120         301
## 573748            GTS_825-24           GTS_655-23 0.489583  114         301
## 573749            GTS_825-24           GTS_983-23 0.500000  117         301
## 573760            GTS_825-24           GTS_630-24 0.490385  118         301
## 573770            GTS_825-24           GTS_665-23 0.490741  119         301
## 573773            GTS_825-24           GTS_997-23 0.490741  119         301
## 573779            GTS_825-24           GTS_644-24 0.481481  119         301
## 573785            GTS_825-24           GTS_818-23 0.490741  119         301
## 573798            GTS_825-24           GTS_677-23 0.500000  118         301
## 573799            GTS_825-24          GTS_1012-23 0.500000  117         301
## 573805            GTS_825-24           GTS_657-24 0.490385  118         301
## 573824            GTS_825-24           GTS_691-23 0.490000  117         301
## 573892            GTS_825-24           GTS_725-23 0.490385  118         301
## 573901            GTS_825-24           GTS_736-23 0.500000  118         301
## 574005            GTS_825-24          GTS_1202-24 0.490741  118         301
## 574023            GTS_825-24           GTS_850-23 0.500000  120         301
## 587026            GTS_935-23 UGA_173101_P019_WE05 0.500000  114         255
## 587051            GTS_935-23 UGA_173101_P019_WA04 0.500000   91         255
## 587092            GTS_935-23 UGA_173101_P016_WC12 0.500000  109         255
## 587093            GTS_935-23 UGA_173101_P017_WC02 0.491379  116         255
## 587100            GTS_935-23 UGA_173101_P018_WB04 0.500000  117         255
## 587103            GTS_935-23 UGA_173101_P016_WF07 0.471154  102         255
## 587147            GTS_935-23 UGA_173101_P019_WG03 0.500000  121         255
## 587152            GTS_935-23 UGA_173101_P016_WE08 0.500000  116         255
## 587181            GTS_935-23 UGA_173101_P017_WC12 0.491071  116         255
## 587201            GTS_935-23 UGA_173101_P019_WA02 0.500000  118         255
## 587227            GTS_935-23 UGA_173101_P018_WD10 0.500000  110         255
## 587284            GTS_935-23 UGA_173101_P016_WH09 0.500000  114         255
## 587317            GTS_935-23 UGA_173101_P017_WD09 0.500000  108         255
## 587336            GTS_935-23 UGA_173101_P019_WA01 0.500000  120         255
## 587363            GTS_935-23 UGA_173101_P018_WB06 0.484375   55         255
## 587387            GTS_935-23 UGA_173101_P018_WC11 0.500000  121         255
## 587399            GTS_935-23 UGA_173101_P023_WF05 0.490000   93         255
## 587406            GTS_935-23 UGA_173101_P023_WB12 0.481481  110         255
## 587414            GTS_935-23 UGA_173101_P023_WA04 0.500000   92         255
## 587460            GTS_935-23 UGA_173101_P023_WA07 0.490741  107         255
## 587479            GTS_935-23 UGA_173101_P022_WD02 0.475000   51         255
## 587554            GTS_935-23 UGA_173101_P020_WH07 0.428571   77         255
## 587602            GTS_935-23 UGA_173101_P021_WF01 0.490385  103         255
## 587641            GTS_935-23 UGA_173101_P022_WA09 0.500000  101         255
## 587677            GTS_935-23           GTS_949-23 0.500000  123         255
## 587726            GTS_935-23           GTS_646-23 0.500000  120         255
## 587740            GTS_935-23           GTS_616-24 0.491071  121         255
## 587749            GTS_935-23           GTS_655-23 0.500000  115         255
```

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## 587750           GTS_935-23          GTS_983-23 0.500000  121         255
## 587761           GTS_935-23          GTS_630-24 0.500000  119         255
## 587771           GTS_935-23          GTS_665-23 0.500000  121         255
## 587774           GTS_935-23          GTS_997-23 0.500000  122         255
## 587780           GTS_935-23          GTS_644-24 0.490741  119         255
## 587786           GTS_935-23          GTS_818-23 0.500000  122         255
## 587799           GTS_935-23          GTS_677-23 0.500000  118         255
## 587800           GTS_935-23         GTS_1012-23 0.500000  121         255
## 587806           GTS_935-23          GTS_657-24 0.491071  120         255
## 587825           GTS_935-23          GTS_691-23 0.500000  118         255
## 587893           GTS_935-23          GTS_725-23 0.500000  119         255
## 587902           GTS_935-23          GTS_736-23 0.480769  119         255
## 588006           GTS_935-23         GTS_1202-24 0.491379  121         255
## 588024           GTS_935-23          GTS_850-23 0.491379  123         255
## 588058           GTS_935-23          GTS_825-24 0.490741  119         255
## 597911          GTS_935-23R UGA_173101_P019_WE05 0.461538  114         289
## 597936          GTS_935-23R UGA_173101_P019_WA04 0.454545   91         289
## 597977          GTS_935-23R UGA_173101_P016_WC12 0.464286  109         289
## 597978          GTS_935-23R UGA_173101_P017_WC02 0.455357  116         289
## 597985          GTS_935-23R UGA_173101_P018_WB04 0.462963  117         289
## 597988          GTS_935-23R UGA_173101_P016_WF07 0.430000  102         289
## 598032          GTS_935-23R UGA_173101_P019_WG03 0.464286  121         289
## 598037          GTS_935-23R UGA_173101_P016_WE08 0.464286  116         289
## 598066          GTS_935-23R UGA_173101_P017_WC12 0.453704  116         289
## 598086          GTS_935-23R UGA_173101_P019_WA02 0.464286  118         289
## 598112          GTS_935-23R UGA_173101_P018_WD10 0.462963  110         289
## 598169          GTS_935-23R UGA_173101_P016_WH09 0.462963  114         289
## 598202          GTS_935-23R UGA_173101_P017_WD09 0.461538  108         289
## 598221          GTS_935-23R UGA_173101_P019_WA01 0.464286  120         289
## 598248          GTS_935-23R UGA_173101_P018_WB06 0.416667   55         289
## 598272          GTS_935-23R UGA_173101_P018_WC11 0.464286  121         289
## 598284          GTS_935-23R UGA_173101_P023_WF05 0.447917   93         289
## 598291          GTS_935-23R UGA_173101_P023_WB12 0.442308  110         289
## 598299          GTS_935-23R UGA_173101_P023_WA04 0.454545   92         289
## 598345          GTS_935-23R UGA_173101_P023_WA07 0.451923  107         289
## 598364          GTS_935-23R UGA_173101_P022_WD02 0.361111   51         289
## 598439          GTS_935-23R UGA_173101_P020_WH07 0.375000   77         289
## 598487          GTS_935-23R UGA_173101_P021_WF01 0.450000  103         289
## 598526          GTS_935-23R UGA_173101_P022_WA09 0.458333  101         289
## 598562          GTS_935-23R          GTS_949-23 0.491379  123         289
## 598611          GTS_935-23R          GTS_646-23 0.500000  119         289
## 598625          GTS_935-23R          GTS_616-24 0.482143  121         289
## 598634          GTS_935-23R          GTS_655-23 0.489583  115         289
## 598635          GTS_935-23R          GTS_983-23 0.500000  121         289
## 598646          GTS_935-23R          GTS_630-24 0.500000  118         289
## 598656          GTS_935-23R          GTS_665-23 0.491071  121         289
## 598659          GTS_935-23R          GTS_997-23 0.491071  122         289
## 598665          GTS_935-23R          GTS_644-24 0.490385  118         289
## 598671          GTS_935-23R          GTS_818-23 0.491071  122         289
## 598684          GTS_935-23R          GTS_677-23 0.490385  119         289
## 598685          GTS_935-23R         GTS_1012-23 0.500000  121         289
## 598691          GTS_935-23R          GTS_657-24 0.490741  119         289
## 598710          GTS_935-23R          GTS_691-23 0.500000  119         289
## 598778          GTS_935-23R          GTS_725-23 0.500000  118         289
```

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## 598787            GTS_935-23R          GTS_736-23 0.472222  120         289
## 598891            GTS_935-23R         GTS_1202-24 0.500000  121         289
## 598909            GTS_935-23R          GTS_850-23 0.491379  123         289
## 598943            GTS_935-23R          GTS_825-24 0.490741  119         289
## 598956            GTS_935-23R          GTS_935-23 0.500000  122         289
## 617756             GTS_855-24 UGA_173101_P019_WE05 0.500000  114         255
## 617781             GTS_855-24 UGA_173101_P019_WA04 0.500000   91         255
## 617822             GTS_855-24 UGA_173101_P016_WC12 0.500000  109         255
## 617823             GTS_855-24 UGA_173101_P017_WC02 0.491379  116         255
## 617830             GTS_855-24 UGA_173101_P018_WB04 0.500000  117         255
## 617833             GTS_855-24 UGA_173101_P016_WF07 0.471154  102         255
## 617877             GTS_855-24 UGA_173101_P019_WG03 0.500000  121         255
## 617882             GTS_855-24 UGA_173101_P016_WE08 0.500000  116         255
## 617911             GTS_855-24 UGA_173101_P017_WC12 0.491071  116         255
## 617931             GTS_855-24 UGA_173101_P019_WA02 0.500000  118         255
## 617957             GTS_855-24 UGA_173101_P018_WD10 0.500000  110         255
## 618014             GTS_855-24 UGA_173101_P016_WH09 0.500000  114         255
## 618047             GTS_855-24 UGA_173101_P017_WD09 0.500000  108         255
## 618066             GTS_855-24 UGA_173101_P019_WA01 0.500000  120         255
## 618093             GTS_855-24 UGA_173101_P018_WB06 0.484375   55         255
## 618117             GTS_855-24 UGA_173101_P018_WC11 0.500000  121         255
## 618129             GTS_855-24 UGA_173101_P023_WF05 0.490000   93         255
## 618136             GTS_855-24 UGA_173101_P023_WB12 0.481481  110         255
## 618144             GTS_855-24 UGA_173101_P023_WA04 0.500000   92         255
## 618190             GTS_855-24 UGA_173101_P023_WA07 0.490741  107         255
## 618209             GTS_855-24 UGA_173101_P022_WD02 0.475000   51         255
## 618284             GTS_855-24 UGA_173101_P020_WH07 0.428571   77         255
## 618332             GTS_855-24 UGA_173101_P021_WF01 0.490385  103         255
## 618371             GTS_855-24 UGA_173101_P022_WA09 0.500000  101         255
## 618407             GTS_855-24           GTS_949-23 0.500000  123         255
## 618456             GTS_855-24           GTS_646-23 0.500000  120         255
## 618470             GTS_855-24           GTS_616-24 0.491071  121         255
## 618479             GTS_855-24           GTS_655-23 0.500000  115         255
## 618480             GTS_855-24           GTS_983-23 0.500000  121         255
## 618491             GTS_855-24           GTS_630-24 0.500000  119         255
## 618501             GTS_855-24           GTS_665-23 0.500000  121         255
## 618504             GTS_855-24           GTS_997-23 0.500000  122         255
## 618510             GTS_855-24           GTS_644-24 0.490741  119         255
## 618516             GTS_855-24           GTS_818-23 0.500000  122         255
## 618529             GTS_855-24           GTS_677-23 0.500000  118         255
## 618530             GTS_855-24          GTS_1012-23 0.500000  121         255
## 618536             GTS_855-24           GTS_657-24 0.491071  120         255
## 618555             GTS_855-24           GTS_691-23 0.500000  118         255
## 618623             GTS_855-24           GTS_725-23 0.500000  119         255
## 618632             GTS_855-24           GTS_736-23 0.480769  119         255
## 618736             GTS_855-24          GTS_1202-24 0.491379  121         255
## 618754             GTS_855-24           GTS_850-23 0.491379  123         255
## 618788             GTS_855-24           GTS_825-24 0.490741  119         255
## 618801             GTS_855-24           GTS_935-23 0.500000  123         255
## 618811             GTS_855-24          GTS_935-23R 0.500000  122         255
## 631166            GTS_1344-24 UGA_173101_P019_WE05 0.500000  115         255
## 631191            GTS_1344-24 UGA_173101_P019_WA04 0.500000   92         255
## 631232            GTS_1344-24 UGA_173101_P016_WC12 0.500000  110         255
## 631233            GTS_1344-24 UGA_173101_P017_WC02 0.491379  117         255
```

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## 631240          GTS_1344-24 UGA_173101_P018_WB04 0.500000  118        255
## 631243          GTS_1344-24 UGA_173101_P016_WF07 0.471154  103        255
## 631287          GTS_1344-24 UGA_173101_P019_WG03 0.500000  122        255
## 631292          GTS_1344-24 UGA_173101_P016_WE08 0.500000  117        255
## 631321          GTS_1344-24 UGA_173101_P017_WC12 0.491071  117        255
## 631341          GTS_1344-24 UGA_173101_P019_WA02 0.500000  119        255
## 631367          GTS_1344-24 UGA_173101_P018_WD10 0.500000  111        255
## 631424          GTS_1344-24 UGA_173101_P016_WH09 0.500000  115        255
## 631457          GTS_1344-24 UGA_173101_P017_WD09 0.500000  109        255
## 631476          GTS_1344-24 UGA_173101_P019_WA01 0.500000  121        255
## 631503          GTS_1344-24 UGA_173101_P018_WB06 0.484375   55        255
## 631527          GTS_1344-24 UGA_173101_P018_WC11 0.500000  122        255
## 631539          GTS_1344-24 UGA_173101_P023_WF05 0.490000   94        255
## 631546          GTS_1344-24 UGA_173101_P023_WB12 0.481481  111        255
## 631554          GTS_1344-24 UGA_173101_P023_WA04 0.500000   93        255
## 631600          GTS_1344-24 UGA_173101_P023_WA07 0.490741  108        255
## 631619          GTS_1344-24 UGA_173101_P022_WD02 0.475000   51        255
## 631694          GTS_1344-24 UGA_173101_P020_WH07 0.428571   77        255
## 631742          GTS_1344-24 UGA_173101_P021_WF01 0.490385  104        255
## 631781          GTS_1344-24 UGA_173101_P022_WA09 0.500000  102        255
## 631817          GTS_1344-24           GTS_949-23 0.500000  123        255
## 631866          GTS_1344-24           GTS_646-23 0.500000  120        255
## 631880          GTS_1344-24           GTS_616-24 0.491071  121        255
## 631889          GTS_1344-24           GTS_655-23 0.500000  115        255
## 631890          GTS_1344-24           GTS_983-23 0.500000  121        255
## 631901          GTS_1344-24           GTS_630-24 0.500000  119        255
## 631911          GTS_1344-24           GTS_665-23 0.500000  121        255
## 631914          GTS_1344-24           GTS_997-23 0.500000  122        255
## 631920          GTS_1344-24           GTS_644-24 0.490741  119        255
## 631926          GTS_1344-24           GTS_818-23 0.500000  122        255
## 631939          GTS_1344-24           GTS_677-23 0.500000  118        255
## 631940          GTS_1344-24          GTS_1012-23 0.500000  121        255
## 631946          GTS_1344-24           GTS_657-24 0.491071  120        255
## 631965          GTS_1344-24           GTS_691-23 0.500000  118        255
## 632033          GTS_1344-24           GTS_725-23 0.500000  119        255
## 632042          GTS_1344-24           GTS_736-23 0.480769  119        255
## 632146          GTS_1344-24          GTS_1202-24 0.491379  121        255
## 632164          GTS_1344-24           GTS_850-23 0.491379  123        255
## 632198          GTS_1344-24           GTS_825-24 0.490741  119        255
## 632211          GTS_1344-24           GTS_935-23 0.500000  123        255
## 632221          GTS_1344-24          GTS_935-23R 0.500000  122        255
## 632239          GTS_1344-24           GTS_855-24 0.500000  123        255
## 673420           GTS_886-24 UGA_173101_P019_WE05 0.490741  115        255
## 673445           GTS_886-24 UGA_173101_P019_WA04 0.489130   92        255
## 673486           GTS_886-24 UGA_173101_P016_WC12 0.491379  110        255
## 673487           GTS_886-24 UGA_173101_P017_WC02 0.483333  117        255
## 673494           GTS_886-24 UGA_173101_P018_WB04 0.491071  118        255
## 673497           GTS_886-24 UGA_173101_P016_WF07 0.462963  103        255
## 673541           GTS_886-24 UGA_173101_P019_WG03 0.491379  122        255
## 673546           GTS_886-24 UGA_173101_P016_WE08 0.491379  117        255
## 673575           GTS_886-24 UGA_173101_P017_WC12 0.482143  117        255
## 673595           GTS_886-24 UGA_173101_P019_WA02 0.491379  119        255
## 673621           GTS_886-24 UGA_173101_P018_WD10 0.491071  111        255
## 673678           GTS_886-24 UGA_173101_P016_WH09 0.491071  115        255
```

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## 673711             GTS_886-24 UGA_173101_P017_WD09 0.490741  109        255
## 673730             GTS_886-24 UGA_173101_P019_WA01 0.491379  121        255
## 673757             GTS_886-24 UGA_173101_P018_WB06 0.468750   56        255
## 673781             GTS_886-24 UGA_173101_P018_WC11 0.491379  122        255
## 673793             GTS_886-24 UGA_173101_P023_WF05 0.480769   94        255
## 673800             GTS_886-24 UGA_173101_P023_WB12 0.473214  111        255
## 673808             GTS_886-24 UGA_173101_P023_WA04 0.489130   93        255
## 673854             GTS_886-24 UGA_173101_P023_WA07 0.481481  108        255
## 673873             GTS_886-24 UGA_173101_P022_WD02 0.450000   52        255
## 673948             GTS_886-24 UGA_173101_P020_WH07 0.416667   78        255
## 673996             GTS_886-24 UGA_173101_P021_WF01 0.481481  104        255
## 674035             GTS_886-24 UGA_173101_P022_WA09 0.490000  102        255
## 674071             GTS_886-24           GTS_949-23 0.500000  124        255
## 674120             GTS_886-24           GTS_646-23 0.500000  120        255
## 674134             GTS_886-24           GTS_616-24 0.491379  122        255
## 674143             GTS_886-24           GTS_655-23 0.500000  116        255
## 674144             GTS_886-24           GTS_983-23 0.500000  121        255
## 674155             GTS_886-24           GTS_630-24 0.500000  119        255
## 674165             GTS_886-24           GTS_665-23 0.500000  122        255
## 674168             GTS_886-24           GTS_997-23 0.500000  123        255
## 674174             GTS_886-24           GTS_644-24 0.490741  119        255
## 674180             GTS_886-24           GTS_818-23 0.500000  123        255
## 674193             GTS_886-24           GTS_677-23 0.500000  119        255
## 674194             GTS_886-24          GTS_1012-23 0.500000  121        255
## 674200             GTS_886-24           GTS_657-24 0.491071  120        255
## 674219             GTS_886-24           GTS_691-23 0.490385  119        255
## 674287             GTS_886-24           GTS_725-23 0.500000  119        255
## 674296             GTS_886-24           GTS_736-23 0.481481  120        255
## 674400             GTS_886-24          GTS_1202-24 0.482759  122        255
## 674418             GTS_886-24           GTS_850-23 0.491667  124        255
## 674452             GTS_886-24           GTS_825-24 0.491071  120        255
## 674465             GTS_886-24           GTS_935-23 0.500000  123        255
## 674475             GTS_886-24          GTS_935-23R 0.491379  123        255
## 674493             GTS_886-24           GTS_855-24 0.500000  123        255
## 674505             GTS_886-24          GTS_1344-24 0.500000  123        255
## 689765            GTS_1403-24 UGA_173101_P019_WE05 0.500000  113        255
## 689790            GTS_1403-24 UGA_173101_P019_WA04 0.500000   90        255
## 689831            GTS_1403-24 UGA_173101_P016_WC12 0.500000  108        255
## 689832            GTS_1403-24 UGA_173101_P017_WC02 0.491071  115        255
## 689839            GTS_1403-24 UGA_173101_P018_WB04 0.500000  116        255
## 689842            GTS_1403-24 UGA_173101_P016_WF07 0.470000  101        255
## 689886            GTS_1403-24 UGA_173101_P019_WG03 0.500000  120        255
## 689891            GTS_1403-24 UGA_173101_P016_WE08 0.500000  115        255
## 689920            GTS_1403-24 UGA_173101_P017_WC12 0.490741  115        255
## 689940            GTS_1403-24 UGA_173101_P019_WA02 0.500000  117        255
## 689966            GTS_1403-24 UGA_173101_P018_WD10 0.500000  109        255
## 690023            GTS_1403-24 UGA_173101_P016_WH09 0.500000  113        255
## 690056            GTS_1403-24 UGA_173101_P017_WD09 0.500000  107        255
## 690075            GTS_1403-24 UGA_173101_P019_WA01 0.500000  119        255
## 690102            GTS_1403-24 UGA_173101_P018_WB06 0.483333   54        255
## 690126            GTS_1403-24 UGA_173101_P018_WC11 0.500000  120        255
## 690138            GTS_1403-24 UGA_173101_P023_WF05 0.489583   92        255
## 690145            GTS_1403-24 UGA_173101_P023_WB12 0.480769  109        255
## 690153            GTS_1403-24 UGA_173101_P023_WA04 0.500000   91        255
```

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## 690199           GTS_1403-24 UGA_173101_P023_WA07 0.490385  106        255
## 690218           GTS_1403-24 UGA_173101_P022_WD02 0.472222   50        255
## 690293           GTS_1403-24 UGA_173101_P020_WH07 0.425000   76        255
## 690341           GTS_1403-24 UGA_173101_P021_WF01 0.490000  102        255
## 690380           GTS_1403-24 UGA_173101_P022_WA09 0.500000  100        255
## 690416           GTS_1403-24           GTS_949-23 0.500000  122        255
## 690465           GTS_1403-24           GTS_646-23 0.500000  119        255
## 690479           GTS_1403-24           GTS_616-24 0.490741  120        255
## 690488           GTS_1403-24           GTS_655-23 0.500000  114        255
## 690489           GTS_1403-24           GTS_983-23 0.500000  121        255
## 690500           GTS_1403-24           GTS_630-24 0.500000  118        255
## 690510           GTS_1403-24           GTS_665-23 0.500000  120        255
## 690513           GTS_1403-24           GTS_997-23 0.500000  121        255
## 690519           GTS_1403-24           GTS_644-24 0.490385  118        255
## 690525           GTS_1403-24           GTS_818-23 0.500000  121        255
## 690538           GTS_1403-24           GTS_677-23 0.500000  118        255
## 690539           GTS_1403-24          GTS_1012-23 0.500000  121        255
## 690545           GTS_1403-24           GTS_657-24 0.490741  119        255
## 690564           GTS_1403-24           GTS_691-23 0.500000  118        255
## 690632           GTS_1403-24           GTS_725-23 0.500000  118        255
## 690641           GTS_1403-24           GTS_736-23 0.480769  119        255
## 690745           GTS_1403-24          GTS_1202-24 0.500000  120        255
## 690763           GTS_1403-24           GTS_850-23 0.500000  122        255
## 690797           GTS_1403-24           GTS_825-24 0.500000  118        255
## 690810           GTS_1403-24           GTS_935-23 0.500000  122        255
## 690820           GTS_1403-24          GTS_935-23R 0.500000  122        255
## 690838           GTS_1403-24           GTS_855-24 0.500000  122        255
## 690850           GTS_1403-24          GTS_1344-24 0.500000  122        255
## 690887           GTS_1403-24           GTS_886-24 0.500000  122        255
## 709876            GTS_931-24 UGA_173101_P019_WE05 0.500000  111        255
## 709901            GTS_931-24 UGA_173101_P019_WA04 0.500000   89        255
## 709942            GTS_931-24 UGA_173101_P016_WC12 0.500000  106        255
## 709943            GTS_931-24 UGA_173101_P017_WC02 0.490741  113        255
## 709950            GTS_931-24 UGA_173101_P018_WB04 0.500000  114        255
## 709953            GTS_931-24 UGA_173101_P016_WF07 0.468750   99        255
## 709997            GTS_931-24 UGA_173101_P019_WG03 0.500000  117        255
## 710002            GTS_931-24 UGA_173101_P016_WE08 0.500000  113        255
## 710031            GTS_931-24 UGA_173101_P017_WC12 0.490385  113        255
## 710051            GTS_931-24 UGA_173101_P019_WA02 0.500000  115        255
## 710077            GTS_931-24 UGA_173101_P018_WD10 0.500000  106        255
## 710134            GTS_931-24 UGA_173101_P016_WH09 0.500000  111        255
## 710167            GTS_931-24 UGA_173101_P017_WD09 0.500000  105        255
## 710186            GTS_931-24 UGA_173101_P019_WA01 0.500000  116        255
## 710213            GTS_931-24 UGA_173101_P018_WB06 0.483333   54        255
## 710237            GTS_931-24 UGA_173101_P018_WC11 0.500000  117        255
## 710249            GTS_931-24 UGA_173101_P023_WF05 0.489130   91        255
## 710256            GTS_931-24 UGA_173101_P023_WB12 0.480000  106        255
## 710264            GTS_931-24 UGA_173101_P023_WA04 0.500000   90        255
## 710310            GTS_931-24 UGA_173101_P023_WA07 0.490000  104        255
## 710329            GTS_931-24 UGA_173101_P022_WD02 0.472222   50        255
## 710404            GTS_931-24 UGA_173101_P020_WH07 0.425000   76        255
## 710452            GTS_931-24 UGA_173101_P021_WF01 0.489583  100        255
## 710491            GTS_931-24 UGA_173101_P022_WA09 0.500000   97        255
## 710527            GTS_931-24           GTS_949-23 0.500000  119        255
```

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## 710576            GTS_931-24           GTS_646-23 0.500000  117        255
## 710590            GTS_931-24           GTS_616-24 0.490741  119        255
## 710599            GTS_931-24           GTS_655-23 0.500000  113        255
## 710600            GTS_931-24           GTS_983-23 0.500000  118        255
## 710611            GTS_931-24           GTS_630-24 0.500000  117        255
## 710621            GTS_931-24           GTS_665-23 0.500000  118        255
## 710624            GTS_931-24           GTS_997-23 0.500000  118        255
## 710630            GTS_931-24           GTS_644-24 0.490385  118        255
## 710636            GTS_931-24           GTS_818-23 0.500000  118        255
## 710649            GTS_931-24           GTS_677-23 0.500000  117        255
## 710650            GTS_931-24          GTS_1012-23 0.500000  118        255
## 710656            GTS_931-24           GTS_657-24 0.500000  118        255
## 710675            GTS_931-24           GTS_691-23 0.500000  116        255
## 710743            GTS_931-24           GTS_725-23 0.500000  117        255
## 710752            GTS_931-24           GTS_736-23 0.490385  118        255
## 710856            GTS_931-24          GTS_1202-24 0.500000  117        255
## 710874            GTS_931-24           GTS_850-23 0.500000  119        255
## 710908            GTS_931-24           GTS_825-24 0.500000  118        255
## 710921            GTS_931-24           GTS_935-23 0.500000  119        255
## 710931            GTS_931-24          GTS_935-23R 0.500000  119        255
## 710949            GTS_931-24           GTS_855-24 0.500000  119        255
## 710961            GTS_931-24          GTS_1344-24 0.500000  119        255
## 710998            GTS_931-24           GTS_886-24 0.500000  119        255
## 711012            GTS_931-24          GTS_1403-24 0.500000  119        255
## 714650           GTS_1428-24 UGA_173101_P019_WE05 0.500000  114        255
## 714675           GTS_1428-24 UGA_173101_P019_WA04 0.500000   91        255
## 714716           GTS_1428-24 UGA_173101_P016_WC12 0.500000  109        255
## 714717           GTS_1428-24 UGA_173101_P017_WC02 0.491379  116        255
## 714724           GTS_1428-24 UGA_173101_P018_WB04 0.500000  117        255
## 714727           GTS_1428-24 UGA_173101_P016_WF07 0.471154  102        255
## 714771           GTS_1428-24 UGA_173101_P019_WG03 0.500000  121        255
## 714776           GTS_1428-24 UGA_173101_P016_WE08 0.500000  116        255
## 714805           GTS_1428-24 UGA_173101_P017_WC12 0.491071  116        255
## 714825           GTS_1428-24 UGA_173101_P019_WA02 0.500000  118        255
## 714851           GTS_1428-24 UGA_173101_P018_WD10 0.500000  110        255
## 714908           GTS_1428-24 UGA_173101_P016_WH09 0.500000  114        255
## 714941           GTS_1428-24 UGA_173101_P017_WD09 0.500000  108        255
## 714960           GTS_1428-24 UGA_173101_P019_WA01 0.500000  120        255
## 714987           GTS_1428-24 UGA_173101_P018_WB06 0.484375   55        255
## 715011           GTS_1428-24 UGA_173101_P018_WC11 0.500000  121        255
## 715023           GTS_1428-24 UGA_173101_P023_WF05 0.490000   93        255
## 715030           GTS_1428-24 UGA_173101_P023_WB12 0.481481  110        255
## 715038           GTS_1428-24 UGA_173101_P023_WA04 0.500000   92        255
## 715084           GTS_1428-24 UGA_173101_P023_WA07 0.490741  107        255
## 715103           GTS_1428-24 UGA_173101_P022_WD02 0.475000   51        255
## 715178           GTS_1428-24 UGA_173101_P020_WH07 0.428571   77        255
## 715226           GTS_1428-24 UGA_173101_P021_WF01 0.490385  103        255
## 715265           GTS_1428-24 UGA_173101_P022_WA09 0.500000  101        255
## 715301           GTS_1428-24           GTS_949-23 0.500000  123        255
## 715350           GTS_1428-24           GTS_646-23 0.500000  120        255
## 715364           GTS_1428-24           GTS_616-24 0.491071  121        255
## 715373           GTS_1428-24           GTS_655-23 0.500000  115        255
## 715374           GTS_1428-24           GTS_983-23 0.500000  121        255
## 715385           GTS_1428-24           GTS_630-24 0.500000  119        255
```

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## 715395            GTS_1428-24           GTS_665-23 0.500000  121         255
## 715398            GTS_1428-24           GTS_997-23 0.500000  122         255
## 715404            GTS_1428-24           GTS_644-24 0.490741  119         255
## 715410            GTS_1428-24           GTS_818-23 0.500000  122         255
## 715423            GTS_1428-24           GTS_677-23 0.500000  118         255
## 715424            GTS_1428-24          GTS_1012-23 0.500000  121         255
## 715430            GTS_1428-24           GTS_657-24 0.491071  120         255
## 715449            GTS_1428-24           GTS_691-23 0.500000  118         255
## 715517            GTS_1428-24           GTS_725-23 0.500000  119         255
## 715526            GTS_1428-24           GTS_736-23 0.480769  119         255
## 715630            GTS_1428-24          GTS_1202-24 0.491379  121         255
## 715648            GTS_1428-24           GTS_850-23 0.491379  123         255
## 715682            GTS_1428-24           GTS_825-24 0.490741  119         255
## 715695            GTS_1428-24           GTS_935-23 0.500000  123         255
## 715705            GTS_1428-24          GTS_935-23R 0.500000  122         255
## 715723            GTS_1428-24           GTS_855-24 0.500000  123         255
## 715735            GTS_1428-24          GTS_1344-24 0.500000  123         255
## 715772            GTS_1428-24           GTS_886-24 0.500000  123         255
## 715786            GTS_1428-24          GTS_1403-24 0.500000  122         255
## 715803            GTS_1428-24           GTS_931-24 0.500000  119         255
## 725450            GTS_3012-24 UGA_173101_P019_WE05 0.500000  114         255
## 725475            GTS_3012-24 UGA_173101_P019_WA04 0.500000   91         255
## 725516            GTS_3012-24 UGA_173101_P016_WC12 0.500000  108         255
## 725517            GTS_3012-24 UGA_173101_P017_WC02 0.491071  115         255
## 725524            GTS_3012-24 UGA_173101_P018_WB04 0.500000  116         255
## 725527            GTS_3012-24 UGA_173101_P016_WF07 0.471154  102         255
## 725571            GTS_3012-24 UGA_173101_P019_WG03 0.500000  120         255
## 725576            GTS_3012-24 UGA_173101_P016_WE08 0.500000  115         255
## 725605            GTS_3012-24 UGA_173101_P017_WC12 0.490741  115         255
## 725625            GTS_3012-24 UGA_173101_P019_WA02 0.500000  117         255
## 725651            GTS_3012-24 UGA_173101_P018_WD10 0.500000  110         255
## 725708            GTS_3012-24 UGA_173101_P016_WH09 0.500000  114         255
## 725741            GTS_3012-24 UGA_173101_P017_WD09 0.500000  108         255
## 725760            GTS_3012-24 UGA_173101_P019_WA01 0.500000  119         255
## 725787            GTS_3012-24 UGA_173101_P018_WB06 0.484375   55         255
## 725811            GTS_3012-24 UGA_173101_P018_WC11 0.500000  120         255
## 725823            GTS_3012-24 UGA_173101_P023_WF05 0.490000   93         255
## 725830            GTS_3012-24 UGA_173101_P023_WB12 0.480769  109         255
## 725838            GTS_3012-24 UGA_173101_P023_WA04 0.500000   92         255
## 725884            GTS_3012-24 UGA_173101_P023_WA07 0.490385  106         255
## 725903            GTS_3012-24 UGA_173101_P022_WD02 0.475000   51         255
## 725978            GTS_3012-24 UGA_173101_P020_WH07 0.425000   76         255
## 726026            GTS_3012-24 UGA_173101_P021_WF01 0.490000  102         255
## 726065            GTS_3012-24 UGA_173101_P022_WA09 0.500000  101         255
## 726101            GTS_3012-24           GTS_949-23 0.500000  122         255
## 726150            GTS_3012-24           GTS_646-23 0.500000  119         255
## 726164            GTS_3012-24           GTS_616-24 0.490741  120         255
## 726173            GTS_3012-24           GTS_655-23 0.500000  114         255
## 726174            GTS_3012-24           GTS_983-23 0.500000  120         255
## 726185            GTS_3012-24           GTS_630-24 0.500000  118         255
## 726195            GTS_3012-24           GTS_665-23 0.500000  120         255
## 726198            GTS_3012-24           GTS_997-23 0.500000  121         255
## 726204            GTS_3012-24           GTS_644-24 0.490385  118         255
## 726210            GTS_3012-24           GTS_818-23 0.500000  121         255
```

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## 726223            GTS_3012-24          GTS_677-23 0.500000  117        255
## 726224            GTS_3012-24         GTS_1012-23 0.500000  120        255
## 726230            GTS_3012-24          GTS_657-24 0.490741  119        255
## 726249            GTS_3012-24          GTS_691-23 0.500000  117        255
## 726317            GTS_3012-24          GTS_725-23 0.500000  118        255
## 726326            GTS_3012-24          GTS_736-23 0.480000  118        255
## 726430            GTS_3012-24         GTS_1202-24 0.491071  120        255
## 726448            GTS_3012-24          GTS_850-23 0.491071  122        255
## 726482            GTS_3012-24          GTS_825-24 0.490385  118        255
## 726495            GTS_3012-24          GTS_935-23 0.500000  122        255
## 726505            GTS_3012-24         GTS_935-23R 0.500000  121        255
## 726523            GTS_3012-24          GTS_855-24 0.500000  122        255
## 726535            GTS_3012-24         GTS_1344-24 0.500000  122        255
## 726572            GTS_3012-24          GTS_886-24 0.500000  122        255
## 726586            GTS_3012-24         GTS_1403-24 0.500000  121        255
## 726603            GTS_3012-24          GTS_931-24 0.500000  118        255
## 726607            GTS_3012-24         GTS_1428-24 0.500000  122        255
## 744850            GTS_1451-24 UGA_173101_P019_WE05 0.500000  114        255
## 744875            GTS_1451-24 UGA_173101_P019_WA04 0.500000   91        255
## 744916            GTS_1451-24 UGA_173101_P016_WC12 0.500000  109        255
## 744917            GTS_1451-24 UGA_173101_P017_WC02 0.491379  116        255
## 744924            GTS_1451-24 UGA_173101_P018_WB04 0.500000  117        255
## 744927            GTS_1451-24 UGA_173101_P016_WF07 0.471154  102        255
## 744971            GTS_1451-24 UGA_173101_P019_WG03 0.500000  121        255
## 744976            GTS_1451-24 UGA_173101_P016_WE08 0.500000  116        255
## 745005            GTS_1451-24 UGA_173101_P017_WC12 0.491071  116        255
## 745025            GTS_1451-24 UGA_173101_P019_WA02 0.500000  118        255
## 745051            GTS_1451-24 UGA_173101_P018_WD10 0.500000  110        255
## 745108            GTS_1451-24 UGA_173101_P016_WH09 0.500000  114        255
## 745141            GTS_1451-24 UGA_173101_P017_WD09 0.500000  108        255
## 745160            GTS_1451-24 UGA_173101_P019_WA01 0.500000  120        255
## 745187            GTS_1451-24 UGA_173101_P018_WB06 0.484375   55        255
## 745211            GTS_1451-24 UGA_173101_P018_WC11 0.500000  121        255
## 745223            GTS_1451-24 UGA_173101_P023_WF05 0.490000   93        255
## 745230            GTS_1451-24 UGA_173101_P023_WB12 0.481481  110        255
## 745238            GTS_1451-24 UGA_173101_P023_WA04 0.500000   92        255
## 745284            GTS_1451-24 UGA_173101_P023_WA07 0.490741  107        255
## 745303            GTS_1451-24 UGA_173101_P022_WD02 0.475000   51        255
## 745378            GTS_1451-24 UGA_173101_P020_WH07 0.428571   77        255
## 745426            GTS_1451-24 UGA_173101_P021_WF01 0.490385  103        255
## 745465            GTS_1451-24 UGA_173101_P022_WA09 0.500000  101        255
## 745501            GTS_1451-24          GTS_949-23 0.500000  123        255
## 745550            GTS_1451-24          GTS_646-23 0.500000  120        255
## 745564            GTS_1451-24          GTS_616-24 0.491071  121        255
## 745573            GTS_1451-24          GTS_655-23 0.500000  115        255
## 745574            GTS_1451-24          GTS_983-23 0.500000  121        255
## 745585            GTS_1451-24          GTS_630-24 0.500000  119        255
## 745595            GTS_1451-24          GTS_665-23 0.500000  121        255
## 745598            GTS_1451-24          GTS_997-23 0.500000  122        255
## 745604            GTS_1451-24          GTS_644-24 0.490741  119        255
## 745610            GTS_1451-24          GTS_818-23 0.500000  122        255
## 745623            GTS_1451-24          GTS_677-23 0.500000  118        255
## 745624            GTS_1451-24         GTS_1012-23 0.500000  121        255
## 745630            GTS_1451-24          GTS_657-24 0.491071  120        255
```

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## 745649            GTS_1451-24           GTS_691-23 0.500000  118         255
## 745717            GTS_1451-24           GTS_725-23 0.500000  119         255
## 745726            GTS_1451-24           GTS_736-23 0.480769  119         255
## 745830            GTS_1451-24          GTS_1202-24 0.491379  121         255
## 745848            GTS_1451-24           GTS_850-23 0.491379  123         255
## 745882            GTS_1451-24           GTS_825-24 0.490741  119         255
## 745895            GTS_1451-24           GTS_935-23 0.500000  123         255
## 745905            GTS_1451-24          GTS_935-23R 0.500000  122         255
## 745923            GTS_1451-24           GTS_855-24 0.500000  123         255
## 745935            GTS_1451-24          GTS_1344-24 0.500000  123         255
## 745972            GTS_1451-24           GTS_886-24 0.500000  123         255
## 745986            GTS_1451-24          GTS_1403-24 0.500000  122         255
## 746003            GTS_1451-24           GTS_931-24 0.500000  119         255
## 746007            GTS_1451-24          GTS_1428-24 0.500000  123         255
## 746016            GTS_1451-24          GTS_3012-24 0.500000  122         255
## 755875             GTS_955-24 UGA_173101_P019_WE05 0.500000  113         255
## 755900             GTS_955-24 UGA_173101_P019_WA04 0.500000   90         255
## 755941             GTS_955-24 UGA_173101_P016_WC12 0.500000  108         255
## 755942             GTS_955-24 UGA_173101_P017_WC02 0.491071  115         255
## 755949             GTS_955-24 UGA_173101_P018_WB04 0.500000  116         255
## 755952             GTS_955-24 UGA_173101_P016_WF07 0.470000  101         255
## 755996             GTS_955-24 UGA_173101_P019_WG03 0.500000  120         255
## 756001             GTS_955-24 UGA_173101_P016_WE08 0.500000  115         255
## 756030             GTS_955-24 UGA_173101_P017_WC12 0.490741  115         255
## 756050             GTS_955-24 UGA_173101_P019_WA02 0.500000  117         255
## 756076             GTS_955-24 UGA_173101_P018_WD10 0.500000  109         255
## 756133             GTS_955-24 UGA_173101_P016_WH09 0.500000  113         255
## 756166             GTS_955-24 UGA_173101_P017_WD09 0.500000  107         255
## 756185             GTS_955-24 UGA_173101_P019_WA01 0.500000  119         255
## 756212             GTS_955-24 UGA_173101_P018_WB06 0.483333   54         255
## 756236             GTS_955-24 UGA_173101_P018_WC11 0.500000  120         255
## 756248             GTS_955-24 UGA_173101_P023_WF05 0.489583   92         255
## 756255             GTS_955-24 UGA_173101_P023_WB12 0.480769  109         255
## 756263             GTS_955-24 UGA_173101_P023_WA04 0.500000   91         255
## 756309             GTS_955-24 UGA_173101_P023_WA07 0.490385  106         255
## 756328             GTS_955-24 UGA_173101_P022_WD02 0.472222   50         255
## 756403             GTS_955-24 UGA_173101_P020_WH07 0.425000   76         255
## 756451             GTS_955-24 UGA_173101_P021_WF01 0.490000  102         255
## 756490             GTS_955-24 UGA_173101_P022_WA09 0.500000  100         255
## 756526             GTS_955-24           GTS_949-23 0.500000  122         255
## 756575             GTS_955-24           GTS_646-23 0.500000  119         255
## 756589             GTS_955-24           GTS_616-24 0.490741  120         255
## 756598             GTS_955-24           GTS_655-23 0.500000  114         255
## 756599             GTS_955-24           GTS_983-23 0.500000  121         255
## 756610             GTS_955-24           GTS_630-24 0.500000  118         255
## 756620             GTS_955-24           GTS_665-23 0.500000  120         255
## 756623             GTS_955-24           GTS_997-23 0.500000  121         255
## 756629             GTS_955-24           GTS_644-24 0.490385  118         255
## 756635             GTS_955-24           GTS_818-23 0.500000  121         255
## 756648             GTS_955-24           GTS_677-23 0.500000  118         255
## 756649             GTS_955-24          GTS_1012-23 0.500000  121         255
## 756655             GTS_955-24           GTS_657-24 0.490741  119         255
## 756674             GTS_955-24           GTS_691-23 0.500000  118         255
## 756742             GTS_955-24           GTS_725-23 0.500000  118         255
```

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## 756751            GTS_955-24           GTS_736-23 0.480769  119        255
## 756855            GTS_955-24          GTS_1202-24 0.500000  120        255
## 756873            GTS_955-24           GTS_850-23 0.500000  122        255
## 756907            GTS_955-24           GTS_825-24 0.500000  118        255
## 756920            GTS_955-24           GTS_935-23 0.500000  122        255
## 756930            GTS_955-24          GTS_935-23R 0.500000  122        255
## 756948            GTS_955-24           GTS_855-24 0.500000  122        255
## 756960            GTS_955-24          GTS_1344-24 0.500000  122        255
## 756997            GTS_955-24           GTS_886-24 0.500000  122        255
## 757011            GTS_955-24          GTS_1403-24 0.500000  122        255
## 757028            GTS_955-24           GTS_931-24 0.500000  119        255
## 757032            GTS_955-24          GTS_1428-24 0.500000  122        255
## 757041            GTS_955-24          GTS_3012-24 0.500000  121        255
## 757057            GTS_955-24          GTS_1451-24 0.500000  122        255
## 789436           GTS_1516-24 UGA_173101_P019_WE05 0.480000  112        276
## 789461           GTS_1516-24 UGA_173101_P019_WA04 0.476190   90        276
## 789502           GTS_1516-24 UGA_173101_P016_WC12 0.481481  107        276
## 789503           GTS_1516-24 UGA_173101_P017_WC02 0.472222  114        276
## 789510           GTS_1516-24 UGA_173101_P018_WB04 0.480769  115        276
## 789513           GTS_1516-24 UGA_173101_P016_WF07 0.447917  100        276
## 789557           GTS_1516-24 UGA_173101_P019_WG03 0.481481  118        276
## 789562           GTS_1516-24 UGA_173101_P016_WE08 0.481481  114        276
## 789591           GTS_1516-24 UGA_173101_P017_WC12 0.471154  114        276
## 789611           GTS_1516-24 UGA_173101_P019_WA02 0.481481  116        276
## 789637           GTS_1516-24 UGA_173101_P018_WD10 0.480769  107        276
## 789694           GTS_1516-24 UGA_173101_P016_WH09 0.480769  112        276
## 789727           GTS_1516-24 UGA_173101_P017_WD09 0.480000  106        276
## 789746           GTS_1516-24 UGA_173101_P019_WA01 0.481481  117        276
## 789773           GTS_1516-24 UGA_173101_P018_WB06 0.464286   54        276
## 789797           GTS_1516-24 UGA_173101_P018_WC11 0.481481  118        276
## 789809           GTS_1516-24 UGA_173101_P023_WF05 0.467391   92        276
## 789816           GTS_1516-24 UGA_173101_P023_WB12 0.460000  107        276
## 789824           GTS_1516-24 UGA_173101_P023_WA04 0.476190   91        276
## 789870           GTS_1516-24 UGA_173101_P023_WA07 0.480769  105        276
## 789889           GTS_1516-24 UGA_173101_P022_WD02 0.444444   51        276
## 789964           GTS_1516-24 UGA_173101_P020_WH07 0.412500   77        276
## 790012           GTS_1516-24 UGA_173101_P021_WF01 0.470000  102        276
## 790051           GTS_1516-24 UGA_173101_P022_WA09 0.478261   98        276
## 790087           GTS_1516-24           GTS_949-23 0.491071  120        276
## 790136           GTS_1516-24           GTS_646-23 0.500000  116        276
## 790150           GTS_1516-24           GTS_616-24 0.490741  119        276
## 790159           GTS_1516-24           GTS_655-23 0.500000  113        276
## 790160           GTS_1516-24           GTS_983-23 0.490385  118        276
## 790171           GTS_1516-24           GTS_630-24 0.500000  117        276
## 790181           GTS_1516-24           GTS_665-23 0.490741  119        276
## 790184           GTS_1516-24           GTS_997-23 0.490741  119        276
## 790190           GTS_1516-24           GTS_644-24 0.490385  118        276
## 790196           GTS_1516-24           GTS_818-23 0.490741  119        276
## 790209           GTS_1516-24           GTS_677-23 0.500000  118        276
## 790210           GTS_1516-24          GTS_1012-23 0.490385  118        276
## 790216           GTS_1516-24           GTS_657-24 0.500000  118        276
## 790235           GTS_1516-24           GTS_691-23 0.480769  118        276
## 790303           GTS_1516-24           GTS_725-23 0.500000  117        276
## 790312           GTS_1516-24           GTS_736-23 0.500000  119        276
```

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## 790416            GTS_1516-24          GTS_1202-24 0.482143  118        276
## 790434            GTS_1516-24           GTS_850-23 0.491071  120        276
## 790468            GTS_1516-24           GTS_825-24 0.500000  119        276
## 790481            GTS_1516-24           GTS_935-23 0.490741  119        276
## 790491            GTS_1516-24          GTS_935-23R 0.482143  120        276
## 790509            GTS_1516-24           GTS_855-24 0.490741  119        276
## 790521            GTS_1516-24          GTS_1344-24 0.490741  119        276
## 790558            GTS_1516-24           GTS_886-24 0.491071  120        276
## 790572            GTS_1516-24          GTS_1403-24 0.490741  119        276
## 790589            GTS_1516-24           GTS_931-24 0.500000  118        276
## 790593            GTS_1516-24          GTS_1428-24 0.490741  119        276
## 790602            GTS_1516-24          GTS_3012-24 0.490385  118        276
## 790618            GTS_1516-24          GTS_1451-24 0.490741  119        276
## 790627            GTS_1516-24           GTS_955-24 0.490741  119        276
## 836611            GTS_2777-24 UGA_173101_P019_WE05 0.500000  114        255
## 836636            GTS_2777-24 UGA_173101_P019_WA04 0.500000   91        255
## 836677            GTS_2777-24 UGA_173101_P016_WC12 0.500000  109        255
## 836678            GTS_2777-24 UGA_173101_P017_WC02 0.491379  116        255
## 836685            GTS_2777-24 UGA_173101_P018_WB04 0.500000  117        255
## 836688            GTS_2777-24 UGA_173101_P016_WF07 0.471154  102        255
## 836732            GTS_2777-24 UGA_173101_P019_WG03 0.500000  120        255
## 836737            GTS_2777-24 UGA_173101_P016_WE08 0.500000  116        255
## 836766            GTS_2777-24 UGA_173101_P017_WC12 0.491071  116        255
## 836786            GTS_2777-24 UGA_173101_P019_WA02 0.500000  118        255
## 836812            GTS_2777-24 UGA_173101_P018_WD10 0.500000  109        255
## 836869            GTS_2777-24 UGA_173101_P016_WH09 0.500000  114        255
## 836902            GTS_2777-24 UGA_173101_P017_WD09 0.500000  108        255
## 836921            GTS_2777-24 UGA_173101_P019_WA01 0.500000  119        255
## 836948            GTS_2777-24 UGA_173101_P018_WB06 0.484375   55        255
## 836972            GTS_2777-24 UGA_173101_P018_WC11 0.500000  120        255
## 836984            GTS_2777-24 UGA_173101_P023_WF05 0.490000   93        255
## 836991            GTS_2777-24 UGA_173101_P023_WB12 0.481481  109        255
## 836999            GTS_2777-24 UGA_173101_P023_WA04 0.500000   92        255
## 837045            GTS_2777-24 UGA_173101_P023_WA07 0.490741  107        255
## 837064            GTS_2777-24 UGA_173101_P022_WD02 0.475000   51        255
## 837139            GTS_2777-24 UGA_173101_P020_WH07 0.428571   77        255
## 837187            GTS_2777-24 UGA_173101_P021_WF01 0.490385  102        255
## 837226            GTS_2777-24 UGA_173101_P022_WA09 0.500000  100        255
## 837262            GTS_2777-24           GTS_949-23 0.500000  121        255
## 837311            GTS_2777-24           GTS_646-23 0.500000  119        255
## 837325            GTS_2777-24           GTS_616-24 0.500000  120        255
## 837334            GTS_2777-24           GTS_655-23 0.500000  115        255
## 837335            GTS_2777-24           GTS_983-23 0.500000  120        255
## 837346            GTS_2777-24           GTS_630-24 0.500000  119        255
## 837356            GTS_2777-24           GTS_665-23 0.500000  121        255
## 837359            GTS_2777-24           GTS_997-23 0.500000  121        255
## 837365            GTS_2777-24           GTS_644-24 0.500000  118        255
## 837371            GTS_2777-24           GTS_818-23 0.500000  121        255
## 837384            GTS_2777-24           GTS_677-23 0.500000  118        255
## 837385            GTS_2777-24          GTS_1012-23 0.500000  120        255
## 837391            GTS_2777-24           GTS_657-24 0.491071  120        255
## 837410            GTS_2777-24           GTS_691-23 0.500000  118        255
## 837478            GTS_2777-24           GTS_725-23 0.500000  119        255
## 837487            GTS_2777-24           GTS_736-23 0.480769  119        255
```

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## 837591           GTS_2777-24          GTS_1202-24 0.491071  119         255
## 837609           GTS_2777-24           GTS_850-23 0.491071  121         255
## 837643           GTS_2777-24           GTS_825-24 0.490385  118         255
## 837656           GTS_2777-24           GTS_935-23 0.500000  121         255
## 837666           GTS_2777-24          GTS_935-23R 0.500000  120         255
## 837684           GTS_2777-24           GTS_855-24 0.500000  121         255
## 837696           GTS_2777-24          GTS_1344-24 0.500000  121         255
## 837733           GTS_2777-24           GTS_886-24 0.500000  121         255
## 837747           GTS_2777-24          GTS_1403-24 0.500000  120         255
## 837764           GTS_2777-24           GTS_931-24 0.500000  118         255
## 837768           GTS_2777-24          GTS_1428-24 0.500000  121         255
## 837777           GTS_2777-24          GTS_3012-24 0.500000  120         255
## 837793           GTS_2777-24          GTS_1451-24 0.500000  121         255
## 837802           GTS_2777-24           GTS_955-24 0.500000  120         255
## 837829           GTS_2777-24          GTS_1516-24 0.490385  118         255
## 874543           GTS_1633-24 UGA_173101_P019_WE05 0.451923  115         289
## 874568           GTS_1633-24 UGA_173101_P019_WA04 0.443182   92         289
## 874609           GTS_1633-24 UGA_173101_P016_WC12 0.455357  110         289
## 874610           GTS_1633-24 UGA_173101_P017_WC02 0.446429  117         289
## 874617           GTS_1633-24 UGA_173101_P018_WB04 0.453704  118         289
## 874620           GTS_1633-24 UGA_173101_P016_WF07 0.420000  103         289
## 874664           GTS_1633-24 UGA_173101_P019_WG03 0.455357  122         289
## 874669           GTS_1633-24 UGA_173101_P016_WE08 0.455357  117         289
## 874698           GTS_1633-24 UGA_173101_P017_WC12 0.446429  117         289
## 874718           GTS_1633-24 UGA_173101_P019_WA02 0.455357  119         289
## 874744           GTS_1633-24 UGA_173101_P018_WD10 0.453704  111         289
## 874801           GTS_1633-24 UGA_173101_P016_WH09 0.453704  115         289
## 874834           GTS_1633-24 UGA_173101_P017_WD09 0.451923  109         289
## 874853           GTS_1633-24 UGA_173101_P019_WA01 0.455357  121         289
## 874880           GTS_1633-24 UGA_173101_P018_WB06 0.406250   56         289
## 874904           GTS_1633-24 UGA_173101_P018_WC11 0.455357  122         289
## 874916           GTS_1633-24 UGA_173101_P023_WF05 0.437500   94         289
## 874923           GTS_1633-24 UGA_173101_P023_WB12 0.432692  111         289
## 874931           GTS_1633-24 UGA_173101_P023_WA04 0.443182   93         289
## 874977           GTS_1633-24 UGA_173101_P023_WA07 0.444444  108         289
## 874996           GTS_1633-24 UGA_173101_P022_WD02 0.350000   52         289
## 875071           GTS_1633-24 UGA_173101_P020_WH07 0.362500   78         289
## 875119           GTS_1633-24 UGA_173101_P021_WF01 0.440000  104         289
## 875158           GTS_1633-24 UGA_173101_P022_WA09 0.447917  102         289
## 875194           GTS_1633-24           GTS_949-23 0.482759  124         289
## 875243           GTS_1633-24           GTS_646-23 0.490385  120         289
## 875257           GTS_1633-24           GTS_616-24 0.473214  122         289
## 875266           GTS_1633-24           GTS_655-23 0.479167  116         289
## 875267           GTS_1633-24           GTS_983-23 0.500000  121         289
## 875278           GTS_1633-24           GTS_630-24 0.490385  119         289
## 875288           GTS_1633-24           GTS_665-23 0.482143  122         289
## 875291           GTS_1633-24           GTS_997-23 0.482143  123         289
## 875297           GTS_1633-24           GTS_644-24 0.481481  119         289
## 875303           GTS_1633-24           GTS_818-23 0.482143  123         289
## 875316           GTS_1633-24           GTS_677-23 0.490385  119         289
## 875317           GTS_1633-24          GTS_1012-23 0.500000  121         289
## 875323           GTS_1633-24           GTS_657-24 0.482143  120         289
## 875342           GTS_1633-24           GTS_691-23 0.500000  119         289
## 875410           GTS_1633-24           GTS_725-23 0.490385  119         289
```

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## 875419           GTS_1633-24          GTS_736-23 0.472222  120        289
## 875523           GTS_1633-24         GTS_1202-24 0.500000  122        289
## 875541           GTS_1633-24          GTS_850-23 0.491379  124        289
## 875575           GTS_1633-24          GTS_825-24 0.490741  120        289
## 875588           GTS_1633-24          GTS_935-23 0.491379  123        289
## 875598           GTS_1633-24         GTS_935-23R 0.500000  123        289
## 875616           GTS_1633-24          GTS_855-24 0.491379  123        289
## 875628           GTS_1633-24         GTS_1344-24 0.491379  123        289
## 875665           GTS_1633-24          GTS_886-24 0.482759  124        289
## 875679           GTS_1633-24         GTS_1403-24 0.500000  122        289
## 875696           GTS_1633-24          GTS_931-24 0.500000  119        289
## 875700           GTS_1633-24         GTS_1428-24 0.491379  123        289
## 875709           GTS_1633-24         GTS_3012-24 0.491071  122        289
## 875725           GTS_1633-24         GTS_1451-24 0.491379  123        289
## 875734           GTS_1633-24          GTS_955-24 0.500000  122        289
## 875761           GTS_1633-24         GTS_1516-24 0.482143  120        289
## 875798           GTS_1633-24         GTS_2777-24 0.491071  121        289
## 886486           GTS_1672-24 UGA_173101_P019_WE05 0.500000  113        255
## 886511           GTS_1672-24 UGA_173101_P019_WA04 0.500000   90        255
## 886552           GTS_1672-24 UGA_173101_P016_WC12 0.500000  108        255
## 886553           GTS_1672-24 UGA_173101_P017_WC02 0.491071  115        255
## 886560           GTS_1672-24 UGA_173101_P018_WB04 0.500000  116        255
## 886563           GTS_1672-24 UGA_173101_P016_WF07 0.470000  101        255
## 886607           GTS_1672-24 UGA_173101_P019_WG03 0.500000  119        255
## 886612           GTS_1672-24 UGA_173101_P016_WE08 0.500000  115        255
## 886641           GTS_1672-24 UGA_173101_P017_WC12 0.490741  115        255
## 886661           GTS_1672-24 UGA_173101_P019_WA02 0.500000  117        255
## 886687           GTS_1672-24 UGA_173101_P018_WD10 0.500000  108        255
## 886744           GTS_1672-24 UGA_173101_P016_WH09 0.500000  113        255
## 886777           GTS_1672-24 UGA_173101_P017_WD09 0.500000  107        255
## 886796           GTS_1672-24 UGA_173101_P019_WA01 0.500000  118        255
## 886823           GTS_1672-24 UGA_173101_P018_WB06 0.483333   54        255
## 886847           GTS_1672-24 UGA_173101_P018_WC11 0.500000  119        255
## 886859           GTS_1672-24 UGA_173101_P023_WF05 0.489583   92        255
## 886866           GTS_1672-24 UGA_173101_P023_WB12 0.480769  108        255
## 886874           GTS_1672-24 UGA_173101_P023_WA04 0.500000   91        255
## 886920           GTS_1672-24 UGA_173101_P023_WA07 0.490385  106        255
## 886939           GTS_1672-24 UGA_173101_P022_WD02 0.472222   50        255
## 887014           GTS_1672-24 UGA_173101_P020_WH07 0.425000   76        255
## 887062           GTS_1672-24 UGA_173101_P021_WF01 0.490000  101        255
## 887101           GTS_1672-24 UGA_173101_P022_WA09 0.500000   99        255
## 887137           GTS_1672-24          GTS_949-23 0.500000  121        255
## 887186           GTS_1672-24          GTS_646-23 0.500000  118        255
## 887200           GTS_1672-24          GTS_616-24 0.490741  120        255
## 887209           GTS_1672-24          GTS_655-23 0.500000  114        255
## 887210           GTS_1672-24          GTS_983-23 0.500000  120        255
## 887221           GTS_1672-24          GTS_630-24 0.500000  118        255
## 887231           GTS_1672-24          GTS_665-23 0.500000  120        255
## 887234           GTS_1672-24          GTS_997-23 0.500000  120        255
## 887240           GTS_1672-24          GTS_644-24 0.490385  118        255
## 887246           GTS_1672-24          GTS_818-23 0.500000  120        255
## 887259           GTS_1672-24          GTS_677-23 0.500000  118        255
## 887260           GTS_1672-24         GTS_1012-23 0.500000  120        255
## 887266           GTS_1672-24          GTS_657-24 0.490741  119        255
```

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## 887285          GTS_1672-24           GTS_691-23 0.500000  118        255
## 887353          GTS_1672-24           GTS_725-23 0.500000  118        255
## 887362          GTS_1672-24           GTS_736-23 0.480769  119        255
## 887466          GTS_1672-24          GTS_1202-24 0.500000  119        255
## 887484          GTS_1672-24           GTS_850-23 0.500000  121        255
## 887518          GTS_1672-24           GTS_825-24 0.500000  118        255
## 887531          GTS_1672-24           GTS_935-23 0.500000  121        255
## 887541          GTS_1672-24          GTS_935-23R 0.500000  121        255
## 887559          GTS_1672-24           GTS_855-24 0.500000  121        255
## 887571          GTS_1672-24          GTS_1344-24 0.500000  121        255
## 887608          GTS_1672-24           GTS_886-24 0.500000  121        255
## 887622          GTS_1672-24          GTS_1403-24 0.500000  121        255
## 887639          GTS_1672-24           GTS_931-24 0.500000  119        255
## 887643          GTS_1672-24          GTS_1428-24 0.500000  121        255
## 887652          GTS_1672-24          GTS_3012-24 0.500000  120        255
## 887668          GTS_1672-24          GTS_1451-24 0.500000  121        255
## 887677          GTS_1672-24           GTS_955-24 0.500000  121        255
## 887704          GTS_1672-24          GTS_1516-24 0.490741  119        255
## 887741          GTS_1672-24          GTS_2777-24 0.500000  120        255
## 887770          GTS_1672-24          GTS_1633-24 0.500000  121        255
## 917375          GTS_2376-24 UGA_173101_P019_WE05 0.490385  114        255
## 917400          GTS_2376-24 UGA_173101_P019_WA04 0.488636   91        255
## 917441          GTS_2376-24 UGA_173101_P016_WC12 0.491071  109        255
## 917442          GTS_2376-24 UGA_173101_P017_WC02 0.482759  116        255
## 917449          GTS_2376-24 UGA_173101_P018_WB04 0.490741  117        255
## 917452          GTS_2376-24 UGA_173101_P016_WF07 0.461538  102        255
## 917496          GTS_2376-24 UGA_173101_P019_WG03 0.491071  121        255
## 917501          GTS_2376-24 UGA_173101_P016_WE08 0.491071  116        255
## 917530          GTS_2376-24 UGA_173101_P017_WC12 0.481481  116        255
## 917550          GTS_2376-24 UGA_173101_P019_WA02 0.491071  118        255
## 917576          GTS_2376-24 UGA_173101_P018_WD10 0.490741  110        255
## 917633          GTS_2376-24 UGA_173101_P016_WH09 0.490741  114        255
## 917666          GTS_2376-24 UGA_173101_P017_WD09 0.490385  108        255
## 917685          GTS_2376-24 UGA_173101_P019_WA01 0.491071  120        255
## 917712          GTS_2376-24 UGA_173101_P018_WB06 0.466667   55        255
## 917736          GTS_2376-24 UGA_173101_P018_WC11 0.491071  121        255
## 917748          GTS_2376-24 UGA_173101_P023_WF05 0.480000   93        255
## 917755          GTS_2376-24 UGA_173101_P023_WB12 0.472222  110        255
## 917763          GTS_2376-24 UGA_173101_P023_WA04 0.488636   92        255
## 917809          GTS_2376-24 UGA_173101_P023_WA07 0.480769  107        255
## 917828          GTS_2376-24 UGA_173101_P022_WD02 0.444444   51        255
## 917903          GTS_2376-24 UGA_173101_P020_WH07 0.412500   77        255
## 917951          GTS_2376-24 UGA_173101_P021_WF01 0.480769  103        255
## 917990          GTS_2376-24 UGA_173101_P022_WA09 0.489583  101        255
## 918026          GTS_2376-24           GTS_949-23 0.500000  123        255
## 918075          GTS_2376-24           GTS_646-23 0.500000  119        255
## 918089          GTS_2376-24           GTS_616-24 0.491071  121        255
## 918098          GTS_2376-24           GTS_655-23 0.500000  115        255
## 918099          GTS_2376-24           GTS_983-23 0.500000  121        255
## 918110          GTS_2376-24           GTS_630-24 0.500000  118        255
## 918120          GTS_2376-24           GTS_665-23 0.500000  121        255
## 918123          GTS_2376-24           GTS_997-23 0.500000  122        255
## 918129          GTS_2376-24           GTS_644-24 0.490385  118        255
## 918135          GTS_2376-24           GTS_818-23 0.500000  122        255
```

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## 918148           GTS_2376-24          GTS_677-23 0.500000  119        255
## 918149           GTS_2376-24         GTS_1012-23 0.500000  121        255
## 918155           GTS_2376-24          GTS_657-24 0.490741  119        255
## 918174           GTS_2376-24          GTS_691-23 0.490385  119        255
## 918242           GTS_2376-24          GTS_725-23 0.500000  118        255
## 918251           GTS_2376-24          GTS_736-23 0.481481  120        255
## 918355           GTS_2376-24         GTS_1202-24 0.491379  121        255
## 918373           GTS_2376-24          GTS_850-23 0.500000  123        255
## 918407           GTS_2376-24          GTS_825-24 0.500000  119        255
## 918420           GTS_2376-24          GTS_935-23 0.500000  122        255
## 918430           GTS_2376-24         GTS_935-23R 0.491379  123        255
## 918448           GTS_2376-24          GTS_855-24 0.500000  122        255
## 918460           GTS_2376-24         GTS_1344-24 0.500000  122        255
## 918497           GTS_2376-24          GTS_886-24 0.500000  123        255
## 918511           GTS_2376-24         GTS_1403-24 0.500000  122        255
## 918528           GTS_2376-24          GTS_931-24 0.500000  119        255
## 918532           GTS_2376-24         GTS_1428-24 0.500000  122        255
## 918541           GTS_2376-24         GTS_3012-24 0.500000  121        255
## 918557           GTS_2376-24         GTS_1451-24 0.500000  122        255
## 918566           GTS_2376-24          GTS_955-24 0.500000  122        255
## 918593           GTS_2376-24         GTS_1516-24 0.491071  120        255
## 918630           GTS_2376-24         GTS_2777-24 0.500000  120        255
## 918659           GTS_2376-24         GTS_1633-24 0.491379  123        255
## 918668           GTS_2376-24         GTS_1672-24 0.500000  121        255
## 947416           GTS_2669-24 UGA_173101_P019_WE05 0.500000  114        255
## 947441           GTS_2669-24 UGA_173101_P019_WA04 0.500000   91        255
## 947482           GTS_2669-24 UGA_173101_P016_WC12 0.500000  109        255
## 947483           GTS_2669-24 UGA_173101_P017_WC02 0.491379  116        255
## 947490           GTS_2669-24 UGA_173101_P018_WB04 0.500000  117        255
## 947493           GTS_2669-24 UGA_173101_P016_WF07 0.471154  102        255
## 947537           GTS_2669-24 UGA_173101_P019_WG03 0.500000  121        255
## 947542           GTS_2669-24 UGA_173101_P016_WE08 0.500000  116        255
## 947571           GTS_2669-24 UGA_173101_P017_WC12 0.491071  116        255
## 947591           GTS_2669-24 UGA_173101_P019_WA02 0.500000  118        255
## 947617           GTS_2669-24 UGA_173101_P018_WD10 0.500000  110        255
## 947674           GTS_2669-24 UGA_173101_P016_WH09 0.500000  114        255
## 947707           GTS_2669-24 UGA_173101_P017_WD09 0.500000  108        255
## 947726           GTS_2669-24 UGA_173101_P019_WA01 0.500000  120        255
## 947753           GTS_2669-24 UGA_173101_P018_WB06 0.484375   55        255
## 947777           GTS_2669-24 UGA_173101_P018_WC11 0.500000  121        255
## 947789           GTS_2669-24 UGA_173101_P023_WF05 0.490000   93        255
## 947796           GTS_2669-24 UGA_173101_P023_WB12 0.481481  110        255
## 947804           GTS_2669-24 UGA_173101_P023_WA04 0.500000   92        255
## 947850           GTS_2669-24 UGA_173101_P023_WA07 0.490741  107        255
## 947869           GTS_2669-24 UGA_173101_P022_WD02 0.475000   51        255
## 947944           GTS_2669-24 UGA_173101_P020_WH07 0.428571   77        255
## 947992           GTS_2669-24 UGA_173101_P021_WF01 0.490385  103        255
## 948031           GTS_2669-24 UGA_173101_P022_WA09 0.500000  101        255
## 948067           GTS_2669-24          GTS_949-23 0.500000  123        255
## 948116           GTS_2669-24          GTS_646-23 0.500000  120        255
## 948130           GTS_2669-24          GTS_616-24 0.491071  121        255
## 948139           GTS_2669-24          GTS_655-23 0.500000  115        255
## 948140           GTS_2669-24          GTS_983-23 0.500000  121        255
## 948151           GTS_2669-24          GTS_630-24 0.500000  119        255
```

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## 948161           GTS_2669-24           GTS_665-23 0.500000  121        255
## 948164           GTS_2669-24           GTS_997-23 0.500000  122        255
## 948170           GTS_2669-24           GTS_644-24 0.490741  119        255
## 948176           GTS_2669-24           GTS_818-23 0.500000  122        255
## 948189           GTS_2669-24           GTS_677-23 0.500000  118        255
## 948190           GTS_2669-24          GTS_1012-23 0.500000  121        255
## 948196           GTS_2669-24           GTS_657-24 0.491071  120        255
## 948215           GTS_2669-24           GTS_691-23 0.500000  118        255
## 948283           GTS_2669-24           GTS_725-23 0.500000  119        255
## 948292           GTS_2669-24           GTS_736-23 0.480769  119        255
## 948396           GTS_2669-24          GTS_1202-24 0.491379  121        255
## 948414           GTS_2669-24           GTS_850-23 0.491379  123        255
## 948448           GTS_2669-24           GTS_825-24 0.490741  119        255
## 948461           GTS_2669-24           GTS_935-23 0.500000  123        255
## 948471           GTS_2669-24          GTS_935-23R 0.500000  122        255
## 948489           GTS_2669-24           GTS_855-24 0.500000  123        255
## 948501           GTS_2669-24          GTS_1344-24 0.500000  123        255
## 948538           GTS_2669-24           GTS_886-24 0.500000  123        255
## 948552           GTS_2669-24          GTS_1403-24 0.500000  122        255
## 948569           GTS_2669-24           GTS_931-24 0.500000  119        255
## 948573           GTS_2669-24          GTS_1428-24 0.500000  123        255
## 948582           GTS_2669-24          GTS_3012-24 0.500000  122        255
## 948598           GTS_2669-24          GTS_1451-24 0.500000  123        255
## 948607           GTS_2669-24           GTS_955-24 0.500000  122        255
## 948634           GTS_2669-24          GTS_1516-24 0.490741  119        255
## 948671           GTS_2669-24          GTS_2777-24 0.500000  121        255
## 948700           GTS_2669-24          GTS_1633-24 0.491379  123        255
## 948709           GTS_2669-24          GTS_1672-24 0.500000  121        255
## 948732           GTS_2669-24          GTS_2376-24 0.500000  122        255
## 958460           GTS_2857-24 UGA_173101_P019_WE05 0.490741  115        255
## 958485           GTS_2857-24 UGA_173101_P019_WA04 0.489130   92        255
## 958526           GTS_2857-24 UGA_173101_P016_WC12 0.491379  110        255
## 958527           GTS_2857-24 UGA_173101_P017_WC02 0.483333  117        255
## 958534           GTS_2857-24 UGA_173101_P018_WB04 0.491071  118        255
## 958537           GTS_2857-24 UGA_173101_P016_WF07 0.462963  103        255
## 958581           GTS_2857-24 UGA_173101_P019_WG03 0.491379  122        255
## 958586           GTS_2857-24 UGA_173101_P016_WE08 0.491379  117        255
## 958615           GTS_2857-24 UGA_173101_P017_WC12 0.482143  117        255
## 958635           GTS_2857-24 UGA_173101_P019_WA02 0.491379  119        255
## 958661           GTS_2857-24 UGA_173101_P018_WD10 0.491071  111        255
## 958718           GTS_2857-24 UGA_173101_P016_WH09 0.491071  115        255
## 958751           GTS_2857-24 UGA_173101_P017_WD09 0.490741  109        255
## 958770           GTS_2857-24 UGA_173101_P019_WA01 0.491379  121        255
## 958797           GTS_2857-24 UGA_173101_P018_WB06 0.468750   56        255
## 958821           GTS_2857-24 UGA_173101_P018_WC11 0.491379  122        255
## 958833           GTS_2857-24 UGA_173101_P023_WF05 0.480769   94        255
## 958840           GTS_2857-24 UGA_173101_P023_WB12 0.473214  111        255
## 958848           GTS_2857-24 UGA_173101_P023_WA04 0.489130   93        255
## 958894           GTS_2857-24 UGA_173101_P023_WA07 0.481481  108        255
## 958913           GTS_2857-24 UGA_173101_P022_WD02 0.450000   52        255
## 958988           GTS_2857-24 UGA_173101_P020_WH07 0.416667   78        255
## 959036           GTS_2857-24 UGA_173101_P021_WF01 0.481481  104        255
## 959075           GTS_2857-24 UGA_173101_P022_WA09 0.490000  102        255
## 959111           GTS_2857-24           GTS_949-23 0.500000  124        255
```

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## 959160            GTS_2857-24           GTS_646-23 0.500000  120        255
## 959174            GTS_2857-24           GTS_616-24 0.491379  122        255
## 959183            GTS_2857-24           GTS_655-23 0.500000  116        255
## 959184            GTS_2857-24           GTS_983-23 0.500000  121        255
## 959195            GTS_2857-24           GTS_630-24 0.500000  119        255
## 959205            GTS_2857-24           GTS_665-23 0.500000  122        255
## 959208            GTS_2857-24           GTS_997-23 0.500000  123        255
## 959214            GTS_2857-24           GTS_644-24 0.490741  119        255
## 959220            GTS_2857-24           GTS_818-23 0.500000  123        255
## 959233            GTS_2857-24           GTS_677-23 0.500000  119        255
## 959234            GTS_2857-24          GTS_1012-23 0.500000  121        255
## 959240            GTS_2857-24           GTS_657-24 0.491071  120        255
## 959259            GTS_2857-24           GTS_691-23 0.490385  119        255
## 959327            GTS_2857-24           GTS_725-23 0.500000  119        255
## 959336            GTS_2857-24           GTS_736-23 0.481481  120        255
## 959440            GTS_2857-24          GTS_1202-24 0.482759  122        255
## 959458            GTS_2857-24           GTS_850-23 0.491667  124        255
## 959492            GTS_2857-24           GTS_825-24 0.491071  120        255
## 959505            GTS_2857-24           GTS_935-23 0.500000  123        255
## 959515            GTS_2857-24          GTS_935-23R 0.491379  123        255
## 959533            GTS_2857-24           GTS_855-24 0.500000  123        255
## 959545            GTS_2857-24          GTS_1344-24 0.500000  123        255
## 959582            GTS_2857-24           GTS_886-24 0.500000  124        255
## 959596            GTS_2857-24          GTS_1403-24 0.500000  122        255
## 959613            GTS_2857-24           GTS_931-24 0.500000  119        255
## 959617            GTS_2857-24          GTS_1428-24 0.500000  123        255
## 959626            GTS_2857-24          GTS_3012-24 0.500000  122        255
## 959642            GTS_2857-24          GTS_1451-24 0.500000  123        255
## 959651            GTS_2857-24           GTS_955-24 0.500000  122        255
## 959678            GTS_2857-24          GTS_1516-24 0.491071  120        255
## 959715            GTS_2857-24          GTS_2777-24 0.500000  121        255
## 959744            GTS_2857-24          GTS_1633-24 0.482759  124        255
## 959753            GTS_2857-24          GTS_1672-24 0.500000  121        255
## 959776            GTS_2857-24          GTS_2376-24 0.500000  123        255
## 959798            GTS_2857-24          GTS_2669-24 0.500000  123        255
## 987755            GTS_2697-24 UGA_173101_P019_WE05 0.490385  114        255
## 987780            GTS_2697-24 UGA_173101_P019_WA04 0.488636   91        255
## 987821            GTS_2697-24 UGA_173101_P016_WC12 0.491071  109        255
## 987822            GTS_2697-24 UGA_173101_P017_WC02 0.482759  116        255
## 987829            GTS_2697-24 UGA_173101_P018_WB04 0.490741  117        255
## 987832            GTS_2697-24 UGA_173101_P016_WF07 0.461538  102        255
## 987876            GTS_2697-24 UGA_173101_P019_WG03 0.491071  120        255
## 987881            GTS_2697-24 UGA_173101_P016_WE08 0.491071  116        255
## 987910            GTS_2697-24 UGA_173101_P017_WC12 0.481481  116        255
## 987930            GTS_2697-24 UGA_173101_P019_WA02 0.491071  118        255
## 987956            GTS_2697-24 UGA_173101_P018_WD10 0.490741  109        255
## 988013            GTS_2697-24 UGA_173101_P016_WH09 0.490741  114        255
## 988046            GTS_2697-24 UGA_173101_P017_WD09 0.490385  108        255
## 988065            GTS_2697-24 UGA_173101_P019_WA01 0.491071  119        255
## 988092            GTS_2697-24 UGA_173101_P018_WB06 0.466667   55        255
## 988116            GTS_2697-24 UGA_173101_P018_WC11 0.491071  120        255
## 988128            GTS_2697-24 UGA_173101_P023_WF05 0.480000   93        255
## 988135            GTS_2697-24 UGA_173101_P023_WB12 0.472222  109        255
## 988143            GTS_2697-24 UGA_173101_P023_WA04 0.488636   92        255
```

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## 988189           GTS_2697-24 UGA_173101_P023_WA07 0.480769  107        255
## 988208           GTS_2697-24 UGA_173101_P022_WD02 0.444444   51        255
## 988283           GTS_2697-24 UGA_173101_P020_WH07 0.412500   77        255
## 988331           GTS_2697-24 UGA_173101_P021_WF01 0.480769  102        255
## 988370           GTS_2697-24 UGA_173101_P022_WA09 0.489583  100        255
## 988406           GTS_2697-24           GTS_949-23 0.500000  122        255
## 988455           GTS_2697-24           GTS_646-23 0.500000  118        255
## 988469           GTS_2697-24           GTS_616-24 0.491071  121        255
## 988478           GTS_2697-24           GTS_655-23 0.500000  115        255
## 988479           GTS_2697-24           GTS_983-23 0.500000  120        255
## 988490           GTS_2697-24           GTS_630-24 0.500000  118        255
## 988500           GTS_2697-24           GTS_665-23 0.500000  121        255
## 988503           GTS_2697-24           GTS_997-23 0.500000  121        255
## 988509           GTS_2697-24           GTS_644-24 0.490385  118        255
## 988515           GTS_2697-24           GTS_818-23 0.500000  121        255
## 988528           GTS_2697-24           GTS_677-23 0.500000  119        255
## 988529           GTS_2697-24          GTS_1012-23 0.500000  120        255
## 988535           GTS_2697-24           GTS_657-24 0.490741  119        255
## 988554           GTS_2697-24           GTS_691-23 0.490385  119        255
## 988622           GTS_2697-24           GTS_725-23 0.500000  118        255
## 988631           GTS_2697-24           GTS_736-23 0.481481  120        255
## 988735           GTS_2697-24          GTS_1202-24 0.491379  120        255
## 988753           GTS_2697-24           GTS_850-23 0.500000  122        255
## 988787           GTS_2697-24           GTS_825-24 0.500000  119        255
## 988800           GTS_2697-24           GTS_935-23 0.500000  121        255
## 988810           GTS_2697-24          GTS_935-23R 0.491379  122        255
## 988828           GTS_2697-24           GTS_855-24 0.500000  121        255
## 988840           GTS_2697-24          GTS_1344-24 0.500000  121        255
## 988877           GTS_2697-24           GTS_886-24 0.500000  122        255
## 988891           GTS_2697-24          GTS_1403-24 0.500000  121        255
## 988908           GTS_2697-24           GTS_931-24 0.500000  119        255
## 988912           GTS_2697-24          GTS_1428-24 0.500000  121        255
## 988921           GTS_2697-24          GTS_3012-24 0.500000  120        255
## 988937           GTS_2697-24          GTS_1451-24 0.500000  121        255
## 988946           GTS_2697-24           GTS_955-24 0.500000  121        255
## 988973           GTS_2697-24          GTS_1516-24 0.491071  120        255
## 989010           GTS_2697-24          GTS_2777-24 0.500000  120        255
## 989039           GTS_2697-24          GTS_1633-24 0.491379  122        255
## 989048           GTS_2697-24          GTS_1672-24 0.500000  121        255
## 989071           GTS_2697-24          GTS_2376-24 0.500000  122        255
## 989093           GTS_2697-24          GTS_2669-24 0.500000  121        255
## 989101           GTS_2697-24          GTS_2857-24 0.500000  122        255
## 1014640          GTS_2906-24 UGA_173101_P019_WE05 0.490741  115        255
## 1014665          GTS_2906-24 UGA_173101_P019_WA04 0.489130   92        255
## 1014706          GTS_2906-24 UGA_173101_P016_WC12 0.491379  110        255
## 1014707          GTS_2906-24 UGA_173101_P017_WC02 0.483333  117        255
## 1014714          GTS_2906-24 UGA_173101_P018_WB04 0.491071  118        255
## 1014717          GTS_2906-24 UGA_173101_P016_WF07 0.462963  103        255
## 1014761          GTS_2906-24 UGA_173101_P019_WG03 0.491379  122        255
## 1014766          GTS_2906-24 UGA_173101_P016_WE08 0.491379  117        255
## 1014795          GTS_2906-24 UGA_173101_P017_WC12 0.482143  117        255
## 1014815          GTS_2906-24 UGA_173101_P019_WA02 0.491379  119        255
## 1014841          GTS_2906-24 UGA_173101_P018_WD10 0.491071  111        255
## 1014898          GTS_2906-24 UGA_173101_P016_WH09 0.491071  115        255
```

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## 1014931          GTS_2906-24 UGA_173101_P017_WD09 0.490741  109        255
## 1014950          GTS_2906-24 UGA_173101_P019_WA01 0.491379  121        255
## 1014977          GTS_2906-24 UGA_173101_P018_WB06 0.468750   56        255
## 1015001          GTS_2906-24 UGA_173101_P018_WC11 0.491379  122        255
## 1015013          GTS_2906-24 UGA_173101_P023_WF05 0.480769   94        255
## 1015020          GTS_2906-24 UGA_173101_P023_WB12 0.473214  111        255
## 1015028          GTS_2906-24 UGA_173101_P023_WA04 0.489130   93        255
## 1015074          GTS_2906-24 UGA_173101_P023_WA07 0.481481  108        255
## 1015093          GTS_2906-24 UGA_173101_P022_WD02 0.450000   52        255
## 1015168          GTS_2906-24 UGA_173101_P020_WH07 0.416667   78        255
## 1015216          GTS_2906-24 UGA_173101_P021_WF01 0.481481  104        255
## 1015255          GTS_2906-24 UGA_173101_P022_WA09 0.490000  102        255
## 1015291          GTS_2906-24           GTS_949-23 0.500000  124        255
## 1015340          GTS_2906-24           GTS_646-23 0.500000  120        255
## 1015354          GTS_2906-24           GTS_616-24 0.491379  122        255
## 1015363          GTS_2906-24           GTS_655-23 0.500000  116        255
## 1015364          GTS_2906-24           GTS_983-23 0.500000  121        255
## 1015375          GTS_2906-24           GTS_630-24 0.500000  119        255
## 1015385          GTS_2906-24           GTS_665-23 0.500000  122        255
## 1015388          GTS_2906-24           GTS_997-23 0.500000  123        255
## 1015394          GTS_2906-24           GTS_644-24 0.490741  119        255
## 1015400          GTS_2906-24           GTS_818-23 0.500000  123        255
## 1015413          GTS_2906-24           GTS_677-23 0.500000  119        255
## 1015414          GTS_2906-24          GTS_1012-23 0.500000  121        255
## 1015420          GTS_2906-24           GTS_657-24 0.491071  120        255
## 1015439          GTS_2906-24           GTS_691-23 0.490385  119        255
## 1015507          GTS_2906-24           GTS_725-23 0.500000  119        255
## 1015516          GTS_2906-24           GTS_736-23 0.481481  120        255
## 1015620          GTS_2906-24          GTS_1202-24 0.482759  122        255
## 1015638          GTS_2906-24           GTS_850-23 0.491667  124        255
## 1015672          GTS_2906-24           GTS_825-24 0.491071  120        255
## 1015685          GTS_2906-24           GTS_935-23 0.500000  123        255
## 1015695          GTS_2906-24          GTS_935-23R 0.491379  123        255
## 1015713          GTS_2906-24           GTS_855-24 0.500000  123        255
## 1015725          GTS_2906-24          GTS_1344-24 0.500000  123        255
## 1015762          GTS_2906-24           GTS_886-24 0.500000  124        255
## 1015776          GTS_2906-24          GTS_1403-24 0.500000  122        255
## 1015793          GTS_2906-24           GTS_931-24 0.500000  119        255
## 1015797          GTS_2906-24          GTS_1428-24 0.500000  123        255
## 1015806          GTS_2906-24          GTS_3012-24 0.500000  122        255
## 1015822          GTS_2906-24          GTS_1451-24 0.500000  123        255
## 1015831          GTS_2906-24           GTS_955-24 0.500000  122        255
## 1015858          GTS_2906-24          GTS_1516-24 0.491071  120        255
## 1015895          GTS_2906-24          GTS_2777-24 0.500000  121        255
## 1015924          GTS_2906-24          GTS_1633-24 0.482759  124        255
## 1015933          GTS_2906-24          GTS_1672-24 0.500000  121        255
## 1015956          GTS_2906-24          GTS_2376-24 0.500000  123        255
## 1015978          GTS_2906-24          GTS_2669-24 0.500000  123        255
## 1015986          GTS_2906-24          GTS_2857-24 0.500000  124        255
## 1016007          GTS_2906-24          GTS_2697-24 0.500000  122        255
## 1094500           GTS_151-24 UGA_173101_P019_WE05 0.500000  112        255
## 1094525           GTS_151-24 UGA_173101_P019_WA04 0.500000   90        255
## 1094566           GTS_151-24 UGA_173101_P016_WC12 0.500000  107        255
## 1094567           GTS_151-24 UGA_173101_P017_WC02 0.490741  114        255
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## 1094574           GTS_151-24 UGA_173101_P018_WB04 0.500000  116        255
## 1094577           GTS_151-24 UGA_173101_P016_WF07 0.468750  101        255
## 1094621           GTS_151-24 UGA_173101_P019_WG03 0.500000  120        255
## 1094626           GTS_151-24 UGA_173101_P016_WE08 0.500000  115        255
## 1094655           GTS_151-24 UGA_173101_P017_WC12 0.490385  115        255
## 1094675           GTS_151-24 UGA_173101_P019_WA02 0.500000  117        255
## 1094701           GTS_151-24 UGA_173101_P018_WD10 0.500000  108        255
## 1094758           GTS_151-24 UGA_173101_P016_WH09 0.500000  112        255
## 1094791           GTS_151-24 UGA_173101_P017_WD09 0.500000  106        255
## 1094810           GTS_151-24 UGA_173101_P019_WA01 0.500000  119        255
## 1094837           GTS_151-24 UGA_173101_P018_WB06 0.483333   54        255
## 1094861           GTS_151-24 UGA_173101_P018_WC11 0.500000  120        255
## 1094873           GTS_151-24 UGA_173101_P023_WF05 0.489130   91        255
## 1094880           GTS_151-24 UGA_173101_P023_WB12 0.480000  109        255
## 1094888           GTS_151-24 UGA_173101_P023_WA04 0.500000   90        255
## 1094934           GTS_151-24 UGA_173101_P023_WA07 0.490000  105        255
## 1094953           GTS_151-24 UGA_173101_P022_WD02 0.472222   50        255
## 1095028           GTS_151-24 UGA_173101_P020_WH07 0.434211   75        255
## 1095076           GTS_151-24 UGA_173101_P021_WF01 0.489583  101        255
## 1095115           GTS_151-24 UGA_173101_P022_WA09 0.500000  100        255
## 1095151           GTS_151-24           GTS_949-23 0.500000  121        255
## 1095200           GTS_151-24           GTS_646-23 0.500000  118        255
## 1095214           GTS_151-24           GTS_616-24 0.490385  119        255
## 1095223           GTS_151-24           GTS_655-23 0.500000  113        255
## 1095224           GTS_151-24           GTS_983-23 0.500000  120        255
## 1095235           GTS_151-24           GTS_630-24 0.500000  117        255
## 1095245           GTS_151-24           GTS_665-23 0.500000  119        255
## 1095248           GTS_151-24           GTS_997-23 0.500000  120        255
## 1095254           GTS_151-24           GTS_644-24 0.490000  117        255
## 1095260           GTS_151-24           GTS_818-23 0.500000  120        255
## 1095273           GTS_151-24           GTS_677-23 0.500000  117        255
## 1095274           GTS_151-24          GTS_1012-23 0.500000  120        255
## 1095280           GTS_151-24           GTS_657-24 0.490385  118        255
## 1095299           GTS_151-24           GTS_691-23 0.500000  117        255
## 1095367           GTS_151-24           GTS_725-23 0.500000  117        255
## 1095376           GTS_151-24           GTS_736-23 0.480000  118        255
## 1095480           GTS_151-24          GTS_1202-24 0.500000  119        255
## 1095498           GTS_151-24           GTS_850-23 0.500000  121        255
## 1095532           GTS_151-24           GTS_825-24 0.500000  117        255
## 1095545           GTS_151-24           GTS_935-23 0.500000  121        255
## 1095555           GTS_151-24          GTS_935-23R 0.500000  121        255
## 1095573           GTS_151-24           GTS_855-24 0.500000  121        255
## 1095585           GTS_151-24          GTS_1344-24 0.500000  121        255
## 1095622           GTS_151-24           GTS_886-24 0.500000  121        255
## 1095636           GTS_151-24          GTS_1403-24 0.500000  121        255
## 1095653           GTS_151-24           GTS_931-24 0.500000  118        255
## 1095657           GTS_151-24          GTS_1428-24 0.500000  121        255
## 1095666           GTS_151-24          GTS_3012-24 0.500000  120        255
## 1095682           GTS_151-24          GTS_1451-24 0.500000  121        255
## 1095691           GTS_151-24           GTS_955-24 0.500000  121        255
## 1095718           GTS_151-24          GTS_1516-24 0.490385  118        255
## 1095755           GTS_151-24          GTS_2777-24 0.500000  119        255
## 1095784           GTS_151-24          GTS_1633-24 0.500000  121        255
## 1095793           GTS_151-24          GTS_1672-24 0.500000  120        255
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## 1095816           GTS_151-24          GTS_2376-24 0.500000  121        255
## 1095838           GTS_151-24          GTS_2669-24 0.500000  121        255
## 1095846           GTS_151-24          GTS_2857-24 0.500000  121        255
## 1095867           GTS_151-24          GTS_2697-24 0.500000  120        255
## 1095886           GTS_151-24          GTS_2906-24 0.500000  121        255
## 1110835           GTS_426-23 UGA_173101_P019_WE05 0.500000  110        255
## 1110860           GTS_426-23 UGA_173101_P019_WA04 0.500000   87        255
## 1110901           GTS_426-23 UGA_173101_P016_WC12 0.500000  105        255
## 1110902           GTS_426-23 UGA_173101_P017_WC02 0.490385  112        255
## 1110909           GTS_426-23 UGA_173101_P018_WB04 0.500000  113        255
## 1110912           GTS_426-23 UGA_173101_P016_WF07 0.467391   99        255
## 1110956           GTS_426-23 UGA_173101_P019_WG03 0.500000  117        255
## 1110961           GTS_426-23 UGA_173101_P016_WE08 0.500000  112        255
## 1110990           GTS_426-23 UGA_173101_P017_WC12 0.490000  112        255
## 1111010           GTS_426-23 UGA_173101_P019_WA02 0.500000  114        255
## 1111036           GTS_426-23 UGA_173101_P018_WD10 0.500000  107        255
## 1111093           GTS_426-23 UGA_173101_P016_WH09 0.500000  110        255
## 1111126           GTS_426-23 UGA_173101_P017_WD09 0.500000  105        255
## 1111145           GTS_426-23 UGA_173101_P019_WA01 0.500000  116        255
## 1111172           GTS_426-23 UGA_173101_P018_WB06 0.482143   53        255
## 1111196           GTS_426-23 UGA_173101_P018_WC11 0.500000  117        255
## 1111208           GTS_426-23 UGA_173101_P023_WF05 0.488636   89        255
## 1111215           GTS_426-23 UGA_173101_P023_WB12 0.479167  106        255
## 1111223           GTS_426-23 UGA_173101_P023_WA04 0.500000   88        255
## 1111269           GTS_426-23 UGA_173101_P023_WA07 0.500000  103        255
## 1111288           GTS_426-23 UGA_173101_P022_WD02 0.472222   50        255
## 1111363           GTS_426-23 UGA_173101_P020_WH07 0.434211   74        255
## 1111411           GTS_426-23 UGA_173101_P021_WF01 0.489583  100        255
## 1111450           GTS_426-23 UGA_173101_P022_WA09 0.500000   97        255
## 1111486           GTS_426-23           GTS_949-23 0.500000  119        255
## 1111535           GTS_426-23           GTS_646-23 0.500000  116        255
## 1111549           GTS_426-23           GTS_616-24 0.490000  117        255
## 1111558           GTS_426-23           GTS_655-23 0.500000  111        255
## 1111559           GTS_426-23           GTS_983-23 0.500000  118        255
## 1111570           GTS_426-23           GTS_630-24 0.500000  116        255
## 1111580           GTS_426-23           GTS_665-23 0.500000  117        255
## 1111583           GTS_426-23           GTS_997-23 0.500000  118        255
## 1111589           GTS_426-23           GTS_644-24 0.490000  116        255
## 1111595           GTS_426-23           GTS_818-23 0.500000  118        255
## 1111608           GTS_426-23           GTS_677-23 0.500000  116        255
## 1111609           GTS_426-23          GTS_1012-23 0.500000  118        255
## 1111615           GTS_426-23           GTS_657-24 0.490000  116        255
## 1111634           GTS_426-23           GTS_691-23 0.500000  116        255
## 1111702           GTS_426-23           GTS_725-23 0.500000  116        255
## 1111711           GTS_426-23           GTS_736-23 0.490000  116        255
## 1111815           GTS_426-23          GTS_1202-24 0.500000  117        255
## 1111833           GTS_426-23           GTS_850-23 0.500000  119        255
## 1111867           GTS_426-23           GTS_825-24 0.500000  116        255
## 1111880           GTS_426-23           GTS_935-23 0.500000  119        255
## 1111890           GTS_426-23          GTS_935-23R 0.500000  119        255
## 1111908           GTS_426-23           GTS_855-24 0.500000  119        255
## 1111920           GTS_426-23          GTS_1344-24 0.500000  119        255
## 1111957           GTS_426-23           GTS_886-24 0.500000  119        255
## 1111971           GTS_426-23          GTS_1403-24 0.500000  119        255
```

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## 1111988           GTS_426-23           GTS_931-24 0.500000  116        255
## 1111992           GTS_426-23          GTS_1428-24 0.500000  119        255
## 1112001           GTS_426-23          GTS_3012-24 0.500000  118        255
## 1112017           GTS_426-23          GTS_1451-24 0.500000  119        255
## 1112026           GTS_426-23           GTS_955-24 0.500000  119        255
## 1112053           GTS_426-23          GTS_1516-24 0.490385  117        255
## 1112090           GTS_426-23          GTS_2777-24 0.500000  117        255
## 1112119           GTS_426-23          GTS_1633-24 0.500000  119        255
## 1112128           GTS_426-23          GTS_1672-24 0.500000  118        255
## 1112151           GTS_426-23          GTS_2376-24 0.500000  119        255
## 1112173           GTS_426-23          GTS_2669-24 0.500000  119        255
## 1112181           GTS_426-23          GTS_2857-24 0.500000  119        255
## 1112202           GTS_426-23          GTS_2697-24 0.500000  118        255
## 1112221           GTS_426-23          GTS_2906-24 0.500000  119        255
## 1112276           GTS_426-23           GTS_151-24 0.500000  119        255
## 1127291          GTS_151-24R UGA_173101_P019_WE05 0.500000  109         33
## 1127316          GTS_151-24R UGA_173101_P019_WA04 0.500000   89         33
## 1127357          GTS_151-24R UGA_173101_P016_WC12 0.500000  104         33
## 1127358          GTS_151-24R UGA_173101_P017_WC02 0.490385  111         33
## 1127365          GTS_151-24R UGA_173101_P018_WB04 0.500000  113         33
## 1127368          GTS_151-24R UGA_173101_P016_WF07 0.478261   99         33
## 1127412          GTS_151-24R UGA_173101_P019_WG03 0.500000  117         33
## 1127417          GTS_151-24R UGA_173101_P016_WE08 0.500000  112         33
## 1127446          GTS_151-24R UGA_173101_P017_WC12 0.490000  112         33
## 1127466          GTS_151-24R UGA_173101_P019_WA02 0.500000  114         33
## 1127492          GTS_151-24R UGA_173101_P018_WD10 0.500000  106         33
## 1127549          GTS_151-24R UGA_173101_P016_WH09 0.500000  109         33
## 1127582          GTS_151-24R UGA_173101_P017_WD09 0.500000  104         33
## 1127601          GTS_151-24R UGA_173101_P019_WA01 0.500000  116         33
## 1127628          GTS_151-24R UGA_173101_P018_WB06 0.482143   53         33
## 1127652          GTS_151-24R UGA_173101_P018_WC11 0.500000  117         33
## 1127664          GTS_151-24R UGA_173101_P023_WF05 0.489130   89         33
## 1127671          GTS_151-24R UGA_173101_P023_WB12 0.479167  106         33
## 1127679          GTS_151-24R UGA_173101_P023_WA04 0.500000   88         33
## 1127725          GTS_151-24R UGA_173101_P023_WA07 0.489583  102         33
## 1127744          GTS_151-24R UGA_173101_P022_WD02 0.472222   49         33
## 1127819          GTS_151-24R UGA_173101_P020_WH07 0.421053   73         33
## 1127867          GTS_151-24R UGA_173101_P021_WF01 0.489130   98         33
## 1127906          GTS_151-24R UGA_173101_P022_WA09 0.500000   97         33
## 1127942          GTS_151-24R           GTS_949-23 0.500000  117         33
## 1127991          GTS_151-24R           GTS_646-23 0.500000  116         33
## 1128005          GTS_151-24R           GTS_616-24 0.500000  115         33
## 1128014          GTS_151-24R           GTS_655-23 0.500000  111         33
## 1128015          GTS_151-24R           GTS_983-23 0.500000  117         33
## 1128026          GTS_151-24R           GTS_630-24 0.500000  114         33
## 1128036          GTS_151-24R           GTS_665-23 0.500000  116         33
## 1128039          GTS_151-24R           GTS_997-23 0.500000  117         33
## 1128045          GTS_151-24R           GTS_644-24 0.500000  113         33
## 1128051          GTS_151-24R           GTS_818-23 0.500000  117         33
## 1128064          GTS_151-24R           GTS_677-23 0.500000  114         33
## 1128065          GTS_151-24R          GTS_1012-23 0.500000  117         33
## 1128071          GTS_151-24R           GTS_657-24 0.490000  115         33
## 1128090          GTS_151-24R           GTS_691-23 0.500000  114         33
## 1128158          GTS_151-24R           GTS_725-23 0.500000  114         33
```

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## 1128167          GTS_151-24R          GTS_736-23 0.479167  115         33
## 1128271          GTS_151-24R         GTS_1202-24 0.500000  115         33
## 1128289          GTS_151-24R          GTS_850-23 0.500000  117         33
## 1128323          GTS_151-24R          GTS_825-24 0.500000  113         33
## 1128336          GTS_151-24R          GTS_935-23 0.500000  117         33
## 1128346          GTS_151-24R         GTS_935-23R 0.500000  117         33
## 1128364          GTS_151-24R          GTS_855-24 0.500000  117         33
## 1128376          GTS_151-24R         GTS_1344-24 0.500000  117         33
## 1128413          GTS_151-24R          GTS_886-24 0.500000  117         33
## 1128427          GTS_151-24R         GTS_1403-24 0.500000  117         33
## 1128444          GTS_151-24R          GTS_931-24 0.500000  114         33
## 1128448          GTS_151-24R         GTS_1428-24 0.500000  117         33
## 1128457          GTS_151-24R         GTS_3012-24 0.500000  116         33
## 1128473          GTS_151-24R         GTS_1451-24 0.500000  117         33
## 1128482          GTS_151-24R          GTS_955-24 0.500000  117         33
## 1128509          GTS_151-24R         GTS_1516-24 0.489583  114         33
## 1128546          GTS_151-24R         GTS_2777-24 0.500000  116         33
## 1128575          GTS_151-24R         GTS_1633-24 0.500000  117         33
## 1128584          GTS_151-24R         GTS_1672-24 0.500000  116         33
## 1128607          GTS_151-24R         GTS_2376-24 0.500000  117         33
## 1128629          GTS_151-24R         GTS_2669-24 0.500000  117         33
## 1128637          GTS_151-24R         GTS_2857-24 0.500000  117         33
## 1128658          GTS_151-24R         GTS_2697-24 0.500000  116         33
## 1128677          GTS_151-24R         GTS_2906-24 0.500000  117         33
## 1128732          GTS_151-24R          GTS_151-24 0.500000  117         33
## 1128743          GTS_151-24R          GTS_426-23 0.500000  115         33
## 1149926           GTS_146-23 UGA_173101_P019_WE05 0.500000  112         33
## 1149951           GTS_146-23 UGA_173101_P019_WA04 0.500000   90         33
## 1149992           GTS_146-23 UGA_173101_P016_WC12 0.500000  107         33
## 1149993           GTS_146-23 UGA_173101_P017_WC02 0.491071  114         33
## 1150000           GTS_146-23 UGA_173101_P018_WB04 0.500000  116         33
## 1150003           GTS_146-23 UGA_173101_P016_WF07 0.470000  102         33
## 1150047           GTS_146-23 UGA_173101_P019_WG03 0.500000  120         33
## 1150052           GTS_146-23 UGA_173101_P016_WE08 0.500000  115         33
## 1150081           GTS_146-23 UGA_173101_P017_WC12 0.490741  115         33
## 1150101           GTS_146-23 UGA_173101_P019_WA02 0.500000  117         33
## 1150127           GTS_146-23 UGA_173101_P018_WD10 0.500000  109         33
## 1150184           GTS_146-23 UGA_173101_P016_WH09 0.500000  112         33
## 1150217           GTS_146-23 UGA_173101_P017_WD09 0.500000  107         33
## 1150236           GTS_146-23 UGA_173101_P019_WA01 0.500000  119         33
## 1150263           GTS_146-23 UGA_173101_P018_WB06 0.483333   54         33
## 1150287           GTS_146-23 UGA_173101_P018_WC11 0.500000  120         33
## 1150299           GTS_146-23 UGA_173101_P023_WF05 0.489583   91         33
## 1150306           GTS_146-23 UGA_173101_P023_WB12 0.480769  109         33
## 1150314           GTS_146-23 UGA_173101_P023_WA04 0.500000   90         33
## 1150360           GTS_146-23 UGA_173101_P023_WA07 0.490385  105         33
## 1150379           GTS_146-23 UGA_173101_P022_WD02 0.472222   50         33
## 1150454           GTS_146-23 UGA_173101_P020_WH07 0.425000   75         33
## 1150502           GTS_146-23 UGA_173101_P021_WF01 0.490000  101         33
## 1150541           GTS_146-23 UGA_173101_P022_WA09 0.500000  100         33
## 1150577           GTS_146-23          GTS_949-23 0.491071  121         33
## 1150626           GTS_146-23          GTS_646-23 0.500000  118         33
## 1150640           GTS_146-23          GTS_616-24 0.462963  119         33
## 1150649           GTS_146-23          GTS_655-23 0.500000  113         33
```

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## 1150650            GTS_146-23          GTS_983-23 0.500000  120         33
## 1150661            GTS_146-23          GTS_630-24 0.500000  117         33
## 1150671            GTS_146-23          GTS_665-23 0.500000  119         33
## 1150674            GTS_146-23          GTS_997-23 0.500000  120         33
## 1150680            GTS_146-23          GTS_644-24 0.461538  117         33
## 1150686            GTS_146-23          GTS_818-23 0.500000  120         33
## 1150699            GTS_146-23          GTS_677-23 0.500000  117         33
## 1150700            GTS_146-23         GTS_1012-23 0.500000  120         33
## 1150706            GTS_146-23          GTS_657-24 0.490741  118         33
## 1150725            GTS_146-23          GTS_691-23 0.500000  117         33
## 1150793            GTS_146-23          GTS_725-23 0.500000  117         33
## 1150802            GTS_146-23          GTS_736-23 0.480769  118         33
## 1150906            GTS_146-23         GTS_1202-24 0.491071  119         33
## 1150924            GTS_146-23          GTS_850-23 0.491071  121         33
## 1150958            GTS_146-23          GTS_825-24 0.490385  117         33
## 1150971            GTS_146-23          GTS_935-23 0.491071  121         33
## 1150981            GTS_146-23         GTS_935-23R 0.491071  121         33
## 1150999            GTS_146-23          GTS_855-24 0.491071  121         33
## 1151011            GTS_146-23         GTS_1344-24 0.491071  121         33
## 1151048            GTS_146-23          GTS_886-24 0.491071  121         33
## 1151062            GTS_146-23         GTS_1403-24 0.491071  121         33
## 1151079            GTS_146-23          GTS_931-24 0.490741  118         33
## 1151083            GTS_146-23         GTS_1428-24 0.491071  121         33
## 1151092            GTS_146-23         GTS_3012-24 0.490741  120         33
## 1151108            GTS_146-23         GTS_1451-24 0.491071  121         33
## 1151117            GTS_146-23          GTS_955-24 0.491071  121         33
## 1151144            GTS_146-23         GTS_1516-24 0.481481  118         33
## 1151181            GTS_146-23         GTS_2777-24 0.500000  119         33
## 1151210            GTS_146-23         GTS_1633-24 0.491071  121         33
## 1151219            GTS_146-23         GTS_1672-24 0.491071  120         33
## 1151242            GTS_146-23         GTS_2376-24 0.491071  121         33
## 1151264            GTS_146-23         GTS_2669-24 0.491071  121         33
## 1151272            GTS_146-23         GTS_2857-24 0.491071  121         33
## 1151293            GTS_146-23         GTS_2697-24 0.491071  120         33
## 1151312            GTS_146-23         GTS_2906-24 0.491071  121         33
## 1151367            GTS_146-23          GTS_151-24 0.490741  121         33
## 1151378            GTS_146-23          GTS_426-23 0.490385  119         33
## 1151389            GTS_146-23         GTS_151-24R 0.500000  118         33
## 1348943            GTS_311-24 UGA_173101_P019_WE05 0.500000  112         33
## 1348968            GTS_311-24 UGA_173101_P019_WA04 0.500000   90         33
## 1349009            GTS_311-24 UGA_173101_P016_WC12 0.500000  107         33
## 1349010            GTS_311-24 UGA_173101_P017_WC02 0.491071  114         33
## 1349017            GTS_311-24 UGA_173101_P018_WB04 0.500000  116         33
## 1349020            GTS_311-24 UGA_173101_P016_WF07 0.470000  101         33
## 1349064            GTS_311-24 UGA_173101_P019_WG03 0.500000  120         33
## 1349069            GTS_311-24 UGA_173101_P016_WE08 0.500000  115         33
## 1349098            GTS_311-24 UGA_173101_P017_WC12 0.490741  115         33
## 1349118            GTS_311-24 UGA_173101_P019_WA02 0.500000  117         33
## 1349144            GTS_311-24 UGA_173101_P018_WD10 0.500000  109         33
## 1349201            GTS_311-24 UGA_173101_P016_WH09 0.500000  112         33
## 1349234            GTS_311-24 UGA_173101_P017_WD09 0.500000  107         33
## 1349253            GTS_311-24 UGA_173101_P019_WA01 0.500000  119         33
## 1349280            GTS_311-24 UGA_173101_P018_WB06 0.483333   53         33
## 1349304            GTS_311-24 UGA_173101_P018_WC11 0.500000  120         33
```

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## 1349316           GTS_311-24 UGA_173101_P023_WF05 0.489583   92         33
## 1349323           GTS_311-24 UGA_173101_P023_WB12 0.480769  109         33
## 1349331           GTS_311-24 UGA_173101_P023_WA04 0.500000   91         33
## 1349377           GTS_311-24 UGA_173101_P023_WA07 0.490385  105         33
## 1349396           GTS_311-24 UGA_173101_P022_WD02 0.472222   50         33
## 1349471           GTS_311-24 UGA_173101_P020_WH07 0.425000   76         33
## 1349519           GTS_311-24 UGA_173101_P021_WF01 0.490000  101         33
## 1349558           GTS_311-24 UGA_173101_P022_WA09 0.500000  100         33
## 1349594           GTS_311-24           GTS_949-23 0.500000  120         33
## 1349643           GTS_311-24           GTS_646-23 0.500000  118         33
## 1349657           GTS_311-24           GTS_616-24 0.500000  118         33
## 1349666           GTS_311-24           GTS_655-23 0.500000  114         33
## 1349667           GTS_311-24           GTS_983-23 0.500000  120         33
## 1349678           GTS_311-24           GTS_630-24 0.500000  117         33
## 1349688           GTS_311-24           GTS_665-23 0.500000  119         33
## 1349691           GTS_311-24           GTS_997-23 0.500000  120         33
## 1349697           GTS_311-24           GTS_644-24 0.500000  116         33
## 1349703           GTS_311-24           GTS_818-23 0.500000  120         33
## 1349716           GTS_311-24           GTS_677-23 0.500000  117         33
## 1349717           GTS_311-24          GTS_1012-23 0.500000  120         33
## 1349723           GTS_311-24           GTS_657-24 0.490741  118         33
## 1349742           GTS_311-24           GTS_691-23 0.500000  117         33
## 1349810           GTS_311-24           GTS_725-23 0.500000  117         33
## 1349819           GTS_311-24           GTS_736-23 0.480769  118         33
## 1349923           GTS_311-24          GTS_1202-24 0.500000  118         33
## 1349941           GTS_311-24           GTS_850-23 0.500000  120         33
## 1349975           GTS_311-24           GTS_825-24 0.500000  116         33
## 1349988           GTS_311-24           GTS_935-23 0.500000  120         33
## 1349998           GTS_311-24          GTS_935-23R 0.500000  120         33
## 1350016           GTS_311-24           GTS_855-24 0.500000  120         33
## 1350028           GTS_311-24          GTS_1344-24 0.500000  120         33
## 1350065           GTS_311-24           GTS_886-24 0.500000  120         33
## 1350079           GTS_311-24          GTS_1403-24 0.500000  120         33
## 1350096           GTS_311-24           GTS_931-24 0.500000  117         33
## 1350100           GTS_311-24          GTS_1428-24 0.500000  120         33
## 1350109           GTS_311-24          GTS_3012-24 0.500000  119         33
## 1350125           GTS_311-24          GTS_1451-24 0.500000  120         33
## 1350134           GTS_311-24           GTS_955-24 0.500000  120         33
## 1350161           GTS_311-24          GTS_1516-24 0.490385  117         33
## 1350198           GTS_311-24          GTS_2777-24 0.500000  119         33
## 1350227           GTS_311-24          GTS_1633-24 0.500000  120         33
## 1350236           GTS_311-24          GTS_1672-24 0.500000  119         33
## 1350259           GTS_311-24          GTS_2376-24 0.500000  120         33
## 1350281           GTS_311-24          GTS_2669-24 0.500000  120         33
## 1350289           GTS_311-24          GTS_2857-24 0.500000  120         33
## 1350310           GTS_311-24          GTS_2697-24 0.500000  119         33
## 1350329           GTS_311-24          GTS_2906-24 0.500000  120         33
## 1350384           GTS_311-24           GTS_151-24 0.500000  120         33
## 1350395           GTS_311-24           GTS_426-23 0.500000  117         33
## 1350406           GTS_311-24          GTS_151-24R 0.500000  117         33
## 1350421           GTS_311-24           GTS_146-23 0.500000  120         33
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## 43996           33          TRUE        TRUE
```

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## 1150217         33          TRUE        TRUE
## 1150236         33          TRUE        TRUE
## 1150263         33          TRUE        TRUE
## 1150287         33          TRUE        TRUE
## 1150299         33          TRUE        TRUE
```

```
## 1150306          33          TRUE         TRUE
## 1150314          33          TRUE         TRUE
## 1150360          33          TRUE         TRUE
## 1150379          33          TRUE         TRUE
## 1150454          33          TRUE         TRUE
## 1150502          33          TRUE         TRUE
## 1150541          33          TRUE         TRUE
## 1150577         255          TRUE        FALSE
## 1150626         255          TRUE        FALSE
## 1150640         276          TRUE        FALSE
## 1150649         255          TRUE        FALSE
## 1150650         255          TRUE        FALSE
## 1150661         255          TRUE        FALSE
## 1150671         255          TRUE        FALSE
## 1150674         255          TRUE        FALSE
## 1150680         276          TRUE        FALSE
## 1150686         255          TRUE        FALSE
## 1150699         255          TRUE        FALSE
## 1150700          33          TRUE         TRUE
## 1150706         276          TRUE        FALSE
## 1150725         289          TRUE        FALSE
## 1150793          33          TRUE         TRUE
## 1150802         276          TRUE        FALSE
## 1150906         289          TRUE        FALSE
## 1150924         301          TRUE        FALSE
## 1150958         301          TRUE        FALSE
## 1150971         255          TRUE        FALSE
## 1150981         289          TRUE        FALSE
## 1150999         255          TRUE        FALSE
## 1151011         255          TRUE        FALSE
## 1151048         255          TRUE        FALSE
## 1151062         255          TRUE        FALSE
## 1151079         255          TRUE        FALSE
## 1151083         255          TRUE        FALSE
## 1151092         255          TRUE        FALSE
## 1151108         255          TRUE        FALSE
## 1151117         255          TRUE        FALSE
## 1151144         276          TRUE        FALSE
## 1151181         255          TRUE        FALSE
## 1151210         289          TRUE        FALSE
## 1151219         255          TRUE        FALSE
## 1151242         255          TRUE        FALSE
## 1151264         255          TRUE        FALSE
## 1151272         255          TRUE        FALSE
## 1151293         255          TRUE        FALSE
## 1151312         255          TRUE        FALSE
## 1151367         255          TRUE        FALSE
## 1151378         255          TRUE        FALSE
## 1151389          33          TRUE         TRUE
## 1348943          33          TRUE         TRUE
## 1348968          33          TRUE         TRUE
## 1349009          33          TRUE         TRUE
## 1349010          33          TRUE         TRUE
## 1349017          33          TRUE         TRUE
```

```
## 1349020          33            TRUE         TRUE
## 1349064          33            TRUE         TRUE
## 1349069          33            TRUE         TRUE
## 1349098          33            TRUE         TRUE
## 1349118          33            TRUE         TRUE
## 1349144          33            TRUE         TRUE
## 1349201          33            TRUE         TRUE
## 1349234          33            TRUE         TRUE
## 1349253          33            TRUE         TRUE
## 1349280          33            TRUE         TRUE
## 1349304          33            TRUE         TRUE
## 1349316          33            TRUE         TRUE
## 1349323          33            TRUE         TRUE
## 1349331          33            TRUE         TRUE
## 1349377          33            TRUE         TRUE
## 1349396          33            TRUE         TRUE
## 1349471          33            TRUE         TRUE
## 1349519          33            TRUE         TRUE
## 1349558          33            TRUE         TRUE
## 1349594         255            TRUE        FALSE
## 1349643         255            TRUE        FALSE
## 1349657         276            TRUE        FALSE
## 1349666         255            TRUE        FALSE
## 1349667         255            TRUE        FALSE
## 1349678         255            TRUE        FALSE
## 1349688         255            TRUE        FALSE
## 1349691         255            TRUE        FALSE
## 1349697         276            TRUE        FALSE
## 1349703         255            TRUE        FALSE
## 1349716         255            TRUE        FALSE
## 1349717          33            TRUE         TRUE
## 1349723         276            TRUE        FALSE
## 1349742         289            TRUE        FALSE
## 1349810          33            TRUE         TRUE
## 1349819         276            TRUE        FALSE
## 1349923         289            TRUE        FALSE
## 1349941         301            TRUE        FALSE
## 1349975         301            TRUE        FALSE
## 1349988         255            TRUE        FALSE
## 1349998         289            TRUE        FALSE
## 1350016         255            TRUE        FALSE
## 1350028         255            TRUE        FALSE
## 1350065         255            TRUE        FALSE
## 1350079         255            TRUE        FALSE
## 1350096         255            TRUE        FALSE
## 1350100         255            TRUE        FALSE
## 1350109         255            TRUE        FALSE
## 1350125         255            TRUE        FALSE
## 1350134         255            TRUE        FALSE
## 1350161         276            TRUE        FALSE
## 1350198         255            TRUE        FALSE
## 1350227         289            TRUE        FALSE
## 1350236         255            TRUE        FALSE
## 1350259         255            TRUE        FALSE
```

```
## 1350281       255         TRUE       FALSE
## 1350289       255         TRUE       FALSE
## 1350310       255         TRUE       FALSE
## 1350329       255         TRUE       FALSE
## 1350384       255         TRUE       FALSE
## 1350395       255         TRUE       FALSE
## 1350406        33         TRUE        TRUE
## 1350421        33         TRUE        TRUE
```

```
#Clone 45 - 5 F, 1 M, direct capture confirms majority
table(smd$sexGT[smd$finalClone == 45])
```

```
##
## F M
## 5 1
```

```
smd[which(smd$finalClone == 45),]
```

```
##                                        extID   ugaID is.SCR is.rep week year
## 49                    UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                   UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                   UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>           <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                       fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                        <NA>
## 1598                                                                        <NA>
## 1819                                                                        <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F     Capture    CGA    892 UGA_173101_P015_WE01 FALSE       LGC
## 492               Hair Snare    CGA    915 UGA_173101_P020_WC06 FALSE       LGC
## 629               Hair Snare    CGA    913 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120           GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123          GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124          GTS_2920-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 49       F       45         45
## 492      F       45         45
## 629      M       45         45
## 1048     F       45         45
## 1598     F       45         45
## 1819     F       45         45
```

```r
unique(smd$COLclone[which(smd$finalClone == 45)])
```

```
## [1] 45
```

```r
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 45)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 45)],]
```

```
##                          id1                  id2  kinship nsnp colclone1
## 20176   UGA_173101_P015_WE01 UGA_173101_P020_WC06 0.492386  890        45
## 176791  UGA_173101_P021_WF11 UGA_173101_P020_WC06 0.464815  908        45
## 176917  UGA_173101_P021_WF11 UGA_173101_P015_WE01 0.461832  888        45
## 340801            GTS_664-24 UGA_173101_P020_WC06 0.483333  113        45
## 340927            GTS_664-24 UGA_173101_P015_WE01 0.481481  101        45
## 341321            GTS_664-24 UGA_173101_P021_WF11 0.474138  109        45
## 793246           GTS_3057-24 UGA_173101_P020_WC06 0.484375  115        45
## 793372           GTS_3057-24 UGA_173101_P015_WE01 0.482143  102        45
## 793766           GTS_3057-24 UGA_173101_P021_WF11 0.475806  112        45
## 793997           GTS_3057-24           GTS_664-24 0.500000  120        45
## 1028971          GTS_2920-24 UGA_173101_P020_WC06 0.453125  116        45
## 1029097          GTS_2920-24 UGA_173101_P015_WE01 0.446429  103        45
## 1029491          GTS_2920-24 UGA_173101_P021_WF11 0.443548  113        45
## 1029722          GTS_2920-24           GTS_664-24 0.500000  120        45
## 1030156          GTS_2920-24          GTS_3057-24 0.500000  123        45
##         colclone2 isclonePLINK iscloneCOL
## 20176          45         TRUE       TRUE
## 176791         45         TRUE       TRUE
## 176917         45         TRUE       TRUE
## 340801         45         TRUE       TRUE
## 340927         45         TRUE       TRUE
## 341321         45         TRUE       TRUE
## 793246         45         TRUE       TRUE
## 793372         45         TRUE       TRUE
## 793766         45         TRUE       TRUE
## 793997         45         TRUE       TRUE
## 1028971        45         TRUE       TRUE
## 1029097        45         TRUE       TRUE
## 1029491        45         TRUE       TRUE
## 1029722        45         TRUE       TRUE
## 1030156        45         TRUE       TRUE
```

```r
#Clone 284 - 23 F, 1 M, direct capture confirms F
table(smd$sexGT[smd$finalClone == 284])
```

```
##
##  F  M
## 23  1
```

```r
smd[which(smd$finalClone == 284),]
```

```
##                               extID   ugaID is.SCR is.rep week year
## 86                UGA_173101_P015_WH02    45-C  FALSE  FALSE <NA> 2024
```

```
## 169                  UGA_173101_P016_WG01  448-24   TRUE  FALSE    5 2024
## 209                  UGA_173101_P017_WB05  497-24   TRUE  FALSE    2 2024
## 426                  UGA_173101_P019_WE12  154-24   TRUE  FALSE    3 2024
## 698                  UGA_173101_P022_WD08  341-24   TRUE  FALSE    3 2024
## 784                  UGA_173101_P023_WC10  205-23   TRUE  FALSE    7 2023
## 1014  GTseek_024_384i5_188_P098.004_649-24  649-24   TRUE  FALSE    2 2024
## 1114  GTseek_024_384i5_288_P098.004_685-24  685-24   TRUE  FALSE    2 2024
## 1194  GTseek_024_384i5_368_P098.003_558-24  558-24   TRUE  FALSE    2 2024
## 1287  GTseek_025_384i5_077_P098.005_781-24  781-24   TRUE  FALSE    2 2024
## 1298 GTseek_025_384i5_088_P098.008_1208-24 1208-24   TRUE  FALSE    2 2024
## 1324 GTseek_025_384i5_114_P098.008_1226-24 1226-24   TRUE  FALSE    2 2024
## 1462 GTseek_025_384i5_252_P098.008_1379-24 1379-24   TRUE  FALSE    3 2024
## 1564 GTseek_025_384i5_354_P098.007_1127-24 1127-24   TRUE  FALSE    2 2024
## 1582 GTseek_025_384i5_372_P098.008_1488-24 1488-24   TRUE  FALSE    5 2024
## 1588 GTseek_025_384i5_378_P098.008_1500-24 1500-24   TRUE  FALSE    3 2024
## 1599 GTseek_026_384i5_005_P098.009_1528-24 1528-24   TRUE  FALSE    3 2024
## 1764 GTseek_026_384i5_261_P098.009_2680-24 2680-24   TRUE  FALSE    3 2024
## 1765 GTseek_026_384i5_263_P098.009_2682-24 2682-24   TRUE  FALSE    4 2024
## 1769 GTseek_026_384i5_271_P098.009_2687-24 2687-24   TRUE  FALSE    6 2024
## 1791 GTseek_026_384i5_315_P098.010_2890-24 2890-24   TRUE  FALSE    2 2024
## 1823 GTseek_026_384i5_379_P098.010_2928-24 2928-24   TRUE  FALSE    6 2024
## 2005 GTseek_028_384i5_180_P098.UGA4_430-24  430-24   TRUE  FALSE    2 2024
## 2059 GTseek_028_384i5_234_P098.UGA3_228-24  228-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 86     <NA>           <NA>
## 169  HS 051 260705x3599596
## 209  HS 048 261267x3598930
## 426  HS 048 261267x3598930
## 698  HS 048 261267x3598930
## 784  HS 051 260705x3599596
## 1014 HS 051 260705x3599596
## 1114 HS 048 261267x3598930
## 1194 HS 051 260705x3599596
## 1287 HS 048 261267x3598930
## 1298 HS 051 260705x3599596
## 1324 HS 048 261267x3598930
## 1462 HS 048 261267x3598930
## 1564 HS 051 260705x3599596
## 1582 HS 051 260705x3599596
## 1588 HS 048 261267x3598930
## 1599 HS 048 261267x3598930
## 1764 HS 048 261267x3598930
## 1765 HS 051 260705x3599596
## 1769 HS 048 261267x3598930
## 1791 HS 051 260705x3599596
## 1823 HS 048 261267x3598930
## 2005 HS 051 260705x3599596
## 2059 HS 051 260705x3599596
##                                                                          fqfile
## 86    RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169  RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209  RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426  RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
## 698   RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
```

```
## 784   RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
## 1014                                                                          <NA>
## 1114                                                                          <NA>
## 1194                                                                          <NA>
## 1287                                                                          <NA>
## 1298                                                                          <NA>
## 1324                                                                          <NA>
## 1462                                                                          <NA>
## 1564                                                                          <NA>
## 1582                                                                          <NA>
## 1588                                                                          <NA>
## 1599                                                                          <NA>
## 1764                                                                          <NA>
## 1765                                                                          <NA>
## 1769                                                                          <NA>
## 1791                                                                          <NA>
## 1823                                                                          <NA>
## 2005                                                                          <NA>
## 2059                                                                          <NA>
##      knownSEX  sampSOURCE region nSNPgt               vcfID  FILT GTservice
## 86          F     Capture    CGA    882 UGA_173101_P015_WH02 FALSE       LGC
## 169           Hair Snare    CGA    621 UGA_173101_P016_WG01 FALSE       LGC
## 209           Hair Snare    CGA    923 UGA_173101_P017_WB05 FALSE       LGC
## 426           Hair Snare    CGA    920 UGA_173101_P019_WE12 FALSE       LGC
## 698           Hair Snare    CGA    862 UGA_173101_P022_WD08 FALSE       LGC
## 784           Hair Snare    CGA    908 UGA_173101_P023_WC10 FALSE       LGC
## 1014     <NA> Hair Sample    CGA    122          GTS_649-24 FALSE    GTSeek
## 1114     <NA> Hair Sample    CGA    120          GTS_685-24 FALSE    GTSeek
## 1194     <NA> Hair Sample    CGA    123          GTS_558-24 FALSE    GTSeek
## 1287     <NA> Hair Sample    CGA    123          GTS_781-24 FALSE    GTSeek
## 1298     <NA> Hair Sample    CGA    124         GTS_1208-24 FALSE    GTSeek
## 1324     <NA> Hair Sample    CGA    116         GTS_1226-24 FALSE    GTSeek
## 1462     <NA> Hair Sample    CGA    125         GTS_1379-24 FALSE    GTSeek
## 1564     <NA> Hair Sample    CGA    120         GTS_1127-24 FALSE    GTSeek
## 1582     <NA> Hair Sample    CGA    122         GTS_1488-24 FALSE    GTSeek
## 1588     <NA> Hair Sample    CGA    123         GTS_1500-24 FALSE    GTSeek
## 1599     <NA> Hair Sample    CGA    122         GTS_1528-24 FALSE    GTSeek
## 1764     <NA> Hair Sample    CGA    124         GTS_2680-24 FALSE    GTSeek
## 1765     <NA> Hair Sample    CGA    124         GTS_2682-24 FALSE    GTSeek
## 1769     <NA> Hair Sample    CGA    122         GTS_2687-24 FALSE    GTSeek
## 1791     <NA> Hair Sample    CGA    123         GTS_2890-24 FALSE    GTSeek
## 1823     <NA> Hair Sample    CGA    124         GTS_2928-24 FALSE    GTSeek
## 2005     <NA> Hair Sample    CGA    122          GTS_430-24 FALSE    GTSeek
## 2059     <NA> Hair Sample    CGA    116          GTS_228-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 86       F       30        284
## 169      F       30        284
## 209      M       30        284
## 426      F       30        284
## 698      F       30        284
## 784      F       30        284
## 1014     F      284        284
## 1114     F      284        284
## 1194     F      284        284
```

```
## 1287     F      284        284
## 1298     F      300        284
## 1324     F      284        284
## 1462     F      300        284
## 1564     F      284        284
## 1582     F      284        284
## 1588     F      284        284
## 1599     F      284        284
## 1764     F      284        284
## 1765     F      284        284
## 1769     F      300        284
## 1791     F      284        284
## 1823     F      284        284
## 2005     F      300        284
## 2059     F      300        284
```

```r
unique(smd$COLclone[which(smd$finalClone == 284)])
```

```
## [1]  30 284 300
```

```r
compare.out[compare.out$colclone1 %in% smd$COLclone[which(smd$finalClone == 284)] &
              compare.out$colclone2 %in% smd$COLclone[which(smd$finalClone == 284)],]
```

```
##                          id1                    id2  kinship nsnp colclone1
## 7296    UGA_173101_P017_WB05 UGA_173101_P019_WE12 0.483062  920        30
## 18372   UGA_173101_P015_WH02 UGA_173101_P019_WE12 0.498599  882        30
## 18458   UGA_173101_P015_WH02 UGA_173101_P017_WB05 0.481894  882        30
## 38262   UGA_173101_P016_WG01 UGA_173101_P019_WE12 0.468496  621        30
## 38348   UGA_173101_P016_WG01 UGA_173101_P017_WB05 0.453252  621        30
## 38419   UGA_173101_P016_WG01 UGA_173101_P015_WH02 0.467573  605        30
## 102414  UGA_173101_P022_WD08 UGA_173101_P019_WE12 0.486607  862        30
## 102500  UGA_173101_P022_WD08 UGA_173101_P017_WB05 0.470238  862        30
## 102571  UGA_173101_P022_WD08 UGA_173101_P015_WH02 0.487082  838        30
## 102656  UGA_173101_P022_WD08 UGA_173101_P016_WG01 0.454060  592        30
## 149367  UGA_173101_P023_WC10 UGA_173101_P019_WE12 0.492466  908        30
## 149453  UGA_173101_P023_WC10 UGA_173101_P017_WB05 0.475342  908        30
## 149524  UGA_173101_P023_WC10 UGA_173101_P015_WH02 0.491549  878        30
## 149609  UGA_173101_P023_WC10 UGA_173101_P016_WG01 0.465447  618        30
## 149785  UGA_173101_P023_WC10 UGA_173101_P022_WD08 0.481399  859        30
## 319636            GTS_649-24 UGA_173101_P019_WE12 0.491379  118       284
## 319722            GTS_649-24 UGA_173101_P017_WB05 0.482759  121       284
## 319793            GTS_649-24 UGA_173101_P015_WH02 0.490385   99       284
## 319878            GTS_649-24 UGA_173101_P016_WG01 0.460000   80       284
## 320054            GTS_649-24 UGA_173101_P022_WD08 0.489130   95       284
## 320148            GTS_649-24 UGA_173101_P023_WC10 0.465517  108       284
## 389439            GTS_685-24 UGA_173101_P019_WE12 0.491379  117       284
## 389525            GTS_685-24 UGA_173101_P017_WB05 0.482759  120       284
## 389596            GTS_685-24 UGA_173101_P015_WH02 0.490385   99       284
## 389681            GTS_685-24 UGA_173101_P016_WG01 0.460000   79       284
## 389857            GTS_685-24 UGA_173101_P022_WD08 0.489130   95       284
## 389951            GTS_685-24 UGA_173101_P023_WC10 0.465517  107       284
## 390204            GTS_685-24           GTS_649-24 0.500000  120       284
## 440427            GTS_558-24 UGA_173101_P019_WE12 0.483333  120       284
```

```
## 440513             GTS_558-24 UGA_173101_P017_WB05 0.475000  123        284
## 440584             GTS_558-24 UGA_173101_P015_WH02 0.490741  100        284
## 440669             GTS_558-24 UGA_173101_P016_WG01 0.461538   81        284
## 440845             GTS_558-24 UGA_173101_P022_WD08 0.489583   96        284
## 440939             GTS_558-24 UGA_173101_P023_WC10 0.466667  109        284
## 441192             GTS_558-24           GTS_649-24 0.500000  121        284
## 441275             GTS_558-24           GTS_685-24 0.500000  120        284
## 512614             GTS_781-24 UGA_173101_P019_WE12 0.491667  119        284
## 512700             GTS_781-24 UGA_173101_P017_WB05 0.483333  122        284
## 512771             GTS_781-24 UGA_173101_P015_WH02 0.490741  100        284
## 512856             GTS_781-24 UGA_173101_P016_WG01 0.461538   81        284
## 513032             GTS_781-24 UGA_173101_P022_WD08 0.489583   96        284
## 513126             GTS_781-24 UGA_173101_P023_WC10 0.466667  109        284
## 513379             GTS_781-24           GTS_649-24 0.500000  122        284
## 513462             GTS_781-24           GTS_685-24 0.500000  120        284
## 513518             GTS_781-24           GTS_558-24 0.500000  122        284
## 522789            GTS_1208-24 UGA_173101_P019_WE12 0.491935  120        300
## 522875            GTS_1208-24 UGA_173101_P017_WB05 0.483871  123        300
## 522946            GTS_1208-24 UGA_173101_P015_WH02 0.490741  100        300
## 523031            GTS_1208-24 UGA_173101_P016_WG01 0.461538   81        300
## 523207            GTS_1208-24 UGA_173101_P022_WD08 0.489583   96        300
## 523301            GTS_1208-24 UGA_173101_P023_WC10 0.466667  109        300
## 523554            GTS_1208-24           GTS_649-24 0.500000  122        300
## 523637            GTS_1208-24           GTS_685-24 0.500000  120        300
## 523693            GTS_1208-24           GTS_558-24 0.491667  123        300
## 523767            GTS_1208-24           GTS_781-24 0.500000  123        300
## 546571            GTS_1226-24 UGA_173101_P019_WE12 0.491379  113        284
## 546657            GTS_1226-24 UGA_173101_P017_WB05 0.491379  116        284
## 546728            GTS_1226-24 UGA_173101_P015_WH02 0.490385   95        284
## 546813            GTS_1226-24 UGA_173101_P016_WG01 0.460000   78        284
## 546989            GTS_1226-24 UGA_173101_P022_WD08 0.489583   92        284
## 547083            GTS_1226-24 UGA_173101_P023_WC10 0.465517  103        284
## 547336            GTS_1226-24           GTS_649-24 0.500000  115        284
## 547419            GTS_1226-24           GTS_685-24 0.500000  114        284
## 547475            GTS_1226-24           GTS_558-24 0.500000  116        284
## 547549            GTS_1226-24           GTS_781-24 0.500000  116        284
## 547559            GTS_1226-24          GTS_1208-24 0.500000  116        284
## 661861            GTS_1379-24 UGA_173101_P019_WE12 0.427419  122        300
## 661947            GTS_1379-24 UGA_173101_P017_WB05 0.419355  125        300
## 662018            GTS_1379-24 UGA_173101_P015_WH02 0.416667  102        300
## 662103            GTS_1379-24 UGA_173101_P016_WG01 0.384615   83        300
## 662279            GTS_1379-24 UGA_173101_P022_WD08 0.406250   98        300
## 662373            GTS_1379-24 UGA_173101_P023_WC10 0.400000  111        300
## 662626            GTS_1379-24           GTS_649-24 0.500000  121        300
## 662709            GTS_1379-24           GTS_685-24 0.500000  120        300
## 662765            GTS_1379-24           GTS_558-24 0.491667  123        300
## 662839            GTS_1379-24           GTS_781-24 0.500000  122        300
## 662849            GTS_1379-24          GTS_1208-24 0.500000  123        300
## 662872            GTS_1379-24          GTS_1226-24 0.500000  116        300
## 757101            GTS_1127-24 UGA_173101_P019_WE12 0.483871  116        284
## 757187            GTS_1127-24 UGA_173101_P017_WB05 0.475806  119        284
## 757258            GTS_1127-24 UGA_173101_P015_WH02 0.490741   98        284
## 757343            GTS_1127-24 UGA_173101_P016_WG01 0.461538   80        284
## 757519            GTS_1127-24 UGA_173101_P022_WD08 0.489583   95        284
```

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## 757613           GTS_1127-24 UGA_173101_P023_WC10 0.466667  106         284
## 757866           GTS_1127-24           GTS_649-24 0.491667  119         284
## 757949           GTS_1127-24           GTS_685-24 0.491379  118         284
## 758005           GTS_1127-24           GTS_558-24 0.491667  119         284
## 758079           GTS_1127-24           GTS_781-24 0.491935  120         284
## 758089           GTS_1127-24          GTS_1208-24 0.491935  120         284
## 758112           GTS_1127-24          GTS_1226-24 0.491379  114         284
## 758217           GTS_1127-24          GTS_1379-24 0.491667  119         284
## 774426           GTS_1488-24 UGA_173101_P019_WE12 0.491667  118         284
## 774512           GTS_1488-24 UGA_173101_P017_WB05 0.483333  121         284
## 774583           GTS_1488-24 UGA_173101_P015_WH02 0.490741   99         284
## 774668           GTS_1488-24 UGA_173101_P016_WG01 0.461538   81         284
## 774844           GTS_1488-24 UGA_173101_P022_WD08 0.489583   95         284
## 774938           GTS_1488-24 UGA_173101_P023_WC10 0.466667  108         284
## 775191           GTS_1488-24           GTS_649-24 0.500000  121         284
## 775274           GTS_1488-24           GTS_685-24 0.500000  119         284
## 775330           GTS_1488-24           GTS_558-24 0.500000  121         284
## 775404           GTS_1488-24           GTS_781-24 0.500000  122         284
## 775414           GTS_1488-24          GTS_1208-24 0.500000  122         284
## 775437           GTS_1488-24          GTS_1226-24 0.500000  116         284
## 775542           GTS_1488-24          GTS_1379-24 0.500000  121         284
## 775622           GTS_1488-24          GTS_1127-24 0.491935  120         284
## 781911           GTS_1500-24 UGA_173101_P019_WE12 0.491667  119         284
## 781997           GTS_1500-24 UGA_173101_P017_WB05 0.483333  122         284
## 782068           GTS_1500-24 UGA_173101_P015_WH02 0.490741  100         284
## 782153           GTS_1500-24 UGA_173101_P016_WG01 0.461538   81         284
## 782329           GTS_1500-24 UGA_173101_P022_WD08 0.489583   96         284
## 782423           GTS_1500-24 UGA_173101_P023_WC10 0.466667  109         284
## 782676           GTS_1500-24           GTS_649-24 0.500000  122         284
## 782759           GTS_1500-24           GTS_685-24 0.500000  120         284
## 782815           GTS_1500-24           GTS_558-24 0.500000  122         284
## 782889           GTS_1500-24           GTS_781-24 0.500000  123         284
## 782899           GTS_1500-24          GTS_1208-24 0.500000  123         284
## 782922           GTS_1500-24          GTS_1226-24 0.500000  116         284
## 783027           GTS_1500-24          GTS_1379-24 0.500000  122         284
## 783107           GTS_1500-24          GTS_1127-24 0.491935  120         284
## 783121           GTS_1500-24          GTS_1488-24 0.500000  122         284
## 794466           GTS_1528-24 UGA_173101_P019_WE12 0.491379  118         284
## 794552           GTS_1528-24 UGA_173101_P017_WB05 0.482759  121         284
## 794623           GTS_1528-24 UGA_173101_P015_WH02 0.490385   99         284
## 794708           GTS_1528-24 UGA_173101_P016_WG01 0.460000   80         284
## 794884           GTS_1528-24 UGA_173101_P022_WD08 0.489130   95         284
## 794978           GTS_1528-24 UGA_173101_P023_WC10 0.465517  108         284
## 795231           GTS_1528-24           GTS_649-24 0.500000  122         284
## 795314           GTS_1528-24           GTS_685-24 0.500000  120         284
## 795370           GTS_1528-24           GTS_558-24 0.500000  121         284
## 795444           GTS_1528-24           GTS_781-24 0.500000  122         284
## 795454           GTS_1528-24          GTS_1208-24 0.500000  122         284
## 795477           GTS_1528-24          GTS_1226-24 0.500000  115         284
## 795582           GTS_1528-24          GTS_1379-24 0.500000  121         284
## 795662           GTS_1528-24          GTS_1127-24 0.491667  119         284
## 795676           GTS_1528-24          GTS_1488-24 0.500000  121         284
## 795682           GTS_1528-24          GTS_1500-24 0.500000  122         284
## 968172           GTS_2680-24 UGA_173101_P019_WE12 0.483333  120         284
```

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## 968258           GTS_2680-24 UGA_173101_P017_WB05 0.475000  123        284
## 968329           GTS_2680-24 UGA_173101_P015_WH02 0.490741  100        284
## 968414           GTS_2680-24 UGA_173101_P016_WG01 0.461538   81        284
## 968590           GTS_2680-24 UGA_173101_P022_WD08 0.489583   96        284
## 968684           GTS_2680-24 UGA_173101_P023_WC10 0.466667  109        284
## 968937           GTS_2680-24           GTS_649-24 0.500000  122        284
## 969020           GTS_2680-24           GTS_685-24 0.500000  120        284
## 969076           GTS_2680-24           GTS_558-24 0.500000  123        284
## 969150           GTS_2680-24           GTS_781-24 0.500000  123        284
## 969160           GTS_2680-24          GTS_1208-24 0.491935  124        284
## 969183           GTS_2680-24          GTS_1226-24 0.500000  116        284
## 969288           GTS_2680-24          GTS_1379-24 0.491667  123        284
## 969368           GTS_2680-24          GTS_1127-24 0.491935  120        284
## 969382           GTS_2680-24          GTS_1488-24 0.500000  122        284
## 969388           GTS_2680-24          GTS_1500-24 0.500000  123        284
## 969398           GTS_2680-24          GTS_1528-24 0.500000  122        284
## 969564           GTS_2682-24 UGA_173101_P019_WE12 0.483333  120        284
## 969650           GTS_2682-24 UGA_173101_P017_WB05 0.475000  123        284
## 969721           GTS_2682-24 UGA_173101_P015_WH02 0.490741  100        284
## 969806           GTS_2682-24 UGA_173101_P016_WG01 0.461538   81        284
## 969982           GTS_2682-24 UGA_173101_P022_WD08 0.489583   96        284
## 970076           GTS_2682-24 UGA_173101_P023_WC10 0.466667  109        284
## 970329           GTS_2682-24           GTS_649-24 0.500000  122        284
## 970412           GTS_2682-24           GTS_685-24 0.500000  120        284
## 970468           GTS_2682-24           GTS_558-24 0.500000  123        284
## 970542           GTS_2682-24           GTS_781-24 0.500000  123        284
## 970552           GTS_2682-24          GTS_1208-24 0.491935  124        284
## 970575           GTS_2682-24          GTS_1226-24 0.500000  116        284
## 970680           GTS_2682-24          GTS_1379-24 0.491667  123        284
## 970760           GTS_2682-24          GTS_1127-24 0.491935  120        284
## 970774           GTS_2682-24          GTS_1488-24 0.500000  122        284
## 970780           GTS_2682-24          GTS_1500-24 0.500000  123        284
## 970790           GTS_2682-24          GTS_1528-24 0.500000  122        284
## 970921           GTS_2682-24          GTS_2680-24 0.500000  124        284
## 975142           GTS_2687-24 UGA_173101_P019_WE12 0.482759  118        300
## 975228           GTS_2687-24 UGA_173101_P017_WB05 0.474138  121        300
## 975299           GTS_2687-24 UGA_173101_P015_WH02 0.480769   99        300
## 975384           GTS_2687-24 UGA_173101_P016_WG01 0.450000   80        300
## 975560           GTS_2687-24 UGA_173101_P022_WD08 0.478261   95        300
## 975654           GTS_2687-24 UGA_173101_P023_WC10 0.456897  108        300
## 975907           GTS_2687-24           GTS_649-24 0.500000  121        300
## 975990           GTS_2687-24           GTS_685-24 0.500000  119        300
## 976046           GTS_2687-24           GTS_558-24 0.491379  121        300
## 976120           GTS_2687-24           GTS_781-24 0.491667  122        300
## 976130           GTS_2687-24          GTS_1208-24 0.491667  122        300
## 976153           GTS_2687-24          GTS_1226-24 0.491071  115        300
## 976258           GTS_2687-24          GTS_1379-24 0.491379  121        300
## 976338           GTS_2687-24          GTS_1127-24 0.483333  119        300
## 976352           GTS_2687-24          GTS_1488-24 0.491667  121        300
## 976358           GTS_2687-24          GTS_1500-24 0.491667  122        300
## 976368           GTS_2687-24          GTS_1528-24 0.500000  121        300
## 976499           GTS_2687-24          GTS_2680-24 0.491667  122        300
## 976500           GTS_2687-24          GTS_2682-24 0.491667  122        300
## 997614           GTS_2890-24 UGA_173101_P019_WE12 0.483333  120        284
```

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## 997700            GTS_2890-24 UGA_173101_P017_WB05 0.475000  123         284
## 997771            GTS_2890-24 UGA_173101_P015_WH02 0.490741  100         284
## 997856            GTS_2890-24 UGA_173101_P016_WG01 0.461538   81         284
## 998032            GTS_2890-24 UGA_173101_P022_WD08 0.489583   96         284
## 998126            GTS_2890-24 UGA_173101_P023_WC10 0.466667  109         284
## 998379            GTS_2890-24           GTS_649-24 0.500000  121         284
## 998462            GTS_2890-24           GTS_685-24 0.500000  120         284
## 998518            GTS_2890-24           GTS_558-24 0.500000  123         284
## 998592            GTS_2890-24           GTS_781-24 0.500000  122         284
## 998602            GTS_2890-24          GTS_1208-24 0.491667  123         284
## 998625            GTS_2890-24          GTS_1226-24 0.500000  116         284
## 998730            GTS_2890-24          GTS_1379-24 0.491667  123         284
## 998810            GTS_2890-24          GTS_1127-24 0.491667  119         284
## 998824            GTS_2890-24          GTS_1488-24 0.500000  121         284
## 998830            GTS_2890-24          GTS_1500-24 0.500000  122         284
## 998840            GTS_2890-24          GTS_1528-24 0.500000  121         284
## 998971            GTS_2890-24          GTS_2680-24 0.500000  123         284
## 998972            GTS_2890-24          GTS_2682-24 0.500000  123         284
## 998976            GTS_2890-24          GTS_2687-24 0.491379  121         284
## 1033239           GTS_2928-24 UGA_173101_P019_WE12 0.483333  120         284
## 1033325           GTS_2928-24 UGA_173101_P017_WB05 0.475000  123         284
## 1033396           GTS_2928-24 UGA_173101_P015_WH02 0.490741  100         284
## 1033481           GTS_2928-24 UGA_173101_P016_WG01 0.461538   81         284
## 1033657           GTS_2928-24 UGA_173101_P022_WD08 0.489583   96         284
## 1033751           GTS_2928-24 UGA_173101_P023_WC10 0.466667  109         284
## 1034004           GTS_2928-24           GTS_649-24 0.500000  122         284
## 1034087           GTS_2928-24           GTS_685-24 0.500000  120         284
## 1034143           GTS_2928-24           GTS_558-24 0.500000  123         284
## 1034217           GTS_2928-24           GTS_781-24 0.500000  123         284
## 1034227           GTS_2928-24          GTS_1208-24 0.491935  124         284
## 1034250           GTS_2928-24          GTS_1226-24 0.500000  116         284
## 1034355           GTS_2928-24          GTS_1379-24 0.491667  123         284
## 1034435           GTS_2928-24          GTS_1127-24 0.491935  120         284
## 1034449           GTS_2928-24          GTS_1488-24 0.500000  122         284
## 1034455           GTS_2928-24          GTS_1500-24 0.500000  123         284
## 1034465           GTS_2928-24          GTS_1528-24 0.500000  122         284
## 1034596           GTS_2928-24          GTS_2680-24 0.500000  124         284
## 1034597           GTS_2928-24          GTS_2682-24 0.500000  124         284
## 1034601           GTS_2928-24          GTS_2687-24 0.491667  122         284
## 1034617           GTS_2928-24          GTS_2890-24 0.500000  123         284
## 1186606            GTS_430-24 UGA_173101_P019_WE12 0.491379  118         300
## 1186692            GTS_430-24 UGA_173101_P017_WB05 0.482759  121         300
## 1186763            GTS_430-24 UGA_173101_P015_WH02 0.490000   97         300
## 1186848            GTS_430-24 UGA_173101_P016_WG01 0.460000   80         300
## 1187024            GTS_430-24 UGA_173101_P022_WD08 0.488636   93         300
## 1187118            GTS_430-24 UGA_173101_P023_WC10 0.464286  106         300
## 1187371            GTS_430-24           GTS_649-24 0.500000  120         300
## 1187454            GTS_430-24           GTS_685-24 0.500000  118         300
## 1187510            GTS_430-24           GTS_558-24 0.491071  120         300
## 1187584            GTS_430-24           GTS_781-24 0.500000  120         300
## 1187594            GTS_430-24          GTS_1208-24 0.500000  121         300
## 1187617            GTS_430-24          GTS_1226-24 0.500000  114         300
## 1187722            GTS_430-24          GTS_1379-24 0.500000  120         300
## 1187802            GTS_430-24          GTS_1127-24 0.491379  118         300
```

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## 1187816           GTS_430-24          GTS_1488-24 0.500000  120        300
## 1187822           GTS_430-24          GTS_1500-24 0.500000  120        300
## 1187832           GTS_430-24          GTS_1528-24 0.500000  120        300
## 1187963           GTS_430-24          GTS_2680-24 0.491379  121        300
## 1187964           GTS_430-24          GTS_2682-24 0.491379  121        300
## 1187968           GTS_430-24          GTS_2687-24 0.500000  119        300
## 1187984           GTS_430-24          GTS_2890-24 0.491071  120        300
## 1188009           GTS_430-24          GTS_2928-24 0.491379  121        300
## 1231701           GTS_228-24 UGA_173101_P019_WE12 0.490741  113        300
## 1231787           GTS_228-24 UGA_173101_P017_WB05 0.481481  116        300
## 1231858           GTS_228-24 UGA_173101_P015_WH02 0.489130   92        300
## 1231943           GTS_228-24 UGA_173101_P016_WG01 0.454545   76        300
## 1232119           GTS_228-24 UGA_173101_P022_WD08 0.487500   88        300
## 1232213           GTS_228-24 UGA_173101_P023_WC10 0.461538  101        300
## 1232466           GTS_228-24           GTS_649-24 0.500000  114        300
## 1232549           GTS_228-24           GTS_685-24 0.500000  113        300
## 1232605           GTS_228-24           GTS_558-24 0.490385  115        300
## 1232679           GTS_228-24           GTS_781-24 0.500000  114        300
## 1232689           GTS_228-24          GTS_1208-24 0.500000  115        300
## 1232712           GTS_228-24          GTS_1226-24 0.500000  109        300
## 1232817           GTS_228-24          GTS_1379-24 0.500000  115        300
## 1232897           GTS_228-24          GTS_1127-24 0.490385  112        300
## 1232911           GTS_228-24          GTS_1488-24 0.500000  114        300
## 1232917           GTS_228-24          GTS_1500-24 0.500000  114        300
## 1232927           GTS_228-24          GTS_1528-24 0.500000  114        300
## 1233058           GTS_228-24          GTS_2680-24 0.490385  115        300
## 1233059           GTS_228-24          GTS_2682-24 0.490385  115        300
## 1233063           GTS_228-24          GTS_2687-24 0.500000  113        300
## 1233079           GTS_228-24          GTS_2890-24 0.490385  115        300
## 1233104           GTS_228-24          GTS_2928-24 0.490385  115        300
## 1233207           GTS_228-24           GTS_430-24 0.500000  115        300
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## 18458          30         TRUE       TRUE
## 38262          30         TRUE       TRUE
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## 38419          30         TRUE       TRUE
## 102414         30         TRUE       TRUE
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## 149524         30         TRUE       TRUE
## 149609         30         TRUE       TRUE
## 149785         30         TRUE       TRUE
## 319636         30         TRUE      FALSE
## 319722         30         TRUE      FALSE
## 319793         30         TRUE      FALSE
## 319878         30         TRUE      FALSE
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## 320148         30         TRUE      FALSE
## 389439         30         TRUE      FALSE
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## 389951         30          TRUE       FALSE
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## 512771         30          TRUE       FALSE
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## 513032         30          TRUE       FALSE
## 513126         30          TRUE       FALSE
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## 513518        284          TRUE        TRUE
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## 522875         30          TRUE       FALSE
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## 523031         30          TRUE       FALSE
## 523207         30          TRUE       FALSE
## 523301         30          TRUE       FALSE
## 523554        284          TRUE       FALSE
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## 523693        284          TRUE       FALSE
## 523767        284          TRUE       FALSE
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## 546657         30          TRUE       FALSE
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## 546989         30          TRUE       FALSE
## 547083         30          TRUE       FALSE
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## 547475        284          TRUE        TRUE
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## 547559        300          TRUE       FALSE
## 661861         30          TRUE       FALSE
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## 662018         30          TRUE       FALSE
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## 662373         30          TRUE       FALSE
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## 662709        284          TRUE       FALSE
## 662765        284          TRUE       FALSE
## 662839        284          TRUE       FALSE
```

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## 662872          284          TRUE       FALSE
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## 757343           30          TRUE       FALSE
## 757519           30          TRUE       FALSE
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## 758079          284          TRUE        TRUE
## 758089          300          TRUE       FALSE
## 758112          284          TRUE        TRUE
## 758217          300          TRUE       FALSE
## 774426           30          TRUE       FALSE
## 774512           30          TRUE       FALSE
## 774583           30          TRUE       FALSE
## 774668           30          TRUE       FALSE
## 774844           30          TRUE       FALSE
## 774938           30          TRUE       FALSE
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## 775274          284          TRUE        TRUE
## 775330          284          TRUE        TRUE
## 775404          284          TRUE        TRUE
## 775414          300          TRUE       FALSE
## 775437          284          TRUE        TRUE
## 775542          300          TRUE       FALSE
## 775622          284          TRUE        TRUE
## 781911           30          TRUE       FALSE
## 781997           30          TRUE       FALSE
## 782068           30          TRUE       FALSE
## 782153           30          TRUE       FALSE
## 782329           30          TRUE       FALSE
## 782423           30          TRUE       FALSE
## 782676          284          TRUE        TRUE
## 782759          284          TRUE        TRUE
## 782815          284          TRUE        TRUE
## 782889          284          TRUE        TRUE
## 782899          300          TRUE       FALSE
## 782922          284          TRUE        TRUE
## 783027          300          TRUE       FALSE
## 783107          284          TRUE        TRUE
## 783121          284          TRUE        TRUE
## 794466           30          TRUE       FALSE
## 794552           30          TRUE       FALSE
## 794623           30          TRUE       FALSE
## 794708           30          TRUE       FALSE
## 794884           30          TRUE       FALSE
## 794978           30          TRUE       FALSE
## 795231          284          TRUE        TRUE
## 795314          284          TRUE        TRUE
## 795370          284          TRUE        TRUE
## 795444          284          TRUE        TRUE
```

```
## 795454           300            TRUE      FALSE
## 795477           284            TRUE       TRUE
## 795582           300            TRUE      FALSE
## 795662           284            TRUE       TRUE
## 795676           284            TRUE       TRUE
## 795682           284            TRUE       TRUE
## 968172            30            TRUE      FALSE
## 968258            30            TRUE      FALSE
## 968329            30            TRUE      FALSE
## 968414            30            TRUE      FALSE
## 968590            30            TRUE      FALSE
## 968684            30            TRUE      FALSE
## 968937           284            TRUE       TRUE
## 969020           284            TRUE       TRUE
## 969076           284            TRUE       TRUE
## 969150           284            TRUE       TRUE
## 969160           300            TRUE      FALSE
## 969183           284            TRUE       TRUE
## 969288           300            TRUE      FALSE
## 969368           284            TRUE       TRUE
## 969382           284            TRUE       TRUE
## 969388           284            TRUE       TRUE
## 969398           284            TRUE       TRUE
## 969564            30            TRUE      FALSE
## 969650            30            TRUE      FALSE
## 969721            30            TRUE      FALSE
## 969806            30            TRUE      FALSE
## 969982            30            TRUE      FALSE
## 970076            30            TRUE      FALSE
## 970329           284            TRUE       TRUE
## 970412           284            TRUE       TRUE
## 970468           284            TRUE       TRUE
## 970542           284            TRUE       TRUE
## 970552           300            TRUE      FALSE
## 970575           284            TRUE       TRUE
## 970680           300            TRUE      FALSE
## 970760           284            TRUE       TRUE
## 970774           284            TRUE       TRUE
## 970780           284            TRUE       TRUE
## 970790           284            TRUE       TRUE
## 970921           284            TRUE       TRUE
## 975142            30            TRUE      FALSE
## 975228            30            TRUE      FALSE
## 975299            30            TRUE      FALSE
## 975384            30            TRUE      FALSE
## 975560            30            TRUE      FALSE
## 975654            30            TRUE      FALSE
## 975907           284            TRUE      FALSE
## 975990           284            TRUE      FALSE
## 976046           284            TRUE      FALSE
## 976120           284            TRUE      FALSE
## 976130           300            TRUE       TRUE
## 976153           284            TRUE      FALSE
## 976258           300            TRUE       TRUE
```

```
## 976338        284          TRUE       FALSE
## 976352        284          TRUE       FALSE
## 976358        284          TRUE       FALSE
## 976368        284          TRUE       FALSE
## 976499        284          TRUE       FALSE
## 976500        284          TRUE       FALSE
## 997614         30          TRUE       FALSE
## 997700         30          TRUE       FALSE
## 997771         30          TRUE       FALSE
## 997856         30          TRUE       FALSE
## 998032         30          TRUE       FALSE
## 998126         30          TRUE       FALSE
## 998379        284          TRUE        TRUE
## 998462        284          TRUE        TRUE
## 998518        284          TRUE        TRUE
## 998592        284          TRUE        TRUE
## 998602        300          TRUE       FALSE
## 998625        284          TRUE        TRUE
## 998730        300          TRUE       FALSE
## 998810        284          TRUE        TRUE
## 998824        284          TRUE        TRUE
## 998830        284          TRUE        TRUE
## 998840        284          TRUE        TRUE
## 998971        284          TRUE        TRUE
## 998972        284          TRUE        TRUE
## 998976        300          TRUE       FALSE
## 1033239        30          TRUE       FALSE
## 1033325        30          TRUE       FALSE
## 1033396        30          TRUE       FALSE
## 1033481        30          TRUE       FALSE
## 1033657        30          TRUE       FALSE
## 1033751        30          TRUE       FALSE
## 1034004       284          TRUE        TRUE
## 1034087       284          TRUE        TRUE
## 1034143       284          TRUE        TRUE
## 1034217       284          TRUE        TRUE
## 1034227       300          TRUE       FALSE
## 1034250       284          TRUE        TRUE
## 1034355       300          TRUE       FALSE
## 1034435       284          TRUE        TRUE
## 1034449       284          TRUE        TRUE
## 1034455       284          TRUE        TRUE
## 1034465       284          TRUE        TRUE
## 1034596       284          TRUE        TRUE
## 1034597       284          TRUE        TRUE
## 1034601       300          TRUE       FALSE
## 1034617       284          TRUE        TRUE
## 1186606        30          TRUE       FALSE
## 1186692        30          TRUE       FALSE
## 1186763        30          TRUE       FALSE
## 1186848        30          TRUE       FALSE
## 1187024        30          TRUE       FALSE
## 1187118        30          TRUE       FALSE
## 1187371       284          TRUE       FALSE
```

```
## 1187454        284          TRUE       FALSE
## 1187510        284          TRUE       FALSE
## 1187584        284          TRUE       FALSE
## 1187594        300          TRUE        TRUE
## 1187617        284          TRUE       FALSE
## 1187722        300          TRUE        TRUE
## 1187802        284          TRUE       FALSE
## 1187816        284          TRUE       FALSE
## 1187822        284          TRUE       FALSE
## 1187832        284          TRUE       FALSE
## 1187963        284          TRUE       FALSE
## 1187964        284          TRUE       FALSE
## 1187968        300          TRUE        TRUE
## 1187984        284          TRUE       FALSE
## 1188009        284          TRUE       FALSE
## 1231701         30          TRUE       FALSE
## 1231787         30          TRUE       FALSE
## 1231858         30          TRUE       FALSE
## 1231943         30          TRUE       FALSE
## 1232119         30          TRUE       FALSE
## 1232213         30          TRUE       FALSE
## 1232466        284          TRUE       FALSE
## 1232549        284          TRUE       FALSE
## 1232605        284          TRUE       FALSE
## 1232679        284          TRUE       FALSE
## 1232689        300          TRUE        TRUE
## 1232712        284          TRUE       FALSE
## 1232817        300          TRUE        TRUE
## 1232897        284          TRUE       FALSE
## 1232911        284          TRUE       FALSE
## 1232917        284          TRUE       FALSE
## 1232927        284          TRUE       FALSE
## 1233058        284          TRUE       FALSE
## 1233059        284          TRUE       FALSE
## 1233063        300          TRUE        TRUE
## 1233079        284          TRUE       FALSE
## 1233104        284          TRUE       FALSE
## 1233207        300          TRUE        TRUE
```

There's little ambiguity in any of these 5 cases.

Region

Hopefully no cross-region recaptures!

```
nRegion2
```

```
##   [1]  1 NA  2  1  1  1  1  1 NA NA  1 NA NA  1  1 NA  1  1  1  1  1  1 NA  1 NA
##  [26] NA NA NA  1 NA NA NA  1  1  1 NA  1  1 NA  1  1  1  2  1  1 NA  1  1  1  1
##  [51]  1  1  1  1 NA  1  1  1  1 NA  1 NA  1 NA  1  1  1  1  1  1  1 NA  1  1 NA
##  [76]  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1 NA  1  1 NA  1  1  1 NA
## [101]  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1  1  1  1  1
## [126]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1
```

```
## [151]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1 NA  1
## [176]  1  1 NA  1  1  1  1  1  1 NA  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1
## [201]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1 NA  1  1  1  1  1  1  1 NA
## [226]  1  1 NA  1 NA  1  1  1  1  1  1  1 NA  1 NA  1 NA  2 NA NA  1  1  1  1 NA
## [251]  1  1 NA  1 NA  1  1 NA NA  1 NA  1 NA NA  1  1  1  1 NA  1 NA  1  1  1 NA
## [276] NA  1  1  1  1 NA NA  1  1  1  1  1  1 NA  1 NA  1  1  1  1  1  1  1  1 NA
## [301] NA  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [326]  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
## [351]  1  1  1  1  1  1
```

```r
which(nRegion2 > 1)
```

```
## [1]   3  43 243
```

```r
smd$region[which(smd$finalClone == 3)]
```

```
##  [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA-H" "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [19] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
```

```r
#Not really 2 regions, just historic sample
smd$region[which(smd$finalClone == 43)]
```

```
##  [1] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA-H"
```

```r
#Not really 2 regions, just historic sample
smd$region[which(smd$finalClone == 243)]
```

```
##  [1] "CGA-H" "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"   "CGA"
## [10] "CGA"
```

```r
#Not really 2 regions, just historic sample
```

Known vs. Genotype-inferred Sex

Finally (again), comparing the known sex of unique genotypes (from direct captures) with that inferred for all samples within a clone…

```r
#check for discrepancies between known and genotyped sex for all clones
discrepancy <- multknown <- c()
#for(i in 1:max(sampleMetadata$cloneID, na.rm = TRUE)) {
for(i in 1:max(smd$finalClone, na.rm=TRUE)) {
  thisclone <- which(smd$finalClone == i)
  if(length(thisclone) > 0) {
    tmp <- smd[thisclone,]
    tmp.knownSEX <- unique(tmp$knownSEX[tmp$knownSEX != "" & !is.na(tmp$knownSEX)])
    if(length(tmp.knownSEX) > 0) {
      if(length(tmp.knownSEX) > 1) {
        multknown[i] <- TRUE
```

```r
        discrepancy[i] <- NA
      } else {
        multknown[i] <- FALSE
        if(sum(tmp$sexGT[tmp$sexGT != "undet"] != tmp.knownSEX) == 0) {
          discrepancy[i] <- FALSE
        } else {
          discrepancy[i] <- TRUE
        }
      }
    }
    rm(tmp, tmp.knownSEX)
  }
  rm(thisclone)
}

sum(discrepancy, na.rm = TRUE) #Still only 3 cases with discrepancies...
```

```
## [1] 3
```

```r
sum(multknown, na.rm = TRUE) #Still zero cases with multiple "known" sexes
```

```
## [1] 0
```

```r
which(discrepancy == TRUE)
```

```
## [1]  45 140 284
```

```r
#CloneIndex 45 - spotted above, direct capture agrees with majority - F
smd[which(smd$finalClone == 45),]
```

```
##                                       extID   ugaID is.SCR is.rep week year
## 49                     UGA_173101_P015_WE01    32-C  FALSE  FALSE <NA> 2024
## 492                    UGA_173101_P020_WC06   57-23   TRUE  FALSE    7 2023
## 629                    UGA_173101_P021_WF11  774-24   TRUE  FALSE    2 2024
## 1048  GTseek_024_384i5_222_P098.004_664-24  664-24   TRUE  FALSE    3 2024
## 1598 GTseek_026_384i5_004_P098.011_3057-24 3057-24   TRUE  FALSE    3 2024
## 1819 GTseek_026_384i5_371_P098.010_2920-24 2920-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 49     <NA>            <NA>
## 492  HS 085 264337x3598727
## 629  HS 084 267164x3597536
## 1048 HS 083 266526x3596862
## 1598 HS 084 267164x3597536
## 1819 HS 083 266526x3596862
##                                                                        fqfile
## 49   RAPiD-Genomics_F534_UGA_173101_P015_WE01_i5-502_i7-48_S14787_L003_R1_001.fastq
## 492    RAPiD-Genomics_F566_UGA_173101_P020_WC06_i5-505_i7-3_S3920_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 1048                                                                         <NA>
## 1598                                                                         <NA>
## 1819                                                                         <NA>
```

```
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 49          F      Capture    CGA    892 UGA_173101_P015_WE01 FALSE       LGC
## 492             Hair Snare    CGA    915 UGA_173101_P020_WC06 FALSE       LGC
## 629             Hair Snare    CGA    913 UGA_173101_P021_WF11 FALSE       LGC
## 1048     <NA> Hair Sample    CGA    120           GTS_664-24 FALSE    GTSeek
## 1598     <NA> Hair Sample    CGA    123          GTS_3057-24 FALSE    GTSeek
## 1819     <NA> Hair Sample    CGA    124          GTS_2920-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 49       F       45         45
## 492      F       45         45
## 629      M       45         45
## 1048     F       45         45
## 1598     F       45         45
## 1819     F       45         45
```

```
#CloneIndex 140 - spotted previously, direct captured cub
smd[which(smd$finalClone == 140),]
```

```
##                    extID ugaID is.SCR is.rep week year snare utmSnare
## 71 UGA_173101_P015_WF11 63-DC  FALSE  FALSE <NA> 2024  <NA>     <NA>
##                                                                       fqfile
## 71 RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
##    knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice sexGT
## 71        M  Den Check    CGA    913 UGA_173101_P015_WF11 FALSE       LGC     F
##    COLclone finalClone
## 71      140        140
```

```
#CloneIndex 284 - spotted above, direct capture agrees with majority - F
smd[which(smd$finalClone == 284),]
```

```
##                                     extID   ugaID is.SCR is.rep week year
## 86                   UGA_173101_P015_WH02    45-C  FALSE  FALSE <NA> 2024
## 169                  UGA_173101_P016_WG01  448-24   TRUE  FALSE    5 2024
## 209                  UGA_173101_P017_WB05  497-24   TRUE  FALSE    2 2024
## 426                  UGA_173101_P019_WE12  154-24   TRUE  FALSE    3 2024
## 698                  UGA_173101_P022_WD08  341-24   TRUE  FALSE    3 2024
## 784                  UGA_173101_P023_WC10  205-23   TRUE  FALSE    7 2023
## 1014  GTseek_024_384i5_188_P098.004_649-24  649-24   TRUE  FALSE    2 2024
## 1114  GTseek_024_384i5_288_P098.004_685-24  685-24   TRUE  FALSE    2 2024
## 1194  GTseek_024_384i5_368_P098.003_558-24  558-24   TRUE  FALSE    2 2024
## 1287  GTseek_025_384i5_077_P098.005_781-24  781-24   TRUE  FALSE    2 2024
## 1298 GTseek_025_384i5_088_P098.008_1208-24 1208-24   TRUE  FALSE    2 2024
## 1324 GTseek_025_384i5_114_P098.008_1226-24 1226-24   TRUE  FALSE    2 2024
## 1462 GTseek_025_384i5_252_P098.008_1379-24 1379-24   TRUE  FALSE    3 2024
## 1564 GTseek_025_384i5_354_P098.007_1127-24 1127-24   TRUE  FALSE    2 2024
## 1582 GTseek_025_384i5_372_P098.008_1488-24 1488-24   TRUE  FALSE    5 2024
## 1588 GTseek_025_384i5_378_P098.008_1500-24 1500-24   TRUE  FALSE    3 2024
## 1599 GTseek_026_384i5_005_P098.009_1528-24 1528-24   TRUE  FALSE    3 2024
## 1764 GTseek_026_384i5_261_P098.009_2680-24 2680-24   TRUE  FALSE    3 2024
## 1765 GTseek_026_384i5_263_P098.009_2682-24 2682-24   TRUE  FALSE    4 2024
## 1769 GTseek_026_384i5_271_P098.009_2687-24 2687-24   TRUE  FALSE    6 2024
## 1791 GTseek_026_384i5_315_P098.010_2890-24 2890-24   TRUE  FALSE    2 2024
## 1823 GTseek_026_384i5_379_P098.010_2928-24 2928-24   TRUE  FALSE    6 2024
```

```
## 2005 GTseek_028_384i5_180_P098.UGA4_430-24  430-24   TRUE  FALSE    2 2024
## 2059 GTseek_028_384i5_234_P098.UGA3_228-24  228-24   TRUE  FALSE    2 2024
##       snare       utmSnare
## 86     <NA>           <NA>
## 169  HS 051 260705x3599596
## 209  HS 048 261267x3598930
## 426  HS 048 261267x3598930
## 698  HS 048 261267x3598930
## 784  HS 051 260705x3599596
## 1014 HS 051 260705x3599596
## 1114 HS 048 261267x3598930
## 1194 HS 051 260705x3599596
## 1287 HS 048 261267x3598930
## 1298 HS 051 260705x3599596
## 1324 HS 048 261267x3598930
## 1462 HS 048 261267x3598930
## 1564 HS 051 260705x3599596
## 1582 HS 051 260705x3599596
## 1588 HS 048 261267x3598930
## 1599 HS 048 261267x3598930
## 1764 HS 048 261267x3598930
## 1765 HS 051 260705x3599596
## 1769 HS 048 261267x3598930
## 1791 HS 051 260705x3599596
## 1823 HS 048 261267x3598930
## 2005 HS 051 260705x3599596
## 2059 HS 051 260705x3599596
##                                                                               fqfile
## 86    RAPiD-Genomics_F534_UGA_173101_P015_WH02_i5-502_i7-32_S14824_L003_R1_001.fastq
## 169  RAPiD-Genomics_F534_UGA_173101_P016_WG01_i5-504_i7-169_S14906_L003_R1_001.fastq
## 209  RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 426  RAPiD-Genomics_F534_UGA_173101_P019_WE12_i5-1102_i7-76_S15165_L003_R1_001.fastq
## 698   RAPiD-Genomics_F566_UGA_173101_P022_WD08_i5-1098_i7-80_S7626_L003_R1_001.fastq
## 784   RAPiD-Genomics_F566_UGA_173101_P023_WC10_i5-1099_i7-47_S7712_L003_R1_001.fastq
## 1014                                                                            <NA>
## 1114                                                                            <NA>
## 1194                                                                            <NA>
## 1287                                                                            <NA>
## 1298                                                                            <NA>
## 1324                                                                            <NA>
## 1462                                                                            <NA>
## 1564                                                                            <NA>
## 1582                                                                            <NA>
## 1588                                                                            <NA>
## 1599                                                                            <NA>
## 1764                                                                            <NA>
## 1765                                                                            <NA>
## 1769                                                                            <NA>
## 1791                                                                            <NA>
## 1823                                                                            <NA>
## 2005                                                                            <NA>
## 2059                                                                            <NA>
##      knownSEX  sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 86          F     Capture    CGA    882 UGA_173101_P015_WH02 FALSE       LGC
```

```
## 169           Hair Snare    CGA    621 UGA_173101_P016_WG01 FALSE       LGC
## 209           Hair Snare    CGA    923 UGA_173101_P017_WB05 FALSE       LGC
## 426           Hair Snare    CGA    920 UGA_173101_P019_WE12 FALSE       LGC
## 698           Hair Snare    CGA    862 UGA_173101_P022_WD08 FALSE       LGC
## 784           Hair Snare    CGA    908 UGA_173101_P023_WC10 FALSE       LGC
## 1014     <NA> Hair Sample   CGA    122           GTS_649-24 FALSE    GTSeek
## 1114     <NA> Hair Sample   CGA    120           GTS_685-24 FALSE    GTSeek
## 1194     <NA> Hair Sample   CGA    123           GTS_558-24 FALSE    GTSeek
## 1287     <NA> Hair Sample   CGA    123           GTS_781-24 FALSE    GTSeek
## 1298     <NA> Hair Sample   CGA    124          GTS_1208-24 FALSE    GTSeek
## 1324     <NA> Hair Sample   CGA    116          GTS_1226-24 FALSE    GTSeek
## 1462     <NA> Hair Sample   CGA    125          GTS_1379-24 FALSE    GTSeek
## 1564     <NA> Hair Sample   CGA    120          GTS_1127-24 FALSE    GTSeek
## 1582     <NA> Hair Sample   CGA    122          GTS_1488-24 FALSE    GTSeek
## 1588     <NA> Hair Sample   CGA    123          GTS_1500-24 FALSE    GTSeek
## 1599     <NA> Hair Sample   CGA    122          GTS_1528-24 FALSE    GTSeek
## 1764     <NA> Hair Sample   CGA    124          GTS_2680-24 FALSE    GTSeek
## 1765     <NA> Hair Sample   CGA    124          GTS_2682-24 FALSE    GTSeek
## 1769     <NA> Hair Sample   CGA    122          GTS_2687-24 FALSE    GTSeek
## 1791     <NA> Hair Sample   CGA    123          GTS_2890-24 FALSE    GTSeek
## 1823     <NA> Hair Sample   CGA    124          GTS_2928-24 FALSE    GTSeek
## 2005     <NA> Hair Sample   CGA    122           GTS_430-24 FALSE    GTSeek
## 2059     <NA> Hair Sample   CGA    116           GTS_228-24 FALSE    GTSeek
##      sexGT COLclone finalClone
## 86       F       30        284
## 169      F       30        284
## 209      M       30        284
## 426      F       30        284
## 698      F       30        284
## 784      F       30        284
## 1014     F      284        284
## 1114     F      284        284
## 1194     F      284        284
## 1287     F      284        284
## 1298     F      300        284
## 1324     F      284        284
## 1462     F      300        284
## 1564     F      284        284
## 1582     F      284        284
## 1588     F      284        284
## 1599     F      284        284
## 1764     F      284        284
## 1765     F      284        284
## 1769     F      300        284
## 1791     F      284        284
## 1823     F      284        284
## 2005     F      300        284
## 2059     F      300        284
```

Assign Final Sex

Using the combined information across samples within a clone, assign sex to each unique genotype.

```r
smd$finalSEX <- rep(NA, nrow(smd))
smd$knownSEX[is.na(smd$knownSEX)] <- ""
for(i in 1:nrow(smd)) {
  tmp <- smd[which(smd$finalClone == smd$finalClone[i]),]
  if(length(tmp$knownSEX[tmp$knownSEX != ""]) > 0) {
    smd$finalSEX[i] <- unique(tmp$knownSEX[tmp$knownSEX != ""])
  } else if(length(tmp$sexGT[tmp$sexGT != "undet"] > 0)) {
    smd$finalSEX[i] <- names(which.max(table(
      tmp$sexGT[tmp$sexGT != "undet"])))
  }
  rm(tmp)
}

table(smd$finalSEX)
```

```
##
##   F   M
## 852 793
```

```r
#Is the final called sex ever different from known sex? Shouldn't be
smd[which(smd$finalSEX != smd$knownSEX & smd$knownSEX != ""),]
```

```
##  [1] extID      ugaID      is.SCR     is.rep     week       year
##  [7] snare      utmSnare   fqfile     knownSEX   sampSOURCE region
## [13] nSNPgt     vcfID      FILT       GTservice  sexGT      COLclone
## [19] finalClone finalSEX
## <0 rows> (or 0-length row.names)
```

```r
#Is the final called sex ever different from genotyped sex - should be a few
# 63-DC: clone 154, den check cub, called a female by GT, M in the field
# 497-24: clone 334, capture in the matches, a F
# 212-23 & 271-23: clone 14, no captures, but majority F
# 774-24: clone 47, capture in the matches, F
# 5-23: clone 8, majority M no captures in matches
# 655-23: clone 307, majority M no captures in matches
smd[which(smd$finalSEX != smd$sexGT & smd$sexGT != "undet"),]
```

```
##                                    extID  ugaID is.SCR is.rep week year  snare
## 71                  UGA_173101_P015_WF11  63-DC  FALSE  FALSE <NA> 2024   <NA>
## 209                 UGA_173101_P017_WB05 497-24   TRUE  FALSE    2 2024 HS 048
## 549                 UGA_173101_P020_WH03 212-23   TRUE  FALSE    6 2023 HS 027
## 550                 UGA_173101_P020_WH04 271-23   TRUE  FALSE    6 2023 HS 025
## 629                 UGA_173101_P021_WF11 774-24   TRUE  FALSE    2 2024 HS 084
## 828   GTseek_024_384i5_001_P098.001_5-23   5-23   TRUE  FALSE    6 2023 HS 012
## 962 GTseek_024_384i5_135_P098.001_655-23 655-23   TRUE  FALSE    6 2023 HS 012
##           utmSnare
## 71            <NA>
## 209 261267x3598930
## 549 264464x3592495
## 550 265276x3591787
## 629 267164x3597536
## 828 258412x3586648
```

```
## 962 258412x3586648
##                                                                           fqfile
## 71   RAPiD-Genomics_F534_UGA_173101_P015_WF11_i5-502_i7-51_S14809_L003_R1_001.fastq
## 209 RAPiD-Genomics_F534_UGA_173101_P017_WB05_i5-512_i7-113_S14946_L003_R1_001.fastq
## 549   RAPiD-Genomics_F566_UGA_173101_P020_WH03_i5-505_i7-88_S3977_L001_R1_001.fastq
## 550   RAPiD-Genomics_F566_UGA_173101_P020_WH04_i5-505_i7-73_S3978_L001_R1_001.fastq
## 629     RAPiD-Genomics_F566_UGA_173101_P021_WF11_i5-501_i7-51_S71_L003_R1_001.fastq
## 828                                                                             <NA>
## 962                                                                             <NA>
##     knownSEX  sampSOURCE region nSNPgt               vcfID  FILT GTservice
## 71         M   Den Check    CGA    913 UGA_173101_P015_WF11 FALSE       LGC
## 209           Hair Snare    CGA    923 UGA_173101_P017_WB05 FALSE       LGC
## 549           Hair Snare    CGA    918 UGA_173101_P020_WH03 FALSE       LGC
## 550           Hair Snare    CGA    916 UGA_173101_P020_WH04 FALSE       LGC
## 629           Hair Snare    CGA    913 UGA_173101_P021_WF11 FALSE       LGC
## 828          Hair Sample    CGA    104             GTS_5-23 FALSE    GTSeek
## 962          Hair Sample    CGA    116           GTS_655-23 FALSE    GTSeek
##     sexGT COLclone finalClone finalSEX
## 71      F      140        140        M
## 209     M       30        284        F
## 549     M       14         14        F
## 550     M       14         14        F
## 629     M       45         45        F
## 828     F      238          8        M
## 962     F      255         33        M
```

Compare kinship coefficients

As one final check of the reliability of our assignments, we'll look at the distribution of KING kinship coefficients for samples from the same vs. different individual. We'll do this in two phases, first for technical replicates (accounting for inferred recaptures among replicated hair samples) and then for all SCR samples.

Technical replicates

First for only the technically replicated samples. Note that it's possible that we have recaptures represented in these, so we'll want to identify unique individuals from this set (as inferred above) before plotting the distribution within / between technical replicates.

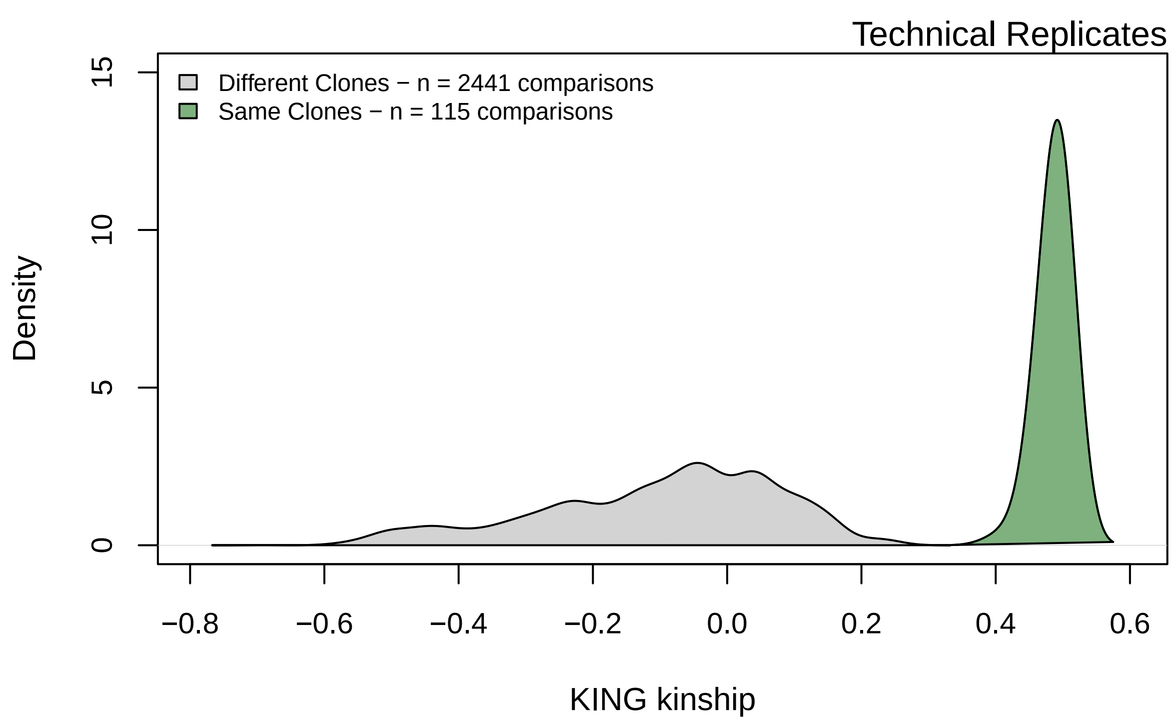
```
#Create an object for within vs. between clone comparisons - technical replicates
king.allREP <- kingrel[
  which(kingrel$id1 %in% smd$vcfID[which(smd$is.SCR == TRUE &
                                          smd$is.rep == TRUE)] &
          kingrel$id2 %in% smd$vcfID[which(smd$is.SCR == TRUE &
                                            smd$is.rep == TRUE)]),]

king.allREP$clone1 <- king.allREP$clone2 <- NA
for(i in 1:nrow(king.allREP)) {
  king.allREP$clone1[i] <- smd$finalClone[which(smd$vcfID == king.allREP$id1[i])]
  king.allREP$clone2[i] <- smd$finalClone[which(smd$vcfID == king.allREP$id2[i])]
}

#Same Clone
range(king.allREP$kinship[which(king.allREP$clone1 == king.allREP$clone2)])
```

```
## [1] 0.395833 0.500000
```

```r
#Different Clone
range(king.allREP$kinship[which(king.allREP$clone1 != king.allREP$clone2)])
```

```
## [1] -0.692308  0.257812
```

```r
#Plot the distributions
library(scales)
same.clone <- density(king.allREP$kinship[which(king.allREP$clone1 == king.allREP$clone2)], bw = 0.025)
diff.clone <- density(king.allREP$kinship[which(king.allREP$clone1 != king.allREP$clone2)], bw = 0.025)
plot(diff.clone, xlim = c(min(king.allREP$kinship)-0.1, max(king.allREP$kinship)+0.1),
     ylim = c(0, 15), lwd = 1, col = "darkgrey", cex.axis = 0.9,
     xlab = "KING kinship",  ylab = "Density", main = "")
polygon(diff.clone, col = alpha("darkgrey", 0.5))
lines(same.clone, col = "darkgreen")
polygon(same.clone, col = alpha("darkgreen", 0.5))
legend("topleft", bty = "n", cex = 0.75,
       fill = c(alpha("darkgrey", 0.5), alpha("darkgreen", 0.5)),
       legend = c(paste("Different Clones - n =",
                        length(which(king.allREP$clone1 != king.allREP$clone2)),
                        "comparisons"),
                  paste("Same Clones - n =",
                        length(which(king.allREP$clone1 == king.allREP$clone2)),
                        "comparisons")))
mtext("Technical Replicates", cex = 1.1, adj = 1)
```



All SCR samples

Now for all SCR samples. Note the *very* large number of pairwise comparisons represented here.

```
## All SCR
#Create an object for all SCR samples (that were assigned to a finalClone)
keepVCFlist <- smd$vcfID[which(smd$is.SCR == TRUE &
                                !is.na(smd$finalClone) &
                                !is.na(smd$finalSEX))]
kingSCR <- kingrel[which(kingrel$id1 %in% keepVCFlist & kingrel$id2 %in% keepVCFlist),]
kingSCR$clone1 <- kingSCR$clone2 <- NA
for(i in 1:nrow(kingSCR)) {
  kingSCR$clone1[i] <- smd$finalClone[which(smd$vcfID == kingSCR$id1[i])]
  kingSCR$clone2[i] <- smd$finalClone[which(smd$vcfID == kingSCR$id2[i])]
}

#Same Clone
range(kingSCR$kinship[which(kingSCR$clone1 == kingSCR$clone2)])
```

```
## [1] 0.242268 0.500000
```

```
sum(kingSCR$kinship[which(kingSCR$clone1 == kingSCR$clone2)] < 0.45)
```

```
## [1] 1875
```

```
#Different Clone
range(kingSCR$kinship[which(kingSCR$clone1 != kingSCR$clone2)])
```

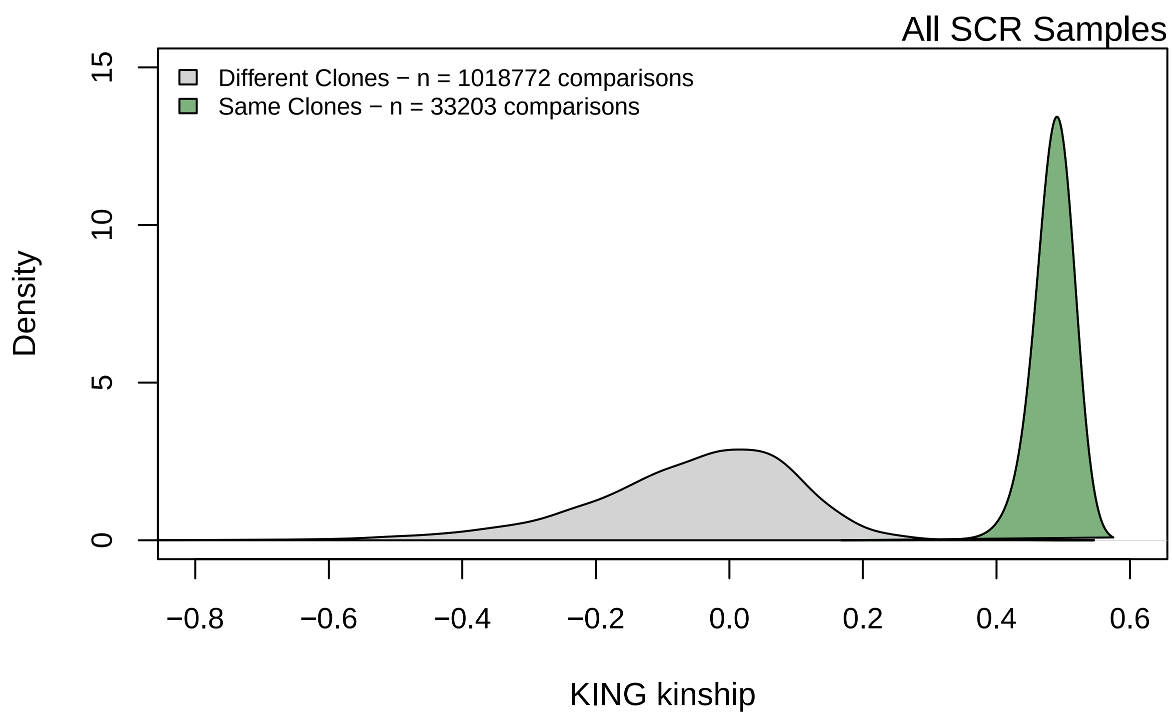
```
## [1] -1.714290  0.471154
```

```
kingSCR$nsnp[which(kingSCR$kinship >= 0.45 & kingSCR$clone1 != kingSCR$clone2)]
```

```
##   [1] 760 762 760 763 761 761 679 756 761 762 763 757 755 762 723 761 751 752
##  [19]  73  91  59  85  85  74  89  88  93  86  89  86  88  89  73  74  64  68
##  [37]  89  73  68  86  89  87  88  74  86  88  74  67  73  59  85  89  89  88
##  [55]  89  89  88  88  88  90  68  73  73  88  89  88  74  89  67  87  91  88
##  [73]  66  88  87  73  88  73  89  73  87  89  74  89  87  89  74  74  74  73
##  [91]  74  87  89  66  89  87  67  66  68  68  57  68  84  91  64
```

```
same.cloneALL <- density(kingSCR$kinship[which(kingSCR$clone1 == kingSCR$clone2)], bw = 0.025)
diff.cloneALL <- density(kingSCR$kinship[which(kingSCR$clone1 != kingSCR$clone2)], bw = 0.025)
plot(diff.cloneALL, xlim = c(-0.8, 0.6),
     ylim = c(0, 15), lwd = 1, col = "darkgrey", cex.axis=0.9,
     xlab = "KING kinship",  ylab = "Density", main = "")
polygon(diff.cloneALL, col = alpha("darkgrey", 0.5))
lines(same.cloneALL, col = "darkgreen")
polygon(same.cloneALL, col = alpha("darkgreen", 0.5))
legend("topleft", bty = "n", cex = 0.75,
       fill = c(alpha("darkgrey", 0.5), alpha("darkgreen", 0.5)),
       legend = c(paste("Different Clones - n =",
                        length(which(kingSCR$clone1 != kingSCR$clone2)),
```

```
                    "comparisons"),
              paste("Same Clones - n =",
                    length(which(kingSCR$clone1 == kingSCR$clone2)),
                    "comparisons")))
mtext("All SCR Samples", cex = 1.1, adj = 1)
```



Excessively high H_o

As one final filter, we also want to check that none of the individuals identified in our sample have excessively high observed heterozygosity. This could be indicative of contamination of the sample. Below, we identify the sample with the least missing data for each individual in the dataset, extract this genotype, and calculate F_{IS} for all individuals in the CGA population. Any individuals with $F_{IS} < -0.5$ will be removed from the dataset.

First, identify the best sample for each individual in CGA.

```
CGAclones <- unique(smd$finalClone[
  which(smd$region %in% c("CGA","CGA-H") &
          !is.na(smd$finalClone))])

keepersCGA <- c()
for(i in 1:length(CGAclones)) {
  tmp <- smd[which(smd$finalClone == CGAclones[i]),]
  keep.me <- which(tmp$nSNPgt == max(tmp$nSNPgt))
  if(length(keep.me) > 1) {
```

```r
    keep.me <- keep.me[1]
  }
  keepersCGA[i] <- tmp$vcfID[keep.me]
  rm(tmp, keep.me)
}

write.table(keepersCGA, "keeperClonesCGA.txt",
            quote = FALSE,row.names = FALSE,col.names = FALSE)
```

Now calculate F_{IS} for these individuals…

```
vcftools --vcf ../snpfiltering/finalSCR.recode.vcf \
         --keep keeperClonesCGA.txt --out CGA_Fcheck --het
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf ../snpfiltering/finalSCR.recode.vcf
    --keep keeperClonesCGA.txt
    --het
    --out CGA_Fcheck

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Keeping individuals in 'keep' list
After filtering, kept 233 out of 1653 Individuals
Outputting Individual Heterozygosity
After filtering, kept 925 out of a possible 925 Sites
Run Time = 0.00 seconds
```

Do we need to remove any for $F_{IS} < -0.5$?

```r
CGA_F <- read.table("CGA_Fcheck.het", header = TRUE)
range(CGA_F$F)
```

```
## [1] -0.83446  0.52540
```

```r
CGA_F[which(CGA_F$F < -0.5),]
```

```
##                      INDV O.HOM. E.HOM. N_SITES        F
## 3   UGA_173101_P020_WB12    297  533.8     915 -0.62100
## 34  UGA_173101_P017_WG11    313  537.2     919 -0.58717
## 65  UGA_173101_P016_WC04    271  464.6     801 -0.57539
## 86  UGA_173101_P016_WH11    285  530.9     910 -0.64864
## 122 UGA_173101_P021_WB05    217  536.3     919 -0.83446
## 128 UGA_173101_P020_WH06    327  535.3     917 -0.54558
## 138 UGA_173101_P020_WE12    320  532.8     914 -0.55836
## 142 UGA_173101_P022_WC06    313  534.3     916 -0.57983
## 147 UGA_173101_P023_WE01    256  430.1     746 -0.55095
## 178             GTS_F3-25     71   87.6     120 -0.51149
## 203             GTS_F9-25     69   90.1     124 -0.62198
```

```r
drop.us <- CGA_F$INDV[which(CGA_F$F < -0.5)]

drop.clones <- smd$finalClone[which(smd$vcfID %in% drop.us)]

table(smd$finalClone[which(smd$finalClone %in% drop.clones)])
```

```
##
##   6  67 132 159 195 200 206 209 215 305 332
##   1   1   1   1   1   1   1   1   1   2   1
```

```r
smd[which(smd$finalClone %in% drop.clones),]
```

```
##                                 extID  ugaID is.SCR is.rep week year   snare
## 124                  UGA_173101_P016_WC04 169-23   TRUE  FALSE    6 2023 HS 025
## 191                  UGA_173101_P016_WH11 213-23   TRUE  FALSE    6 2023 HS 066
## 275                  UGA_173101_P017_WG11 457-23   TRUE  FALSE    6 2023 HS 066
## 486                  UGA_173101_P020_WB12 110-23   TRUE  FALSE    5 2023 HS 066
## 522                  UGA_173101_P020_WE12 340-23   TRUE  FALSE    5 2023 HS 027
## 552                  UGA_173101_P020_WH06 496-23   TRUE  FALSE    6 2023 HS 027
## 575                  UGA_173101_P021_WB05 281-23   TRUE  FALSE    4 2023 HS 051
## 684                  UGA_173101_P022_WC06 203-23   TRUE  FALSE    4 2023 HS 068
## 799                  UGA_173101_P023_WE01 257-23   TRUE  FALSE    6 2023 HS 030
## 1369 GTseek_025_384i5_159_P098.006_F3-25  F3-25  FALSE  FALSE <NA> 2025    <NA>
## 1387 GTseek_025_384i5_177_P098.006_F4-25  F4-25  FALSE  FALSE <NA> 2025    <NA>
## 1676 GTseek_026_384i5_106_P098.011_F9-25  F9-25  FALSE  FALSE  N/A 2025    <NA>
##            utmSnare
## 124  265276x3591787
## 191  268064x3590555
## 275  268064x3590555
## 486  268064x3590555
## 522  264464x3592495
## 552  264464x3592495
## 575  260705x3599596
## 684  268186x3591620
## 799  260871x3593893
## 1369           <NA>
## 1387           <NA>
```

```
## 1676          <NA>
##                                                                                 fqfile
## 124  RAPiD-Genomics_F534_UGA_173101_P016_WC04_i5-504_i7-124_S14862_L003_R1_001.fastq
## 191  RAPiD-Genomics_F534_UGA_173101_P016_WH11_i5-504_i7-191_S14928_L003_R1_001.fastq
## 275  RAPiD-Genomics_F534_UGA_173101_P017_WG11_i5-512_i7-179_S15012_L003_R1_001.fastq
## 486    RAPiD-Genomics_F566_UGA_173101_P020_WB12_i5-505_i7-75_S3914_L001_R1_001.fastq
## 522    RAPiD-Genomics_F566_UGA_173101_P020_WE12_i5-505_i7-76_S3950_L001_R1_001.fastq
## 552    RAPiD-Genomics_F566_UGA_173101_P020_WH06_i5-505_i7-49_S3980_L001_R1_001.fastq
## 575      RAPiD-Genomics_F566_UGA_173101_P021_WB05_i5-501_i7-90_S17_L003_R1_001.fastq
## 684    RAPiD-Genomics_F566_UGA_173101_P022_WC06_i5-1098_i7-3_S7612_L003_R1_001.fastq
## 799   RAPiD-Genomics_F566_UGA_173101_P023_WE01_i5-1099_i7-48_S7727_L003_R1_001.fastq
## 1369                                                                              <NA>
## 1387                                                                              <NA>
## 1676                                                                              <NA>
##      knownSEX sampSOURCE region nSNPgt                vcfID  FILT GTservice
## 124           Hair Snare    CGA    802 UGA_173101_P016_WC04 FALSE       LGC
## 191           Hair Snare    CGA    911 UGA_173101_P016_WH11 FALSE       LGC
## 275           Hair Snare    CGA    920 UGA_173101_P017_WG11 FALSE       LGC
## 486           Hair Snare    CGA    917 UGA_173101_P020_WB12 FALSE       LGC
## 522           Hair Snare    CGA    914 UGA_173101_P020_WE12 FALSE       LGC
## 552           Hair Snare    CGA    917 UGA_173101_P020_WH06 FALSE       LGC
## 575           Hair Snare    CGA    919 UGA_173101_P021_WB05 FALSE       LGC
## 684           Hair Snare    CGA    916 UGA_173101_P022_WC06 FALSE       LGC
## 799           Hair Snare    CGA    746 UGA_173101_P023_WE01 FALSE       LGC
## 1369              Foster    CGA    122             GTS_F3-25 FALSE    GTSeek
## 1387              Foster    CGA    122             GTS_F4-25 FALSE    GTSeek
## 1676              Foster    CGA    126             GTS_F9-25 FALSE    GTSeek
##      sexGT COLclone finalClone finalSEX
## 124  undet      132        132     <NA>
## 191      M      159        159        M
## 275      M       67         67        M
## 486      M        6          6        M
## 522      F      206        206        F
## 552      M      200        200        M
## 575      M      195        195        M
## 684      F      209        209        F
## 799      M      215        215        M
## 1369     M      305        305        M
## 1387     M      305        305        M
## 1676     F      332        332        F
```

```
smd$FILT[which(smd$finalClone %in% drop.clones)] <- TRUE
smd$finalClone[which(smd$finalClone %in% drop.clones)] <- NA
```

Final Summary

How many total genotypes do we see in the dataset? How many from each region? How many male and female genotypes in our SCR datasets?

```
#Loop through all clones and record clone ID, size of clone, final sex, etc.
sumtab <- data.frame(cloneID=unique(smd$finalClone[!is.na(smd$finalClone)]),
                     size=NA, finalSEX=NA, region=NA)
for(i in 1:nrow(sumtab)) {
```

```r
  sumtab$size[i] <- length(which(smd$finalClone == sumtab$cloneID[i]))
  sumtab$finalSEX[i] <- smd$finalSEX[which(smd$finalClone == sumtab$cloneID[i])[1]]
  sumtab$region[i] <- smd$region[which(smd$finalClone == sumtab$cloneID[i])[1]]
}

#Total unique genotypes
nrow(sumtab)
```

```
## [1] 284
```

```r
#By sex
table(sumtab$finalSEX)
```

```
##
##   F   M
## 141 140
```

```r
#By region
table(sumtab$region)
```

```
##
##   CGA CGA-H   NGA   SGA
##   208    14    35    25
```

```r
#CGA hair samples from this study only
hair.smd <- smd[which(smd$is.SCR == TRUE & smd$region == "CGA" & smd$year > 2020),]
#Total hair samples
nrow(hair.smd)
```

```
## [1] 1958
```

```r
#Total hair samples successfully assigned to a clone / not filtered
nrow(hair.smd[which(hair.smd$FILT == FALSE),])
```

```
## [1] 1446
```

```r
nrow(hair.smd[!is.na(hair.smd$finalClone),])
```

```
## [1] 1446
```

```r
nrow(hair.smd[which(!is.na(hair.smd$finalClone) & hair.smd$FILT == FALSE),])
```

```
## [1] 1446
```

```r
#Total unique genotypes from hair samples
length(unique(hair.smd$finalClone[!is.na(hair.smd$finalClone)]))
```

```
## [1] 150
```

```r
#Total unique female genotypes from hair samples
length(unique(hair.smd$finalClone[which(hair.smd$finalSEX == "F" &
                                          !is.na(hair.smd$finalClone))]))
```

```
## [1] 81
```

```r
#Total male genotypes from hair samples
length(unique(hair.smd$finalClone[which(hair.smd$finalSEX == "M" &
                                          !is.na(hair.smd$finalClone))]))
```

```
## [1] 66
```

Save the output!

Write the sampleMetaData file out to a csv for use in building the capture history!

```r
write.csv(smd, "../sampleinfo/sampleMetaData_cloneID.csv", quote = FALSE, row.names = FALSE)
```